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. Then, we propose an RLRS framework with four components, i.e., state representation, policy optimization,
 reward formulation, and environment building, and survey RLRS algorithms accordingly. We highlight emergin
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elated algorithms, which for us turns into a recommendations problem as well. We explain the motivations be
hind and review the approach that we use to improve the recommendation algorithms, combining A/B testing fo
cused on improving member retention and medium term engagement, as well as offline experimentation using hi
storical member engagement data. We discuss some of the issues in designing and interpreting A/B tests. Fin
ally, we describe some current areas of focused innovation, which include making our recommender system glo
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ge\"}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Recom
mendation systems are the most important components in the world of technology today, especially Social Med
ia, Netflix, Spotify and wherever there is a need to show users what to interact with and reduce their deci
sion-making efforts. Today’s recommendation systems have overgrown from simple filtering systems to showing
 and recommending users as to what they want and what they will like to interact with so that they don’t or
 to the very least search for their preferred content. Some examples of platforms that use recommendation s
ystems are: Netflix, Amazon, Spotify, YouTube and many more. The purpose of the recommendation system has a
lso overgrown from just recommending users personalized content to retaining them on the platform for as lo
ng as possible and keep them engaged with the platform so that they use the platform more and more. While p
latforms like Netflix and Spotify use recommendation systems to retain the user into their platform and kee
p them engaged keep them busy, social media on the other hand use recommendation systems to show users cont
ent for maximum likes comments shares Etc. (Afsar et al., 2023; Gomez-Uribe \u0026 Hunt, 2016; Raza et al.,
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n\"} There is something called “paradox of choice”, which says that for user to choose from options it is b
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To what extent and in which ways recommender systems create\\n   business value\\n   is, however, much less clear, and the literature on the topic is scattered. In this research commentary, we review existing publications on field tests of recommender systems and report which business-related performance measures were used in such real-world deployments. We summarize common challenges of measuring the business value in practice and critically discuss the value of algorithmic improvements and offline experiments as commonly done in academic environments. Overall, our review indicates that various open questions remain both regarding the realistic quantification of the business effects of recommenders and the performance assessment of recommendation algorithms in academia.\",\"container-title\":\"ACM Transactions on Management Information Systems\",\"DOI\":\"10.1145/3370082\",\"ISSN\":\"2158-656X, 2158-6578\",\"issue\":\"4\",\"journalAbbreviation\":\"ACM Trans. Manage. Inf. 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Because the more user engages with the platform, the more engagement with the sponsors or the advertisements or the affiliates will happen and that will help the platform to generate even more money. So for example, in platforms like Instagram and Facebook if the user happens to spend time more than previous day then these platforms have a business model of showing ads and sponsored content. So it is more likely that the user will click on some advertisement or some sponsored content which will lead to platform making more money.. We can quote another example of recommendation system in Amazon which is called frequently bought together. This feature recommends related items which user is interested in. so if user buys an item X and the user is being shown item a b and c.It is likely that user may be interested in at least one of the a b and c and may end up buying any of the recommended items which the user did not intend or did not think of buying previously. so that results in the sellers making more money and Amazon making more money. (Hostler et al., 2011, pp. 336–343; Jannach \u0026 Jugovac, 2019; R. Tang et al., 2025, pp. 282–309)",
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\"itemData\":{\"id\":110,\"type\":\"article-journal\",\"abstract\":\"Recommender Systems are nowadays succe
ssfully used by all major web sites—from e-commerce to social media—to filter content and make suggestions
in a personalized way. Academic research largely focuses on the value of recommenders for consumers, e.g.,
in terms of reduced information overload. To what extent and in which ways recommender systems create\\n
business value\\n is, however, much less clear, and the literature on the topic is

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scattered. In this research commentary, we review existing publications on field tests of recommender syste
ms and report which business-related performance measures were used in such real-world deployments. We summ
arize common challenges of measuring the business value in practice and critically discuss the value of alg
orithmic improvements and offline experiments as commonly done in academic environments. Overall, our revie
w indicates that various open questions remain both regarding the realistic quantification of the business 
effects of recommenders and the performance assessment of recommendation algorithms in academia.\",\"contai
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henzhen Finance Institute, School of Management and Economics, The Chinese University of Hong Kong, Shenzhe
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}},\"locator\":\"282-309\",\"label\":\"page\"}],\"schema\":\"https://github.com/citation-style-language/sch
ema/raw/master/csl-citation.json\"} The second purpose of recommendation system is to generate revenue for 
the platform. Because the more user engages with the platform, the more engagement with the sponsors or the
 advertisements or the affiliates will happen and that will help the platform to generate even more money. 
So for example,  in platforms like Instagram and Facebook if the user happens to spend time more than previ
ous day then these platforms have a business model of showing ads and sponsored content. So it is more like
ly that the user will click on some advertisement or some sponsored content which will lead to platform mak
ing more money.. We can quote another example of recommendation system in Amazon which is called frequently
 bought together. This feature recommends related items which user is interested in. so if user buys an ite
m X  and the user is being shown item a b and c.It is likely that user may be interested in at least one of
 the a b and c and may end up buying any of the recommended items which the user did not intend or did not 
think of buying previously.  so that results in the sellers making more money and Amazon making more money.
 (Hostler et al., 2011, pp. 336–343; Jannach \u0026 Jugovac, 2019; R. Tang et al., 2025, pp. 282–309)",
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mp; Economics, Nanjing 210023, China",
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ms/YKS945QC\"],\"itemData\":{\"id\":127,\"type\":\"article\",\"abstract\":\"We investigate the application 
of classification techniques to utility elicitation. In a decision problem, two sets of parameters must gen
erally be elicited: the probabilities and the utilities. While the prior and conditional probabilities in t
he model do not change from user to user, the utility models do. Thus it is necessary to elicit a utility m
odel separately for each new user. Elicitation is long and tedious, particularly if the outcome space is la
rge and not decomposable. There are two common approaches to utility function elicitation. The first is to 
base the determination of the users utility function solely ON elicitation OF qualitative preferences.The s
econd makes assumptions about the form AND decomposability OF the utility function.Here we take a different
 approach: we attempt TO identify the new USERs utility function based on classification relative to a data
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base of previously collected utility functions. We do this by identifying clusters of utility functions tha
t minimize an appropriate distance measure. Having identified the clusters, we develop a classification sch
eme that requires many fewer and simpler assessments than full utility elicitation and is more robust than
utility elicitation based solely on preferences. We have tested our algorithm on a small database of utilit
y functions in a prenatal diagnosis domain and the results are quite promising.\",\"DOI\":\"10.48550/arXiv.
1301.7367\",\"note\":\"arXiv:1301.7367 [cs]\",\"number\":\"arXiv:1301.7367\",\"publisher\":\"arXiv\",\"sour
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cle\",\"abstract\":\"Generalized linear models with nonlinear feature transformations are widely used for l
arge-scale regression and classification problems with sparse inputs. Memorization of feature interactions
through a wide set of cross-product feature transformations are effective and interpretable, while generali
zation requires more feature engineering effort. With less feature engineering, deep neural networks can ge
neralize better to unseen feature combinations through low-dimensional dense embeddings learned for the spa
rse features. However, deep neural networks with embeddings can over-generalize and recommend less relevant
 items when the user-item interactions are sparse and high-rank. In this paper, we present Wide \u0026 Deep
 learning---jointly trained wide linear models and deep neural networks---to combine the benefits of memori
zation and generalization for recommender systems. We productionized and evaluated the system on Google Pla
y, a commercial mobile app store with over one billion active users and over one million apps. Online exper
iment results show that Wide \u0026 Deep significantly increased app acquisitions compared with wide-only a
nd deep-only models. We have also open-sourced our implementation in TensorFlow.\",\"DOI\":\"10.48550/arXiv
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"abstract\":\"Low-rank matrix approximations, such as the truncated singular value decomposition and the ra
nk-revealing QR decomposition, play a central role in data analysis and scientific computing. This work sur
veys and extends recent research which demonstrates that randomization offers a powerful tool for performin
g low-rank matrix approximation. These techniques exploit modern computational architectures more fully tha
n classical methods and open the possibility of dealing with truly massive data sets. This paper presents a
 modular framework for constructing randomized algorithms that compute partial matrix decompositions. These
 methods use random sampling to identify a subspace that captures most of the action of a matrix. The input
 matrix is then compressed---either explicitly or implicitly---to this subspace, and the reduced matrix is
manipulated deterministically to obtain the desired low-rank factorization. In many cases, this approach be
ats its classical competitors in terms of accuracy, speed, and robustness. These claims are supported by ex
tensive numerical experiments and a detailed error analysis.\",\"DOI\":\"10.48550/arXiv.0909.4061\",\"note\
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",\"title\":\"Finding structure with randomness: Probabilistic algorithms for constructing approximate matr
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09.4061\",\"author\":[{\"family\":\"Halko\",\"given\":\"Nathan\"},{\"family\":\"Martinsson\",\"given\":\"Pe
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wYg/items/HBUKBJ59\"],\"itemData\":{\"id\":120,\"type\":\"article\",\"abstract\":\"In recent years, deep ne
ural networks have yielded immense success on speech recognition, computer vision and natural language proc
essing. However, the exploration of deep neural networks on recommender systems has received relatively les
s scrutiny. In this work, we strive to develop techniques based on neural networks to tackle the key proble
m in recommendation -- collaborative filtering -- on the basis of implicit feedback. Although some recent w
ork has employed deep learning for recommendation, they primarily used it to model auxiliary information, s
uch as textual descriptions of items and acoustic features of musics. When it comes to model the key factor
 in collaborative filtering -- the interaction between user and item features, they still resorted to matri
x factorization and applied an inner product on the latent features of users and items. By replacing the in
ner product with a neural architecture that can learn an arbitrary function from data, we present a general
 framework named NCF, short for Neural network-based Collaborative Filtering. NCF is generic and can expres
s and generalize matrix factorization under its framework. To supercharge NCF modelling with non-linearitie
s, we propose to leverage a multi-layer perceptron to learn the user-item interaction function. Extensive e
xperiments on two real-world datasets show significant improvements of our proposed NCF framework over the
state-of-the-art methods. Empirical evidence shows that using deeper layers of neural networks offers bette
r recommendation performance.\",\"DOI\":\"10.48550/arXiv.1708.05031\",\"note\":\"arXiv:1708.05031 [cs]\",\"
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pe\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e

.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a centr
al server (e.g. service provider), while keeping the training data decentralized. FL embodies the principle
s of focused data collection and minimization, and can mitigate many of the systemic privacy risks and cost
s resulting from traditional, centralized machine learning and data science approaches. Motivated by the ex
plosive growth in FL research, this paper discusses recent advances and presents an extensive collection of
 open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",\"note\":\"arXiv:1912.04977 [cs]\",\
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Applying large pre-trained Vision-Language Models to recommendation is a burgeoning field, a direction we t
erm Vision-Language-Recommendation (VLR). Bringing VLR to user-oriented on-device intelligence within a fed
erated learning framework is a crucial step for enhancing user privacy and delivering personalized experien
ces. This paper introduces FedVLR, a federated VLR framework specially designed for user-specific personali
zed fusion of vision-language representations. At its core is a novel bi-level fusion mechanism: The server
-side multi-view fusion module first generates a diverse set of pre-fused multimodal views. Subsequently, e
ach client employs a user-specific mixture-of-expert mechanism to adaptively integrate these views based on
 individual user interaction history. This designed lightweight personalized fusion module provides an effi
cient solution to implement a federated VLR system. The effectiveness of our proposed FedVLR has been valid
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\":131,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/N28PRJ4U\"],\"itemData\":{\"id\":131,\"type
\":\"article\",\"abstract\":\"Most data for evaluating and training recommender systems is subject to selec
tion biases, either through self-selection by the users or through the actions of the recommendation system
 itself. In this paper, we provide a principled approach to handling selection biases, adapting models and
estimation techniques from causal inference. The approach leads to unbiased performance estimators despite
biased data, and to a matrix factorization method that provides substantially improved prediction performan
ce on real-world data. We theoretically and empirically characterize the robustness of the approach, findin
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bstract\":\"With the ever-growing volume of online information, recommender systems have been an effective
strategy to overcome such information overload. The utility of recommender systems cannot be overstated, gi
ven its widespread adoption in many web applications, along with its potential impact to ameliorate many pr
oblems related to over-choice. In recent years, deep learning has garnered considerable interest in many re
search fields such as computer vision and natural language processing, owing not only to stellar performanc
e but also the attractive property of learning feature representations from scratch. The influence of deep
learning is also pervasive, recently demonstrating its effectiveness when applied to information retrieval
and recommender systems research. Evidently, the field of deep learning in recommender system is flourishin
g. This article aims to provide a comprehensive review of recent research efforts on deep learning based re

commender systems. More concretely, we provide and devise a taxonomy of deep learning based recommendation
models, along with providing a comprehensive summary of the state-of-the-art. Finally, we expand on current
 trends and provide new perspectives pertaining to this new exciting development of the field.\",\"containe
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,\"title-short\":\"Deep Learning based Recommender System\",\"URL\":\"http://arxiv.org/abs/1707.07435\",\"v
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"},{\"family\":\"Sun\",\"given\":\"Aixin\"},{\"family\":\"Tay\",\"given\":\"Yi\"}],\"accessed\":{\"date-par
ts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\",1,31]]}}}],\"schema\":\"https://github.com/ci
tation-style-language/schema/raw/master/csl-citation.json\"} Let’s talk a little bit about how the recommen
dation system actually works. The conventional way is that the users data the data which is for example use
rs likes, user comments, users favourite things, users preferred bought items ACC. These data has to be tak
en to a central server where the server processes this data and recommends the user based upon the data tha
t the user generates. The user generates a lot of data depending upon the time and interaction. Even though
 from user standpoint it is just a simple like or it is just a simple comment which does not do anything to
 the user but it is a very useful data to the platform that the user is interacting with. So this all data
has to be taken to central server in order to recommend more things and more content and more items to the
user. This is how most of the company’s work and most of the platforms work. If we name, if we start to nam
e, let’s say Facebook or Instagram, they all collect data and take them to their central servers, which are
 far away from the place where user is using it or at least far away from the users own device. Tender cent
ral servers or central platforms, they process this data, they process all this users likes and comments an
d shares etc, and they applied algorithm to the data to generate more recommendations that is then brought
back to the users device. The user then sees his recommendations and most likely interacts with the recomme
ndations. So for example in the frequently bought items of Amazon user may buy one of the recommended items
 and the and the bought item is another data which then goes to Amazon central servers and then it adds to
the already users data that is present in the central server and the algorithm adapts to the new data that
has come and the cycle goes on. The new data then generates a new recommendation content and the cycle goes
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ural networks have yielded immense success on speech recognition, computer vision and natural language proc
essing. However, the exploration of deep neural networks on recommender systems has received relatively les
s scrutiny. In this work, we strive to develop techniques based on neural networks to tackle the key proble
m in recommendation -- collaborative filtering -- on the basis of implicit feedback. Although some recent w
ork has employed deep learning for recommendation, they primarily used it to model auxiliary information, s
uch as textual descriptions of items and acoustic features of musics. When it comes to model the key factor
 in collaborative filtering -- the interaction between user and item features, they still resorted to matri
x factorization and applied an inner product on the latent features of users and items. By replacing the in
ner product with a neural architecture that can learn an arbitrary function from data, we present a general
 framework named NCF, short for Neural network-based Collaborative Filtering. NCF is generic and can expres
s and generalize matrix factorization under its framework. To supercharge NCF modelling with non-linearitie
s, we propose to leverage a multi-layer perceptron to learn the user-item interaction function. Extensive e
xperiments on two real-world datasets show significant improvements of our proposed NCF framework over the
state-of-the-art methods. Empirical evidence shows that using deeper layers of neural networks offers bette
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pe\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e
.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a centr
al server (e.g. service provider), while keeping the training data decentralized. FL embodies the principle
s of focused data collection and minimization, and can mitigate many of the systemic privacy risks and cost
s resulting from traditional, centralized machine learning and data science approaches. Motivated by the ex
plosive growth in FL research, this paper discusses recent advances and presents an extensive collection of
 open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",\"note\":\"arXiv:1912.04977 [cs]\",\
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Applying large pre-trained Vision-Language Models to recommendation is a burgeoning field, a direction we t
erm Vision-Language-Recommendation (VLR). Bringing VLR to user-oriented on-device intelligence within a fed
erated learning framework is a crucial step for enhancing user privacy and delivering personalized experien
ces. This paper introduces FedVLR, a federated VLR framework specially designed for user-specific personali
zed fusion of vision-language representations. At its core is a novel bi-level fusion mechanism: The server
-side multi-view fusion module first generates a diverse set of pre-fused multimodal views. Subsequently, e
ach client employs a user-specific mixture-of-expert mechanism to adaptively integrate these views based on
 individual user interaction history. This designed lightweight personalized fusion module provides an effi
cient solution to implement a federated VLR system. The effectiveness of our proposed FedVLR has been valid
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tion biases, either through self-selection by the users or through the actions of the recommendation system
 itself. In this paper, we provide a principled approach to handling selection biases, adapting models and
estimation techniques from causal inference. The approach leads to unbiased performance estimators despite
biased data, and to a matrix factorization method that provides substantially improved prediction performan
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bstract\":\"With the ever-growing volume of online information, recommender systems have been an effective
strategy to overcome such information overload. The utility of recommender systems cannot be overstated, gi
ven its widespread adoption in many web applications, along with its potential impact to ameliorate many pr
oblems related to over-choice. In recent years, deep learning has garnered considerable interest in many re
search fields such as computer vision and natural language processing, owing not only to stellar performanc
e but also the attractive property of learning feature representations from scratch. The influence of deep
learning is also pervasive, recently demonstrating its effectiveness when applied to information retrieval
and recommender systems research. Evidently, the field of deep learning in recommender system is flourishin
g. This article aims to provide a comprehensive review of recent research efforts on deep learning based re
commender systems. More concretely, we provide and devise a taxonomy of deep learning based recommendation
models, along with providing a comprehensive summary of the state-of-the-art. Finally, we expand on current
 trends and provide new perspectives pertaining to this new exciting development of the field.\",\"containe
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"},{\"family\":\"Sun\",\"given\":\"Aixin\"},{\"family\":\"Tay\",\"given\":\"Yi\"}],\"accessed\":{\"date-par
ts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\",1,31]]}}}],\"schema\":\"https://github.com/ci
tation-style-language/schema/raw/master/csl-citation.json\"} Let’s talk a little bit about how the recommendation system actually works. The conventional way is that the users data the data which is for example users likes, user comments, users favourite things, users preferred bought items ACC. These data has to be taken to a central server where the server processes this data and recommends the user based upon the data that the user generates. The user generates a lot of data depending upon the time and interaction. Even though from user standpoint it is just a simple like or it is just a simple comment which does not do anything to the user but it is a very useful data to the platform that the user is interacting with. So this all data has to be taken to central server in order to recommend more things and more content and more items to the user. This is how most of the company’s work and most of the platforms work. If we name, if we start to name, let’s say Facebook or Instagram, they all collect data and take them to their central servers, which are far away from the place where user is using it or at least far away from the users own device. Tender central servers or central platforms, they process this data, they process all this users likes and comments an

d shares etc, and they applied algorithm to the data to generate more recommendations that is then brought
back to the users device. The user then sees his recommendations and most likely interacts with the recomme
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Applying large pre-trained Vision-Language Models to recommendation is a burgeoning field, a direction we t
erm Vision-Language-Recommendation (VLR). Bringing VLR to user-oriented on-device intelligence within a fed
erated learning framework is a crucial step for enhancing user privacy and delivering personalized experien
ces. This paper introduces FedVLR, a federated VLR framework specially designed for user-specific personali
zed fusion of vision-language representations. At its core is a novel bi-level fusion mechanism: The server
-side multi-view fusion module first generates a diverse set of pre-fused multimodal views. Subsequently, e
ach client employs a user-specific mixture-of-expert mechanism to adaptively integrate these views based on
 individual user interaction history. This designed lightweight personalized fusion module provides an effi
cient solution to implement a federated VLR system. The effectiveness of our proposed FedVLR has been valid
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tion biases, either through self-selection by the users or through the actions of the recommendation system
 itself. In this paper, we provide a principled approach to handling selection biases, adapting models and
estimation techniques from causal inference. The approach leads to unbiased performance estimators despite
biased data, and to a matrix factorization method that provides substantially improved prediction performan
ce on real-world data. We theoretically and empirically characterize the robustness of the approach, findin
g that it is highly practical and scalable.\",\"DOI\":\"10.48550/arXiv.1602.05352\",\"note\":\"arXiv:1602.0
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bstract\":\"With the ever-growing volume of online information, recommender systems have been an effective
strategy to overcome such information overload. The utility of recommender systems cannot be overstated, gi
ven its widespread adoption in many web applications, along with its potential impact to ameliorate many pr
oblems related to over-choice. In recent years, deep learning has garnered considerable interest in many re
search fields such as computer vision and natural language processing, owing not only to stellar performanc
e but also the attractive property of learning feature representations from scratch. The influence of deep
learning is also pervasive, recently demonstrating its effectiveness when applied to information retrieval
and recommender systems research. Evidently, the field of deep learning in recommender system is flourishin
g. This article aims to provide a comprehensive review of recent research efforts on deep learning based re
commender systems. More concretely, we provide and devise a taxonomy of deep learning based recommendation
models, along with providing a comprehensive summary of the state-of-the-art. Finally, we expand on current
 trends and provide new perspectives pertaining to this new exciting development of the field.\",\"containe
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ts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\",1,31]]}}}],\"schema\":\"https://github.com/ci
tation-style-language/schema/raw/master/csl-citation.json\"} Let’s talk a little bit about how the recommen
dation system actually works. The conventional way is that the users data the data which is for example use
rs likes, user comments, users favourite things, users preferred bought items ACC. These data has to be tak
en to a central server where the server processes this data and recommends the user based upon the data tha
t the user generates. The user generates a lot of data depending upon the time and interaction. Even though
 from user standpoint it is just a simple like or it is just a simple comment which does not do anything to
 the user but it is a very useful data to the platform that the user is interacting with. So this all data
has to be taken to central server in order to recommend more things and more content and more items to the
user. This is how most of the company’s work and most of the platforms work. If we name, if we start to nam
e, let’s say Facebook or Instagram, they all collect data and take them to their central servers, which are
 far away from the place where user is using it or at least far away from the users own device. Tender cent
ral servers or central platforms, they process this data, they process all this users likes and comments an
d shares etc, and they applied algorithm to the data to generate more recommendations that is then brought
back to the users device. The user then sees his recommendations and most likely interacts with the recomme
ndations. So for example in the frequently bought items of Amazon user may buy one of the recommended items
 and the and the bought item is another data which then goes to Amazon central servers and then it adds to
the already users data that is present in the central server and the algorithm adapts to the new data that
has come and the cycle goes on. The new data then generates a new recommendation content and the cycle goes
 on and the recommendation improves centrally. (Chajewska et al., 2013; H.-T. Cheng et al., 2016; Halko et
al., 2010; X. He et al., 2017; Kairouz et al., 2021a; Z. Li et al., 2025; Schnabel et al., 2016; S. Zhang e
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of classification techniques to utility elicitation. In a decision problem, two sets of parameters must gen
erally be elicited: the probabilities and the utilities. While the prior and conditional probabilities in t
he model do not change from user to user, the utility models do. Thus it is necessary to elicit a utility m
odel separately for each new user. Elicitation is long and tedious, particularly if the outcome space is la
rge and not decomposable. There are two common approaches to utility function elicitation. The first is to
base the determination of the users utility function solely ON elicitation OF qualitative preferences.The s
econd makes assumptions about the form AND decomposability OF the utility function.Here we take a different
 approach: we attempt TO identify the new USERs utility function based on classification relative to a data
base of previously collected utility functions. We do this by identifying clusters of utility functions tha
t minimize an appropriate distance measure. Having identified the clusters, we develop a classification sch
eme that requires many fewer and simpler assessments than full utility elicitation and is more robust than
utility elicitation based solely on preferences. We have tested our algorithm on a small database of utilit
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cle\",\"abstract\":\"Generalized linear models with nonlinear feature transformations are widely used for l
arge-scale regression and classification problems with sparse inputs. Memorization of feature interactions
through a wide set of cross-product feature transformations are effective and interpretable, while generali
zation requires more feature engineering effort. With less feature engineering, deep neural networks can ge
neralize better to unseen feature combinations through low-dimensional dense embeddings learned for the spa
rse features. However, deep neural networks with embeddings can over-generalize and recommend less relevant
 items when the user-item interactions are sparse and high-rank. In this paper, we present Wide \u0026 Deep
 learning---jointly trained wide linear models and deep neural networks---to combine the benefits of memori
zation and generalization for recommender systems. We productionized and evaluated the system on Google Pla
y, a commercial mobile app store with over one billion active users and over one million apps. Online exper
iment results show that Wide \u0026 Deep significantly increased app acquisitions compared with wide-only a
nd deep-only models. We have also open-sourced our implementation in TensorFlow.\",\"DOI\":\"10.48550/arXiv
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"abstract\":\"Low-rank matrix approximations, such as the truncated singular value decomposition and the ra
nk-revealing QR decomposition, play a central role in data analysis and scientific computing. This work sur
veys and extends recent research which demonstrates that randomization offers a powerful tool for performin
g low-rank matrix approximation. These techniques exploit modern computational architectures more fully tha
n classical methods and open the possibility of dealing with truly massive data sets. This paper presents a
 modular framework for constructing randomized algorithms that compute partial matrix decompositions. These
 methods use random sampling to identify a subspace that captures most of the action of a matrix. The input
 matrix is then compressed---either explicitly or implicitly---to this subspace, and the reduced matrix is
manipulated deterministically to obtain the desired low-rank factorization. In many cases, this approach be
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09.4061\",\"author\":[{\"family\":\"Halko\",\"given\":\"Nathan\"},{\"family\":\"Martinsson\",\"given\":\"Pe
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of classification techniques to utility elicitation. In a decision problem, two sets of parameters must gen
erally be elicited: the probabilities and the utilities. While the prior and conditional probabilities in t
he model do not change from user to user, the utility models do. Thus it is necessary to elicit a utility m
odel separately for each new user. Elicitation is long and tedious, particularly if the outcome space is la
rge and not decomposable. There are two common approaches to utility function elicitation. The first is to
base the determination of the users utility function solely ON elicitation OF qualitative preferences.The s
econd makes assumptions about the form AND decomposability OF the utility function.Here we take a different
 approach: we attempt TO identify the new USERs utility function based on classification relative to a data

base of previously collected utility functions. We do this by identifying clusters of utility functions that minimize an appropriate distance measure. Having identified the clusters, we develop a classification scheme that requires many fewer and simpler assessments than full utility elicitation and is more robust than utility elicitation based solely on preferences. We have tested our algorithm on a small database of utility functions in a prenatal diagnosis domain and the results are quite promising.\",\"DOI\":\"10.48550/arXiv.1301.7367\",\"note\":\"arXiv:1301.7367 [cs]\",\"number\":\"arXiv:1301.7367\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Utility Elicitation as a Classification Problem\",\"URL\":\"http://arxiv.org/abs/1301.7367\",\"author\":[{\"family\":\"Chajewska\",\"given\":\"Urszula\"},{\"family\":\"Getoor\",\"given\":\"Lise\"},{\"family\":\"Norman\",\"given\":\"Joseph\"},{\"family\":\"Shahar\",\"given\":\"Yuval\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2013\",1,30]]}}},{\"id\":124,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/IXSJIXLL\"],\"itemData\":{\"id\":124,\"type\":\"article\",\"abstract\":\"Generalized linear models with nonlinear feature transformations are widely used for large-scale regression and classification problems with sparse inputs. Memorization of feature interactions through a wide set of cross-product feature transformations are effective and interpretable, while generalization requires more feature engineering effort. With less feature engineering, deep neural networks can generalize better to unseen feature combinations through low-dimensional dense embeddings learned for the sparse features. However, deep neural networks with embeddings can over-generalize and recommend less relevant items when the user-item interactions are sparse and high-rank. In this paper, we present Wide \u0026 Deep learning---jointly trained wide linear models and deep neural networks---to combine the benefits of memorization and generalization for recommender systems. We productionized and evaluated the system on Google Play, a commercial mobile app store with over one billion active users and over one million apps. Online experiment results show that Wide \u0026 Deep significantly increased app acquisitions compared with wide-only and deep-only models. We have also open-sourced our implementation in TensorFlow.\",\"DOI\":\"10.48550/arXiv.1606.07792\",\"note\":\"arXiv:1606.07792 [cs]\",\"number\":\"arXiv:1606.07792\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Wide \u0026 Deep Learning for Recommender Systems\",\"URL\":\"http://arxiv.org/abs/1606.07792\",\"author\":[{\"family\":\"Cheng\",\"given\":\"Heng-Tze\"},{\"family\":\"Koc\",\"given\":\"Levent\"},{\"family\":\"Harmsen\",\"given\":\"Jeremiah\"},{\"family\":\"Shaked\",\"given\":\"Tal\"},{\"family\":\"Chandra\",\"given\":\"Tushar\"},{\"family\":\"Aradhye\",\"given\":\"Hrishi\"},{\"family\":\"Anderson\",\"given\":\"Glen\"},{\"family\":\"Corrado\",\"given\":\"Greg\"},{\"family\":\"Chai\",\"given\":\"Wei\"},{\"family\":\"Ispir\",\"given\":\"Mustafa\"},{\"family\":\"Anil\",\"given\":\"Rohan\"},{\"family\":\"Haque\",\"given\":\"Zakaria\"},{\"family\":\"Hong\",\"given\":\"Lichan\"},{\"family\":\"Jain\",\"given\":\"Vihan\"},{\"family\":\"Liu\",\"given\":\"Xiaobing\"},{\"family\":\"Shah\",\"given\":\"Hemal\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2016\",6,24]]}}},{\"id\":117,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/L4VQ4CFL\"],\"itemData\":{\"id\":117,\"type\":\"article\",\"abstract\":\"Low-rank matrix approximations, such as the truncated singular value decomposition and the rank-revealing QR decomposition, play a central role in data analysis and scientific computing. This work surveys and extends recent research which demonstrates that randomization offers a powerful tool for performing low-rank matrix approximation. These techniques exploit modern computational architectures more fully than classical methods and open the possibility of dealing with truly massive data sets. This paper presents a modular framework for constructing randomized algorithms that compute partial matrix decompositions. These methods use random sampling to identify a subspace that captures most of the action of a matrix. The input matrix is then compressed---either explicitly or implicitly---to this subspace, and the reduced matrix is manipulated deterministically to obtain the desired low-rank factorization. In many cases, this approach beats its classical competitors in terms of accuracy, speed, and robustness. These claims are supported by extensive numerical experiments and a detailed error analysis.\",\"DOI\":\"10.48550/arXiv.0909.4061\",\"note\":\"arXiv:0909.4061 [math]\",\"number\":\"arXiv:0909.4061\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions\",\"title-short\":\"Finding structure with randomness\",\"URL\":\"http://arxiv.org/abs/0909.4061\",\"author\":[{\"family\":\"Halko\",\"given\":\"Nathan\"},{\"family\":\"Martinsson\",\"given\":\"Per-Gunnar\"},{\"family\":\"Tropp\",\"given\":\"Joel A.\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2010\",12,14]]}}},{\"id\":120,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/HBUKBJ59\"],\"itemData\":{\"id\":120,\"type\":\"article\",\"abstract\":\"In recent years, deep neural networks have yielded immense success on speech recognition, computer vision and natural language processing. However, the exploration of deep neural networks on recommender systems has received relatively less scrutiny. In this work, we strive to develop techniques based on neural networks to tackle the key problem in recommendation -- collaborative filtering -- on the basis of implicit feedback. Although some recent work has employed deep learning for recommendation, they primarily used it to model auxiliary information, such as textual descriptions of items and acoustic features of musics. When it comes to model the key factor in collaborative filtering -- the interaction between user and item features, they still resorted to matrix factorization and applied an inner product on the latent features of users and items. By replacing the inner product with a neural architecture that can learn an arbitrary function from data, we present a general framework named NCF, short for Neural network-based Collaborative Filtering. NCF is generic and can express and generalize matrix factorization under its framework. To supercharge NCF modelling with non-linearities, we propose to leverage a multi-layer perceptron to learn the user-item interaction function. Extensive experiments on two real-world datasets show significant improvements of our proposed NCF framework over the state-of-the-art methods. Empirical evidence shows that using deeper layers of neural networks offers better recommendation performance.\",\"DOI\":\"10.48550/arXiv.1708.05031\",\"note\":\"arXiv:1708.05031 [cs]\",\"number\":\"arXiv:1708.05031\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Neural Collaborative Filtering\",\"URL\":\"http://arxiv.org/abs/1708.05031\",\"author\":[{\"family\":\"He\",\"given\":\"Xiangnan\"},{\"family\":\"Liao\",\"given\":\"Lizi\"},{\"family\":\"Zhang\",\"given\":\"Hanwang\"},{\"family\":\"Nie\",\"given\":\"Liqiang\"},{\"family\":\"Hu\",\"given\":\"Xia\"},{\"family\":\"Chua\",\"given\":\"Tat-Seng\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2017\",8,26]]}}},{\"id\":135,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/4ZXSSBS7\"],\"itemData\":{\"id\":135,\"type\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e

.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a centr
al server (e.g. service provider), while keeping the training data decentralized. FL embodies the principle
s of focused data collection and minimization, and can mitigate many of the systemic privacy risks and cost
s resulting from traditional, centralized machine learning and data science approaches. Motivated by the ex
plosive growth in FL research, this paper discusses recent advances and presents an extensive collection of
 open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",\"note\":\"arXiv:1912.04977 [cs]\",\
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wYg/items/HBUKBJ59\"],\"itemData\":{\"id\":120,\"type\":\"article\",\"abstract\":\"In recent years, deep ne
ural networks have yielded immense success on speech recognition, computer vision and natural language proc
essing. However, the exploration of deep neural networks on recommender systems has received relatively les
s scrutiny. In this work, we strive to develop techniques based on neural networks to tackle the key proble
m in recommendation -- collaborative filtering -- on the basis of implicit feedback. Although some recent w
ork has employed deep learning for recommendation, they primarily used it to model auxiliary information, s
uch as textual descriptions of items and acoustic features of musics. When it comes to model the key factor
 in collaborative filtering -- the interaction between user and item features, they still resorted to matri
x factorization and applied an inner product on the latent features of users and items. By replacing the in
ner product with a neural architecture that can learn an arbitrary function from data, we present a general
 framework named NCF, short for Neural network-based Collaborative Filtering. NCF is generic and can expres
s and generalize matrix factorization under its framework. To supercharge NCF modelling with non-linearitie
s, we propose to leverage a multi-layer perceptron to learn the user-item interaction function. Extensive e
xperiments on two real-world datasets show significant improvements of our proposed NCF framework over the
state-of-the-art methods. Empirical evidence shows that using deeper layers of neural networks offers bette
r recommendation performance.\",\"DOI\":\"10.48550/arXiv.1708.05031\",\"note\":\"arXiv:1708.05031 [cs]\",\"
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pe\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e
.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a centr
al server (e.g. service provider), while keeping the training data decentralized. FL embodies the principle
s of focused data collection and minimization, and can mitigate many of the systemic privacy risks and cost
s resulting from traditional, centralized machine learning and data science approaches. Motivated by the ex
plosive growth in FL research, this paper discusses recent advances and presents an extensive collection of
 open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",\"note\":\"arXiv:1912.04977 [cs]\",\
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Applying large pre-trained Vision-Language Models to recommendation is a burgeoning field, a direction we t
erm Vision-Language-Recommendation (VLR). Bringing VLR to user-oriented on-device intelligence within a fed
erated learning framework is a crucial step for enhancing user privacy and delivering personalized experien
ces. This paper introduces FedVLR, a federated VLR framework specially designed for user-specific personali
zed fusion of vision-language representations. At its core is a novel bi-level fusion mechanism: The server
-side multi-view fusion module first generates a diverse set of pre-fused multimodal views. Subsequently, e

ach client employs a user-specific mixture-of-expert mechanism to adaptively integrate these views based on individual user interaction history. This designed lightweight personalized fusion module provides an efficient solution to implement a federated VLR system. The effectiveness of our proposed FedVLR has been validated on seven benchmark datasets.\",\"DOI\":\"10.48550/arXiv.2410.08478\",\"note\":\"arXiv:2410.08478 [cs]\",\"number\":\"arXiv:2410.08478\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Federated Vision-Language-Recommendation with Personalized Fusion\",\"URL\":\"http://arxiv.org/abs/2410.08478\",\"author\":[{\"family\":\"Li\",\"given\":\"Zhiwei\"},{\"family\":\"Long\",\"given\":\"Guodong\"},{\"family\":\"Jiang\",\"given\":\"Jing\"},{\"family\":\"Zhang\",\"given\":\"Chengqi\"},{\"family\":\"Yang\",\"given\":\"Qiang\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2025\",11,2]]}}},{\"id\":131,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/N28PRJ4U\"],\"itemData\":{\"id\":131,\"type\":\"article\",\"abstract\":\"Most data for evaluating and training recommender systems is subject to selection biases, either through self-selection by the users or through the actions of the recommendation system itself. In this paper, we provide a principled approach to handling selection biases, adapting models and estimation techniques from causal inference. The approach leads to unbiased performance estimators despite biased data, and to a matrix factorization method that provides substantially improved prediction performance on real-world data. We theoretically and empirically characterize the robustness of the approach, finding that it is highly practical and scalable.\",\"DOI\":\"10.48550/arXiv.1602.05352\",\"note\":\"arXiv:1602.05352 [cs]\",\"number\":\"arXiv:1602.05352\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Recommendations as Treatments: Debiasing Learning and Evaluation\",\"title-short\":\"Recommendations as Treatments\",\"URL\":\"http://arxiv.org/abs/1602.05352\",\"author\":[{\"family\":\"Schnabel\",\"given\":\"Tobias\"},{\"family\":\"Swaminathan\",\"given\":\"Adith\"},{\"family\":\"Singh\",\"given\":\"Ashudeep\"},{\"family\":\"Chandak\",\"given\":\"Navin\"},{\"family\":\"Joachims\",\"given\":\"Thorsten\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2016\",5,27]]}}},{\"id\":139,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/GAXKR4YJ\"],\"itemData\":{\"id\":139,\"type\":\"article-journal\",\"abstract\":\"With the ever-growing volume of online information, recommender systems have been an effective strategy to overcome such information overload. The utility of recommender systems cannot be overstated, given its widespread adoption in many web applications, along with its potential impact to ameliorate many problems related to over-choice. In recent years, deep learning has garnered considerable interest in many research fields such as computer vision and natural language processing, owing not only to stellar performance but also the attractive property of learning feature representations from scratch. The influence of deep learning is also pervasive, recently demonstrating its effectiveness when applied to information retrieval and recommender systems research. Evidently, the field of deep learning in recommender system is flourishing. This article aims to provide a comprehensive review of recent research efforts on deep learning based recommender systems. More concretely, we provide and devise a taxonomy of deep learning based recommendation models, along with providing a comprehensive summary of the state-of-the-art. Finally, we expand on current trends and provide new perspectives pertaining to this new exciting development of the field.\",\"container-title\":\"ACM Computing Surveys\",\"DOI\":\"10.1145/3285029\",\"ISSN\":\"0360-0300, 1557-7341\",\"issue\":\"1\",\"journalAbbreviation\":\"ACM Comput. Surv.\",\"note\":\"arXiv:1707.07435 [cs]\",\"page\":\"1-38\",\"source\":\"arXiv.org\",\"title\":\"Deep Learning based Recommender System: A Survey and New Perspectives\",\"title-short\":\"Deep Learning based Recommender System\",\"URL\":\"http://arxiv.org/abs/1707.07435\",\"volume\":\"52\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Shuai\"},{\"family\":\"Yao\",\"given\":\"Lina\"},{\"family\":\"Sun\",\"given\":\"Aixin\"},{\"family\":\"Tay\",\"given\":\"Yi\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\",1,31]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Let’s talk a little bit about how the recommendation system actually works. The conventional way is that the users data the data which is for example users likes, user comments, users favourite things, users preferred bought items ACC. These data has to be taken to a central server where the server processes this data and recommends the user based upon the data that the user generates. The user generates a lot of data depending upon the time and interaction. Even though from user standpoint it is just a simple like or it is just a simple comment which does not do anything to the user but it is a very useful data to the platform that the user is interacting with. So this all data has to be taken to central server in order to recommend more things and more content and more items to the user. This is how most of the company’s work and most of the platforms work. If we name, if we start to name, let’s say Facebook or Instagram, they all collect data and take them to their central servers, which are far away from the place where user is using it or at least far away from the users own device. Tender central servers or central platforms, they process this data, they process all this users likes and comments and shares etc, and they applied algorithm to the data to generate more recommendations that is then brought back to the users device. The user then sees his recommendations and most likely interacts with the recommendations. So for example in the frequently bought items of Amazon user may buy one of the recommended items and the and the bought item is another data which then goes to Amazon central servers and then it adds to the already users data that is present in the central server and the algorithm adapts to the new data that has come and the cycle goes on. The new data then generates a new recommendation content and the cycle goes on and the recommendation improves centrally. (Chajewska et al., 2013; H.-T. Cheng et al., 2016; Halko et al., 2010; X. He et al., 2017; Kairouz et al., 2021a; Z. Li et al., 2025; Schnabel et al., 2016; S. Zhang et al., 2020)",

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number\":\"arXiv:1708.05031\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Neural Collabora
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Applying large pre-trained Vision-Language Models to recommendation is a burgeoning field, a direction we t
erm Vision-Language-Recommendation (VLR). Bringing VLR to user-oriented on-device intelligence within a fed
erated learning framework is a crucial step for enhancing user privacy and delivering personalized experien
ces. This paper introduces FedVLR, a federated VLR framework specially designed for user-specific personali
zed fusion of vision-language representations. At its core is a novel bi-level fusion mechanism: The server
-side multi-view fusion module first generates a diverse set of pre-fused multimodal views. Subsequently, e
ach client employs a user-specific mixture-of-expert mechanism to adaptively integrate these views based on
 individual user interaction history. This designed lightweight personalized fusion module provides an effi
cient solution to implement a federated VLR system. The effectiveness of our proposed FedVLR has been valid
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\":\"article\",\"abstract\":\"Most data for evaluating and training recommender systems is subject to selec
tion biases, either through self-selection by the users or through the actions of the recommendation system
 itself. In this paper, we provide a principled approach to handling selection biases, adapting models and
estimation techniques from causal inference. The approach leads to unbiased performance estimators despite
biased data, and to a matrix factorization method that provides substantially improved prediction performan
ce on real-world data. We theoretically and empirically characterize the robustness of the approach, findin
g that it is highly practical and scalable.\",\"DOI\":\"10.48550/arXiv.1602.05352\",\"note\":\"arXiv:1602.0
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bstract\":\"With the ever-growing volume of online information, recommender systems have been an effective
strategy to overcome such information overload. The utility of recommender systems cannot be overstated, gi
ven its widespread adoption in many web applications, along with its potential impact to ameliorate many pr
oblems related to over-choice. In recent years, deep learning has garnered considerable interest in many re
search fields such as computer vision and natural language processing, owing not only to stellar performanc
e but also the attractive property of learning feature representations from scratch. The influence of deep
learning is also pervasive, recently demonstrating its effectiveness when applied to information retrieval
and recommender systems research. Evidently, the field of deep learning in recommender system is flourishin
g. This article aims to provide a comprehensive review of recent research efforts on deep learning based re
commender systems. More concretely, we provide and devise a taxonomy of deep learning based recommendation
models, along with providing a comprehensive summary of the state-of-the-art. Finally, we expand on current
 trends and provide new perspectives pertaining to this new exciting development of the field.\",\"containe
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ts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\",1,31]]}}}],\"schema\":\"https://github.com/ci
tation-style-language/schema/raw/master/csl-citation.json\"} Let’s talk a little bit about how the recommen
dation system actually works. The conventional way is that the users data the data which is for example use
rs likes, user comments, users favourite things, users preferred bought items ACC. These data has to be tak
en to a central server where the server processes this data and recommends the user based upon the data tha
t the user generates. The user generates a lot of data depending upon the time and interaction. Even though
 from user standpoint it is just a simple like or it is just a simple comment which does not do anything to
 the user but it is a very useful data to the platform that the user is interacting with. So this all data
has to be taken to central server in order to recommend more things and more content and more items to the
user. This is how most of the company’s work and most of the platforms work. If we name, if we start to nam
e, let’s say Facebook or Instagram, they all collect data and take them to their central servers, which are
 far away from the place where user is using it or at least far away from the users own device. Tender cent
ral servers or central platforms, they process this data, they process all this users likes and comments an
d shares etc, and they applied algorithm to the data to generate more recommendations that is then brought
back to the users device. The user then sees his recommendations and most likely interacts with the recomme
ndations. So for example in the frequently bought items of Amazon user may buy one of the recommended items
 and the and the bought item is another data which then goes to Amazon central servers and then it adds to
the already users data that is present in the central server and the algorithm adapts to the new data that
has come and the cycle goes on. The new data then generates a new recommendation content and the cycle goes
 on and the recommendation improves centrally. (Chajewska et al., 2013; H.-T. Cheng et al., 2016; Halko et
al., 2010; X. He et al., 2017; Kairouz et al., 2021a; Z. Li et al., 2025; Schnabel et al., 2016; S. Zhang e
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ms/YKS945QC\"],\"itemData\":{\"id\":127,\"type\":\"article\",\"abstract\":\"We investigate the application
of classification techniques to utility elicitation. In a decision problem, two sets of parameters must gen
erally be elicited: the probabilities and the utilities. While the prior and conditional probabilities in t
he model do not change from user to user, the utility models do. Thus it is necessary to elicit a utility m
odel separately for each new user. Elicitation is long and tedious, particularly if the outcome space is la
rge and not decomposable. There are two common approaches to utility function elicitation. The first is to
base the determination of the users utility function solely ON elicitation OF qualitative preferences.The s
econd makes assumptions about the form AND decomposability OF the utility function.Here we take a different
 approach: we attempt TO identify the new USERs utility function based on classification relative to a data
base of previously collected utility functions. We do this by identifying clusters of utility functions tha
t minimize an appropriate distance measure. Having identified the clusters, we develop a classification sch
eme that requires many fewer and simpler assessments than full utility elicitation and is more robust than
utility elicitation based solely on preferences. We have tested our algorithm on a small database of utilit
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cle\",\"abstract\":\"Generalized linear models with nonlinear feature transformations are widely used for l
arge-scale regression and classification problems with sparse inputs. Memorization of feature interactions
through a wide set of cross-product feature transformations are effective and interpretable, while generali
zation requires more feature engineering effort. With less feature engineering, deep neural networks can ge
neralize better to unseen feature combinations through low-dimensional dense embeddings learned for the spa
rse features. However, deep neural networks with embeddings can over-generalize and recommend less relevant
 items when the user-item interactions are sparse and high-rank. In this paper, we present Wide \u0026 Deep
 learning---jointly trained wide linear models and deep neural networks---to combine the benefits of memori
zation and generalization for recommender systems. We productionized and evaluated the system on Google Pla
y, a commercial mobile app store with over one billion active users and over one million apps. Online exper
iment results show that Wide \u0026 Deep significantly increased app acquisitions compared with wide-only a
nd deep-only models. We have also open-sourced our implementation in TensorFlow.\",\"DOI\":\"10.48550/arXiv
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"abstract\":\"Low-rank matrix approximations, such as the truncated singular value decomposition and the ra
nk-revealing QR decomposition, play a central role in data analysis and scientific computing. This work sur
veys and extends recent research which demonstrates that randomization offers a powerful tool for performin
g low-rank matrix approximation. These techniques exploit modern computational architectures more fully tha
n classical methods and open the possibility of dealing with truly massive data sets. This paper presents a
 modular framework for constructing randomized algorithms that compute partial matrix decompositions. These
 methods use random sampling to identify a subspace that captures most of the action of a matrix. The input
 matrix is then compressed---either explicitly or implicitly---to this subspace, and the reduced matrix is
manipulated deterministically to obtain the desired low-rank factorization. In many cases, this approach be
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Applying large pre-trained Vision-Language Models to recommendation is a burgeoning field, a direction we t
erm Vision-Language-Recommendation (VLR). Bringing VLR to user-oriented on-device intelligence within a fed
erated learning framework is a crucial step for enhancing user privacy and delivering personalized experien
ces. This paper introduces FedVLR, a federated VLR framework specially designed for user-specific personali
zed fusion of vision-language representations. At its core is a novel bi-level fusion mechanism: The server
-side multi-view fusion module first generates a diverse set of pre-fused multimodal views. Subsequently, e
ach client employs a user-specific mixture-of-expert mechanism to adaptively integrate these views based on
 individual user interaction history. This designed lightweight personalized fusion module provides an effi
cient solution to implement a federated VLR system. The effectiveness of our proposed FedVLR has been valid
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tion biases, either through self-selection by the users or through the actions of the recommendation system
 itself. In this paper, we provide a principled approach to handling selection biases, adapting models and
estimation techniques from causal inference. The approach leads to unbiased performance estimators despite
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bstract\":\"With the ever-growing volume of online information, recommender systems have been an effective
strategy to overcome such information overload. The utility of recommender systems cannot be overstated, gi
ven its widespread adoption in many web applications, along with its potential impact to ameliorate many pr
oblems related to over-choice. In recent years, deep learning has garnered considerable interest in many re
search fields such as computer vision and natural language processing, owing not only to stellar performanc
e but also the attractive property of learning feature representations from scratch. The influence of deep
learning is also pervasive, recently demonstrating its effectiveness when applied to information retrieval
and recommender systems research. Evidently, the field of deep learning in recommender system is flourishin
g. This article aims to provide a comprehensive review of recent research efforts on deep learning based re
commender systems. More concretely, we provide and devise a taxonomy of deep learning based recommendation
models, along with providing a comprehensive summary of the state-of-the-art. Finally, we expand on current
 trends and provide new perspectives pertaining to this new exciting development of the field.\",\"containe
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ts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\",1,31]]}}}],\"schema\":\"https://github.com/ci
tation-style-language/schema/raw/master/csl-citation.json\"} Let’s talk a little bit about how the recommen
dation system actually works. The conventional way is that the users data the data which is for example use
rs likes, user comments, users favourite things, users preferred bought items ACC. These data has to be tak
en to a central server where the server processes this data and recommends the user based upon the data tha
t the user generates. The user generates a lot of data depending upon the time and interaction. Even though
 from user standpoint it is just a simple like or it is just a simple comment which does not do anything to
 the user but it is a very useful data to the platform that the user is interacting with. So this all data
has to be taken to central server in order to recommend more things and more content and more items to the
user. This is how most of the company’s work and most of the platforms work. If we name, if we start to nam
e, let’s say Facebook or Instagram, they all collect data and take them to their central servers, which are
 far away from the place where user is using it or at least far away from the users own device. Tender cent
ral servers or central platforms, they process this data, they process all this users likes and comments an
d shares etc, and they applied algorithm to the data to generate more recommendations that is then brought
back to the users device. The user then sees his recommendations and most likely interacts with the recomme
ndations. So for example in the frequently bought items of Amazon user may buy one of the recommended items
 and the and the bought item is another data which then goes to Amazon central servers and then it adds to
the already users data that is present in the central server and the algorithm adapts to the new data that
has come and the cycle goes on. The new data then generates a new recommendation content and the cycle goes
 on and the recommendation improves centrally. (Chajewska et al., 2013; H.-T. Cheng et al., 2016; Halko et
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entrally has a problem. The problem is that all the user data is open to the company and the there is no pr
ivacy in the user data. In a sense, the company knows what user likes, the company knows what user favourit
es, and the company knows what user is going to click next. Privacy is completely jeopardised. And the user
 has no control over his content and over his data that is being sent to the central server. Not only does
the company know the user history, users like history, user watch history, the data since it is open to the
 company and is and privacy is jeopardised, The data can be easily misused by the company or anybody in bet
ween who might be reading the data. Based upon the user’s history and the user’s recommendation lot of wron
g Moves can be made by The company itself as well as anybody in between, Which could dramatically harm the
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FL embodies the principles of focused data collection and minimization, and can mitigate many of the systemic privacy risks and costs resulting from traditional, centralized machine learning and data science approaches. Motivated by the explosive growth in FL research, this monograph discusses recent advances and presents an extensive collection of open problems and challenges.\",\"container-title\":\"Foundations and Trends in Machine Learning\",\"DOI\":\"10.1561/2200000083\",\"ISSN\":\"1935-8237, 1935-8245\",\"issue\":\"1-2\",\"language\":\"en\",\"page\":\"1-210\",\"source\":\"DOI.org (Crossref)\",\"title\":\"Advances and Open Problems in Federated Learning\",\"URL\":\"https://www.emerald.com/ftmal/article/14/1-2/1/1332154/Advances-and-Open-Problems-in-Federated-Learning\",\"volume\":\"14\",\"author\":[{\"family\":\"Kairouz\",\"given\":\"Peter\"},{\"family\":\"McMahan\",\"given\":\"H. 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 many of the systemic privacy risks and costs resulting from traditional, centralized machine learning and
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The nuclear-spin lattice relaxation rate $1/T_1$ shows T^3 behavior below the superconducting transition temperature T_c without a coherence peak. The $T_1T=$ const. behavior, indicative of the Fermi liquid state, can be seen in a wide temperature range above T_c. The superconductivity in $α$-FeSe is also an unconventional one as well as LaFeAsO$_{1-x}$F$_x$ and related materials. The FeAs layer is not essential for the occurrence of the unconventional superconductivity.\",\"container-title\":\"Journal of the Physical Society of Japan\",\"DOI\":\"10.1143/JPSJ.77.113703\",\"ISSN\":\"0031-9015, 1347-4073\",\"issue\":\"11\",\"journalAbbreviation\":\"J. Phys. Soc. 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The paper begins with a brief introduction into informetrics studies and highlights drawbacks of the contemporary approaches to modeling the citation process as a product of social interactions. An alternative modeling framework based on results obtained in cognitive psychology is then introduced and applied in an experiment to investigate properties of the citation process, as they are revealed by a large collection of citation statistics. Major research findings are discussed, and a summary is given.\",\"DOI\":\"10.48550/arXiv.cs/0607070\",\"note\":\"arXiv:cs/0607070\",\"number\":\"arXiv:cs/0607070\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Citation as a Representation Process\",\"URL\":\"http://arxiv.org/abs/cs/0607070\",\"author\":[{\"family\":\"Kryssanov\",\"given\":\"V. V.\"},{\"family\":\"Rinaldo\",\"given\":\"F. 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We then show that the TBA and the renormalized TCSA match each other to a very high precision for a large range of the volume parameter, providing both a further verification of the TBA system and a demonstration of the efficiency of the TCSA renormalization procedure. We also discuss the lessons learned from our results concerning recent developments regarding the low-energy scattering of quasi-particles in the quantum Potts spin chain.\",\"container-title\":\"Journal of High Energy Physics\",\"DOI\":\"10.1007/JHEP09(2014)052\",\"ISSN\":\"1029-8479\",\"issue\":\"9\",\"journalAbbreviation\":\"J. High Energ. Phys.\",\"note\":\"arXiv:1405.3157 [hep-th]\",\"page\":\"52\",\"source\":\"arXiv.org\",\"title\":\"Excited state TBA and renormalized TCSA in the scaling Potts model\",\"URL\":\"http://arxiv.org/abs/1405.3157\",\"volume\":\"2014\",\"author\":[{\"family\":\"Lencses\",\"given\":\"M.\"},{\"family\":\"Takacs\",\"given\":\"G.\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2014\",9]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Now companies come with a loophole that they include a lengthy terms and condition and disclaimers in their applications and platforms that user has to cheque before signing up. Legally this is binding but nobody reads these lengthy terms and conditions and disclaimers. Show in every sense they work against the user in the matter of selection of data that has to be sent to the central server. 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tps://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Now companies come with a l
oophole that they include a lengthy terms and condition and disclaimers in their applications and platforms
 that user has to cheque before signing up. Legally this is binding but nobody reads these lengthy terms an
d conditions and disclaimers. Show in every sense they work against the user in the matter of selection of
data that has to be sent to the central server. Clearly this is completely correct but ethically this is no
t correct because even companies are aware of the fact that users are not going to read the whole terms and
 conditions and disclaimers of the platform. So if a user happens to suffer some kind of loss because of th
is data breach or data leak then companies have this upper hand in the legal aspect that they can blame the
 user themselves as to because they clicked or they checked the box where it says that I agree to terms and
 conditions or I agree to the disclaimer. Obviously legally user is on the wrong side but ethically the com
pany is on the wrong side. So the loophole is that legally the user consented to the terms and conditions t
he user consented to the use of the platform despite not knowing what data will be sent and what data will
not be sent to the central server.(Cano et al., 2015a; Kotegawa et al., 2008; Kryssanov et al., 2006; Lencs
es \u0026 Takacs, 2014)",
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integer quantum Hall state can support Majorana zero modes at domain walls between its two different stable
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support gapless fermionic excitations; all gapless excitations are bosonic in this edge phase. Majorana fer
mion zero modes occur at a domain wall between this edge phase and the more conventional one that does supp
ort gapless fermions. Remarkably, due to the chirality of the system, the topological degeneracy of these z
ero modes has exponential protection, as a function of the relevant length scales, in spite of the presence
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 but do not require it. We discuss generalizations to other integer and fractional quantum Hall states, and
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served in amorphous solids. Structurally, they exhibit a power-law scaling in their probability distributio
n of weak forces and small interparticle gaps as well as a flat density of vibrational states. Dynamically,
 they display anomalous aging above a characteristic pressure. Quantitatively this disordered crystal phase
 has much in common with the Gardner-like phase seen in maximally disordered solids. Near crystals should b
e amenable to experimental realizations in commercially-available particulate systems and are to be indispe
nsable in verifying the theory of amorphous materials.\",\"container-title\":\"Physical Review E\",\"DOI\":
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We present AdFisher, an automated tool that explores how user behaviors, Google\u0027s ads, and Ad Settings
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nce tests. Our tool uses a rigorous experimental design and statistical analysis to ensure the statistical
soundness of our results. We use AdFisher to find that the Ad Settings was opaque about some features of a
user\u0027s profile, that it does provide some choice on ads, and that these choices can lead to seemingly
discriminatory ads. In particular, we found that visiting webpages associated with substance abuse changed
the ads shown but not the settings page. We also found that setting the gender to female resulted in gettin
g fewer instances of an ad related to high paying jobs than setting it to male. We cannot determine who cau
sed these findings due to our limited visibility into the ad ecosystem, which includes Google, advertisers,
 websites, and users. Nevertheless, these results can form the starting point for deeper investigations by
either the companies themselves or by regulatory bodies.\",\"DOI\":\"10.48550/arXiv.1408.6491\",\"note\":\"
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 neural networks, consistently misclassify adversarial examples---inputs formed by applying small but inten
tionally worst-case perturbations to examples from the dataset, such that the perturbed input results in th
e model outputting an incorrect answer with high confidence. Early attempts at explaining this phenomenon f
ocused on nonlinearity and overfitting. We argue instead that the primary cause of neural networks\u0027 vu
lnerability to adversarial perturbation is their linear nature. This explanation is supported by new quanti
tative results while giving the first explanation of the most intriguing fact about them: their generalizat
ion across architectures and training sets. Moreover, this view yields a simple and fast method of generati
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arning setting where many clients (e.g. mobile devices or whole organizations) collaboratively train a mode
l under the orchestration of a central server (e.g. service provider), while keeping the training data dece
ntralized. FL embodies the principles of focused data collection and minimization, and can mitigate many of
 the systemic privacy risks and costs resulting from traditional, centralized machine learning and data sci
ence approaches. Motivated by the explosive growth in FL research, this paper discusses recent advances and
 presents an extensive collection of open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",
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high-dimensional micro-data, such as individual preferences, recommendations, transaction records and so on
. Our techniques are robust to perturbation in the data and tolerate some mistakes in the adversary\u0027s
background knowledge. We apply our de-anonymization methodology to the Netflix Prize dataset, which contain
s anonymous movie ratings of 500,000 subscribers of Netflix, the world\u0027s largest online movie rental s
ervice. We demonstrate that an adversary who knows only a little bit about an individual subscriber can eas
ily identify this subscriber\u0027s record in the dataset. Using the Internet Movie Database as the source
of background knowledge, we successfully identified the Netflix records of known users, uncovering their ap
parent political preferences and other potentially sensitive information.\",\"DOI\":\"10.48550/arXiv.cs/061
0105\",\"note\":\"arXiv:cs/0610105\",\"number\":\"arXiv:cs/0610105\",\"publisher\":\"arXiv\",\"source\":\"a
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:184,\"type\":\"article\",\"abstract\":\"We quantitatively investigate how machine learning models leak inf
ormation about the individual data records on which they were trained. We focus on the basic membership inf
erence attack: given a data record and black-box access to a model, determine if the record was in the mode
l\u0027s training dataset. To perform membership inference against a target model, we make adversarial use
of machine learning and train our own inference model to recognize differences in the target model\u0027s p
redictions on the inputs that it trained on versus the inputs that it did not train on. We empirically eval
uate our inference techniques on classification models trained by commercial \\\"machine learning as a serv
ice\\\" providers such as Google and Amazon. Using realistic datasets and classification tasks, including a
 hospital discharge dataset whose membership is sensitive from the privacy perspective, we show that these
models can be vulnerable to membership inference attacks. We then investigate the factors that influence th
is leakage and evaluate mitigation strategies.\",\"DOI\":\"10.48550/arXiv.1610.05820\",\"note\":\"arXiv:161
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} One hack to the central server and the
complete database is vulnerable to the hacker. Once that happens imagine the financial fraud imagine the le
gal fraud that can be done because of this data breach and the vulnerability of the central server. Also it
 is not just about the ill-intentional use of the data once the central server is hacked. Let us suppose th
at the central server is not hacked and the company is using the data as they intend, however the way they
are using the data may not be favourable to the users and may be ethically wrong when it comes to users poi
nt of view. There can also be behavioural manipulation because of the central server data that is collected
 at one place. For example if the central server knows from reading the data that the user is depressed at
a certain period of time in a day the central server can recommend the user things like junk food or maybe
items of suicide or items that can be used as suicide etc. So the fact is that they are not just recommendi
ng or showing the user content or items they are basically nudging the users behaviour toward a certain con
sequence which is not right. (Cano et al., 2015b; Charbonneau et al., 2019; Datta et al., 2015a; Goodfellow
 et al., 2015a; Kairouz et al., 2021c; Narayanan \u0026 Shmatikov, 2007a; Shokri et al., 2017b)",
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 chiral edge phases without superconductivity. This is due to the existence of an edge phase that does not
support gapless fermionic excitations; all gapless excitations are bosonic in this edge phase. Majorana fer
mion zero modes occur at a domain wall between this edge phase and the more conventional one that does supp
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 but do not require it. We discuss generalizations to other integer and fractional quantum Hall states, and
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Structurally, they exhibit a power-law scaling in their probability distribution of weak forces and small interparticle gaps as well as a flat density of vibrational states. Dynamically, they display anomalous aging above a characteristic pressure. Quantitatively this disordered crystal phase has much in common with the Gardner-like phase seen in maximally disordered solids. Near crystals should be amenable to experimental realizations in commercially-available particulate systems and are to be indispensable in verifying the theory of amorphous materials.\",\"container-title\":\"Physical Review E\",\"DOI\":\"10.1103/PhysRevE.99.020901\",\"ISSN\":\"2470-0045, 2470-0053\",\"issue\":\"2\",\"journalAbbreviation\":\"Phys. Rev. E\",\"note\":\"arXiv:1802.07391 [cond-mat]\",\"page\":\"020901\",\"source\":\"arXiv.org\",\"title\":\"Glassy, Gardner-like Phenomenology in Minimally Polydisperse Crystalline Systems\",\"URL\":\"http://arxiv.org/abs/1802.07391\",\"volume\":\"99\",\"author\":[{\"family\":\"Charbonneau\",\"given\":\"Patrick\"},{\"family\":\"Corwin\",\"given\":\"Eric I.\"},{\"family\":\"Fu\",\"given\":\"Lin\"},{\"family\":\"Tsekenis\",\"given\":\"Georgios\"},{\"family\":\"Naald\",\"given\":\"Michael\",\"dropping-particle\":\"van der\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2019\",2,13]]}}},{\"id\":199,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/8F9SW3DD\"],\"itemData\":{\"id\":199,\"type\":\"article\",\"abstract\":\"To partly address people\u0027s concerns over web tracking, Google has created the Ad Settings webpage to provide information about and some choice over the profiles Google creates on users. 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Early attempts at explaining this phenomenon focused on nonlinearity and overfitting. We argue instead that the primary cause of neural networks\u0027 vulnerability to adversarial perturbation is their linear nature. This explanation is supported by new quantitative results while giving the first explanation of the most intriguing fact about them: their generalization across architectures and training sets. Moreover, this view yields a simple and fast method of generating adversarial examples. Using this approach to provide examples for adversarial training, we reduce the test set error of a maxout network on the MNIST dataset.\",\"DOI\":\"10.48550/arXiv.1412.6572\",\"note\":\"arXiv:1412.6572 [stat]\",\"number\":\"arXiv:1412.6572\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Explaining and Harnessing Adversarial Examples\",\"URL\":\"http://arxiv.org/abs/1412.6572\",\"author\":[{\"family\":\"Goodfellow\",\"given\":\"Ian J.\"},{\"family\":\"Shlens\",\"given\":\"Jonathon\"},{\"family\":\"Szegedy\",\"given\":\"Christian\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2015\",3,20]]}}},{\"id\":202,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/HYUINALC\"],\"itemData\":{\"id\":202,\"type\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a central server (e.g. service provider), while keeping the training data decentralized. FL embodies the principles of focused data collection and minimization, and can mitigate many of the systemic privacy risks and costs resulting from traditional, centralized machine learning and data science approaches. Motivated by the explosive growth in FL research, this paper discusses recent advances and presents an extensive collection of open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",\"note\":\"arXiv:1912.04977 [cs]\",\"number\":\"arXiv:1912.04977\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Advances and Open Problems in Federated Learning\",\"URL\":\"http://arxiv.org/abs/1912.04977\",\"author\":[{\"family\":\"Kairouz\",\"given\":\"Peter\"},{\"family\":\"McMahan\",\"given\":\"H. 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.com/citation-style-language/schema/raw/master/csl-citation.json\"} One hack to the central server and the
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We discuss generalizations to other integer and fractional quantum Hall states, and classify possible mechanisms for appearance of Majorana zero modes at domain walls.\",\"container-title\":\"Physical Review B\",\"DOI\":\"10.1103/PhysRevB.92.195152\",\"ISSN\":\"1098-0121, 1550-235X\",\"issue\":\"19\",\"journalAbbreviation\":\"Phys. Rev. B\",\"note\":\"arXiv:1505.07825 [cond-mat]\",\"page\":\"195152\",\"source\":\"arXiv.org\",\"title\":\"Chirality-Protected Majorana Zero Modes at the Gapless Edge of Abelian Quantum Hall States\",\"URL\":\"http://arxiv.org/abs/1505.07825\",\"volume\":\"92\",\"author\":[{\"family\":\"Cano\",\"given\":\"Jennifer\"},{\"family\":\"Cheng\",\"given\":\"Meng\"},{\"family\":\"Barkeshli\",\"given\":\"Maissam\"},{\"family\":\"Clarke\",\"given\":\"David J.\"},{\"family\":\"Nayak\",\"given\":\"Chetan\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2015\",11,30]]}}},{\"id\":191,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/E56AYUY5\"],\"itemData\":{\"id\":191,\"type\":\"article-journal\",\"abstract\":\"We report on a non-equilibrium phase of matter, the minimally disordered crystal phase, which we find exists between the maximally amorphous glasses and the ideal crystal. Even though these near crystals appear highly ordered, they display glassy and jamming features akin to those observed in amorphous solids. Structurally, they exhibit a power-law scaling in their probability distribution of weak forces and small interparticle gaps as well as a flat density of vibrational states. Dynamically, they display anomalous aging above a characteristic pressure. Quantitatively this disordered crystal phase has much in common with the Gardner-like phase seen in maximally disordered solids. Near crystals should be amenable to experimental realizations in commercially-available particulate systems and are to be indispensable in verifying the theory of amorphous materials.\",\"container-title\":\"Physical Review E\",\"DOI\":\"10.1103/PhysRevE.99.020901\",\"ISSN\":\"2470-0045, 2470-0053\",\"issue\":\"2\",\"journalAbbreviation\":\"Phys. Rev. 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We present AdFisher, an automated tool that explores how user behaviors, Google\u0027s ads, and Ad Settings interact. AdFisher can run browser-based experiments and analyze data using machine learning and significance tests. Our tool uses a rigorous experimental design and statistical analysis to ensure the statistical soundness of our results. We use AdFisher to find that the Ad Settings was opaque about some features of a user\u0027s profile, that it does provide some choice on ads, and that these choices can lead to seemingly discriminatory ads. In particular, we found that visiting webpages associated with substance abuse changed the ads shown but not the settings page. We also found that setting the gender to female resulted in getting fewer instances of an ad related to high paying jobs than setting it to male. We cannot determine who caused these findings due to our limited visibility into the ad ecosystem, which includes Google, advertisers, websites, and users. Nevertheless, these results can form the starting point for deeper investigations by either the companies themselves or by regulatory bodies.\",\"DOI\":\"10.48550/arXiv.1408.6491\",\"note\":\"arXiv:1408.6491 [cs]\",\"number\":\"arXiv:1408.6491\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Automated Experiments on Ad Privacy Settings: A Tale of Opacity, Choice, and Discrimination\",\"title-short\":\"Automated Experiments on Ad Privacy Settings\",\"URL\":\"http://arxiv.org/abs/1408.6491\",\"author\":[{\"family\":\"Datta\",\"given\":\"Amit\"},{\"family\":\"Tschantz\",\"given\":\"Michael Carl\"},{\"family\":\"Datta\",\"given\":\"Anupam\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2015\",3,17]]}}},{\"id\":188,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/BAGGHN8S\"],\"itemData\":{\"id\":188,\"type\":\"article\",\"abstract\":\"Several machine learning models, including neural networks, consistently misclassify adversarial examples---inputs formed by applying small but intentionally worst-case perturbations to examples from the dataset, such that the perturbed input results in the model outputting an incorrect answer with high confidence. Early attempts at explaining this phenomenon focused on nonlinearity and overfitting. We argue instead that the primary cause of neural networks\u0027 vulnerability to adversarial perturbation is their linear nature. This explanation is supported by new quantitative results while giving the first explanation of the most intriguing fact about them: their generalization across architectures and training sets. Moreover, this view yields a simple and fast method of generating adversarial examples. Using this approach to provide examples for adversarial training, we reduce the test set error of a maxout network on the MNIST dataset.\",\"DOI\":\"10.48550/arXiv.1412.6572\",\"note\":\"arXiv:1412.6572 [stat]\",\"number\":\"arXiv:1412.6572\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Explaining and Harnessing Adversarial Examples\",\"URL\":\"http://arxiv.org/abs/1412.6572\",\"author\":[{\"family\":\"Goodfellow\",\"given\":\"Ian J.\"},{\"family\":\"Shlens\",\"given\":\"Jonathon\"},{\"family\":\"Szegedy\",\"given\":\"Christian\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2015\",3,20]]}}},{\"id\":202,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/HYUINALC\"],\"itemData\":{\"id\":202,\"type\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a central server (e.g. service provider), while keeping the training data decentralized. FL embodies the principles of focused data collection and minimization, and can mitigate many of the systemic privacy risks and costs resulting from traditional, centralized machine learning and data sci

ence approaches. Motivated by the explosive growth in FL research, this paper discusses recent advances and
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background knowledge. We apply our de-anonymization methodology to the Netflix Prize dataset, which contain
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} One hack to the central server and the

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complete database is vulnerable to the hacker. Once that happens imagine the financial fraud imagine the le
gal fraud that can be done because of this data breach and the vulnerability of the central server. Also it
 is not just about the ill-intentional use of the data once the central server is hacked. Let us suppose th
at the central server is not hacked and the company is using the data as they intend, however the way they
are using the data may not be favourable to the users and may be ethically wrong when it comes to users poi
nt of view. There can also be behavioural manipulation because of the central server data that is collected
 at one place. For example if the central server knows from reading the data that the user is depressed at
a certain period of time in a day the central server can recommend the user things like junk food or maybe
items of suicide or items that can be used as suicide etc. So the fact is that they are not just recommendi
ng or showing the user content or items they are basically nudging the users behaviour toward a certain con
sequence which is not right. (Cano et al., 2015b; Charbonneau et al., 2019; Datta et al., 2015a; Goodfellow
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integer quantum Hall state can support Majorana zero modes at domain walls between its two different stable
 chiral edge phases without superconductivity. This is due to the existence of an edge phase that does not
support gapless fermionic excitations; all gapless excitations are bosonic in this edge phase. Majorana fer
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ort gapless fermions. Remarkably, due to the chirality of the system, the topological degeneracy of these z
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 but do not require it. We discuss generalizations to other integer and fractional quantum Hall states, and
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served in amorphous solids. Structurally, they exhibit a power-law scaling in their probability distributio
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 they display anomalous aging above a characteristic pressure. Quantitatively this disordered crystal phase
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either the companies themselves or by regulatory bodies.\",\"DOI\":\"10.48550/arXiv.1408.6491\",\"note\":\"
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ntralized. FL embodies the principles of focused data collection and minimization, and can mitigate many of
 the systemic privacy risks and costs resulting from traditional, centralized machine learning and data sci
ence approaches. Motivated by the explosive growth in FL research, this paper discusses recent advances and
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background knowledge. We apply our de-anonymization methodology to the Netflix Prize dataset, which contain
s anonymous movie ratings of 500,000 subscribers of Netflix, the world\u0027s largest online movie rental s
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ily identify this subscriber\u0027s record in the dataset. Using the Internet Movie Database as the source
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of machine learning and train our own inference model to recognize differences in the target model\u0027s p
redictions on the inputs that it trained on versus the inputs that it did not train on. We empirically eval
uate our inference techniques on classification models trained by commercial \\\"machine learning as a serv
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models can be vulnerable to membership inference attacks. We then investigate the factors that influence th
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Early attempts at explaining this phenomenon focused on nonlinearity and overfitting. We argue instead that the primary cause of neural networks\u0027 vulnerability to adversarial perturbation is their linear nature. This explanation is supported by new quantitative results while giving the first explanation of the most intriguing fact about them: their generalization across architectures and training sets. Moreover, this view yields a simple and fast method of generating adversarial examples. Using this approach to provide examples for adversarial training, we reduce the test set error of a maxout network on the MNIST dataset.\",\"DOI\":\"10.48550/arXiv.1412.6572\",\"note\":\"arXiv:1412.6572 [stat]\",\"number\":\"arXiv:1412.6572\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Explaining and Harnessing Adversarial Examples\",\"URL\":\"http://arxiv.org/abs/1412.6572\",\"author\":[{\"family\":\"Goodfellow\",\"given\":\"Ian J.\"},{\"family\":\"Shlens\",\"given\":\"Jonathon\"},{\"family\":\"Szegedy\",\"given\":\"Christian\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2015\",3,20]]}}},{\"id\":202,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/HYUINALC\"],\"itemData\":{\"id\":202,\"type\":\"article\",\"abstract\":\"Federated learning (FL) is a machine learning setting where many clients (e.g. mobile devices or whole organizations) collaboratively train a model under the orchestration of a central server (e.g. service provider), while keeping the training data decentralized. FL embodies the principles of focused data collection and minimization, and can mitigate many of the systemic privacy risks and costs resulting from traditional, centralized machine learning and data science approaches. Motivated by the explosive growth in FL research, this paper discusses recent advances and presents an extensive collection of open problems and challenges.\",\"DOI\":\"10.48550/arXiv.1912.04977\",\"note\":\"arXiv:1912.04977 [cs]\",\"number\":\"arXiv:1912.04977\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Advances and Open Problems in Federated Learning\",\"URL\":\"http://arxiv.org/abs/1912.04977\",\"author\":[{\"family\":\"Kairouz\",\"given\":\"Peter\"},{\"family\":\"McMahan\",\"given\":\"H. 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Our techniques are robust to perturbation in the data and tolerate some mistakes in the adversary\u0027s background knowledge. We apply our de-anonymization methodology to the Netflix Prize dataset, which contains anonymous movie ratings of 500,000 subscribers of Netflix, the world\u0027s largest online movie rental service. We demonstrate that an adversary who knows only a little bit about an individual subscriber can easily identify this subscriber\u0027s record in the dataset. Using the Internet Movie Database as the source of background knowledge, we successfully identified the Netflix records of known users, uncovering their apparent political preferences and other potentially sensitive information.\",\"DOI\":\"10.48550/arXiv.cs/0610105\",\"note\":\"arXiv:cs/0610105\",\"number\":\"arXiv:cs/0610105\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"How To Break Anonymity of the Netflix Prize Dataset\",\"URL\":\"http://arxiv.org/abs/cs/0610105\",\"author\":[{\"family\":\"Narayanan\",\"given\":\"Arvind\"},{\"family\":\"Shmatikov\",\"given\":\"Vitaly\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2007\",11,22]]}}},{\"id\":184,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/TZANIIT3\"],\"itemData\":{\"id\":184,\"type\":\"article\",\"abstract\":\"We quantitatively investigate how machine learning models leak information about the individual data records on which they were trained. We focus on the basic membership inference attack: given a data record and black-box access to a model, determine if the record was in the mode

l\u0027s training dataset. To perform membership inference against a target model, we make adversarial use of machine learning and train our own inference model to recognize differences in the target model\u0027s predictions on the inputs that it trained on versus the inputs that it did not train on. We empirically evaluate our inference techniques on classification models trained by commercial \\\"machine learning as a service\\\" providers such as Google and Amazon. Using realistic datasets and classification tasks, including a hospital discharge dataset whose membership is sensitive from the privacy perspective, we show that these models can be vulnerable to membership inference attacks. We then investigate the factors that influence this leakage and evaluate mitigation strategies.\",\"DOI\":\"10.48550/arXiv.1610.05820\",\"note\":\"arXiv:1610.05820 [cs]\",\"number\":\"arXiv:1610.05820\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Membership Inference Attacks against Machine Learning Models\",\"URL\":\"http://arxiv.org/abs/1610.05820\",\"author\":[{\"family\":\"Shokri\",\"given\":\"Reza\"},{\"family\":\"Stronati\",\"given\":\"Marco\"},{\"family\":\"Song\",\"given\":\"Congzheng\"},{\"family\":\"Shmatikov\",\"given\":\"Vitaly\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2017\",3,31]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} One hack to the central server and the complete database is vulnerable to the hacker. Once that happens imagine the financial fraud imagine the legal fraud that can be done because of this data breach and the vulnerability of the central server. Also it is not just about the ill-intentional use of the data once the central server is hacked. Let us suppose that the central server is not hacked and the company is using the data as they intend, however the way they are using the data may not be favourable to the users and may be ethically wrong when it comes to users point of view. There can also be behavioural manipulation because of the central server data that is collected at one place. For example if the central server knows from reading the data that the user is depressed at a certain period of time in a day the central server can recommend the user things like junk food or maybe items of suicide or items that can be used as suicide etc. So the fact is that they are not just recommending or showing the user content or items they are basically nudging the users behaviour toward a certain consequence which is not right. (Cano et al., 2015b; Charbonneau et al., 2019; Datta et al., 2015a; Goodfellow et al., 2015a; Kairouz et al., 2021c; Narayanan \u0026 Shmatikov, 2007a; Shokri et al., 2017b)",
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d security. We propose a possible solution to these challenges: secure federated learning. Beyond the feder
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ated learning framework first proposed by Google in 2016, we introduce a comprehensive secure federated lea
rning framework, which includes horizontal federated learning, vertical federated learning and federated tr
ansfer learning. We provide definitions, architectures and applications for the federated learning framewor
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utilizes the abundance of client information, and is dependent on the providers of the service despite the
profile itself and this reduces the chances of manipulation of behaviour, specific advertisement, or commod
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and contain sensitive information. The models should not expose private information in these datasets. Addr
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eters optimized in decentralized fashion by multiple clients. The resulting model is then distributed back
to all clients, ultimately converging to a joint representative model without explicitly having to share th
e data. However, the protocol is vulnerable to differential attacks, which could originate from any party c
ontributing during federated optimization. In such an attack, a client\u0027s contribution during training
and information about their data set is revealed through analyzing the distributed model. We tackle this pr
oblem and propose an algorithm for client sided differential privacy preserving federated optimization. The
 aim is to hide clients\u0027 contributions during training, balancing the trade-off between privacy loss a
nd model performance. Empirical studies suggest that given a sufficiently large number of participating cli
ents, our proposed procedure can maintain client-level differential privacy at only a minor cost in model p
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ted learning is a recent advance in privacy protection. In this context, a trusted curator aggregates param
eters optimized in decentralized fashion by multiple clients. The resulting model is then distributed back
to all clients, ultimately converging to a joint representative model without explicitly having to share th
e data. However, the protocol is vulnerable to differential attacks, which could originate from any party c
ontributing during federated optimization. In such an attack, a client\u0027s contribution during training
and information about their data set is revealed through analyzing the distributed model. We tackle this pr
oblem and propose an algorithm for client sided differential privacy preserving federated optimization. The
 aim is to hide clients\u0027 contributions during training, balancing the trade-off between privacy loss a
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 learning (FL) is a machine learning setting where many clients (e.g. mobile devices or whole organizations
) collaboratively train a model under the orchestration of a central server (e.g. service provider), while
keeping the training data decentralized. FL embodies the principles of focused data collection and minimiza
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ch. The Federated Learning System also comes with a compensation. This compensation comes as the accuracy l
evel of the models and the recommendation system as compared to the centralised learning systems. In other
words, the centralised recommendation systems because they intake raw data from the devices have greater ac
curacy as compared to the Federated Learning Systems because the federated learning system do not give out
raw data to the centralised server and hence centralised server does not know the use and patterns of each
user individually. Hence the Federated learning system only averages the aggregated results of models recei
ved from individual devices and not the raw data itself. Had they received the raw data themselves the accu
racy would be greater but they do not. But on the other hand, the privacy is maintained. This paper tries t
o establish a sweet spot between accuracy and privacy of Federated learning system in mobile phones (refer
to Figure 1). Since as we discussed that there is no free lunch so obviously there is a trade-off between a
ccuracy and privacy when it comes to Federated learning system. So, we have to find a sweet spot where both
 accuracy and privacy work the best and work in the most optimal way possible. This does not mean that the
accuracy will match the accuracy of a centralised learning system or the privacy level will be the maximum
this is impossible. Our goal is to find the optimal point where both accuracy and privacy are at their best
 when it comes to Federated learning system and we are not concerned with comparing the accuracy and privac
y with centralised learning system because obviously centralised learning system have far greater accuracy
but far lesser privacy so we are not comparing that here. We’re only trying to find the optimal point where
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and contain sensitive information. The models should not expose private information in these datasets. Addr
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We first show that the accuracy of federated learning reduces significantly, by up to 55% for neural networks trained for highly skewed non-IID data, where each client device trains only on a single class of data. We further show that this accuracy reduction can be explained by the weight divergence, which can be quantified by the earth mover\u0027s distance (EMD) between the distribution over classes on each device and the population distribution. As a solution, we propose a strategy to improve training on non-IID data by creating a small subset of data which is globally shared between all the edge devices. Experiments show that accuracy can be increased by 30% for the CIFAR-10 dataset with only 5% globally shared data.\",\"DOI\":\"10.48550/arXiv.1806.00582\",\"note\":\"arXiv:1806.00582 [cs]\",\"source\":\"arXiv.org\",\"title\":\"Federated Learning with Non-IID

Data\",\"URL\":\"http://arxiv.org/abs/1806.00582\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Yue\"},{\"family\":\"Li\",\"given\":\"Meng\"},{\"family\":\"Lai\",\"given\":\"Liangzhen\"},{\"family\":\"Suda\",\"given\":\"Naveen\"},{\"family\":\"Civin\",\"given\":\"Damon\"},{\"family\":\"Chandra\",\"given\":\"Vikas\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2018\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} However, there is no free lunch. The Federated Learning System also comes with a compensation. This compensation comes as the accuracy level of the models and the recommendation system as compared to the centralised learning systems. In other words, the centralised recommendation systems because they intake raw data from the devices have greater accuracy as compared to the Federated Learning Sysems because the federated learning system do not give out raw data to the centralised server and hence centralised server does not know the use and patterns of each user individually. Hence the Federated learning system only averages the aggregated results of models received from individual devices and not the raw data itself. Had they received the raw data themselves the accuracy would be greater but they do not. But on the other hand, the privacy is maintained. This paper tries to establish a sweet spot between accuracy and privacy of Federated learning system in mobile phones (refer to Figure 1). Since as we discussed that there is no free lunch so obviously there is a trade-off between accuracy and privacy when it comes to Federated learning system. So, we have to find a sweet spot where both accuracy and privacy work the best and work in the most optimal way possible. This does not mean that the accuracy will match the accuracy of a centralised learning system or the privacy level will be the maximum this is impossible. Our goal is to find the optimal point where both accuracy and privacy are at their best when it comes to Federated learning system and we are not concerned with comparing the accuracy and privacy with centralised learning system because obviously centralised learning system have far greater accuracy but far lesser privacy so we are not comparing that here. We’re only trying to find the optimal point where Federated learning will work best in terms of both accuracy and privacy. (Abadi et al., 2016a; Geyer et al., 2018a; Kairouz et al., 2021f; T. Li, Sahu, et al., 2020; McMahan et al., 2023c; Y. Zhao et al., 2018)",
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Security Problems with Improper Implementations of Improved FEA-M\",\"URL\":\"http://arxiv.org/abs/cs/05090
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i-Party Computation: This method works in such a way that the central server only sees the aggregated resul
t and never the individual data. This is done through adding noise to the individual data point in such a w
ay that when the aggregation occur in the central server, the noises cancel out and an aggregated result is
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iques based on neural networks are achieving remarkable results in a wide variety of domains. Often, the tr
aining of models requires large, representative datasets, which may be crowdsourced and contain sensitive i
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end on sharing a key (public or private) among the peers involved in exchanging an encrypted message. Howev
er, this approach poses privacy concerns. Especially with popular cloud services, the control over the priv
acy of the sensitive data is lost. Even when the keys are not shared, the encrypted material is shared with
 a third party that does not necessarily need to access the content. Moreover, untrusted servers, providers
, and cloud operators can keep identifying elements of users long after users end the relationship with the
 services. Indeed, Homomorphic Encryption (HE), a special kind of encryption scheme, can address these conc
erns as it allows any third party to operate on the encrypted data without decrypting it in advance. Althou
gh this extremely useful feature of the HE scheme has been known for over 30 years, the first plausible and
 achievable Fully Homomorphic Encryption (FHE) scheme, which allows any computable function to perform on t
he encrypted data, was introduced by Craig Gentry in 2009. Even though this was a major achievement, differ
ent implementations so far demonstrated that FHE still needs to be improved significantly to be practical o
n every platform. First, we present the basics of HE and the details of the well-known Partially Homomorphi
c Encryption (PHE) and Somewhat Homomorphic Encryption (SWHE), which are important pillars of achieving FHE
. Then, the main FHE families, which have become the base for the other follow-up FHE schemes are presented
. Furthermore, the implementations and recent improvements in Gentry-type FHE schemes are also surveyed. Fi
nally, further research directions are discussed. This survey is intended to give a clear knowledge and fou
ndation to researchers and practitioners interested in knowing, applying, as well as extending the state of
 the art HE, PHE, SWHE, and FHE systems.\",\"DOI\":\"10.48550/arXiv.1704.03578\",\"note\":\"arXiv:1704.0357
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those values without revealing the values themselves. We consider training a deep neural network in the Fed
erated Learning model, using distributed stochastic gradient descent across user-held training data on mobi
le devices, wherein Secure Aggregation protects each user\u0027s model gradient. We design a novel, communi
cation-efficient Secure Aggregation protocol for high-dimensional data that tolerates up to 1/3 users faili
ng to complete the protocol. For 16-bit input values, our protocol offers 1.73x communication expansion for
 $2^{10}$ users and $2^{20}$-dimensional vectors, and 1.98x expansion for $2^{14}$ users and $2^{24}$ dimen
sional vectors.\",\"DOI\":\"10.48550/arXiv.1611.04482\",\"note\":\"arXiv:1611.04482 [cs]\",\"number\":\"arX
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iques based on neural networks are achieving remarkable results in a wide variety of domains. Often, the tr
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g is to collaboratively train a neural network on a server. Each user receives the current weights of the n
etwork and in turns sends parameter updates (gradients) based on local data. This protocol has been designe
d not only to train neural networks data-efficiently, but also to provide privacy benefits for users, as th
eir input data remains on device and only parameter gradients are shared. But how secure is sharing paramet
er gradients? Previous attacks have provided a false sense of security, by succeeding only in contrived set
tings - even for a single image. However, by exploiting a magnitude-invariant loss along with optimization
strategies based on adversarial attacks, we show that is is actually possible to faithfully reconstruct ima
ges at high resolution from the knowledge of their parameter gradients, and demonstrate that such a break o
f privacy is possible even for trained deep networks. We analyze the effects of architecture as well as par
ameters on the difficulty of reconstructing an input image and prove that any input to a fully connected la
yer can be reconstructed analytically independent of the remaining architecture. Finally we discuss setting
s encountered in practice and show that even averaging gradients over several iterations or several images
does not protect the user\u0027s privacy in federated learning applications in computer vision.\",\"DOI\":\
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t advance in privacy protection. In this context, a trusted curator aggregates parameters optimized in dece
ntralized fashion by multiple clients. The resulting model is then distributed back to all clients, ultimat
ely converging to a joint representative model without explicitly having to share the data. However, the pr
otocol is vulnerable to differential attacks, which could originate from any party contributing during fede
rated optimization. In such an attack, a client\u0027s contribution during training and information about t
heir data set is revealed through analyzing the distributed model. We tackle this problem and propose an al
gorithm for client sided differential privacy preserving federated optimization. The aim is to hide clients
\u0027 contributions during training, balancing the trade-off between privacy loss and model performance. E
mpirical studies suggest that given a sufficiently large number of participating clients, our proposed proc
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 including neural networks, consistently misclassify adversarial examples---inputs formed by applying small
 but intentionally worst-case perturbations to examples from the dataset, such that the perturbed input res
ults in the model outputting an incorrect answer with high confidence. Early attempts at explaining this ph
enomenon focused on nonlinearity and overfitting. We argue instead that the primary cause of neural network
s\u0027 vulnerability to adversarial perturbation is their linear nature. This explanation is supported by
new quantitative results while giving the first explanation of the most intriguing fact about them: their g
eneralization across architectures and training sets. Moreover, this view yields a simple and fast method o
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 how machine learning models leak information about the individual data records on which they were trained.
 We focus on the basic membership inference attack: given a data record and black-box access to a model, de
termine if the record was in the model\u0027s training dataset. To perform membership inference against a t
arget model, we make adversarial use of machine learning and train our own inference model to recognize dif
ferences in the target model\u0027s predictions on the inputs that it trained on versus the inputs that it
did not train on. We empirically evaluate our inference techniques on classification models trained by comm
ercial \\\"machine learning as a service\\\" providers such as Google and Amazon. Using realistic datasets
and classification tasks, including a hospital discharge dataset whose membership is sensitive from the pri
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ate training data in distributed learning systems such as Collaborative Learning and Federated Learning, et
c. Recently, Zhu et al. presented an approach which shows the possibility to obtain private training data f
rom the publicly shared gradients. In their Deep Leakage from Gradient (DLG) method, they synthesize the du
mmy data and corresponding labels with the supervision of shared gradients. However, DLG has difficulty in
convergence and discovering the ground-truth labels consistently. In this paper, we find that sharing gradi
ents definitely leaks the ground-truth labels. We propose a simple but reliable approach to extract accurat
e data from the gradients. Particularly, our approach can certainly extract the ground-truth labels as oppo
sed to DLG, hence we name it Improved DLG (iDLG). Our approach is valid for any differentiable model traine
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 how machine learning models leak information about the individual data records on which they were trained.
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:\"Exchanging gradients is a widely used method in modern multi-node machine learning system (e.g., distrib
uted training, collaborative learning). For a long time, people believed that gradients are safe to share:
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t of the book \\\"Differential Privacy in Artificial Intelligence: From Theory to Practice\\\" and provides
 an introduction to Differential Privacy. It starts by illustrating various attempts to protect data privac
y, emphasizing where and why they failed, and providing the key desiderata of a robust privacy definition.
It then defines the key actors, tasks, and scopes that make up the domain of privacy-preserving data analys
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iques based on neural networks are achieving remarkable results in a wide variety of domains. Often, the tr
aining of models requires large, representative datasets, which may be crowdsourced and contain sensitive i
nformation. The models should not expose private information in these datasets. Addressing this goal, we de
velop new algorithmic techniques for learning and a refined analysis of privacy costs within the framework
of differential privacy. Our implementation and experiments demonstrate that we can train deep neural netwo
rks with non-convex objectives, under a modest privacy budget, and at a manageable cost in software complex
ity, training efficiency, and model quality.\",\"container-title\":\"Proceedings of the 2016 ACM SIGSAC Con
ference on Computer and Communications Security\",\"DOI\":\"10.1145/2976749.2978318\",\"note\":\"arXiv:1607
.00133 [stat]\",\"page\":\"308-318\",\"source\":\"arXiv.org\",\"title\":\"Deep Learning with Differential P
rivacy\",\"URL\":\"http://arxiv.org/abs/1607.00133\",\"author\":[{\"family\":\"Abadi\",\"given\":\"Martín\"
},{\"family\":\"Chu\",\"given\":\"Andy\"},{\"family\":\"Goodfellow\",\"given\":\"Ian\"},{\"family\":\"McMah
an\",\"given\":\"H. Brendan\"},{\"family\":\"Mironov\",\"given\":\"Ilya\"},{\"family\":\"Talwar\",\"given\"
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ed\":{\"date-parts\":[[\"2016\",10,24]]}}},{\"id\":334,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/i
tems/NJ6EM485\"],\"itemData\":{\"id\":334,\"type\":\"article\",\"abstract\":\"The idea of federated learnin
g is to collaboratively train a neural network on a server. Each user receives the current weights of the n
etwork and in turns sends parameter updates (gradients) based on local data. This protocol has been designe
d not only to train neural networks data-efficiently, but also to provide privacy benefits for users, as th
eir input data remains on device and only parameter gradients are shared. But how secure is sharing paramet
er gradients? Previous attacks have provided a false sense of security, by succeeding only in contrived set
tings - even for a single image. However, by exploiting a magnitude-invariant loss along with optimization
strategies based on adversarial attacks, we show that is is actually possible to faithfully reconstruct ima
ges at high resolution from the knowledge of their parameter gradients, and demonstrate that such a break o
f privacy is possible even for trained deep networks. We analyze the effects of architecture as well as par
ameters on the difficulty of reconstructing an input image and prove that any input to a fully connected la
yer can be reconstructed analytically independent of the remaining architecture. Finally we discuss setting
s encountered in practice and show that even averaging gradients over several iterations or several images
does not protect the user\u0027s privacy in federated learning applications in computer vision.\",\"DOI\":\
"10.48550/arXiv.2003.14053\",\"note\":\"arXiv:2003.14053 [cs]\",\"number\":\"arXiv:2003.14053\",\"publisher
\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Inverting Gradients -- How easy is it to break privacy in
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ah\"},{\"family\":\"Moeller\",\"given\":\"Michael\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"iss
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tems/47ZF3Q7E\"],\"itemData\":{\"id\":345,\"type\":\"article\",\"abstract\":\"Federated learning is a recen
t advance in privacy protection. In this context, a trusted curator aggregates parameters optimized in dece
ntralized fashion by multiple clients. The resulting model is then distributed back to all clients, ultimat
ely converging to a joint representative model without explicitly having to share the data. However, the pr
otocol is vulnerable to differential attacks, which could originate from any party contributing during fede
rated optimization. In such an attack, a client\u0027s contribution during training and information about t
heir data set is revealed through analyzing the distributed model. We tackle this problem and propose an al
gorithm for client sided differential privacy preserving federated optimization. The aim is to hide clients
\u0027 contributions during training, balancing the trade-off between privacy loss and model performance. E
mpirical studies suggest that given a sufficiently large number of participating clients, our proposed proc
edure can maintain client-level differential privacy at only a minor cost in model performance.\",\"DOI\":\
"10.48550/arXiv.1712.07557\",\"note\":\"arXiv:1712.07557 [cs]\",\"number\":\"arXiv:1712.07557\",\"publisher
\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Differentially Private Federated Learning: A Client Level
 Perspective\",\"title-short\":\"Differentially Private Federated Learning\",\"URL\":\"http://arxiv.org/abs
/1712.07557\",\"author\":[{\"family\":\"Geyer\",\"given\":\"Robin C.\"},{\"family\":\"Klein\",\"given\":\"T
assilo\"},{\"family\":\"Nabi\",\"given\":\"Moin\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issue
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s/TF5PCCFI\"],\"itemData\":{\"id\":338,\"type\":\"article\",\"abstract\":\"This paper presents the TermScie
nces portal, which deals with the implementation of a conceptual model that uses the recent ISO 16642 stand
ard (Terminological Markup Framework). This standard turns out to be suitable for concept modeling since it
 allowed for organizing the original resources by concepts and to associate the various terms for a given c
oncept. Additional structuring is produced by sharing conceptual relationships, that is, cross-linking of r
esource results through the introduction of semantic relations which may have initially be missing.\",\"DOI
\":\"10.48550/arXiv.cs/0604027\",\"note\":\"arXiv:cs/0604027\",\"number\":\"arXiv:cs/0604027\",\"publisher\
":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Unification of multi-lingual scientific terminological res
ources using the ISO 16642 standard. The TermSciences initiative\",\"URL\":\"http://arxiv.org/abs/cs/060402
7\",\"author\":[{\"family\":\"Khayari\",\"given\":\"Majid\"},{\"family\":\"Schneider\",\"given\":\"Stéphane
\"},{\"family\":\"Kramer\",\"given\":\"Isabelle\"},{\"family\":\"Romary\",\"given\":\"Laurent\"},{\"family\
":\"Collaboration\",\"given\":\"the\",\"dropping-particle\":\"termsciences\"}],\"accessed\":{\"date-parts\"
:[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2006\",4,7]]}}},{\"id\":328,\"uris\":[\"http://zotero.or
g/users/local/M0u71wYg/items/5TCXCTZG\"],\"itemData\":{\"id\":328,\"type\":\"article\",\"abstract\":\"We pr
esent a new class of statistical de-anonymization attacks against high-dimensional micro-data, such as indi
vidual preferences, recommendations, transaction records and so on. Our techniques are robust to perturbati
on in the data and tolerate some mistakes in the adversary\u0027s background knowledge. We apply our de-ano
nymization methodology to the Netflix Prize dataset, which contains anonymous movie ratings of 500,000 subs
cribers of Netflix, the world\u0027s largest online movie rental service. We demonstrate that an adversary
who knows only a little bit about an individual subscriber can easily identify this subscriber\u0027s recor
d in the dataset. Using the Internet Movie Database as the source of background knowledge, we successfully
identified the Netflix records of known users, uncovering their apparent political preferences and other po
tentially sensitive information.\",\"DOI\":\"10.48550/arXiv.cs/0610105\",\"note\":\"arXiv:cs/0610105\",\"nu
mber\":\"arXiv:cs/0610105\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"How To Break Anony
mity of the Netflix Prize Dataset\",\"URL\":\"http://arxiv.org/abs/cs/0610105\",\"author\":[{\"family\":\"N
arayanan\",\"given\":\"Arvind\"},{\"family\":\"Shmatikov\",\"given\":\"Vitaly\"}],\"accessed\":{\"date-part
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Exchanging gradients is a widely used method in modern multi-node machine learning system (e.g., distribute
d training, collaborative learning). For a long time, people believed that gradients are safe to share: i.e
., the training data will not be leaked by gradient exchange. However, we show that it is possible to obtai
n the private training data from the publicly shared gradients. We name this leakage as Deep Leakage from G
radient and empirically validate the effectiveness on both computer vision and natural language processing
tasks. Experimental results show that our attack is much stronger than previous approaches: the recovery is
 pixel-wise accurate for images and token-wise matching for texts. We want to raise people\u0027s awareness

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 to rethink the gradient\u0027s safety. Finally, we discuss several possible strategies to prevent such dee
p leakage. The most effective defense method is gradient pruning.\",\"DOI\":\"10.48550/arXiv.1906.08935\",\
"note\":\"arXiv:1906.08935 [cs]\",\"number\":\"arXiv:1906.08935\",\"publisher\":\"arXiv\",\"source\":\"arXi
v.org\",\"title\":\"Deep Leakage from Gradients\",\"URL\":\"http://arxiv.org/abs/1906.08935\",\"author\":[{
\"family\":\"Zhu\",\"given\":\"Ligeng\"},{\"family\":\"Liu\",\"given\":\"Zhijian\"},{\"family\":\"Han\",\"g
iven\":\"Song\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2019\",12,
19]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The
 anonymity of data points (gradient) which can be reverse engineered to arrive back at the original data po
int is solved by making the data point a probability data point instead of deterministic data point. This k
ind of adds a layer of mathematical fog so that no single tile in a Mosaic provide enough contrast to ident
ify a specific individual who put the tile there. The tile is basically an analogy, that let’s assume there
 is a portrait or there is a Mosaic made up of many different tiles and each tile is placed by different in
dividual. Also, assume that each child has a meta data which does not directly name the person who put the
tile up there. However, it details the other information about the person like their address, their phone n
umber and the date at which that I was placed up there. An intruder can reverse engineer this information b
y making sure that particular address, phone number, et cetera Max, their neighbours and that the neighbour
 was absent in their house on that particular day on which the tile was placed, so this clearly indicate th
e person was, even though their name was not taken. This is how the gradient cause problem and the attacks
and the intruders can benefit out of it. Adding a mathematical fog and a proper fog helps a bit. (Abadi et
al., 2016d; Geiping et al., 2020b; Geyer et al., 2018c; Khayari et al., 2006c; Narayanan \u0026 Shmatikov,
2007b; L. Zhu et al., 2019b) (refer to Figure 2)",
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lainCitation\":\"(Acar et al., 2017b; García-Álvarez et al., 2017; Hill, 2009b; McLean, 2019; Murray, 2016)
\",\"noteIndex\":0},\"citationItems\":[{\"id\":353,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items
/6U363UT2\"],\"itemData\":{\"id\":353,\"type\":\"article\",\"abstract\":\"Legacy encryption systems depend
on sharing a key (public or private) among the peers involved in exchanging an encrypted message. However,
this approach poses privacy concerns. Especially with popular cloud services, the control over the privacy
of the sensitive data is lost. Even when the keys are not shared, the encrypted material is shared with a t
hird party that does not necessarily need to access the content. Moreover, untrusted servers, providers, an
d cloud operators can keep identifying elements of users long after users end the relationship with the ser
vices. Indeed, Homomorphic Encryption (HE), a special kind of encryption scheme, can address these concerns
 as it allows any third party to operate on the encrypted data without decrypting it in advance. Although t
his extremely useful feature of the HE scheme has been known for over 30 years, the first plausible and ach
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ncrypted data, was introduced by Craig Gentry in 2009. Even though this was a major achievement, different
implementations so far demonstrated that FHE still needs to be improved significantly to be practical on ev
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cryption (PHE) and Somewhat Homomorphic Encryption (SWHE), which are important pillars of achieving FHE. Th
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rthermore, the implementations and recent improvements in Gentry-type FHE schemes are also surveyed. Finall
y, further research directions are discussed. This survey is intended to give a clear knowledge and foundat
ion to researchers and practitioners interested in knowing, applying, as well as extending the state of the
 art HE, PHE, SWHE, and FHE systems.\",\"DOI\":\"10.48550/arXiv.1704.03578\",\"note\":\"arXiv:1704.03578 [c
s]\",\"number\":\"arXiv:1704.03578\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"A Survey
on Homomorphic Encryption Schemes: Theory and Implementation\",\"title-short\":\"A Survey on Homomorphic En
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"Abbas\"},{\"family\":\"Aksu\",\"given\":\"Hidayet\"},{\"family\":\"Uluagac\",\"given\":\"A. Selcuk\"},{\"f
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parts\":[[\"2017\",10,6]]}}},{\"id\":365,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/44M2LC39\
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 motion can generate entanglement between artificial atoms and protect them from spontaneous emission. We c
onsider a pair of superconducting qubits coupled to a resonator mode, where the modulation of the coupling
strength can mimic the harmonic motion of the qubits at relativistic speeds, generating acceleration radiat
ion. We find the optimal feasible conditions for generating a stationary entangled state between the qubits
 when they are initially prepared in their ground state. Furthermore, we analyze the effects of motion on t
he probability of spontaneous emission in the standard scenarios of single-atom and two-atom superradiance,
 where one or two excitations are initially present. Finally, we show that relativistic motion induces sub-
radiance and can generate a Zeno-like effect, preserving the excitations from radiative decay.\",\"containe
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1\",\"journalAbbreviation\":\"Sci Rep\",\"note\":\"arXiv:1606.06170 [quant-ph]\",\"page\":\"657\",\"source\
":\"arXiv.org\",\"title\":\"Entanglement of superconducting qubits via acceleration radiation\",\"URL\":\"h
ttp://arxiv.org/abs/1606.06170\",\"volume\":\"7\",\"author\":[{\"family\":\"García-Álvarez\",\"given\":\"La
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e present a descent style, Bockstein spectral sequence computing Ext over the motivic Steenrod algebra over
 $\\\\R$ and related sub-Hopf algebras. We demonstrate the workings of this spectral sequence in several ex
amples, providing motivic analogues to the classical computations related to BP\u003cn\u003e and ko.\",\"DO
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sher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Ext and the Motivic Steenrod Algebra over $\\\\R$\",\
"URL\":\"http://arxiv.org/abs/0904.1998\",\"author\":[{\"family\":\"Hill\",\"given\":\"Michael A.\"}],\"acc
essed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2009\",4,13]]}}},{\"id\":361,\"ur
is\":[\"http://zotero.org/users/local/M0u71wYg/items/7F48IIIV\"],\"itemData\":{\"id\":361,\"type\":\"articl
e-journal\",\"abstract\":\"Let f be a polynomial over the complex numbers with an isolated singularity at 0
. We show that the multiplicity and the log canonical threshold of f at 0 are invariants of the link of f v
iewed as a contact submanifold of the sphere. This is done by first constructing a spectral sequence conver
ging to the fixed point Floer cohomology of any iterate of the Milnor monodromy map whose E^1 page is expli
citly described in terms of a log resolution of f. This spectral sequence is a generalization of a formula
by A\u0027Campo. By looking at this spectral sequence, we get a purely Floer theoretic description of the m
ultiplicity and log canonical threshold of f.\",\"container-title\":\"Geometry \u0026 Topology\",\"DOI\":\"
10.2140/gt.2019.23.957\",\"ISSN\":\"1364-0380, 1465-3060\",\"issue\":\"2\",\"journalAbbreviation\":\"Geom.
Topol.\",\"note\":\"arXiv:1608.07541 [math]\",\"page\":\"957-1056\",\"source\":\"arXiv.org\",\"title\":\"Fl
oer Cohomology, Multiplicity and the Log Canonical Threshold\",\"URL\":\"http://arxiv.org/abs/1608.07541\",
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rs/local/M0u71wYg/items/UGIK3S8L\"],\"itemData\":{\"id\":369,\"type\":\"article\",\"abstract\":\"We review
strategies for differentiating matrix-based computations, and derive symbolic and algorithmic update rules
for differentiating expressions containing the Cholesky decomposition. We recommend new `blocked\u0027 algo
rithms, based on differentiating the Cholesky algorithm DPOTRF in the LAPACK library, which uses `Level 3\u
0027 matrix-matrix operations from BLAS, and so is cache-friendly and easy to parallelize. For large matric
es, the resulting algorithms are the fastest way to compute Cholesky derivatives, and are an order of magni
tude faster than the algorithms in common usage. In some computing environments, symbolically-derived updat
es are faster for small matrices than those based on differentiating Cholesky algorithms. The symbolic and
algorithmic approaches can be combined to get the best of both worlds.\",\"DOI\":\"10.48550/arXiv.1602.0752
7\",\"note\":\"arXiv:1602.07527 [stat]\",\"number\":\"arXiv:1602.07527\",\"publisher\":\"arXiv\",\"source\"
:\"arXiv.org\",\"title\":\"Differentiation of the Cholesky decomposition\",\"URL\":\"http://arxiv.org/abs/1
602.07527\",\"author\":[{\"family\":\"Murray\",\"given\":\"Iain\"}],\"accessed\":{\"date-parts\":[[\"2026\"
,2,16]]},\"issued\":{\"date-parts\":[[\"2016\",2,24]]}}}],\"schema\":\"https://github.com/citation-style-la
nguage/schema/raw/master/csl-citation.json\"} Homomorphic encryption solve this problem by performing calcu
lations and computation on encrypted data instead of exposing the data to the processor and the operating s
ystem. The result which is obtained by computing on the encrypted data is also encrypted. However, when dec
rypted the decrypted data is almost same as the result obtained by performing the same computation on the u
nencrypted data. So in other words, let’s say we have two cases in which we are performing same competition
 on two unencrypted data and performing the same using homomorphic encryption. The result obtained from the
 homomorphic. Encryption computation will be similar to the one which we obtain from the unencrypted data.
However, hundred percent accuracy is not achievable. (Acar et al., 2017b; García-Álvarez et al., 2017; Hill
, 2009b; McLean, 2019; Murray, 2016)",
        "title":  "A Survey on Homomorphic Encryption Schemes: Theory and Implementation",
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lainCitation\":\"(Acar et al., 2017b; García-Álvarez et al., 2017; Hill, 2009b; McLean, 2019; Murray, 2016)
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/6U363UT2\"],\"itemData\":{\"id\":353,\"type\":\"article\",\"abstract\":\"Legacy encryption systems depend
on sharing a key (public or private) among the peers involved in exchanging an encrypted message. However,
this approach poses privacy concerns. Especially with popular cloud services, the control over the privacy
of the sensitive data is lost. Even when the keys are not shared, the encrypted material is shared with a t
hird party that does not necessarily need to access the content. Moreover, untrusted servers, providers, an
d cloud operators can keep identifying elements of users long after users end the relationship with the ser
vices. Indeed, Homomorphic Encryption (HE), a special kind of encryption scheme, can address these concerns
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his extremely useful feature of the HE scheme has been known for over 30 years, the first plausible and ach
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ncrypted data, was introduced by Craig Gentry in 2009. Even though this was a major achievement, different
implementations so far demonstrated that FHE still needs to be improved significantly to be practical on ev
ery platform. First, we present the basics of HE and the details of the well-known Partially Homomorphic En
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y, further research directions are discussed. This survey is intended to give a clear knowledge and foundation to researchers and practitioners interested in knowing, applying, as well as extending the state of the art HE, PHE, SWHE, and FHE systems.\",\"DOI\":\"10.48550/arXiv.1704.03578\",\"note\":\"arXiv:1704.03578 [cs]\",\"number\":\"arXiv:1704.03578\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"A Survey on Homomorphic Encryption Schemes: Theory and Implementation\",\"title-short\":\"A Survey on Homomorphic Encryption Schemes\",\"URL\":\"http://arxiv.org/abs/1704.03578\",\"author\":[{\"family\":\"Acar\",\"given\":\"Abbas\"},{\"family\":\"Aksu\",\"given\":\"Hidayet\"},{\"family\":\"Uluagac\",\"given\":\"A. Selcuk\"},{\"family\":\"Conti\",\"given\":\"Mauro\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2017\",10,6]]}}},{\"id\":365,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/44M2LC39\"],\"itemData\":{\"id\":365,\"type\":\"article-journal\",\"abstract\":\"We show that simulated relativistic motion can generate entanglement between artificial atoms and protect them from spontaneous emission. We consider a pair of superconducting qubits coupled to a resonator mode, where the modulation of the coupling strength can mimic the harmonic motion of the qubits at relativistic speeds, generating acceleration radiation. We find the optimal feasible conditions for generating a stationary entangled state between the qubits when they are initially prepared in their ground state. Furthermore, we analyze the effects of motion on the probability of spontaneous emission in the standard scenarios of single-atom and two-atom superradiance, where one or two excitations are initially present. Finally, we show that relativistic motion induces sub-radiance and can generate a Zeno-like effect, preserving the excitations from radiative decay.\",\"container-title\":\"Scientific Reports\",\"DOI\":\"10.1038/s41598-017-00770-z\",\"ISSN\":\"2045-2322\",\"issue\":\"1\",\"journalAbbreviation\":\"Sci Rep\",\"note\":\"arXiv:1606.06170 [quant-ph]\",\"page\":\"657\",\"source\":\"arXiv.org\",\"title\":\"Entanglement of superconducting qubits via acceleration radiation\",\"URL\":\"http://arxiv.org/abs/1606.06170\",\"volume\":\"7\",\"author\":[{\"family\":\"García-Álvarez\",\"given\":\"Laura\"},{\"family\":\"Felicetti\",\"given\":\"Simone\"},{\"family\":\"Rico\",\"given\":\"Enrique\"},{\"family\":\"Solano\",\"given\":\"Enrique\"},{\"family\":\"Sabín\",\"given\":\"Carlos\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2017\",4,6]]}}},{\"id\":357,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ECAUE6XP\"],\"itemData\":{\"id\":357,\"type\":\"article\",\"abstract\":\"We present a descent style, Bockstein spectral sequence computing Ext over the motivic Steenrod algebra over $\\\\R$ and related sub-Hopf algebras. We demonstrate the workings of this spectral sequence in several examples, providing motivic analogues to the classical computations related to BP\u003cn\u003e and ko.\",\"DOI\":\"10.48550/arXiv.0904.1998\",\"note\":\"arXiv:0904.1998 [math]\",\"number\":\"arXiv:0904.1998\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Ext and the Motivic Steenrod Algebra over $\\\\R$\",\"URL\":\"http://arxiv.org/abs/0904.1998\",\"author\":[{\"family\":\"Hill\",\"given\":\"Michael A.\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2009\",4,13]]}}},{\"id\":361,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7F48IIIV\"],\"itemData\":{\"id\":361,\"type\":\"article-journal\",\"abstract\":\"Let f be a polynomial over the complex numbers with an isolated singularity at 0. We show that the multiplicity and the log canonical threshold of f at 0 are invariants of the link of f viewed as a contact submanifold of the sphere. 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Finally, we show that relativistic motion induces sub-radiance and can generate a Zeno-like effect, preserving the excitations from radiative decay.\",\"container-title\":\"Scientific Reports\",\"DOI\":\"10.1038/s41598-017-00770-z\",\"ISSN\":\"2045-2322\",\"issue\":\"1\",\"journalAbbreviation\":\"Sci Rep\",\"note\":\"arXiv:1606.06170 [quant-ph]\",\"page\":\"657\",\"source\":\"arXiv.org\",\"title\":\"Entanglement of superconducting qubits via acceleration radiation\",\"URL\":\"http://arxiv.org/abs/1606.06170\",\"volume\":\"7\",\"author\":[{\"family\":\"García-Álvarez\",\"given\":\"Laura\"},{\"family\":\"Felicetti\",\"given\":\"Simone\"},{\"family\":\"Rico\",\"given\":\"Enrique\"},{\"family\":\"Solano\",\"given\":\"Enrique\"},{\"family\":\"Sabín\",\"given\":\"Carlos\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2017\",4,6]]}}},{\"id\":357,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ECAUE6XP\"],\"itemData\":{\"id\":357,\"type\":\"article\",\"abstract\":\"We present a descent style, Bockstein spectral sequence computing Ext over the motivic Steenrod algebra over $\\\\R$ and related sub-Hopf algebras. We demonstrate the workings of this spectral sequence in several examples, providing motivic analogues to the classical computations related to BP\u003cn\u003e and ko.\",\"DOI\":\"10.48550/arXiv.0904.1998\",\"note\":\"arXiv:0904.1998 [math]\",\"number\":\"arXiv:0904.1998\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Ext and the Motivic Steenrod Algebra over $\\\\R$\",\"URL\":\"http://arxiv.org/abs/0904.1998\",\"author\":[{\"family\":\"Hill\",\"given\":\"Michael A.\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2009\",4,13]]}}},{\"id\":361,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7F48IIIV\"],\"itemData\":{\"id\":361,\"type\":\"article-journal\",\"abstract\":\"Let f be a polynomial over the complex numbers with an isolated singularity at 0. We show that the multiplicity and the log canonical threshold of f at 0 are invariants of the link of f viewed as a contact submanifold of the sphere. This is done by first constructing a spectral sequence converging to the fixed point Floer cohomology of any iterate of the Milnor monodromy map whose E^1 page is explicitly described in terms of a log resolution of f. This spectral sequence is a generalization of a formula by A\u0027Campo. By looking at this spectral sequence, we get a purely Floer theoretic description of the multiplicity and log canonical threshold of f.\",\"container-title\":\"Geometry \u0026 Topology\",\"DOI\":\"10.2140/gt.2019.23.957\",\"ISSN\":\"1364-0380, 1465-3060\",\"issue\":\"2\",\"journalAbbreviation\":\"Geom. Topol.\",\"note\":\"arXiv:1608.07541 [math]\",\"page\":\"957-1056\",\"source\":\"arXiv.org\",\"title\":\"Floer Cohomology, Multiplicity and the Log Canonical Threshold\",\"URL\":\"http://arxiv.org/abs/1608.07541\",\"volume\":\"23\",\"author\":[{\"family\":\"McLean\",\"given\":\"Mark\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2019\",4,8]]}}},{\"id\":369,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/UGIK3S8L\"],\"itemData\":{\"id\":369,\"type\":\"article\",\"abstract\":\"We review strategies for differentiating matrix-based computations, and derive symbolic and algorithmic update rules for differentiating expressions containing the Cholesky decomposition. We recommend new `blocked\u0027 algorithms, based on differentiating the Cholesky algorithm DPOTRF in the LAPACK library, which uses `Level 3\u0027 matrix-matrix operations from BLAS, and so is cache-friendly and easy to parallelize. For large matrices, the resulting algorithms are the fastest way to compute Cholesky derivatives, and are an order of magnitude faster than the algorithms in common usage. In some computing environments, symbolically-derived updates are faster for small matrices than those based on differentiating Cholesky algorithms. The symbolic and algorithmic approaches can be combined to get the best of both worlds.\",\"DOI\":\"10.48550/arXiv.1602.07527\",\"note\":\"arXiv:1602.07527 [stat]\",\"number\":\"arXiv:1602.07527\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Differentiation of the Cholesky decomposition\",\"URL\":\"http://arxiv.org/abs/1602.07527\",\"author\":[{\"family\":\"Murray\",\"given\":\"Iain\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2016\",2,24]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Homomorphic encryption solve this problem by performing calculations and computation on encrypted data instead of exposing the data to the processor and the operating system. The result which is obtained by computing on the encrypted data is also encrypted. However, when decrypted the decrypted data is almost same as the result obtained by performing the same computation on the unencrypted data. So in other words, let’s say we have two cases in which we are performing same competition on two unencrypted data and performing the same using homomorphic encryption. The result obtained from the homomorphic. Encryption computation will be similar to the one which we obtain from the unencrypted data. However, hundred percent accuracy is not achievable. (Acar et al., 2017b; García-Álvarez et al., 2017; Hill, 2009b; McLean, 2019; Murray, 2016)",
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l security parameter for an encrypted control system under a network eavesdropper and a malicious controller server who attempt to identify system parameters using a least squares method. The security parameter design is achieved based on a modification of conventional homomorphic encryption for improving a sample deciphering time and a novel sample identifying complexity, characterized by controllability Gramians and the variance ratio of identification input to system noise. The effectiveness of the proposed design method for a security parameter is demonstrated through numerical simulations.\",\"DOI\":\"10.48550/arXiv.2303.13059\",\"note\":\"arXiv:2303.13059 [eess]\",\"number\":\"arXiv:2303.13059\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Optimal Security Parameter for Encrypted Control Systems Against Eavesdropper and Malicious Server\",\"URL\":\"http://arxiv.org/abs/2303.13059\",\"author\":[{\"family\":\"Teranishi\",\"given\":\"Kaoru\"},{\"family\":\"Kogiso\",\"given\":\"Kiminao\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2023\",3,23]]}},\"locator\":\"3\",\"label\":\"page\"}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} This equation states that performing the operation on the ciphertexts and then decrypting is equivalent to performing the operation on the plaintexts. (Teranishi \u0026 Kogiso, 2023, p. 3)",
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Our novel protocol achieves an online speedup of 17$\\\\times$ in nonlinear operations compared to state-of-the-art solutions while reducing communication overhead. To further enhance performance, SecONNds employs Number Theoretic Transform (NTT) preprocessing and leverages GPU acceleration for homomorphic encryption operations, resulting in speedups of 1.6$\\\\times$ on CPU and 2.2$\\\\times$ on GPU for linear operations. We also present SecONNds-P, a bit-exact variant that ensures verifiable full-precision results in secure computation, matching the results of plaintext computations. Evaluated on a 37-bit quantized SqueezeNet model, SecONNds achieves an end-to-end inference time of 2.8 s on GPU and 3.6 s on CPU, with a total communication of just 420 MiB. SecONNds\u0027 efficiency and reduced computational load make it well-suited for deploying privacy-sensitive applications in resource-constrained environments. SecONNds is open source and can be accessed from: https://github.com/shashankballa/SecONNds.\",\"DOI\":\"10.48550/arXiv.2506.11586\",\"note\":\"arXiv:2506.11586 [cs]\",\"number\":\"arXiv:2506.11586\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"SecONNds: Secure Outsourced Neural Network Inference on ImageNet\",\"title-short\":\"SecONNds\",\"URL\":\"http://arxiv.org/abs/2506.11586\",\"author\":[{\"family\":\"Balla\",\"given\":\"Shashank\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2025\",6,13]]}},\"locator\":\"6\",\"label\":\"page\"},{\"id\":384,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JCJS33WC\"],\"itemData\":{\"id\":384,\"type\":\"article\",\"abstract\":\"In this work, we provide an industry research view for approaching the design, deployment, and operation of trustworthy Artificial Intelligence (AI) inference systems. Such systems provide customers with timely, informed, and customized inferences to aid their decision, while at the same time utilizing appropriate security protection mechanisms for AI models. Additionally, such systems should also use Privacy-Enhancing Technologies (PETs) to protect customers\u0027 data at any time. To approach the subject, we start by introducing current trends in AI inference systems. We continue by elaborating on the relationship between Intellectual Property (IP) and private data protection in such systems. Regarding the protection mechanisms, we survey the security and privacy building blocks instrumental in designing, building, deploying, and operating private AI inference systems. For example, we highlight opportunities and challenges in AI systems using trusted execution environments combined with more recent advances in cryptographic techniques to protect data in use. Finally, we outline areas of further development that require the global collective attention of industry, academia, and government researchers to sustain the operation of trustworthy AI inference systems.\",\"DOI\":\"10.48550/arXiv.2008.04449\",\"note\":\"arXiv:2008.04449 [cs]\",\"number\":\"arXiv:2008.04449\",\"publisher\":\"arXiv\",\"source\":\"arXiv.org\",\"title\":\"Trustworthy AI Inference Systems: An Industry Research View\",\"title-short\":\"Trustworthy AI Inference Systems\",\"URL\":\"http://arxiv.org/abs/2008.04449\",\"author\":[{\"family\":\"Cammarota\",\"given\":\"Rosario\"},{\"family\":\"Schunter\",\"given\":\"Matthias\"},{\"family\":\"Rajan\",\"given\":\"Anand\"},{\"family\":\"Boemer\",\"given\":\"Fabian\"},{\"family\":\"Kiss\",\"given\":\"Ágnes\"},{\"family\":\"Treiber\",\"given\":\"Amos\"},{\"family\":\"Weinert\",\"given\":\"Christian\"},{\"family\":\"Schneider\",\"given\":\"Thomas\"},{\"family\":\"Stapf\",\"given\":\"Emmanuel\"},{\"family\":\"Sadeghi\",\"given\":\"Ahmad-Reza\"},{\"family\":\"Demmler\",\"given\":\"Daniel\"},{\"family\":\"Stock\",\"given\":\"Joshua\"},{\"family\":\"Chen\",\"given\":\"Huili\"},{\"family\":\"Hussain\",\"given\":\"Siam Umar\"},{\"family\":\"Riazi\",\"given\":\"Sadegh\"},{\"family\":\"Koushanfar\",\"given\":\"Farinaz\"},{\"family\":\"Gupta\",\"given\":\"Saransh\"},{\"family\":\"Rosing\",\"given\":\"Tajan Simunic\"},{\"family\":\"Chaudhuri\",\"given\":\"Kamalika\"},{\"family\":\"Nejatollahi\",\"given\":\"Hamid\"},{\"family\":\"Dutt\",\"given\":\"Nikil\"},{\"family\":\"Imani\",\"given\":\"Mohsen\"},{\"family\":\"Laine\",\"given\":\"Kim\"},{\"family\":\"Dubey\",\"given\":\"Anuj\"},{\"family\":\"Aysu\",\"

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 subject, we start by introducing current trends in AI inference systems. We continue by elaborating on the
 relationship between Intellectual Property (IP) and private data protection in such systems. Regarding the
 protection mechanisms, we survey the security and privacy building blocks instrumental in designing, build
ing, deploying, and operating private AI inference systems. For example, we highlight opportunities and cha
llenges in AI systems using trusted execution environments combined with more recent advances in cryptograp
hic techniques to protect data in use. Finally, we outline areas of further development that require the gl
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We show that TPACAS, unlike previous works, preserves the following privacies relevant to an auction: agent privacy, the identities of the losing bidders must not be revealed to any other agent except the auctioneer (AU), bid privacy, the bid values must be hidden from the other agents as well as the AU and bid-topology privacy, the items for which the agents are bidding must be hidden from the other agents as well as the AU. We demonstrate the practicality of TPACAS through simulations. Lastly, we also look at TPACAS\u0027 implementation over a publicly distributed ledger, such as the Ethereum blockchain.\",\"DOI\":\"10.48550/ARXIV.1906.06567\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 1\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"A Practical Solution to Yao\u0027s Millionaires\u0027 Problem and Its Application in Designing Secure Combinatorial Auction\",\"URL\":\"https://arxiv.org/abs/1906.06567\",\"author\":[{\"family\":\"Damle\",\"given\":\"Sankarshan\"},{\"family\":\"Faltings\",\"given\":\"Boi\"},{\"family\":\"Gujar\",\"given\":\"Sujit\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2019\"]]}},\"locator\":\"2\",\"label\":\"page\"},{\"id\":392,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YJ5SBNLM\"],\"itemData\":{\"id\":392,\"type\":\"article\",\"abstract\":\"Increasing use of computers and networks in business, government, recreation, and almost all aspects of daily life has led to a proliferation of online sensitive data about individuals and organizations. Consequently, concern about the privacy of these data has become a top priority, particularly those data that are created and used in electronic commerce. There have been many formulations of privacy and, unfortunately, many negative results about the feasibility of maintaining privacy of sensitive data in realistic networked environments. We formulate communication-complexity-based definitions, both worst-case and average-case, of a problem\u0027s privacy-approximation ratio. We use our definitions to investigate the extent to which approximate privacy is achievable in two standard problems: the second-price Vickrey auction and the millionaires problem of Yao. For both the second-price Vickrey auction and the millionaires problem, we show that not only is perfect privacy impossible or infeasibly costly to achieve, but even close approximations of perfect privacy suffer from the same lower bounds. 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Thus, the parties may conspire to send all their processed results to a trusted third party (perhaps the data provider) at the conclusion of the computations, with only the trusted third party being able to view the final results. Secure multiparty computations for privacy-preserving machine-learning turn out to be possible using solely standard floating-point arithmetic, at least with a carefully controlled leakage of information less than the loss of accuracy due to roundoff, all backed by rigorous mathematical proofs of worst-case bounds on information loss and numerical stability in finite-precision arithmetic. Numerical examples illustrate the high performance attained on commodity off-the-shelf hardware for generalized linear models, including ordinary linear least-squares regression, binary and multinomial logistic regression, probit regression, and Poisson regression.\",\"DOI\":\"10.48550/ARXIV.2001.03192\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 1\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"Secure multiparty computations in floating-point arithmetic\",\"URL\":\"https://arxiv.org/abs/2001.03192\",\"author\":[{\"family\":\"Guo\",\"given\":\"Chuan\"},{\"family\":\"Hannun\",\"given\":\"Awni\"},{\"family\":\"Knott\",\"given\":\"Brian\"},{\"family\":\"Maaten\",\"given\":\"Laurens\",\"non-dropping-particle\":\"van der\"},{\"family\":\"Tygert\",\"given\":\"Mark\"},{\"family\":\"Zhu\",\"given\":\"Ruiyu\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2020\"]]}},\"locator\":\"1\",\"label\":\"page\"},{\"id\":391,\"uris\":[\"http://zotero

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all parties to put back together all the shares). Thus, the parties may conspire to send all their processe
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only the trusted third party being able to view the final results. Secure multiparty computations for priva
cy-preserving machine-learning turn out to be possible using solely standard floating-point arithmetic, at
least with a carefully controlled leakage of information less than the loss of accuracy due to roundoff, al
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 suggest two new methodologies for the design of efficient secure protocols, that differ with respect to th
eir underlying computational models. In one methodology we utilize the communication complexity tree (or br
anching for f and transform it into a secure protocol. In other words, \\\"any function f that can be compu
ted using communication complexity c can be can be computed securely using communication complexity that is
 polynomial in c and a security parameter\\\". The second methodology uses the circuit computing f, enhance
d with look-up tables as its underlying computational model. It is possible to simulate any RAM machine in
this model with polylogarithmic blowup. Hence it is possible to start with a computation of f on a RAM mach
ine and transform it into a secure protocol. We show many applications of these new methodologies resulting
 in protocols efficient either in communication or in computation. In particular, we exemplify a protocol f
or the \\\"millionaires problem\\\", where two participants want to compare their values but reveal no othe
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-party or TTP. So in an ideal world, this TTP is a trusted party, which computes the function faithfully. A
ll the parties sent their inputs to this TTP to compute the function, the TTP then computes the function wi
thout revealing anything to the parties, and then it sends back the output to the parties without reviewing
 the inputs of any party to any other party. The only input that a party knows their own inputs and nobody
else’s, and the other thing that they know is the output that TTP has given to them, that way they come to
know who is the richest millionaire among all the parties. (Ah-Fat \u0026 Huth, 2018, p. 3; Damle et al., 2
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text of multi-party computation (MPC); Furthermore, an instance of FL protocol along Kairouz et al.\u0027s
definition can be viewed as an implementation of the defined $m$-ary functionality. As such, an instance of
 FL procedure is also an instance of MPC protocol. In short, FL is a subset of MPC. To privately computing
the defined FL (m-ary) functionality, various techniques such as homomorphic encryption (HE), secure multi-
party computation (SMPC) and differential privacy (DP) have been deployed. In the second part, we are able
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derated recommender systems are still many miles to be completely safe and effective. Such systems that all
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 of raw information concerning users are likely to fall victim to sophisticated attacks such as inference a
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by every learning round to decode sensitive user information, e.g. personal preferences or viewing history,
 which should be regarded as a breach of the privacy principles of the framework itself. (Q. Hu \u0026 Song
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri

vacy-preserving way. Experiments on two real-world datasets show that our method can reduce the computation and communication cost on clients while keep promising model performance.\",\"DOI\":\"10.48550/ARXIV.2109.05446\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 3\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"Efficient-FedRec: Efficient Federated Learning Framework for Privacy-Preserving News Recommendation\",\"title-short\":\"Efficient-FedRec\",\"URL\":\"https://arxiv.org/abs/2109.05446\",\"author\":[{\"family\":\"Yi\",\"given\":\"Jingwei\"},{\"family\":\"Wu\",\"given\":\"Fangzhao\"},{\"family\":\"Wu\",\"given\":\"Chuhan\"},{\"family\":\"Liu\",\"given\":\"Ruixuan\"},{\"family\":\"Sun\",\"given\":\"Guangzhong\"},{\"family\":\"Xie\",\"given\":\"Xing\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2021\"]]}},{\"id\":502,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3LGAHGWS\"],\"itemData\":{\"id\":502,\"type\":\"article-journal\",\"container-title\":\"Acm Transactions on Information Systems\",\"DOI\":\"10.1145/3779442\",\"issue\":\"2\",\"journalAbbreviation\":\"Acm Transactions on Information Systems\",\"page\":\"1-28\",\"title\":\"Beyond Similarity: Personalized Federated Recommendation With Composite Aggregation\",\"volume\":\"44\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Honglei\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}},{\"id\":495,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LGPYCZ4M\"],\"itemData\":{\"id\":495,\"type\":\"article-journal\",\"container-title\":\"Computer Science and Information Systems\",\"DOI\":\"10.2298/csis250221057z\",\"issue\":\"00\",\"journalAbbreviation\":\"Computer Science and Information Systems\",\"page\":\"57-57\",\"title\":\"VSAF: Verifiable and Secure Aggregation Scheme for Federated Learning in Edge Computing\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Shiwen\"},{\"family\":\"Ren\",\"given\":\"Feixiang\"},{\"family\":\"Liang\",\"given\":\"Wei\"},{\"family\":\"Li\",\"given\":\"Kuanching\"},{\"family\":\"Pathan\",\"given\":\"Al■Sakib K.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":506,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XVM7LVUQ\"],\"itemData\":{\"id\":506,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2209.07807\",\"title\":\"Model Inversion Attacks Against Graph Neural Networks\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Zaixi\"},{\"family\":\"Liu\",\"given\":\"Qi\"},{\"family\":\"Huang\",\"given\":\"Zhenya\"},{\"family\":\"Wang\",\"given\":\"Hao\"},{\"family\":\"Lee\",\"given\":\"Chee■Kong\"},{\"family\":\"Chen\",\"given\":\"Enhong\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Moreover, the information of the users is not homogenous, and this is a significant threat. In practice user data in any realistic context are usually not expected to be typically independent and identically distributed (non-IID), and the tendency of the distributions of preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur because of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the biased models of the world, a reduced rate and quality of convergence, and such models are local and trained on independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali et al., 2024; Briouya et al., 2025a; Z. Chen et al., 2024a; Y. Deng et al., 2025; Fowl et al., 2021; Hussien et al., 2023a; Lam et al., 2021; Lebrun et al., 2022; H. Li \u0026 Han, 2019; LI et al., 2022; T. Liu et al., 2023; Mehta, 2025; Qi et al., 2020; Rodio, 2025; Shalabi, 2025; Singh et al., 2025; H. Sun et al., 2020; Wan et al., 2022; J. Wang et al., 2022; Wei et al., 2021a; xue et al., 2022; Yi et al., 2021a; H. Zhang, 2026a; S. Zhang, Ren, et al., 2025; Z. Zhang, Liu, et al., 2022)",
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Most existing news recommendation methods rely on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concerns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaboratively train models without sharing their private data. However, the computation and communication cost of directly learning many existing news recommendation models in a federated way are unacceptable for user clients. In this paper, we propose an efficient federated learning framework for privacy-preserving news recommendation. Instead of training and communicating the whole model, we decompose the news recommendation model into a large news model maintained in the server and a light-weight user model shared on both server and clients, where news representations and user model are communicated between server and clients. More specifically, the clients request the user model and news representations from the server, and send their locally computed gradients to the server for aggregation. The server updates its global user model with the aggregated gradients, and further updates its news model to infer updated news representations. Since the local gradients may contain private information, we propose a secure aggregation method to aggregate gradients in a privacy-preserving way. Experiments on two real-world datasets show that our method can reduce the computation and communication cost on clients while keep promising model performance.\",\"DOI\":\"10.48550/ARXIV.2109.05446\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 3\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"Efficient-FedRec: Efficient Federated Learning Framework for Privacy-Preserving News Recommendation\",\"title-short\":\"Efficient-FedRec\",\"URL\":\"https://arxiv.org/abs/2109.05446\",\"author\":[{\"family\":\"Yi\",\"given\":\"Jingwei\"},{\"family\":\"Wu\",\"given\":\"Fangzhao\"},{\"family\":\"Wu\",\"given\":\"Chuhan\"},{\"family\":\"Liu\",\"given\":\"Ruixuan\"},{\"family\":\"Sun\",\"given\":\"Guangzhong\"},{\"family\":\"Xie\",\"given\":\"Xing\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":502,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3LGAHGWS\"],\"itemData\":{\"id\":502,\"type\":\"article-journal\",\"container-title\":\"Acm Transactions on Information Systems\",\"DOI\":\"10.1145/3779442\",\"issue\":\"2\",\"journalAbbreviation\":\"Acm Transactions on Information Systems\",\"page\":\"1-28\",\"title\":\"Beyond Similarity: Personalized Federated Recommendation With Composite Aggregation\",\"volume\":\"44\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Honglei\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}}},{\"id\":495,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LGPYCZ4M\"],\"itemData\":{\"id\":495,\"type\":\"article-journal\",\"container-title\":\"Computer Science and Information Systems\",\"DOI\":\"10.2298/csis250221057z\",\"issue\":\"00\",\"journalAbbreviation\":\"Computer Science and Information Systems\",\"page\":\"57-57\",\"title\":\"VSAF: Verifiable and Secure Aggregation Scheme for Federated Learning in Edge Computing\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Shiwen\"},{\"family\":\"Ren\",\"given\":\"Feixiang\"},{\"family\":\"Liang\",\"given\":\"Wei\"},{\"family\":\"Li\",\"given\":\"Kuanching\"},{\"family\":\"Pathan\",\"given\":\"Al■Sakib K.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":506,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XVM7LVUQ\"],\"itemData\":{\"id\":506,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2209.07807\",\"title\":\"Model Inversion Attacks Against Graph Neural Networks\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Zaixi\"},{\"family\":\"Liu\",\"given\":\"Qi\"},{\"family\":\"Huang\",\"given\":\"Zhenya\"},{\"family\":\"Wang\",\"given\":\"Hao\"},{\"family\":\"Lee\",\"given\":\"Chee■Kong\"},{\"family\":\"Chen\",\"given\":\"Enhong\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Moreover, the information of the users is not homogenous, and this is a significant threat. In practice user data in any realistic context are usually not expected to be typically independent and identically distributed (non-IID), and the tendency of the distributions of preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur because of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the biased models of the world, a reduced rate and quality of convergence, and such models are local and trained on independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali et al., 2024; Briouya et al., 2025a; Z. Chen et al., 2024a; Y. Deng et al., 2025; Fowl et al., 2021; Hussien et al., 2023a; Lam et al., 2021; Lebrun et al., 2022; H. Li \u0026 Han, 2019; LI et al., 2022; T. Liu et al., 2023; Mehta, 2025; Qi et al., 2020; Rodio, 2025; Shalabi, 2025; Singh et al., 2025; H. Sun et al., 2020; Wan et al., 2022; J. 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recommendation is critical for personalized news access. Most existing news recommendation methods rely on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concerns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaboratively train models without sharing their private data. However, the computation and communication cost of directly learning many existing news recommendation models in a federated way are unacceptable for user clients. In this paper, we propose an efficient federated learning framework for privacy-preserving news recommendation. Instead of training and communicating the whole model, we decompose the news recommendation model into a large news model maintained in the server and a light-weight user model shared on both server and clients, where news representations and user model are communicated between server and clients. More specifically, the clients request the user model and news representations from the server, and send their locally computed gradients to the server for aggregation. The server updates its global user model with the aggregated gradients, and further updates its news model to infer updated news representations. Since the local gradients may contain private information, we propose a secure aggregation method to aggregate gradients in a privacy-preserving way. Experiments on two real-world datasets show that our method can reduce the computation and communication cost on clients while keep promising model performance.\",\"DOI\":\"10.48550/ARXIV.2109.05446\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 3\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"Efficient-FedRec: Efficient Federated Learning Framework for Privacy-Preserving News Recommendation\",\"title-short\":\"Efficient-FedRec\",\"URL\":\"https://arxiv.org/abs/2109.05446\",\"author\":[{\"family\":\"Yi\",\"given\":\"Jingwei\"},{\"family\":\"Wu\",\"given\":\"Fangzhao\"},{\"family\":\"Wu\",\"given\":\"Chuhan\"},{\"family\":\"Liu\",\"given\":\"Ruixuan\"},{\"family\":\"Sun\",\"given\":\"Guangzhong\"},{\"family\":\"Xie\",\"given\":\"Xing\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":502,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3LGAHGWS\"],\"itemData\":{\"id\":502,\"type\":\"article-journal\",\"container-title\":\"Acm Transactions on Information Systems\",\"DOI\":\"10.1145/3779442\",\"issue\":\"2\",\"journalAbbreviation\":\"Acm Transactions on Information Systems\",\"page\":\"1-28\",\"title\":\"Beyond Similarity: Person

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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien

ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
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Most existing news recommendation methods rely on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concerns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaboratively train models without sharing their private data. However, the computation and communication cost of directly learning many existing news recommendation models in a federated way are unacceptable for user clients. In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda

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tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated 
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of 
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
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on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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recommendation is critical for personalized news access. Most existing news recommendation methods rely on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concerns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaboratively train models without sharing their private data. However, the computation and communication cost of directly learning many existing news recommendation models in a federated way are unacceptable for user clients. In this paper, we propose an efficient federated learning framework for privacy-preserving news recommendation. Instead of training and communicating the whole model, we decompose the news recommendation model into a large news model maintained in the server and a light-weight user model shared on both server and clients, where news representations and user model are communicated between server and clients. More specifically, the clients request the user model and news representations from the server, and send their locally computed gradients to the server for aggregation. The server updates its global user model with the aggregated gradients, and further updates its news model to infer updated news representations. Since the local gradients may contain private information, we propose a secure aggregation method to aggregate gradients in a privacy-preserving way. Experiments on two real-world datasets show that our method can reduce the computation and communication cost on clients while keep promising model performance.\",\"DOI\":\"10.48550/ARXIV.2109.05446\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 3\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"Efficient-FedRec: Efficient Federated Learning Framework for Privacy-Preserving News Recommendation\",\"title-short\":\"Efficient-FedRec\",\"URL\":\"https://arxiv.org/abs/2109.05446\",\"author\":[{\"family\":\"Yi\",\"given\":\"Jingwei\"},{\"family\":\"Wu\",\"given\":\"Fangzhao\"},{\"family\":\"Wu\",\"given\":\"Chuhan\"},{\"family\":\"Liu\",\"given\":\"Ruixuan\"},{\"family\":\"Sun\",\"given\":\"Guangzhong\"},{\"family\":\"Xie\",\"given\":\"Xing\"}],\"accessed\":{\"date-parts\":[[\"2026\",2,16]]},\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":502,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3LGAHGWS\"],\"itemData\":{\"id\":502,\"type\":\"article-journal\",\"container-title\":\"Acm Transactions on Information Systems\",\"DOI\":\"10.1145/3779442\",\"issue\":\"2\",\"journalAbbreviation\":\"Acm Transactions on Information Systems\",\"page\":\"1-28\",\"title\":\"Beyond Similarity: Person

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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda

tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
vacy-preserving way. Experiments on two real-world datasets show that our method can reduce the computation
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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en et al., 2023a; Lam et al., 2021; Lebrun et al., 2022; H. Li \u0026 Han, 2019; LI et al., 2022; T. Liu et
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recommendation is critical for personalized news access. Most existing news recommendation methods rely on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concerns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaboratively train models without sharing their private data. However, the computation and communication cost of directly learning many existing news recommendation models in a federated way are unacceptable for user clients. In this paper, we propose an efficient federated learning framework for privacy-preserving news recommendation. Instead of training and communicating the whole model, we decompose the news recommendation model into a large news model maintained in the server and a light-weight user model shared on both server and clients, where news representations and user model are communicated between server and clients. More specifically, the clients request the user model and news representations from the server, and send their locally computed gradients to the server for aggregation. The server updates its global user model with the aggregated gradients, and further updates its news model to infer updated news representations. Since the local gradients may contain private information, we propose a secure aggregation method to aggregate gradients in a privacy-preserving way. Experiments on two real-world datasets show that our method can reduce the computation and communication cost on clients while keep promising model performance.\",\"DOI\":\"10.48550/ARXIV.2109.05446\",\"license\":\"arXiv.org perpetual, non-exclusive license\",\"note\":\"version: 3\",\"publisher\":\"arXiv\",\"source\":\"DOI.org (Datacite)\",\"title\":\"Efficient-FedRec: Efficient Federated Learning Framework for Privacy-Preserving News Recommendation\",\"title-short\":\"Efficient-FedRec\",\"URL\":\"https://arx

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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of
preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
 In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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ws recommendation is critical for personalized news access. Most existing news recommendation methods rely
on centralized storage of users\u0027 historical news click behavior data, which may lead to privacy concer
ns and hazards. Federated Learning is a privacy-preserving framework for multiple clients to collaborativel
y train models without sharing their private data. However, the computation and communication cost of direc
tly learning many existing news recommendation models in a federated way are unacceptable for user clients.
In this paper, we propose an efficient federated learning framework for privacy-preserving news recommenda
tion. Instead of training and communicating the whole model, we decompose the news recommendation model int
o a large news model maintained in the server and a light-weight user model shared on both server and clien
ts, where news representations and user model are communicated between server and clients. More specificall
y, the clients request the user model and news representations from the server, and send their locally comp
uted gradients to the server for aggregation. The server updates its global user model with the aggregated
gradients, and further updates its news model to infer updated news representations. Since the local gradie
nts may contain private information, we propose a secure aggregation method to aggregate gradients in a pri
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and this is a significant threat. In practice user data in any realistic context are usually not expected t
o be typically independent and identically distributed (non-IID), and the tendency of the distributions of

preferences, the distributions of interaction dissipating across a wide spectrum are likely to occur becaus
e of the conditions of a circumstance or demographic usage. Such a non-IID character can result in the bias
ed models of the world, a reduced rate and quality of convergence, and such models are local and trained on
 independently specific data, which should not be aggregated to perform an adequate generalization. (W. Ali
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene

ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
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ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically, 
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically, 
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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In evaluating the effectiveness of federated movie recommenders using varying privacy settings we do also consider their effectiveness in terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically, we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budget and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as recommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). The ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise addition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z. Cheng et al., 2023a; Chourasia et al., 2022; Y. Han, 2024; H. He \u0026 He, 2023; Z. 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In evaluating the effectiveness of federated movie recommenders using varying privacy settings we do also consider their effectiveness in terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically, we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budget and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as recommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). The ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise addition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z. Cheng et al., 2023a; Chourasia et al., 2022; Y. Han, 2024; H. He \u0026 He, 2023; Z. Liang \u0026 Chen, 2025a; F. Liu et al., 2022; R. Liu et al., 2022a; Z. Liu et al., 2019; Parra■Arnau et al., 2014; Rao et al., 2021; T \u0026 Tan, 2017; TAN, 2024; Tanveer et al., 2025a; Valdez \u0026 Ziefle, 2019; XUE, 2025a; Z. Yang et al., 2023a; zeng, 2024a; Zhan et al., 2023; X. Zhang et al., 2022; J. Zhao et al., 2021; Q. Zheng et al., 2021; Y. Zhu, 2025a)",
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
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e problems of trade-offs between accuracy and privacy in the federated learning are referred to in the prov
ided paper using the case of the movie recommender systems in mobile devices. In evaluating the effectivene
ss of federated movie recommenders using varying privacy settings we do also consider their effectiveness i
n terms of local training epochs and also client sampling plan and aggregation plan on data. Specifically,
we examine implications of such a privacy-related trade-offs as lossless noise multipliers to privacy budge
t and privacy level indicator (e.g., specifies privacy level 5) does affect such valuable indicators as rec
ommendation precision, recall, NDCG and user experience indicators (e.g., rate of session satisfaction). Th
e ancillary costs we also measure (in terms of CPU or GPU cycles used to perform an encryption or noise add
ition, network overheads (also called secure aggregations), battery overheads, and latency spurts) are then
 tested in actual mobile hardware (Android phones). (Ahmadzai \u0026 Nguyen, 2024; Z. Chen et al., 2022a; Z
. Cheng et al., 2023a; Chourasia et al., 2022; Y. Han, 2024; H. He \u0026 He, 2023; Z. Liang \u0026 Chen, 2
025a; F. Liu et al., 2022; R. Liu et al., 2022a; Z. Liu et al., 2019; Parra■Arnau et al., 2014; Rao et al.,
 2021; T \u0026 Tan, 2017; TAN, 2024; Tanveer et al., 2025a; Valdez \u0026 Ziefle, 2019; XUE, 2025a; Z. Yan
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Are the Ways to Learn: Identifying Multi-Modal Behavioral Profiles of Collaborative Learning in Constructiv
ist Activities\",\"volume\":\"20\",\"author\":[{\"family\":\"Nasir\",\"given\":\"Jauwairia\"},{\"family\":\
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"given\":\"Pierre\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} In a collaborative machine learning innovation, federated systems such as FedSlate develop these recommendations systems to be designed in such a way that more than two parties collaboratively learn a model without the agencies having to share user information (although eventually they might share it indirectly). However, the advantages of such approaches are potentially negative impacts on uneven distribution of benefits among parties involved. More specifically, organisations or sites with fewer users (probably due to limited numbers or reduced analysability) are more likely to see increased accuracy of the recommendation than the ones with stronger sources of feedback. The rationale of this relationship is that privacy-saving techniques and, in particular, privacy introduce noise to user data to the extent of blurring personal identities. With less information entities, it can be used to have a positive mask to fill the gaps of information and more or less equalises the playing field. Conversely, the user feedback is abundant and high quality on participants making them, it may easily become diluted by the same noise and reduces the accuracy of the recommendations thereby causing a lowered user satisfaction. (Marx \u0026 Fuegi, 2020; Nasir et al., 2022, 2022)",
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Then we study users\u0027 privacy decision making under different data disclosure mechanisms and recommendation models, and how their data disclosure decisions affect the recommender system\u0027s performance.To avoid the high cost of real-world experiments, we apply simulations to study the effects of our proposed framework. Specifically, we propose a reinforcement learning algorithm to simulate users\u0027 decisions (with various sensitivities) under three proposed platform mechanisms on two datasets with three representative recommendation models. The simulation results show that the platform mechanisms with finer split granularity and more unrestrained disclosure strategy can bring better results for both end users and platforms than the \\\"all or nothing\\\" binary mechanism adopted by most real-world applications. It also shows that our proposed framework can effectively protect users\u0027 privacy since they can obtain comparable or even better results with much less disclosed data.\",\"DOI\":\"10.48550/arxiv.2204.00279\",\"title\":\"Studying the Impact of Data Disclosure Mechanism in Recommender Systems via Simulation\",\"author\":[{\"family\":\"Chen\",\"given\":\"Ziqian\"},{\"family\":\"Sun\",\"given\":\"Fei\"},{\"family\":\"Tang\",\"given\":\"Yifan\"},{\"family\":\"Chen\",\"given\":\"Haokun\"},{\"family\":\"Gao\",\"given\":\"Jinyang\"},{\"family\":\"Ding\",\"given\":\"Bolin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":544,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5GQ9B68L\"],\"itemData\":{\"id\":544,\"type\":\"article-journal\",\"abstract\":\"Federated Semi-supervised Learning (FSSL) combines techniques from both fields of federated and semi- supervised learning to improve the accuracy and performance of models in a distributed environment by using a small fraction of labeled data and a large amount of unlabeled data. Without the need tocentralize all data in one place for training, it collect updates of model training after devices train models at local, and thus can protect the privacy of user data. However, during the federal training process, some of the devices fail to collect enough data for local training, while new devices will be included to the group training. This leads to an unbalanced global data distribution and thus af ect the performance of the global model training. Most of the current research is focusing on class imbalance with a fixednumber of classes, while little attention is paid to data imbalance with a variable number of classes. Therefore, in this paper, we propose Federated Semi-supervised Learning for Class Variable Imbalance (FCVI) to solve class variable imbalance. The class-variable learning algorithm is used to mitigate the data imbalance due to changes of the number of classes. Our scheme is proved to be significantly better than baseline methods, while maintaining client privacy.\",\"container-title\":\"Computer Science, Engineering and Applications\",\"DOI\":\"10.5121/csit.2023.130522\",\"journalAbbreviation\":\"Computer Science, Engineering and Applications\",\"page\":\"257-272\",\"title\":\"Addressing Class Variable Imbalance in Federated Semi-Supervised Learning\",\"author\":[{\"family\":\"Dong\",\"given\":\"Zehui\"},{\"family\":\"Liu\",\"given\":\"Wenjing\"},{\"family\":\"Liu\",\"given\":\"Siyuan\"},{\"family\":\"Chen\",\"given\":\"Xingzhi\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":539,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ASC3IKVI\"],\"itemData\":{\"id\":539,\"type\":\"article-journal\",\"abstract\":\"As big data technologies continue to evolve, recommendation systems have found broad application in domains such as online retail and social networking platforms. However, centralized recommendation systems raise numerous data privacy concerns. Federated learning addresses these concerns by allowing model training on client devices and aggregating model parameters without sharing raw data. Nevertheless, federated learning faces critical challenges related to feature extraction efficiency and noise sensitivity, limiting its application in e-commerce recommendation systems where data heterogeneity and high-dimensional features are prevalent. To address these gaps, this paper introduces a novel multi-view fede

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ion overhead. Experimental results on four real-world recommendation datasets demonstrate the effectiveness
 of FedRAP.\",\"DOI\":\"10.48550/arxiv.2301.09109\",\"title\":\"Federated Recommendation With Additive Pers
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This approach yields a 12%-18% increase in recommendation accuracy across various performance metrics compared to single-view and other multi-view methods. By addressing data heterogeneity and optimizing feature utilization through dynamic rescaling, Fed-FR-MVD effectively mitigates the impact of noisy data, with performance maintained across noise levels of 5%-15%. Experimental results demonstrate that Fed-FR-MVD fills a key research gap by providing a more resilient and efficient framework for federated recommendation systems in privacy-sensitive and data-diverse e-commerce environments.\",\"container-title\":\"Scientific Reports\",\"DOI\":\"10.1038/s41598-024-81278-1\",\"issue\":\"1\",\"journalAbbreviation\":\"Scientific Reports\",\"title\":\"Federated Cross-View E-Commerce Recommendation Based on Feature Rescaling\",\"volume\":\"14\",\"author\":[{\"family\":\"Li\",\"given\":\"Ruiheng\"},{\"family\":\"Shu\",\"given\":\"Yuhang\"},{\"family\":\"Cao\",\"given\":\"Yue\"},{\"family\":\"Luo\",\"given\":\"Yiming\"},{\"family\":\"Zuo\",\"given\":\"Qiankun\"},{\"family\":\"Wu\",\"given\":\"Xuan\"},{\"family\":\"Yu\",\"given\":\"Jiaojiao\"},{\"family\":\"Zhang\",\"given\":\"Wenxin\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":543,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/FKLR62K3\"],\"itemData\":{\"id\":543,\"type\":\"article-journal\",\"abstract\":\"With rising concerns about privacy, developing recommendation systems in a federated setting become a new paradigm to develop next-generation Internet service architecture. However, existing approaches are usually derived from a distributed recommendation framework with an additional mechanism for privacy protection, thus most of them fail to fully exploit personalization in the new context of federated recommendation settings. In this paper, we propose a novel approach called Federated Recommendation with Additive Personalization (FedRAP) to enhance recommendation by learning user embedding and the user\u0027s personal view of item embeddings. Specifically, the proposed additive personalization is to add a personalized item embedding to a spars
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e global item embedding aggregated from all users. Moreover, a curriculum learning mechanism has been appli
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cy-preserving technologies such as federated learning and differential privacy, we explore another way to m
itigate users\u0027 privacy concerns, giving them control over their own data. For this goal, we propose a
privacy aware recommendation framework that gives users delicate control over their personal data, includin
g implicit behaviors, e.g., clicks and watches. In this new framework, users can proactively control which
data to disclose based on the trade-off between anticipated privacy risks and potential utilities. Then we
study users\u0027 privacy decision making under different data disclosure mechanisms and recommendation mod
els, and how their data disclosure decisions affect the recommender system\u0027s performance.To avoid the
high cost of real-world experiments, we apply simulations to study the effects of our proposed framework. S
pecifically, we propose a reinforcement learning algorithm to simulate users\u0027 decisions (with various
sensitivities) under three proposed platform mechanisms on two datasets with three representative recommend
ation models. The simulation results show that the platform mechanisms with finer split granularity and mor
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tion of labeled data and a large amount of unlabeled data. Without the need tocentralize all data in one pl
ace for training, it collect updates of model training after devices train models at local, and thus can pr
otect the privacy of user data. However, during the federal training process, some of the devices fail to c
ollect enough data for local training, while new devices will be included to the group training. This leads
 to an unbalanced global data distribution and thus af ect the performance of the global model training. Mo
st of the current research is focusing on class imbalance with a fixednumber of classes, while little atten
tion is paid to data imbalance with a variable number of classes. Therefore, in this paper, we propose Fede
rated Semi-supervised Learning for Class Variable Imbalance (FCVI) to solve class variable imbalance. The c
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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury 
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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury

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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury
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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury
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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury 
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}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Privacy and utility: this is tricky contact which requires application of a multidimensional observation of federated recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency because it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance that privacy protecting practices are favourable to all the categories of user, regardless of the extent of their involvement, and the depth of information they consume no matter how minimal. This is essential in making sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are present. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the same time, the utility and equity of recommendations are not undermined to the stakeholders of the federated ecosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury et al., 2024a; Z. Dai et al., 2021; Feretzakis et al., 2024a; Q. Hu \u0026 Song, 2024b; Karami, 2025; Kim et al., 2024a; Y. Li et al., 2025; Y. Lian et al., 2025; Naresh et al., 2025; Oyekan, 2024; H. Peng, 2021; P

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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury
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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury
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nd utility: this is tricky contact which requires application of a multidimensional observation of federate
d recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau
se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury
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Alhakbani\",\"given\":\"Haya Abdullah\"},{\"family\":\"Ahmad\",\"given\":\"Awais\"},{\"family\":\"Khan\",\"given\":\"Fakhri Alam\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":591,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/DC43632B\"],\"itemData\":{\"id\":591,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2204.05157\",\"title\":\"SF-PATE: Scalable, Fair, and Private Aggregation of Teacher Ensembles\",\"author\":[{\"family\":\"Tran\",\"given\":\"Cuong\"},{\"family\":\"Zhü\",\"given\":\"Kèyù\"},{\"family\":\"Fioretto\",\"given\":\"Ferdinando\"},{\"family\":\"Hentenryck\",\"given\":\"Pascal Van\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":574,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AR24GNLU\"],\"itemData\":{\"id\":574,\"type\":\"article-journal\",\"DOI\":\"10.18653/v1/2021.emnlp-main.223\",\"title\":\"Efficient-FedRec: Efficient Federated Learning Framework for Privacy-Preserving News Recommendation\",\"author\":[{\"family\":\"Yi\",\"given\":\"Jingwei\"},{\"family\":\"Wu\",\"given\":\"Fangzhao\"},{\"family\":\"Wu\",\"given\":\"Chuhan\"},{\"family\":\"Liu\",\"given\":\"Ruixuan\"},{\"family\":\"Sun\",\"given\":\"Guangzhong\"},{\"family\":\"Xie\",\"given\":\"Xing\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":582,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/PYMQEM5F\"],\"itemData\":{\"id\":582,\"type\":\"article-journal\",\"container-title\":\"International Journal of Environmental Research and Public Health\",\"DOI\":\"10.3390/ijerph191911876\",\"title\":\"Privacy-by-Design Environments for Large-Scale Health Research and Federated Learning From Data\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Peng\"},{\"family\":\"Kamel Boulos\",\"given\":\"Maged N.\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":588,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/4WQLF67C\"],\"itemData\":{\"id\":588,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2205.11857\",\"title\":\"Comprehensive Privacy Analysis on Federated Recommender System Against Attribute Inference Attacks\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Shijie\"},{\"family\":\"Yin\",\"given\":\"Hongzhi\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":572,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/26E67RY5\"],\"itemData\":{\"id\":572,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2102.10109\",\"title\":\"When Crowdsensing Meets Federated Learning: Privacy-Preserving Mobile Crowdsensing System\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Bowen\"},{\"family\":\"Liu\",\"given\":\"Ximeng\"},{\"family\":\"Chen\",\"given\":\"Wei–Neng\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":594,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/U9UXMNQR\"],\"itemData\":{\"id\":594,\"type\":\"article-journal\",\"container-title\":\"Ieee Internet of Things Journal\",\"DOI\":\"10.1109/jiot.2023.3309132\",\"title\":\"PPMM-DA: Privacy-Preserving Multidimensional and Multisubset Data Aggregation With Differential Privacy for Fog-Based Smart Grids\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Shuai\"},{\"family\":\"Shu-hua\",\"given\":\"XU\"},{\"family\":\"Han\",\"given\":\"Song\"},{\"family\":\"Ren\",\"given\":\"Siqi\"},{\"family\":\"Wang\",\"given\":\"Yu\"},{\"family\":\"Chen\",\"given\":\"Zhixian\"},{\"family\":\"Chen\",\"given\":\"Xiaoli\"},{\"family\":\"Lin\",\"given\":\"Jianhong\"},{\"family\":\"Liu\",\"given\":\"Weinan\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Privacy and utility: this is tricky contact which requires application of a multidimensional observation of federated recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency because it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance that privacy protecting practices are favourable to all the categories of user, regardless of the extent of their involvement, and the depth of information they consume no matter how minimal. This is essential in making sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are present. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the same time, the utility and equity of recommendations are not undermined to the stakeholders of the federated ecosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury et al., 2024a; Z. Dai et al., 2021; Feretzakis et al., 2024a; Q. Hu \u0026 Song, 2024b; Karami, 2025; Kim et al., 2024a; Y. Li et al., 2025; Y. Lian et al., 2025; Naresh et al., 2025; Oyekan, 2024; H. Peng, 2021; Pentyala et al., 2022a; J. A. Sun et al., 2023a; K. Sun, 2025a; Tanveer et al., 2025b; C. Tran et al., 2022; Yi et al., 2021b; P. Zhang \u0026 Kamel Boulos, 2022; S. Zhang \u0026 Yin, 2022a; B. Zhao et al., 2021; S. Zhao et al., 2024)",
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The privacy mechanisms cannot be simply judged in terms of their efficiency because it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance that privacy protecting practices are favourable to all the categories of user, regardless of the extent of their involvement, and the depth of information they consume no matter how minimal. This is essential in making sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are present. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the same time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e

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}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Privacy and utility: this is tricky contact which requires application of a multidimensional observation of federated recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency becau

se it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance tha
t privacy protecting practices are favourable to all the categories of user, regardless of the extent of th
eir involvement, and the depth of information they consume no matter how minimal. This is essential in maki
ng sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are pre
sent. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the sam
e time, the utility and equity of recommendations are not undermined to the stakeholders of the federated e
cosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury
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author\":[{\"family\":\"Zhao\",\"given\":\"Shuai\"},{\"family\":\"Shu-hua\",\"given\":\"XU\"},{\"family\":\"Han\",\"given\":\"Song\"},{\"family\":\"Ren\",\"given\":\"Siqi\"},{\"family\":\"Wang\",\"given\":\"Yu\"},{\"family\":\"Chen\",\"given\":\"Zhixian\"},{\"family\":\"Chen\",\"given\":\"Xiaoli\"},{\"family\":\"Lin\",\"given\":\"Jianhong\"},{\"family\":\"Liu\",\"given\":\"Weinan\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Privacy and utility: this is tricky contact which requires application of a multidimensional observation of federated recommendation systems. The privacy mechanisms cannot be simply judged in terms of their efficiency because it is as well crucial to deeply examine them in the area of fairness. It has to target the assurance that privacy protecting practices are favourable to all the categories of user, regardless of the extent of their involvement, and the depth of information they consume no matter how minimal. This is essential in making sure that no cases of over-benefits or wrongful harming of some groups of users or organisations are present. In other words, the challenge lies in ensuring that the privacy assurance is provided, and at the same time, the utility and equity of recommendations are not undermined to the stakeholders of the federated ecosystem. (M. Ali et al., 2025; Arafat, 2025a; Arcolezi et al., 2025a; Baumgartner et al., 2024; Choudhury et al., 2024a; Z. Dai et al., 2021; Feretzakis et al., 2024a; Q. Hu \u0026 Song, 2024b; Karami, 2025; Kim et al., 2024a; Y. Li et al., 2025; Y. Lian et al., 2025; Naresh et al., 2025; Oyekan, 2024; H. Peng, 2021; Pentyala et al., 2022a; J. A. Sun et al., 2023a; K. Sun, 2025a; Tanveer et al., 2025b; C. Tran et al., 2022; Yi et al., 2021b; P. Zhang \u0026 Kamel Boulos, 2022; S. Zhang \u0026 Yin, 2022a; B. Zhao et al., 2021; S. Zhao et al., 2024)",
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
 Hafeez et al., 2021; Y. Hu et al., 2023; Jarin \u0026 Eshete, 2023; Jayaraman \u0026 Evans, 2019; Kserawi
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rch et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023;
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} To successfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the application of the static technique where privacy budget is equally applied to everyone participating in the system, the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust privacy settings in response to real-time system feedback and learned performance deviations. Such processes are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to maximize the privacy settings of individual players or groups. An example of this is that, in case one particular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can

be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendations are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be calibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that will unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019; Hafeez et al., 2021; Y. Hu et al., 2023; Jarin \u0026 Eshete, 2023; Jayaraman \u0026 Evans, 2019; Kserawi et al., 2022; Z. Liang \u0026 Chen, 2025b; Z. Lu, 2022; Minto et al., 2021; Nanayakkara et al., 2022; Research et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023; Tanveer et al., 2025c; Vinterbo et al., 2012; XUE, 2025b; X. Zhang, Zhang, et al., 2025; Y. Zhang, 2025; Z. Zhao et al., 2024; G. Zhou, 2025)",
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With Implicit Feedback\",\"author\":[{\"family\":\"Minto\",\"given\":\"Lorenzo\"},{\"family\":\"Haller\",\"given\":\"Moritz\"},{\"family\":\"Haddadi\",\"given\":\"Hamed\"},{\"family\":\"Livshits\",\"given\":\"Benjamin\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}},{\"id\":604,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/2VHCMWZ4\"],\"itemData\":{\"id\":604,\"type\":\"article-journal\",\"container-title\":\"Proceedings on Privacy Enhancing Technologies\",\"DOI\":\"10.2478/popets-2022-0058\",\"title\":\"Visualizing Privacy-Utility Trade-Offs in Differentially Private Data Releases\",\"author\":[{\"family\":\"Nanayakkara\",\"given\":\"Priyanka\"},{\"family\":\"Bater\",\"given\":\"Johes\"},{\"family\":\"He\",\"given\":\"Xi\"},{\"family\":\"Hullman\",\"given\":\"Jessica\"},{\"family\":\"Rogers\",\"given\":\"Jennie\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}},{\"id\":620,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5GM8IG8F\"],\"itemData\":{\"id\":620,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-4121481/v1\",\"title\":\"PMNBARL: Enhancing Efficiency of Privacy Management in Digital Networks Through Blockchain and Adaptive Reinforcement Learning\",\"author\":[{\"family\":\"Research\",\"given\":\"Hiralal Solunke\"},{\"family\":\"Bhaladhare\",\"given\":\"Pawan\"},{\"family\":\"Potgantwar\",\"given\":\"Amol\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}},{\"id\":613,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/K9FBJNYM\"],\"itemData\":{\"id\":613,\"type\":\"article-journal\",\"container-title\":\"Statistical Science\",\"DOI\":\"10.1214/17-sts641\",\"title\":\"Confidentiality and Differential Privacy in the Dissemination of Frequency Tables\",\"author\":[{\"family\":\"Rinott\",\"given\":\"Yosef\"},{\"family\":\"O’Keefe\",\"given\":\"Christine M.\"},{\"family\":\"Shlomo\",\"given\":\"Natalie\"},{\"family\":\"Skinner\",\"given\":\"Chris\"}],\"issued\":{\"date-parts\":[[\"2018\"]]}},{\"id\":606,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/M8Q4KPCZ\"],\"itemData\":{\"id\":606,\"type\":\"article-journal\",\"DOI\":\"10.4018/978-1-7998-5728-0.ch019\",\"title\":\"Learning With Differential Privacy\",\"author\":[{\"family\":\"Sengupta\",\"given\":\"Poushali\"},{\"family\":\"Paul\",\"given\":\"Sudipta\"},{\"family\":\"Mishra\",\"given\":\"Subhankar\"}],\"issued\":{\"date-parts\":[[\"2020\"]]}},{\"id\":618,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7PPKNE58\"],\"itemData\":{\"id\":618,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-1847248/v1\",\"title\":\"Have the Cake and Eat It Too: Differential Privacy Enables Privacy and Precise Analytics\",\"author\":[{\"family\":\"Subramanian\",\"given\":\"Rishabh\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}},{\"id\":612,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7CUD6XWI\"],\"itemData\":{\"id\":612,\"type\":\"article-journal\",\"DOI\":\"10.1145/3576842.3582325\",\"title\":\"adaPARL: Adaptive Privacy-Aware Reinforcement Learning for Sequential Decision Making Human-in-the-Loop Systems\",\"author\":[{\"family\":\"Taherisadr\",\"given\":\"Mojtaba\"},{\"family\":\"Stavroulakis\",\"given\":\"Stelios\"},{\"family\":\"Elmalaki\",\"given\":\"Salma\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}},{\"id\":616,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/WVJS8DXA\"],\"itemData\":{\"id\":616,\"type\":\"article-journal\",\"container-title\":\"Digital Health\",\"DOI\":\"10.1177/20552076251381769\",\"title\":\"Balancing Privacy and Performance in Healthcare: A Federated Learning Framework for Sensitive Data\",\"author\":[{\"family\":\"Tanveer\",\"given\":\"Fatima\"},{\"family\":\"Iradat\",\"given\":\"Faisal\"},{\"family\":\"Iqbal\",\"given\":\"Waseem\"},{\"family\":\"AlSagri\",\"given\":\"Hatoon S.\"},{\"family\":\"A. Alhakbani\",\"given\":\"Haya Abdullah\"},{\"family\":\"Ahmad\",\"given\":\"Awais\"},{\"family\":\"Khan\",\"given\":\"Fakhri Alam\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":611,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YMGAQAYN\"],\"itemData\":{\"id\":611,\"type\":\"article-journal\",\"container-title\":\"Journal of the American Medical Informatics Association\",\"DOI\":\"10.1136/amiajnl-2011-000459\",\"title\":\"Protecting Count Queries in Study Design\",\"author\":[{\"family\":\"Vinterbo\",\"given\":\"Staal A.\"},{\"family\":\"Sarwate\",\"given\":\"Anand D.\"},{\"family\":\"Boxwala\",\"given\":\"Aziz A.\"}],\"issued\":{\"date-parts\":[[\"2012\"]]}},{\"id\":597,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/2W5PMGH9\"],\"itemData\":{\"id\":597,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-7839869/v1\",\"title\":\"Federated Learning Under the Uncertainty Principle of Machine Learning\",\"author\":[{\"family\":\"XUE\",\"given\":\"Zhonghui\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":617,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AJ4KGLXR\"],\"itemData\":{\"id\":617,\"type\":\"article-journal\",\"container-title\":\"Caai Transactions on Intelligence Technology\",\"DOI\":\"10.1049/cit2.70008\",\"title\":\"Layer■Level Adaptive Gradient Perturbation Protecting Deep Learning Based on Differential Privacy\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Xiangfei\"},{\"family\":\"Zhang\",\"given\":\"Qingchen\"},{\"family\":\"Liming\",\"given\":\"Jiang\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":605,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/BHUWD5EX\"],\"itemData\":{\"id\":605,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3092511\",\"title\":\"Privacy-Preserving Continuous Medical Data Stream Analysis With Contextual Pattern Awareness\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yao\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":599,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/M29ZCZ9B\"],\"itemData\":{\"id\":599,\"type\":\"article-journal\",\"container-title\":\"Frontiers in Big Data\",\"DOI\":\"10.3389/fdata.2023.1296508\",\"title\":\"CTAB-GAN+: Enhancing Tabular Data Synthesis\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zilong\"},{\"family\":\"Kunar\",\"given\":\"Aditya\"},{\"family\":\"Birke\",\"given\":\"Robert\"},{\"family\":\"Scheer\",\"given\":\"Hiek Van\",\"non-dropping-particle\":\"der\"},{\"family\":\"Chen\",\"given\":\"Lydia Y.\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}},{\"id\":610,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/9G6DVFQS\"],\"itemData\":{\"id\":610,\"type\":\"article-journal\",\"container-title\":\"Aip Advances\",\"DOI\":\"10.1063/5.0302915\",\"title\":\"Privacy Optimization Method for Federated Learning Based on Evolutionary Game Theory\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Guorong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} To successfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the application of the static technique where privacy budget is equally applied to everyone participating in the system, the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust privacy settings in response to real-time system feedback and learned performance deviations. Such processes are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to maximize the privacy settings of individual players or groups. An example of this is that, in case one particular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendations are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be calibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi

ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019; Hafeez et al., 2021; Y. Hu et al., 2023; Jarin \u0026 Eshete, 2023; Jayaraman \u0026 Evans, 2019; Kserawi et al., 2022; Z. Liang \u0026 Chen, 2025b; Z. Lu, 2022; Minto et al., 2021; Nanayakkara et al., 2022; Research et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023; Tanveer et al., 2025c; Vinterbo et al., 2012; XUE, 2025b; X. Zhang, Zhang, et al., 2025; Y. Zhang, 2025; Z. Zhao et al., 2024; G. Zhou, 2025)",
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Access\",\"DOI\":\"10.1109/access.2019.2940096\",\"title\":\"A Differential Game Model for Data Utility and Privacy-Preserving in Mobile Crowdsensing\",\"author\":[{\"family\":\"Gao\",\"given\":\"Hongjie\"},{\"family\":\"Xu\",\"given\":\"Haitao\"},{\"family\":\"Zhang\",\"given\":\"Long\"},{\"family\":\"Zhou\",\"given\":\"Xianwei\"}],\"issued\":{\"date-parts\":[[\"2019\"]]}}},{\"id\":608,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5PI3AQPH\"],\"itemData\":{\"id\":608,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2102.09458\",\"title\":\"DPNCT: A Differential Private Noise Cancellation Scheme for Load Monitoring and Billing for Smart Meters\",\"author\":[{\"family\":\"Hafeez\",\"given\":\"Khadija\"},{\"family\":\"Rehmani\",\"given\":\"Mubashir 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Unlike with the application of the static technique where privacy budget is equally applied to everyone participating in the system, the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust privacy settings in response to real-time system feedback and learned performance deviations. Such processes are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to maximize the privacy settings of individual players or groups. An example of this is that, in case one particular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati

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ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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rch et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023;
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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Conversely, the participants possessing a more abundant data can be calibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that will unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019; Hafeez et al., 2021; Y. Hu et al., 2023; Jarin \u0026 Eshete, 2023; Jayaraman \u0026 Evans, 2019; Kserawi et al., 2022; Z. Liang \u0026 Chen, 2025b; Z. Lu, 2022; Minto et al., 2021; Nanayakkara et al., 2022; Research et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023; Tanveer et al., 2025c; Vinterbo et al., 2012; XUE, 2025b; X. Zhang, Zhang, et al., 2025; Y. Zhang, 2025; Z. Zhao et al., 2024; G. Zhou, 2025)",
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
 Hafeez et al., 2021; Y. Hu et al., 2023; Jarin \u0026 Eshete, 2023; Jayaraman \u0026 Evans, 2019; Kserawi
et al., 2022; Z. Liang \u0026 Chen, 2025b; Z. Lu, 2022; Minto et al., 2021; Nanayakkara et al., 2022; Resea
rch et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023;
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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rch et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023;
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uccessfully go through such challenges, adaptive privacy mechanism must be used. Unlike with the applicatio
n of the static technique where privacy budget is equally applied to everyone participating in the system,
the adaptive mechanisms are developed or modelled similarly, in that they are able to dynamically adjust pr
ivacy settings in response to real-time system feedback and learned performance deviations. Such processes
are constantly monitoring the privacy and utility trade-offs, which are evolving, to enable the system to m
aximize the privacy settings of individual players or groups. An example of this is that, in case one parti
cular entity turns out to be particularly affected by the noise of privacy, then the adaptive mechanism can
 be used to ensure that they have a lower appropriate privacy budget so that their accuracy of recommendati
ons are not impaired unintentionally. Conversely, the participants possessing a more abundant data can be c
alibrated through the mechanism of noise with a lower likelihood of increasing the privacy paranoia that wi
ll unreasonably reduce utility. (Arcolezi et al., 2025b; Cao et al., 2019; Farokhi, 2024; Gao et al., 2019;
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rch et al., 2024a; Rinott et al., 2018; Sengupta et al., 2020; Subramanian, 2022; Taherisadr et al., 2023;
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csl-citation.json\"} It is shown through empirical literature that adaptive privacy approaches result in a huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly smarter than the traditional privacy settings in the manner in which they allocate privacy resources in those areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise and trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the developments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets are provided to fit local demands and requirements of each of the participants. With this, adaptive privacy budgeting must be built and communicated to activate fair results and to make the system not favour the ones that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025b; Are et al., 2025; Badar et al., 2024a; S. Chen \u0026 Qi, 2025; Y. Chen et al., 2023; Z. Chen et al., 2024b; Z. Cheng et al., 2023b; Dipika Birari, 2024; Farzana et al., 2024; Ganji et al., 2025; Geng et al., 2024a; Ju et al., 2023a; Kim et al., 2024b; Y. Li, 2025; Z. Liang \u0026 Chen, 2025c; P. J. Lu et al., 2023; M.Saraswathi, 2025; K. I. Wang et al., 2022; S. Wang, 2024; Wei et al., 2021b; D. Xia et al., 2025; D. Xu et al., 2020; XUE, 2025c; Yaghini et al., 2023a; B. Zheng et al., 2024)",

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csl-citation.json\"} It is shown through empirical literature that adaptive privacy approaches result in a
huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly
 smarter than the traditional privacy settings in the manner in which they allocate privacy resources in th
ose areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise a
nd trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the dev
elopments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets
 are provided to fit local demands and requirements of each of the participants. With this, adaptive privac
y budgeting must be built and communicated to activate fair results and to make the system not favour the o
nes that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025
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csl-citation.json\"} It is shown through empirical literature that adaptive privacy approaches result in a
huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly
 smarter than the traditional privacy settings in the manner in which they allocate privacy resources in th
ose areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise a
nd trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the dev
elopments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets
 are provided to fit local demands and requirements of each of the participants. With this, adaptive privac
y budgeting must be built and communicated to activate fair results and to make the system not favour the o
nes that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025
b; Are et al., 2025; Badar et al., 2024a; S. Chen \u0026 Qi, 2025; Y. Chen et al., 2023; Z. Chen et al., 20
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huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly smarter than the traditional privacy settings in the manner in which they allocate privacy resources in those areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise and trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the developments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets are provided to fit local demands and requirements of each of the participants. With this, adaptive privacy budgeting must be built and communicated to activate fair results and to make the system not favour the ones that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025b; Are et al., 2025; Badar et al., 2024a; S. Chen \u0026 Qi, 2025; Y. Chen et al., 2023; Z. Chen et al., 2024b; Z. Cheng et al., 2023b; Dipika Birari, 2024; Farzana et al., 2024; Ganji et al., 2025; Geng et al., 2024a; Ju et al., 2023a; Kim et al., 2024b; Y. Li, 2025; Z. Liang \u0026 Chen, 2025c; P. J. Lu et al., 2023; M.Saraswathi, 2025; K. I. Wang et al., 2022; S. Wang, 2024; Wei et al., 2021b; D. Xia et al., 2025; D. Xu et al., 2020; XUE, 2025c; Yaghini et al., 2023a; B. Zheng et al., 2024)",
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Learning\",\"author\":[{\"family\":\"Zheng\",\"given\":\"Bowen\"},{\"family\":\"Wu\",\"given\":\"Dinghui\"},{\"family\":\"Fan\",\"given\":\"Junyan\"},{\"family\":\"Du\",\"given\":\"Kangning\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} It is shown through empirical literature that adaptive privacy approaches result in a huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly smarter than the traditional privacy settings in the manner in which they allocate privacy resources in those areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise and trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the developments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets are provided to fit local demands and requirements of each of the participants. With this, adaptive privac

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nes that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025
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24a; Ju et al., 2023a; Kim et al., 2024b; Y. Li, 2025; Z. Liang \u0026 Chen, 2025c; P. J. Lu et al., 2023;
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Learning\",\"author\":[{\"family\":\"Zheng\",\"given\":\"Bowen\"},{\"family\":\"Wu\",\"given\":\"Dinghui\"},{\"family\":\"Fan\",\"given\":\"Junyan\"},{\"family\":\"Du\",\"given\":\"Kangning\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} It is shown through empirical literature that adaptive privacy approaches result in a huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly smarter than the traditional privacy settings in the manner in which they allocate privacy resources in those areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise and trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the developments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets are provided to fit local demands and requirements of each of the participants. With this, adaptive privacy budgeting must be built and communicated to activate fair results and to make the system not favour the ones that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025b; Are et al., 2025; Badar et al., 2024a; S. Chen \u0026 Qi, 2025; Y. Chen et al., 2023; Z. Chen et al., 2024b; Z. Cheng et al., 2023b; Dipika Birari, 2024; Farzana et al., 2024; Ganji et al., 2025; Geng et al., 2024a; Ju et al., 2023a; Kim et al., 2024b; Y. Li, 2025; Z. Liang \u0026 Chen, 2025c; P. J. Lu et al., 2023; M.Saraswathi, 2025; K. I. Wang et al., 2022; S. Wang, 2024; Wei et al., 2021b; D. Xia et al., 2025; D. Xu et al., 2020; XUE, 2025c; Yaghini et al., 2023a; B. Zheng et al., 2024)",
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 are provided to fit local demands and requirements of each of the participants. With this, adaptive privac
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nes that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025
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They are able to be significantly smarter than the traditional privacy settings in the manner in which they allocate privacy resources in those areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise and trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the developments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets are provided to fit local demands and requirements of each of the participants. With this, adaptive privacy budgeting must be built and communicated to activate fair results and to make the system not favour the ones that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025b; Are et al., 2025; Badar et al., 2024a; S. Chen \u0026 Qi, 2025; Y. Chen et al., 2023; Z. Chen et al., 2024b; Z. 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They are able to be significantly smarter than the traditional privacy settings in the manner in which they allocate privacy resources in those areas where they are most needed, which would ensure a maintenance of a privacy at minimal compromise and trade-off of the accuracy and fairness. However, there are still difficulties that occur despite the developments. Even federated networks like FedSlate will yield disproportionate returns unless privacy budgets are provided to fit local demands and requirements of each of the participants. With this, adaptive privacy budgeting must be built and communicated to activate fair results and to make the system not favour the ones that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025b; Are et al., 2025; Badar et al., 2024a; S. Chen \u0026 Qi, 2025; Y. Chen et al., 2023; Z. Chen et al., 2024b; Z. Cheng et al., 2023b; Dipika Birari, 2024; Farzana et al., 2024; Ganji et al., 2025; Geng et al., 2024a; Ju et al., 2023a; Kim et al., 2024b; Y. Li, 2025; Z. Liang \u0026 Chen, 2025c; P. J. Lu et al., 2023; M.Saraswathi, 2025; K. I. Wang et al., 2022; S. Wang, 2024; Wei et al., 2021b; D. Xia et al., 2025; D. Xu et al., 2020; XUE, 2025c; Yaghini et al., 2023a; B. Zheng et al., 2024)",
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csl-citation.json\"} It is shown through empirical literature that adaptive privacy approaches result in a
huge loss in utility and simultaneously, they result in targeted privacy. They are able to be significantly
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nes that receive the least of users and penalize the ones that belong to the stronger dataset.(Arafat, 2025
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Blockchain-Enabled, Federated Learning Platform for Algorithmic Fairness in Predictive Health Care: Design Science Study\",\"author\":[{\"family\":\"Liang\",\"given\":\"Xueping\"},{\"family\":\"Zhao\",\"given\":\"Juan\"},{\"family\":\"Chen\",\"given\":\"Yan\"},{\"family\":\"Bandara\",\"given\":\"Eranga\"},{\"family\":\"Shetty\",\"given\":\"Sachin\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":659,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/MSGSTQGH\"],\"itemData\":{\"id\":659,\"type\":\"article-journal\",\"container-title\":\"Frontiers in Big Data\",\"DOI\":\"10.3389/fdata.2023.1249997\",\"title\":\"Differential Privacy in Collaborative Filtering Recommender Systems: A 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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma

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Jerry\"},{\"family\":\"Reddi\",\"given\":\"Sashank J.\"},{\"family\":\"Richtárik\",\"given\":\"Peter\"},{\"family\":\"Singhal\",\"given\":\"Karan\"},{\"family\":\"Smith\",\"given\":\"Virginia\"},{\"family\":\"Soltanolkotabi\",\"given\":\"Mahdi\"},{\"family\":\"Song\",\"given\":\"Weikang\"},{\"family\":\"Suresh\",\"given\":\"Ananda Theertha\"},{\"family\":\"Stich\",\"given\":\"Sebastian U.\"},{\"family\":\"Talwalkar\",\"given\":\"Ameet\"},{\"family\":\"Wang\",\"given\":\"Hongyi\"},{\"family\":\"Woodworth\",\"given\":\"Blake\"},{\"family\":\"Wu\",\"given\":\"Shanshan\"},{\"family\":\"Yu\",\"given\":\"Felix X.\"},{\"family\":\"Yuan\",\"given\":\"Honglin\"},{\"family\":\"Zaheer\",\"given\":\"Manzil\"},{\"family\":\"Zhang\",\"given\":\"Mi\"},{\"family\":\"Zhang\",\"given\":\"Tong\"},{\"family\":\"Zheng\",\"given\":\"Chunxiang\"},{\"family\":\"Zhu\",\"given\":\"Chen\"},{\"family\":\"Zhu\",\"given\":\"Wennan\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":650,\"uris\

":[\"http://zotero.org/users/local/M0u71wYg/items/W6WNYZ79\"],\"itemData\":{\"id\":650,\"type\":\"article-journal\",\"DOI\":\"10.1145/3543873.3587577\",\"title\":\"Fairness-Aware Differentially Private Collaborative Filtering\",\"author\":[{\"family\":\"Yang\",\"given\":\"Zhenhuan\"},{\"family\":\"Ge\",\"given\":\"Yingqiang\"},{\"family\":\"Su\",\"given\":\"Congzhe\"},{\"family\":\"Wang\",\"given\":\"Dingxian\"},{\"family\":\"Zhao\",\"given\":\"Xiaoting\"},{\"family\":\"Ying\",\"given\":\"Yiming\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":660,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/QNNRLT8A\"],\"itemData\":{\"id\":660,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics13112023\",\"title\":\"A Federated-Learning Algorithm Based on Client Sampling and Gradient Projection for the Smart Grid\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Ruifeng\"},{\"family\":\"Lu\",\"given\":\"Jiangang\"},{\"family\":\"Liu\",\"given\":\"Zewei\"},{\"family\":\"Wang\",\"given\":\"Tianqi\"},{\"family\":\"Guo\",\"given\":\"Wenxin\"},{\"family\":\"Lan\",\"given\":\"Tian\"},{\"family\":\"Hu\",\"given\":\"Chunqiang\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":673,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/HKBKNYD2\"],\"itemData\":{\"id\":673,\"type\":\"article-journal\",\"container-title\":\"Expert Systems\",\"DOI\":\"10.1111/exsy.70173\",\"title\":\"Personalised Recommendation With Federated Lightweight Graph Convolutional Networks\",\"author\":[{\"family\":\"Zhu\",\"given\":\"Yansong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":654,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/MAC94JPG\"],\"itemData\":{\"id\":654,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2205.13121\",\"title\":\"Cali3F: Calibrated Fast Fair Federated Recommendation System\",\"author\":[{\"family\":\"Zhu\",\"given\":\"Zhitao\"},{\"family\":\"Si\",\"given\":\"Shijing\"},{\"family\":\"Wang\",\"given\":\"Jianzong\"},{\"family\":\"Xiao\",\"given\":\"Jing\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dynamic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness should be one of the research directions in the future. It is noteworthy especially in heterogeneous user population and complex data environment where a one-size-fits-all approach is most likely to fail. In addition, in federated learning systems where the interaction information between the users is sparse it is dire to apply data augmentation, as well as skilled spreading of information across richer datasets in federated learning. Such strategies will have the capability of enhancing both strength and effectiveness of models of recommendations which will be capable of operating in a dependable manner even during data paucity. Last but not the least, federated recommendation system development requires the ability to balance between the maintenance of privacy and high utility and equity such that every one of the users experiences safe, efficient and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H. Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini et al., 2023a; Ju et al., 2023b; Kim et al., 2024c; Lee \u0026 Gargroetzi, 2023; X. Liang et al., 2023; Müllner et al., 2023; Odera, 2023; Padala et al., 2021a; Rodríguez-Gálvez et al., 2021a; Shiranthika et al., 2023a; Siniosoglou et al., 2023; J. A. Sun et al., 2023b; K. Sun, 2025b; J. Wang et al., 2021b; Z. Yang et al., 2023b; R. Zhao et al., 2024a; Y. Zhu, 2025b; Z. Zhu et al., 2022a)",
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It is noteworthy especially in heterogeneous user population and complex data environment where a one-size-fits-all approach is most likely to fail. In addition, in federated learning systems where the interaction information between the users is sparse it is dire to apply data augmentation, as well as skilled spreading of information across richer datasets in federated learning. Such strategies will have the capability of enhancing both strength and effectiveness of models of recommendations which will be capable of operating in a dependable manner even during data paucity. Last but not the least, federated recommendation system development requires the ability to balance between the maintenance of privacy and high utility and equity such that every one of the users experiences safe, efficient and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H. Chen et al., 2023; W. 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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
 Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini
 et al., 2023a; Ju et al., 2023b; Kim et al., 2024c; Lee \u0026 Gargroetzi, 2023; X. Liang et al., 2023; Mü
llner et al., 2023; Odera, 2023; Padala et al., 2021a; Rodríguez-Gálvez et al., 2021a; Shiranthika et al.,
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It is noteworthy especially in heterogeneous user population and complex data environment where a one-size-fits-all approach is most likely to fail. In addition, in federated learning systems where the interaction information between the users is sparse it is dire to apply data augmentation, as well as skilled spreading of information across richer datasets in federated learning. Such strategies will have the capability of enhancing both strength and effectiveness of models of recommendations which will be capable of operating in a dependable manner even during data paucity. Last but not the least, federated recommendation system development requires the ability to balance between the maintenance of privacy and high utility and equity such that every one of the users experiences safe, efficient and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H. Chen et al., 2023; W. 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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
 Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini
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It is noteworthy especially in heterogeneous user population and complex data environment where a one-size-fits-all approach is most likely to fail. In addition, in federated learning systems where the interaction information between the users is sparse it is dire to apply data augmentation, as well as skilled spreading of information across richer datasets in federated learning. Such strategies will have the capability of enhancing both strength and effectiveness of models of recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu

t not the least, federated recommendation system development requires the ability to balance between the maintenance of privacy and high utility and equity such that every one of the users experiences safe, efficient and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H. Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini et al., 2023a; Ju et al., 2023b; Kim et al., 2024c; Lee \u0026 Gargroetzi, 2023; X. Liang et al., 2023; Müllner et al., 2023; Odera, 2023; Padala et al., 2021a; Rodríguez-Gálvez et al., 2021a; Shiranthika et al., 2023a; Siniosoglou et al., 2023; J. A. Sun et al., 2023b; K. Sun, 2025b; J. Wang et al., 2021b; Z. Yang et al., 2023b; R. Zhao et al., 2024a; Y. Zhu, 2025b; Z. Zhu et al., 2022a)",
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
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It is noteworthy especially in heterogeneous user population and complex data environment where a one-size-fits-all approach is most likely to fail. In addition, in federated learning systems where the interaction information between the users is sparse it is dire to apply data augmentation, as well as skilled spreading of information across richer datasets in federated learning. Such strategies will have the capability of enhancing both strength and effectiveness of models of recommendations which will be capable of operating in a dependable manner even during data paucity. Last but not the least, federated recommendation system development requires the ability to balance between the maintenance of privacy and high utility and equity such that every one of the users experiences safe, efficient and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H. Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini et al., 2023a; Ju et al., 2023b; Kim et al., 2024c; Lee \u0026 Gargroetzi, 2023; X. Liang et al., 2023; Müllner et al., 2023; Odera, 2023; Padala et al., 2021a; Rodríguez-Gálvez et al., 2021a; Shiranthika et al., 2023a; Siniosoglou et al., 2023; J. A. Sun et al., 2023b; K. Sun, 2025b; J. Wang et al., 2021b; Z. Yang et al., 2023b; R. Zhao et al., 2024a; Y. Zhu, 2025b; Z. Zhu et al., 2022a)",
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dynamic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness should be one of the research directions in the future. It is noteworthy especially in heterogeneous user population and complex data environment where a one-size-fits-all approach is most likely to fail. In addition, in federated learning systems where the interaction information between the users is sparse it is dire to apply data augmentation, as well as skilled spreading of information across richer datasets in federated learning. Such strategies will have the capability of enhancing both strength and effectiveness of models of recommendations which will be capable of operating in a dependable manner even during data paucity. Last but not the least, federated recommendation system development requires the ability to balance between the maintenance of privacy and high utility and equity such that every one of the users experiences safe, efficient and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H. Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini et al., 2023a; Ju et al., 2023b; Kim et al., 2024c; Lee \u0026 Gargroetzi, 2023; X. Liang et al., 2023; Müllner et al., 2023; Odera, 2023; Padala et al., 2021a; Rodríguez-Gálvez et al., 2021a; Shiranthika et al., 2023a; Siniosoglou et al., 2023; J. A. Sun et al., 2023b; K. Sun, 2025b; J. Wang et al., 2021b; Z. Yang et al., 2023b; R. Zhao et al., 2024a; Y. Zhu, 2025b; Z. Zhu et al., 2022a)",
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
 Chen et al., 2023; W. Chen, 2025; Chu et al., 2021a; Cong et al., 2023; Desai \u0026 Verma, 2019; Hosseini
 et al., 2023a; Ju et al., 2023b; Kim et al., 2024c; Lee \u0026 Gargroetzi, 2023; X. Liang et al., 2023; Mü
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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on-style-language/schema/raw/master/csl-citation.json\"} A research direction that is aimed at improving dy
namic privacy systems, which maximize the trade-off which exists between privacy, utility, and fairness sho
uld be one of the research directions in the future. It is noteworthy especially in heterogeneous user popu
lation and complex data environment where a one-size-fits-all approach is most likely to fail. In addition,
 in federated learning systems where the interaction information between the users is sparse it is dire to
apply data augmentation, as well as skilled spreading of information across richer datasets in federated le
arning. Such strategies will have the capability of enhancing both strength and effectiveness of models of
recommendations which will be capable of operating in a dependable manner even during data paucity. Last bu
t not the least, federated recommendation system development requires the ability to balance between the ma
intenance of privacy and high utility and equity such that every one of the users experiences safe, efficie
nt and personalised recommendation in their everyday lives online. (Agrawal et al., 2024; Boratto, 2020; H.
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hub.com/citation-style-language/schema/raw/master/csl-citation.json\"} The section under consideration eval
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hub.com/citation-style-language/schema/raw/master/csl-citation.json\"} The section under consideration eval
uates the literature content related to federated learning in reference to recommendation systems with part
icular focus on privacy-preserving tools and their impact on the model results. It reviews as well a number
 of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in
 federated recommender systems and how the rating bias is effectively handled by the models. The latest inn
ovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused o
n matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmiss
ion of the gradients. Regardless of such developments, the majority of the existing literature is inclined
to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integ
rity of the systems that play a fundamental role in generating effective and robust recommender systems. Ev
en though some of the literature attempts to enhance effectiveness of federated recommender system by sugge
sting mechanisms through which convergence can be expedited or lightweight communication based on lightweig
ht communication is offered, a holistic understanding of veracity and faithfulness of federated recommender
 system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportu
nities of higher performance and privacy facilitated by the adoption of foundation models in federated reco
mmendation systems is the fact that the technology is still in its infancy. More specifically, they apply t
heir massive pre-training knowledge in the models to improve the quality of the recommendations and federat
ed learning is used to make the sensitive user interaction data fed to fine-tune the models local. However,
 to fully incorporate the opportunities suggested by the foundation models in federated recommendation syst
ems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabili
ties can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model ag
gregation. Paradigm shift in this federated environment suggested in the recent literature using large lang
uage model will entail simplicity of data integration and complete utilization of large language model to a
ttain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier
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ht communication is offered, a holistic understanding of veracity and faithfulness of federated recommender
 system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportu
nities of higher performance and privacy facilitated by the adoption of foundation models in federated reco
mmendation systems is the fact that the technology is still in its infancy. More specifically, they apply t
heir massive pre-training knowledge in the models to improve the quality of the recommendations and federat
ed learning is used to make the sensitive user interaction data fed to fine-tune the models local. However,
 to fully incorporate the opportunities suggested by the foundation models in federated recommendation syst
ems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabili
ties can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model ag
gregation. Paradigm shift in this federated environment suggested in the recent literature using large lang
uage model will entail simplicity of data integration and complete utilization of large language model to a
ttain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier
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nities of higher performance and privacy facilitated by the adoption of foundation models in federated recommendation systems is the fact that the technology is still in its infancy. More specifically, they apply their massive pre-training knowledge in the models to improve the quality of the recommendations and federated learning is used to make the sensitive user interaction data fed to fine-tune the models local. However, to fully incorporate the opportunities suggested by the foundation models in federated recommendation systems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabilities can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model aggregation. Paradigm shift in this federated environment suggested in the recent literature using large language model will entail simplicity of data integration and complete utilization of large language model to attain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier et al., 2020; Bo, 2025; H. Chen et al., 2024; D. Guo \u0026 Choo, 2025; Hartmann et al., 2025; Hosseinalipour et al., 2020; A. N. Khan et al., 2024; Mohammadi et al., 2022; R. Song et al., 2022; S. Wang et al., 2021; X. Wu et al., 2020; G. Yan, 2025a; D. Yao et al., 2021; Yi et al., 2021c; X. Yu et al., 2024; D. Zhang et al., 2023; J. Zhang, Hu, et al., 2023; A. Zhou et al., 2024; Z. Zhu et al., 2022b)",
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It reviews as well a number of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in federated recommender systems and how the rating bias is effectively handled by the models. The latest innovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused on matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmission of the gradients. Regardless of such developments, the majority of the existing literature is inclined to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integrity of the systems that play a fundamental role in generating effective and robust recommender systems. 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Even though some of the literature attempts to enhance effectiveness of federated recommender system by suggesting mechanisms through which convergence can be expedited or lightweight communication based on lightweight communication is offered, a holistic understanding of veracity and faithfulness of federated recommender system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportunities of higher performance and privacy facilitated by the adoption of foundation models in federated recommendation systems is the fact that the technology is still in its infancy. More specifically, they apply t

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J.\"},{\"family\":\"Li\",\"given\":\"Pu\"},{\"family\":\"Zhou\",\"given\":\"Shuaishuai\"},{\"family\":\"Peng\",\"given\":\"Hao\"},{\"family\":\"Sun\",\"given\":\"Li\"},{\"family\":\"Zhu\",\"given\":\"Liehuang\"},{\"family\":\"Yu\",\"given\":\"Philip S.\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":692,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/GVBNYN4C\"],\"itemData\":{\"id\":692,\"type\":\"article-journal\",\"container-title\":\"Journal of Cloud Computing Advances Systems and Applications\",\"DOI\":\"10.1186/s13677-023-00515-6\",\"title\":\"Adaptive Device Sampling and Deadline Determination for Cloud-Based Heterogeneous Federated Learning\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Deyu\"},{\"family\":\"Sun\",\"given\":\"Wang\"},{\"family\":\"Zheng\",\"given\":\"Zi-Ang\"},{\"family\":\"Chen\",\"given\":\"Wenxin\"},{\"family\":\"He\",\"given\":\"Shiwen\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":698,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7I8S96G4\"],\"itemData\":{\"id\":698,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics12214476\",\"title\":\"Federated Learning Based on Mutual Information Clustering for Wireless Traffic Prediction\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jianwei\"},{\"family\":\"Hu\",\"given\":\"Xin■Hua\"},{\"family\":\"Cai\",\"given\":\"Zengyu\"},{\"family\":\"Zhu\",\"given\":\"Liang\"},{\"family\":\"Feng\",\"given\":\"Yuan\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":694,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/492495GH\"],\"itemData\":{\"id\":694,\"type\":\"article-journal\",\"container-title\":\"World Electric Vehicle Journal\",\"DOI\":\"10.3390/wevj15010018\",\"title\":\"Asynchronous Robust Aggregation Method With Privacy Protection for IoV Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Antong\"},{\"family\":\"Jiang\",\"given\":\"Ning\"},{\"family\":\"Tang\",\"given\":\"Tong\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":684,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/4N9RV6QM\"],\"itemData\":{\"id\":684,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2205.13121\",\"title\":\"Cali3F: Calibrated Fast Fair Federated Recommendation System\",\"author\":[{\"family\":\"Zhu\",\"given\":\"Zhitao\"},{\"family\":\"Si\",\"given\":\"Shijing\"},{\"family\":\"Wang\",\"given\":\"Jianzong\"},{\"family\":\"Xiao\",\"given\":\"Jing\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The section under consideration evaluates the literature content related to federated learning in reference to recommendation systems with particular focus on privacy-preserving tools and their impact on the model results. It reviews as well a number of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in federated recommender systems and how the rating bias is effectively handled by the models. The latest innovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused on matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmission of the gradients. Regardless of such developments, the majority of the existing literature is inclined to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integrity of the systems that play a fundamental role in generating effective and robust recommender systems. Even though some of the literature attempts to enhance effectiveness of federated recommender system by suggesting mechanisms through which convergence can be expedited or lightweight communication based on lightweight communication is offered, a holistic understanding of veracity and faithfulness of federated recommender system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportunities of higher performance and privacy facilitated by the adoption of foundation models in federated recommendation systems is the fact that the technology is still in its infancy. More specifically, they apply their massive pre-training knowledge in the models to improve the quality of the recommendations and federated learning is used to make the sensitive user interaction data fed to fine-tune the models local. However, to fully incorporate the opportunities suggested by the foundation models in federated recommendation systems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabilities can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model aggregation. Paradigm shift in this federated environment suggested in the recent literature using large language model will entail simplicity of data integration and complete utilization of large language model to a

ttain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier
 et al., 2020; Bo, 2025; H. Chen et al., 2024; D. Guo \u0026 Choo, 2025; Hartmann et al., 2025; Hosseinalip
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hub.com/citation-style-language/schema/raw/master/csl-citation.json\"} The section under consideration evaluates the literature content related to federated learning in reference to recommendation systems with particular focus on privacy-preserving tools and their impact on the model results. It reviews as well a number of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in federated recommender systems and how the rating bias is effectively handled by the models. The latest innovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused on matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmission of the gradients. Regardless of such developments, the majority of the existing literature is inclined to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integrity of the systems that play a fundamental role in generating effective and robust recommender systems. Even though some of the literature attempts to enhance effectiveness of federated recommender system by suggesting mechanisms through which convergence can be expedited or lightweight communication based on lightweight communication is offered, a holistic understanding of veracity and faithfulness of federated recommender system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportunities of higher performance and privacy facilitated by the adoption of foundation models in federated recommendation systems is the fact that the technology is still in its infancy. More specifically, they apply their massive pre-training knowledge in the models to improve the quality of the recommendations and federated learning is used to make the sensitive user interaction data fed to fine-tune the models local. However, to fully incorporate the opportunities suggested by the foundation models in federated recommendation systems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabilities can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model aggregation. Paradigm shift in this federated environment suggested in the recent literature using large language model will entail simplicity of data integration and complete utilization of large language model to attain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier et al., 2020; Bo, 2025; H. Chen et al., 2024; D. Guo \u0026 Choo, 2025; Hartmann et al., 2025; Hosseinalipour et al., 2020; A. N. Khan et al., 2024; Mohammadi et al., 2022; R. Song et al., 2022; S. Wang et al., 2021; X. Wu et al., 2020; G. Yan, 2025a; D. Yao et al., 2021; Yi et al., 2021c; X. Yu et al., 2024; D. Zhang et al., 2023; J. Zhang, Hu, et al., 2023; A. Zhou et al., 2024; Z. Zhu et al., 2022b)",

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hub.com/citation-style-language/schema/raw/master/csl-citation.json\"} The section under consideration eval
uates the literature content related to federated learning in reference to recommendation systems with part
icular focus on privacy-preserving tools and their impact on the model results. It reviews as well a number
 of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in
 federated recommender systems and how the rating bias is effectively handled by the models. The latest inn
ovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused o
n matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmiss
ion of the gradients. Regardless of such developments, the majority of the existing literature is inclined
 to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integ
rity of the systems that play a fundamental role in generating effective and robust recommender systems. Ev
en though some of the literature attempts to enhance effectiveness of federated recommender system by sugge
sting mechanisms through which convergence can be expedited or lightweight communication based on lightweig
ht communication is offered, a holistic understanding of veracity and faithfulness of federated recommender
 system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportu
nities of higher performance and privacy facilitated by the adoption of foundation models in federated reco
mmendation systems is the fact that the technology is still in its infancy. More specifically, they apply t
heir massive pre-training knowledge in the models to improve the quality of the recommendations and federat
ed learning is used to make the sensitive user interaction data fed to fine-tune the models local. However,
 to fully incorporate the opportunities suggested by the foundation models in federated recommendation syst
ems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabili
ties can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model ag
gregation. Paradigm shift in this federated environment suggested in the recent literature using large lang
uage model will entail simplicity of data integration and complete utilization of large language model to a
ttain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier
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hub.com/citation-style-language/schema/raw/master/csl-citation.json\"} The section under consideration eval
uates the literature content related to federated learning in reference to recommendation systems with part
icular focus on privacy-preserving tools and their impact on the model results. It reviews as well a number
 of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in
 federated recommender systems and how the rating bias is effectively handled by the models. The latest inn
ovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused o
n matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmiss
ion of the gradients. Regardless of such developments, the majority of the existing literature is inclined
to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integ
rity of the systems that play a fundamental role in generating effective and robust recommender systems. Ev
en though some of the literature attempts to enhance effectiveness of federated recommender system by sugge
sting mechanisms through which convergence can be expedited or lightweight communication based on lightweig
ht communication is offered, a holistic understanding of veracity and faithfulness of federated recommender
 system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportu
nities of higher performance and privacy facilitated by the adoption of foundation models in federated reco
mmendation systems is the fact that the technology is still in its infancy. More specifically, they apply t
heir massive pre-training knowledge in the models to improve the quality of the recommendations and federat
ed learning is used to make the sensitive user interaction data fed to fine-tune the models local. However,
 to fully incorporate the opportunities suggested by the foundation models in federated recommendation syst
ems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabili
ties can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model ag
gregation. Paradigm shift in this federated environment suggested in the recent literature using large lang
uage model will entail simplicity of data integration and complete utilization of large language model to a
ttain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier
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to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integ
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heir massive pre-training knowledge in the models to improve the quality of the recommendations and federat
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 to fully incorporate the opportunities suggested by the foundation models in federated recommendation syst
ems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabili
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ttain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier
 et al., 2020; Bo, 2025; H. Chen et al., 2024; D. Guo \u0026 Choo, 2025; Hartmann et al., 2025; Hosseinalip
our et al., 2020; A. N. Khan et al., 2024; Mohammadi et al., 2022; R. Song et al., 2022; S. Wang et al., 20
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It reviews as well a number of strategies of how to solve the challenges of data heterogeneity, communication efficiency and equity in federated recommender systems and how the rating bias is effectively handled by the models. The latest innovations in federated learning as a recommendation system, such as FedRec and FCF, have primarily focused on matrix-factorization-based collaborative filtering, and FedMF also has a homomorphic encryption transmission of the gradients. Regardless of such developments, the majority of the existing literature is inclined to disapprove of the multifacetedness of the challenges of fairness, prejudice, and the threat to the integrity of the systems that play a fundamental role in generating effective and robust recommender systems. Even though some of the literature attempts to enhance effectiveness of federated recommender system by suggesting mechanisms through which convergence can be expedited or lightweight communication based on lightweight communication is offered, a holistic understanding of veracity and faithfulness of federated recommender system with various model of attacks is yet to be accomplished as a research. Moreover, one of the opportunities of higher performance and privacy facilitated by the adoption of foundation models in federated recommendation systems is the fact that the technology is still in its infancy. More specifically, they apply their massive pre-training knowledge in the models to improve the quality of the recommendations and federated learning is used to make the sensitive user interaction data fed to fine-tune the models local. However, to fully incorporate the opportunities suggested by the foundation models in federated recommendation systems, a novel area of research is required to explore the phenomenon of how their pre-training, and capabilities can be used to mitigate the problem of heterogeneity in data and improve the effectiveness of model aggregation. Paradigm shift in this federated environment suggested in the recent literature using large language model will entail simplicity of data integration and complete utilization of large language model to attain better results, but, at the expense of augmented communication due to large parameter sizes. (Béguier et al., 2020; Bo, 2025; H. Chen et al., 2024; D. Guo \u0026 Choo, 2025; Hartmann et al., 2025; Hosseinalipour et al., 2020; A. N. Khan et al., 2024; Mohammadi et al., 2022; R. Song et al., 2022; S. Wang et al., 2021; X. Wu et al., 2020; G. Yan, 2025a; D. Yao et al., 2021; Yi et al., 2021c; X. Yu et al., 2024; D. Zhang et al., 2023; J. 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Convolutional Networks\",\"author\":[{\"family\":\"Zhu\",\"given\":\"Yansong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and their impact on the performance of the model are discussed. It also dwells into various mechanisms of coping with such problems as heterogeneity of data, efficiency of communication and equity of federated recommender systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent trends in federated learning as a recommendation system which include FedRec and FCF, has largely focused on collaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encoding aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing studies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, which play a vital role in the formulation of effective and trustworthy recommender systems. However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by accelerating convergence or by using light-weight communication schemes. In addition to that, the foundation models supported by federated recommendation systems represent a shining light to high-performance and privacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better the quality of the recommendations in which also federated learning maintains the sensitive user interaction data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to the federated recommendation system is achieved with the perception of how their pre-training capabilities can be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model aggregation. Whereas the recent literature suggests as an application of large language models, the idea of large language models in such a federated structure suggests rather a change in the thinking of data integration, which is not necessitating a complete utilization of the large language models but suggests that it will be driven by a paradigm shift to the complete utilization of the large parameters in data integration at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.

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mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu

dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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Viet\"},{\"family\":\"Cui\",\"given\":\"Lizhen\"},{\"family\":\"He\",\"given\":\"Tieke\"},{\"family\":\"Yin\",\"given\":\"Hongzhi\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":707,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JGF5UX6J\"],\"itemData\":{\"id\":707,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3034770\",\"title\":\"D2PDRF: A Differential Privacy Recommendation Algorithm Based on Federated Learning\",\"author\":[{\"family\":\"zeng\",\"given\":\"jiaqi\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":720,\"uris\":[\"http://zotero.org/users/

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However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by accelerating convergence or by using light-weight communication schemes. In addition to that, the foundation models supported by federated recommendation systems represent a shining light to high-performance and privacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better the quality of the recommendations in which also federated learning maintains the sensitive user interaction data utilised in a fine-tuning process localised. 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n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning 
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender 
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite 
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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25; Qiu et al., 2022; Ribero et al., 2020; X. Wang et al., 2022b; Z. Xu et al., 2024; L. Yang et al., 2021;
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite

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of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by accelerating convergence or by using light-weight communication schemes. In addition to that, the foundation models supported by federated recommendation systems represent a shining light to high-performance and privacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better the quality of the recommendations in which also federated learning maintains the sensitive user interaction data utilised in a fine-tuning process localised. 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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and their impact on the performance of the model are discussed. It also dwells into various mechanisms of coping with such problems as heterogeneity of data, efficiency of communication and equity of federated recommender systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent trends in federated learning as a recommendation system which include FedRec and FCF, has largely focused on collaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encoding aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing studies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, which play a vital role in the formulation of effective and trustworthy recommender systems. However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite

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of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning 
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender 
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite 
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning 
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender 
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite 
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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Convolutional Networks\",\"author\":[{\"family\":\"Zhu\",\"given\":\"Yansong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and their impact on the performance of the model are discussed. It also dwells into various mechanisms of coping with such problems as heterogeneity of data, efficiency of communication and equity of federated recommender systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent trends in federated learning as a recommendation system which include FedRec and FCF, has largely focused on collaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encoding aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing studies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, which play a vital role in the formulation of effective and trustworthy recommender systems. However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by accelerating convergence or by using light-weight communication schemes. In addition to that, the foundation models supported by federated recommendation systems represent a shining light to high-performance and privacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better the quality of the recommendations in which also federated learning maintains the sensitive user interaction data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to the federated recommendation system is achieved with the perception of how their pre-training capabilities c

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an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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Convolutional Networks\",\"author\":[{\"family\":\"Zhu\",\"given\":\"Yansong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and their impact on the performance of the model are discussed. It also dwells into various mechanisms of coping with such problems as heterogeneity of data, efficiency of communication and equity of federated recommender systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent trends in federated learning as a recommendation system which include FedRec and FCF, has largely focused on collaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encoding aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing studies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, which play a vital role in the formulation of effective and trustworthy recommender systems. However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by accelerating convergence or by using light-weight communication schemes. In addition to that, the foundation models supported by federated recommendation systems represent a shining light to high-performance and privacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models , on the one hand, take advantages of their widely computerised pre-training knowledge in order to better the quality of the recommendations in which also federated learning maintains the sensitive user interaction data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to the federated recommendation system is achieved with the perception of how their pre-training capabilities can be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model aggregation. Whereas the recent literature suggests as an application of large language models, the idea of large language models in such a federated structure suggests rather a change in the thinking of data integration, which is not necessitating a complete utilization of the large language models but suggests that it will be driven by a paradigm shift to the complete utilization of the large parameters in data integration at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.

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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning

mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and their impact on the performance of the model are discussed. It also dwells into various mechanisms of coping with such problems as heterogeneity of data, efficiency of communication and equity of federated recommender systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent trends in federated learning as a recommendation system which include FedRec and FCF, has largely focused on collaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encoding aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing studies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, which play a vital role in the formulation of effective and trustworthy recommender systems. However, the research of fairness, integrity in the circumstances of various attacks is an active research question in spite of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by accelerating convergence or by using light-weight communication schemes. In addition to that, the foundation models supported by federated recommendation systems represent a shining light to high-performance and privacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better the quality of the recommendations in which also federated learning maintains the sensitive user interaction data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to the federated recommendation system is achieved with the perception of how their pre-training capabilities can be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model aggregation. Whereas the recent literature suggests as an application of large language models, the idea of large language models in such a federated structure suggests rather a change in the thinking of data integration, which is not necessitating a complete utilization of the large language models but suggests that it will be driven by a paradigm shift to the complete utilization of the large parameters in data integration at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H. Hu et al., 2021; Q. Hu \u0026 Song, 2024c; R. Li et al., 2024b; T. Li, Song, et al., 2020; Z. Li et al., 2023b; Y. Lin et al., 2020; L. Liu et al., 2024; R. Liu et al., 2022b; Z. Liu et al., 2022; Otari et al., 2025; Qiu et al., 2022; Ribero et al., 2020; X. Wang et al., 2022b; Z. Xu et al., 2024; L. Yang et al., 2021; Yi et al., 2021d, 2022; Yuan et al., 2023; zeng, 2024b; H. Zhang et al., 2022; S. Zhang \u0026 Yin, 2022b; Y. Zhu, 2025c)",
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/raw/master/csl-citation.json\"} In this part, the available literature on the topic of federated learning 
mechanism in recommendation systems is presented and, in particular, the privacy-saving mechanisms and thei
r impact on the performance of the model are discussed. It also dwells into various mechanisms of coping wi
th such problems as heterogeneity of data, efficiency of communication and equity of federated recommender 
systems and how the models can be utilized to effectively deal with the problem of rating bias. Recent tren
ds in federated learning as a recommendation system which include FedRec and FCF, has largely focused on co
llaborative filtering and matrix factorization of the matrices FedMF also incorporates a homomorphic encodi
ng aspect to transmit gradients in a secure manner. In spite of such developments, many of the existing stu
dies are inclined to dwell on the primitive aspects of equity, bigotry, and risks to system integrity, whic
h play a vital role in the formulation of effective and trustworthy recommender systems. However, the resea
rch of fairness, integrity in the circumstances of various attacks is an active research question in spite 
of the particularity of other studies aimed at enhancing the efficiency of federated recommender systems by
 accelerating convergence or by using light-weight communication schemes. In addition to that, the foundati
on models supported by federated recommendation systems represent a shining light to high-performance and p
rivacy despite the fact that the sphere is still at its initial developmental phases. Precisely, the models
, on the one hand, take advantages of their widely computerised pre-training knowledge in order to better t
he quality of the recommendations in which also federated learning maintains the sensitive user interaction
 data utilised in a fine-tuning process localised. However, the complete capacity of foundation models to t
he federated recommendation system is achieved with the perception of how their pre-training capabilities c
an be applied to address the problem of data heterogeneity and serve to optimize the effectiveness of model
 aggregation. Whereas the recent literature suggests as an application of large language models, the idea o
f large language models in such a federated structure suggests rather a change in the thinking of data inte
gration, which is not necessitating a complete utilization of the large language models but suggests that i
t will be driven by a paradigm shift to the complete utilization of the large parameters in data integratio
n at an increased communication cost. (Z. Chen et al., 2022c; H. Dai et al., 2024; Y. Ding et al., 2024; H.
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ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th
e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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Wagh, 2025; X. Wang, 2022; B. Wu et al., 2025; Xiang, 2025; Y. Zhang et al., 2017)\",\"noteIndex\":0},\"cit
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M.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":747,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LJH98IMD\"],\"itemData\":{\"id\":747,\"type\":\"article-journal\",\"container-title\":\"Expert Systems\",\"DOI\":\"10.1111/exsy.70053\",\"title\":\"Deep Learning■Based Privacy Preserving Multimodal Biometrics Recognition for Cross■Silo Datasets\",\"author\":[{\"family\":\"Kansal\",\"given\":\"Isha\"},{\"family\":\"Khuallar\",\"given\":\"Vikas\"},{\"family\":\"Gupta\",\"given\":\"Gifty\"},{\"family\":\"Gupta\",\"given\":\"Deepali\"},{\"family\":\"Juneja\",\"given\":\"Sapna\"},{\"family\":\"Nauman\",\"given\":\"Ali\"},{\"family\":\"Muhammad\",\"given\":\"Ghulam\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":735,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/24RGEQIZ\"],\"itemData\":{\"id\":735,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics12183867\",\"title\":\"Trustworthy Anti-Collusion Federated Learning 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Detection\",\"author\":[{\"family\":\"Samee\",\"given\":\"Nagwan Abdel\"},{\"family\":\"Khan\",\"given\":\"Umair\"},{\"family\":\"Khan\",\"given\":\"Salabat\"},{\"family\":\"Jamjo

om\",\"given\":\"Mona\"},{\"family\":\"Sharif\",\"given\":\"Muhammad\"},{\"family\":\"Kim\",\"given\":\"Do Hyuen\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":746,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/I9Q6AAG8\"],\"itemData\":{\"id\":746,\"type\":\"article-journal\",\"container-title\":\"Ieee Access\",\"DOI\":\"10.1109/access.2023.3281832\",\"title\":\"Decentralized Learning in Healthcare: A Review of Emerging Techniques\",\"author\":[{\"family\":\"Shiranthika\",\"given\":\"Chamani\"},{\"family\":\"Saedi\",\"given\":\"Parvaneh\"},{\"family\":\"Baji■\",\"given\":\"Ivan V.\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":743,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/82ZA4L7D\"],\"itemData\":{\"id\":743,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3052237\",\"title\":\"Federated Learning-Based Intrusion Detection System for Industrial Internet of Things: Enhancing Security and Efficiency\",\"author\":[{\"family\":\"Sun\",\"given\":\"Yan\"},{\"family\":\"Liu\",\"given\":\"Caiyun\"},{\"family\":\"Weng\",\"given\":\"Ying\"},{\"family\":\"Liu\",\"given\":\"Yitong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":728,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3JNWYCPQ\"],\"itemData\":{\"id\":728,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3036891\",\"title\":\"Current Research and Prospects of Federated Language Large Models in the Medical Field\",\"author\":[{\"family\":\"Tang\",\"given\":\"Yumeng\"},{\"family\":\"Deng\",\"given\":\"Yuanpeng\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":737,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NYX7TZGI\"],\"itemData\":{\"id\":737,\"type\":\"article-journal\",\"container-title\":\"Sensors\",\"DOI\":\"10.3390/s25103083\",\"title\":\"Introducing a Quality-Driven Approach for Federated Learning\",\"author\":[{\"family\":\"Usman\",\"given\":\"Muhammad\"},{\"family\":\"Bernardi\",\"given\":\"Mario Luca\"},{\"family\":\"Cimitile\",\"given\":\"Marta\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":727,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NWMF6F7A\"],\"itemData\":{\"id\":727,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.02055\",\"title\":\"Private Map-Secure Reduce: Infrastructure for Efficient AI Data Markets\",\"author\":[{\"family\":\"Wagh\",\"given\":\"Sameer\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":750,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JU3XNPKP\"],\"itemData\":{\"id\":750,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.2640939\",\"title\":\"An Efficient Federated Learning Optimization Algorithm on Non-Iid Data\",\"author\":[{\"family\":\"Wang\",\"given\":\"Xue\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":748,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/74XPZY7U\"],\"itemData\":{\"id\":748,\"type\":\"article-journal\",\"container-title\":\"International Journal of Imaging Systems and Technology\",\"DOI\":\"10.1002/ima.70046\",\"title\":\"Brain Tumors Classification in \u003cscp\u003eMRIs\u003c/Scp\u003e Based on Personalized Federated Distillation Learning With Similarity■Preserving\",\"author\":[{\"family\":\"Wu\",\"given\":\"Bo\"},{\"family\":\"Shi\",\"given\":\"Donghui\"},{\"family\":\"Aguilar\",\"given\":\"José\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":745,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/UFSKEDQV\"],\"itemData\":{\"id\":745,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.09025\",\"title\":\"FLAD: Federated Learning for LLM-based Autonomous Driving in Vehicle-Edge-Cloud Networks\",\"author\":[{\"family\":\"Xiang\",\"given\":\"Tianao\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":738,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/C2UHNLLR\"],\"itemData\":{\"id\":738,\"type\":\"article-journal\",\"DOI\":\"10.1007/978-3-319-69035-3_39\",\"title\":\"Personalized Quality Centric Service Recommendation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yiwen\"},{\"family\":\"Ai\",\"given\":\"Xiaofei\"},{\"family\":\"He\",\"given\":\"Qiang\"},{\"family\":\"Zhang\",\"given\":\"Xuyun\"},{\"family\":\"Dou\",\"given\":\"Wanchun\"},{\"family\":\"Chen\",\"given\":\"Feifei\"},{\"family\":\"Chen\",\"given\":\"Liang\"},{\"family\":\"Yang\",\"given\":\"Yun\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integration, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution management issues that are situation in federated settings. Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramírez, 2024; J. Chen, 2021; Z. Chen et al., 2025; F. X. Fan et al., 2025; Hou et al., 2023; Jimenez-Gutierrez, 2025; Kansal et al., 2025; Q. Li et al., 2023; S. Lin et al., 2020; Y. Liu et al., 2023a, 2024; Raheem, 2025; Samee et al., 2023a; Shiranthika et al., 2023b; Y. Sun et al., 2025; Y. Tang \u0026 Deng, 2024; Usman et al., 2025; Wagh, 2025; X. Wang, 2022; B. Wu et al., 2025; Xiang, 2025; Y. Zhang et al., 2017)",
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integrat
ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th
e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integrat
ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th
e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re

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.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integrat
ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th
e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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Prefinetuning to Better Match User Data in Federated Learning\",\"author\":[{\"family\":\"Hou\",\"given\":\"Charlie\"},{\"family\":\"Zhan\",\"given\":\"Hongyuan\"},{\"family\":\"Shrivastava\",\"given\":\"Akshat\"},{\"family\":\"Wang\",\"given\":\"Sid\"},{\"family\":\"Livshits\",\"given\":\"Sasha\"},{\"family\":\"Fanti\",\"given\":\"Giulia\"},{\"family\":\"Lazar\",\"given\":\"Daniel A.\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":744,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/EG9YN36K\"],\"itemData\":{\"id\":744,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.11714\",\"title\":\"Federated Learning for Pediatric Pneumonia Detection: Enabling Collaborative Diagnosis Without Sharing Patient Data\",\"author\":[{\"family\":\"Jimenez-Gutierrez\",\"given\":\"Daniel M.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":747,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LJH98IMD\"],\"itemData\":{\"id\":747,\"type\":\"article-journal\",\"container-title\":\"Expert Systems\",\"DOI\":\"10.1111/exsy.70053\",\"title\":\"Deep Learning■Based Privacy Preserving Multimodal Biometrics Recognition for Cross■Silo Datasets\",\"author\":[{\"family\":\"Kansal\",\"given\":\"Isha\"},{\"family\":\"Khuallar\",\"given\":\"Vikas\"},{\"family\":\"Gupta\",\"given\":\"Gifty\"},{\"family\":\"Gupta\",\"given\":\"Deepali\"},{\"family\":\"Juneja\",\"given\":\"Sapna\"},{\"family\":\"Nauman\",\"given\":\"Ali\"},{\"family\":\"Muhammad\",\"given\":\"Ghulam\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":735,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/24RGEQIZ\"],\"itemData\":{\"id\":735,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics12183867\",\"title\":\"Trustworthy Anti-Collusion Federated Learning Scheme Optimized by Game Theory\",\"author\":[{\"family\":\"Li\",\"given\":\"Qiuxian\"},{\"family\":\"Zhou\",\"given\":\"Quanxing\"},{\"family\":\"Li\",\"given\":\"Mingyang\"},{\"family\":\"Wang\",\"given\":\"Zhenlong\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":736,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/G2K7AFDF\"],\"itemData\":{\"id\":736,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2009.01867\",\"title\":\"ESMFL: Efficient and Secure Models for Federated Learning\",\"author\":[{\"family\":\"Lin\",\"given\":\"Sheng\"},{\"family\":\"Wang\",\"given\":\"Chenghong\"},{\"family\":\"Li\",\"given\":\"Hongjia\"},{\"family\":\"Deng\",\"given\":\"Jieren\"},{\"family\":\"Wang\",\"given\":\"Yanzhi\"},{\"family\":\"Ding\",\"given\":\"Caiwen\"}],\"issued\":{\"date-parts\":[[\"2020\"]]}}},{\"id\":751,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/KAKYQJBS\"],\"itemData\":{\"id\":751,\"type\":\"article-journal\",\"container-title\":\"Peer-to-Peer 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Detection\",\"author\":[{\"family\":\"Samee\",\"given\":\"Nagwan Abdel\"},{\"family\":\"Khan\",\"given\":\"Umair\"},{\"family\":\"Khan\",\"given\":\"Salabat\"},{\"family\":\"Jamjoom\",\"given\":\"Mona\"},{\"family\":\"Sharif\",\"given\":\"Muhammad\"},{\"family\":\"Kim\",\"given\":\"Do

Hyuen\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":746,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/I9Q6AAG8\"],\"itemData\":{\"id\":746,\"type\":\"article-journal\",\"container-title\":\"Ieee Access\",\"DOI\":\"10.1109/access.2023.3281832\",\"title\":\"Decentralized Learning in Healthcare: A Review of Emerging Techniques\",\"author\":[{\"family\":\"Shiranthika\",\"given\":\"Chamani\"},{\"family\":\"Saeedi\",\"given\":\"Parvaneh\"},{\"family\":\"Baji■\",\"given\":\"Ivan V.\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":743,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/82ZA4L7D\"],\"itemData\":{\"id\":743,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3052237\",\"title\":\"Federated Learning-Based Intrusion Detection System for Industrial Internet of Things: Enhancing Security and Efficiency\",\"author\":[{\"family\":\"Sun\",\"given\":\"Yan\"},{\"family\":\"Liu\",\"given\":\"Caiyun\"},{\"family\":\"Weng\",\"given\":\"Ying\"},{\"family\":\"Liu\",\"given\":\"Yitong\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":728,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3JNWYCPQ\"],\"itemData\":{\"id\":728,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3036891\",\"title\":\"Current Research and Prospects of Federated Language Large Models in the Medical Field\",\"author\":[{\"family\":\"Tang\",\"given\":\"Yumeng\"},{\"family\":\"Deng\",\"given\":\"Yuanpeng\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":737,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NYX7TZGI\"],\"itemData\":{\"id\":737,\"type\":\"article-journal\",\"container-title\":\"Sensors\",\"DOI\":\"10.3390/s25103083\",\"title\":\"Introducing a Quality-Driven Approach for Federated Learning\",\"author\":[{\"family\":\"Usman\",\"given\":\"Muhammad\"},{\"family\":\"Bernardi\",\"given\":\"Mario Luca\"},{\"family\":\"Cimitile\",\"given\":\"Marta\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":727,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NWMF6F7A\"],\"itemData\":{\"id\":727,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.02055\",\"title\":\"Private Map-Secure Reduce: Infrastructure for Efficient AI Data Markets\",\"author\":[{\"family\":\"Wagh\",\"given\":\"Sameer\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":750,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JU3XNPKP\"],\"itemData\":{\"id\":750,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.2640939\",\"title\":\"An Efficient Federated Learning Optimization Algorithm on Non-Iid Data\",\"author\":[{\"family\":\"Wang\",\"given\":\"Xue\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":748,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/74XPZY7U\"],\"itemData\":{\"id\":748,\"type\":\"article-journal\",\"container-title\":\"International Journal of Imaging Systems and Technology\",\"DOI\":\"10.1002/ima.70046\",\"title\":\"Brain Tumors Classification in \u003cscp\u003eMRIs\u003c/Scp\u003e Based on Personalized Federated Distillation Learning With Similarity■Preserving\",\"author\":[{\"family\":\"Wu\",\"given\":\"Bo\"},{\"family\":\"Shi\",\"given\":\"Donghui\"},{\"family\":\"Aguilar\",\"given\":\"José\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":745,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/UFSKEDQV\"],\"itemData\":{\"id\":745,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.09025\",\"title\":\"FLAD: Federated Learning for LLM-based Autonomous Driving in Vehicle-Edge-Cloud Networks\",\"author\":[{\"family\":\"Xiang\",\"given\":\"Tianao\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":738,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/C2UHNLLR\"],\"itemData\":{\"id\":738,\"type\":\"article-journal\",\"DOI\":\"10.1007/978-3-319-69035-3_39\",\"title\":\"Personalized Quality Centric Service Recommendation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yiwen\"},{\"family\":\"Ai\",\"given\":\"Xiaofei\"},{\"family\":\"He\",\"given\":\"Qiang\"},{\"family\":\"Zhang\",\"given\":\"Xuyun\"},{\"family\":\"Dou\",\"given\":\"Wanchun\"},{\"family\":\"Chen\",\"given\":\"Feifei\"},{\"family\":\"Chen\",\"given\":\"Liang\"},{\"family\":\"Yang\",\"given\":\"Yun\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integration, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution management issues that are situation in federated settings. Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramírez, 2024; J. Chen, 2021; Z. Chen et al., 2025; F. X. Fan et al., 2025; Hou et al., 2023; Jimenez-Gutierrez, 2025; Kansal et al., 2025; Q. Li et al., 2023; S. Lin et al., 2020; Y. Liu et al., 2023a, 2024; Raheem, 2025; Samee et al., 2023a; Shiranthika et al., 2023b; Y. Sun et al., 2025; Y. 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Zhang et al., 2017)\",\"noteIndex\":0},\"citationItems\":[{\"id\":733,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/TJVI4S6M\"],\"itemData\":{\"id\":733,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.01113\",\"title\":\"Privacy Preserved Federated Learning With Attention-Based Aggregation for Biometric Recognition\",\"author\":[{\"family\":\"Azezew\",\"given\":\"Kassahun\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":730,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/4FLKXH7N\"],\"itemData\":{\"id\":730,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-7917089/v1\",\"title\":\"Clifti-Gpt: Privacy-Preserving Federated Fine-Tuning and Transferable Inference of Foundation Models on Clinical Single-Cell Data\",\"author\":[{\"family\":\"Bakhtiari\",\"given\":\"Mohammad\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":731,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/SKQFVKA8\"],\"itemData\":{\"id\":731,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-8369921/v1\",\"title\":\"Federated CT Foundation Models for Multi-Center Detection of Lymph Node Metastasis in Pancreatic Cancer\",\"author\":[{\"family\":\"Bhalla\",\"given\":\"Parinishtha\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":739,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/6NXMVNNR\"],\"itemData\":{\"id\":739,\"type\":\"article-journal\",\"container-title\":\"Jaigs\",\"DOI\":\"10.60087/jaigs.v3i1.114\",\"title\":\"Constructing Executing and Overcoming Challenges in Distributed AI Systems: A Study of Federated Learning Framework\",\"author\":[{\"family\":\"Carrasco Ramírez\",\"given\":\"José Gabriel\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":734,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/6Q4BJBS4\"],\"itemData\":{\"id\":734,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2108.05069\",\"title\":\"FedMatch: Federated Learning Over Heterogeneous Question Answering Data\",\"author\":[{\"family\":\"Chen\",\"given\":\"Jiangui\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":729,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/SSPDZKKC\"],\"itemData\":{\"id\":729,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.00543\",\"title\":\"Flow of Knowledge: Federated Fine-Tuning of LLMs in Healthcare Under Non-Iid Conditions\",\"author\":[{\"family\":\"Chen\",\"given\":\"Zeyu\"},{\"family\":\"Ji\",\"given\":\"Y. 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Networks\",\"author\":[{\"family\":\"Xiang\",\"given\":\"Tianao\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":738,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/C2UHNLLR\"],\"itemData\":{\"id\":738,\"type\":\"article-journal\",\"DOI\":\"10.1007/978-3-319-69035-3_39\",\"title\":\"Personalized Quality Centric Service Recommendation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yiwen\"},{\"family\":\"Ai\",\"given\":\"Xiaofei\"},{\"family\":\"He\",\"given\":\"Qiang\"},{\"family\":\"Zhang\",\"given\":\"Xuyun\"},{\"family\":\"Dou\",\"given\":\"Wanchun\"},{\"family\":\"Chen\",\"given\":\"Feifei\"},{\"family\":\"Chen\",\"given\":\"Liang\"},{\"family\":\"Yang\",\"given\":\"Yun\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integration, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution management issues that are situation in federated settings. Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramírez, 2024; J. Chen, 2021; Z. Chen et al., 2025; F. X. Fan et al., 2025; Hou et al., 2023; Jimenez-Gutierrez, 2025; Kansal et al., 2025; Q. Li et al., 2023; S. Lin et al., 2020; Y. Liu et al., 2023a, 2024; Raheem, 2025; Samee et al., 2023a; Shiranthika et al., 2023b; Y. Sun et al., 2025; Y. 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Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí

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Zhang et al., 2017)\",\"noteIndex\":0},\"citationItems\":[{\"id\":733,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/TJVI4S6M\"],\"itemData\":{\"id\":733,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.01113\",\"title\":\"Privacy Preserved Federated Learning With Attention-Based Aggregation for Biometric Recognition\",\"author\":[{\"family\":\"Azezew\",\"given\":\"Kassahun\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":730,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/4FLKXH7N\"],\"itemData\":{\"id\":730,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-7917089/v1\",\"title\":\"Clifti-Gpt: Privacy-Preserving Federated Fine-Tuning and Transferable Inference of Foundation Models on Clinical Single-Cell Data\",\"author\":[{\"family\":\"Bakhtiari\",\"given\":\"Mohammad\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":731,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/SKQFVKA8\"],\"itemData\":{\"id\":731,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-8369921/v1\",\"title\":\"Federated CT Foundation Models for Multi-Center Detection of Lymph Node Metastasis in Pancreatic Cancer\",\"author\":[{\"family\":\"Bhalla\",\"given\":\"Parinishtha\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":739,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/6NXMVNNR\"],\"itemData\":{\"id\":739,\"type\":\"article-journal\",\"container-title\":\"Jaigs\",\"DOI\":\"10.60087/jaigs.v3i1.114\",\"title\":\"Constructing Executing and Overcoming Challenges in Distributed AI Systems: A Study of Federated Learning Framework\",\"author\":[{\"family\":\"Carrasco Ramírez\",\"given\":\"José Gabriel\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":734,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/6Q4BJBS4\"],\"itemData\":{\"id\":734,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2108.05069\",\"title\":\"FedMatch: Federated Learning Over Heterogeneous Question Answering Data\",\"author\":[{\"family\":\"Chen\",\"given\":\"Jiangui\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":729,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/SSPDZKKC\"],\"itemData\":{\"id\":729,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.00543\",\"title\":\"Flow of Knowledge: Federated Fine-Tuning of LLMs in Healthcare Under Non-Iid Conditions\",\"author\":[{\"family\":\"Chen\",\"given\":\"Zeyu\"},{\"family\":\"Ji\",\"given\":\"Y. 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urnal\",\"DOI\":\"10.1007/978-3-319-69035-3_39\",\"title\":\"Personalized Quality Centric Service Recommend

ation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yiwen\"},{\"family\":\"Ai\",\"given\":\"Xiaofei\"},{\"family\":\"He\",\"given\":\"Qiang\"},{\"family\":\"Zhang\",\"given\":\"Xuyun\"},{\"family\":\"Dou\",\"given\":\"Wanchun\"},{\"family\":\"Chen\",\"given\":\"Feifei\"},{\"family\":\"Chen\",\"given\":\"Liang\"},{\"family\":\"Yang\",\"given\":\"Yun\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integration, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution management issues that are situation in federated settings. Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramírez, 2024; J. Chen, 2021; Z. Chen et al., 2025; F. X. Fan et al., 2025; Hou et al., 2023; Jimenez-Gutierrez, 2025; Kansal et al., 2025; Q. Li et al., 2023; S. Lin et al., 2020; Y. Liu et al., 2023a, 2024; Raheem, 2025; Samee et al., 2023a; Shiranthika et al., 2023b; Y. Sun et al., 2025; Y. Tang \u0026 Deng, 2024; Usman et al., 2025; Wagh, 2025; X. Wang, 2022; B. Wu et al., 2025; Xiang, 2025; Y. Zhang et al., 2017)",
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:[\"http://zotero.org/users/local/M0u71wYg/items/74XPZY7U\"],\"itemData\":{\"id\":748,\"type\":\"article-journal\",\"container-title\":\"International Journal of Imaging Systems and Technology\",\"DOI\":\"10.1002/ima.70046\",\"title\":\"Brain Tumors Classification in \u003cscp\u003eMRIs\u003c/Scp\u003e Based on Personalized Federated Distillation Learning With Similarity■Preserving\",\"author\":[{\"family\":\"Wu\",\"given\":\"Bo\"},{\"family\":\"Shi\",\"given\":\"Donghui\"},{\"family\":\"Aguilar\",\"given\":\"José\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":745,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/UFSKEDQV\"],\"itemData\":{\"id\":745,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.09025\",\"title\":\"FLAD: Federated Learning for LLM-based Autonomous Driving in Vehicle-Edge-Cloud Networks\",\"author\":[{\"family\":\"Xiang\",\"given\":\"Tianao\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":738,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/C2UHNLLR\"],\"itemData\":{\"id\":738,\"type\":\"article-journal\",\"DOI\":\"10.1007/978-3-319-69035-3_39\",\"title\":\"Personalized Quality Centric Service Recommendation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yiwen\"},{\"family\":\"Ai\",\"given\":\"Xiaofei\"},{\"family\":\"He\",\"given\":\"Qiang\"},{\"family\":\"Zhang\",\"given\":\"Xuyun\"},{\"family\":\"Dou\",\"given\":\"Wanchun\"},{\"family\":\"Chen\",\"given\":\"Feifei\"},{\"family\":\"Chen\",\"given\":\"Liang\"},{\"family\":\"Yang\",\"given\":\"Yun\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integration, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution management issues that are situation in federated settings. Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramírez, 2024; J. Chen, 2021; Z. Chen et al., 2025; F. X. Fan et al., 2025; Hou et al., 2023; Jimenez-Gutierrez, 2025; Kansal et al., 2025; Q. Li et al., 2023; S. Lin et al., 2020; Y. Liu et al., 2023a, 2024; Raheem, 2025; Samee et al., 2023a; Shiranthika et al., 2023b; Y. Sun et al., 2025; Y. Tang \u0026 Deng, 2024; Usman et al., 2025; Wagh, 2025; X. Wang, 2022; B. Wu et al., 2025; Xiang, 2025; Y. Zhang et al., 2017)",
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integrat
ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th
e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integrat
ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th
e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the quality of its recommendations, which are limited by the classical approaches to collaborative filtering. It is through this integration that the foundation models possess vast pre-trained knowledge that allows it to be capable of solving data sources sparsity and cold-start problems and federated learning provides privacy reassuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramírez, 2024; J. Chen, 2021; Z. Chen et al., 2025; F. X. Fan et al., 2025; Hou et al., 2023; Jimenez-Gutierrez, 2025; Kansal et al., 2025; Q. Li et al., 2023; S. Lin et al., 2020; Y. Liu et al., 2023a, 2024; Raheem, 2025; Samee et al., 2023a; Shiranthika et al., 2023b; Y. Sun et al., 2025; Y. 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Recommendation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yiwen\"},{\"family\":\"Ai\",\"given\":\"Xiaofei\"},{\"family\":\"He\",\"given\":\"Qiang\"},{\"family\":\"Zhang\",\"given\":\"Xuyun\"},{\"family\":\"Dou\",\"given\":\"Wanchun\"},{\"family\":\"Chen\",\"given\":\"Feifei\"},{\"family\":\"Chen\",\"given\":\"Liang\"},{\"family\":\"Yang\",\"given\":\"Yun\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integration, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution management issues that are situation in federated settings. Furthermore, in spite of the fact that federated learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in their abilities, such as the impossibility of working with non-IID data distributions, and reduced the performance of the models under the influence of privacy-saving processes. As a way of managing those problems, the necessity to shift to more developed federated learning models and secure a high rank of dealing with different data characteristics and high-quality recommendations and high privacy policies. Development of frameworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and disclosure of the outcomes is significant in improving the future recommender systems. In particular, the introduction of large language models and foundation models to federated recommendation systems introduces the new paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit

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y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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.com/citation-style-language/schema/raw/master/csl-citation.json\"} Other issues, raised with this integrat
ion, include unfamiliar issues regarding privacy-security trade-offs as well as non-IID data distribution m
anagement issues that are situation in federated settings. Furthermore, in spite of the fact that federated
 learning has privacy-friendly properties, nowadays the federated recommendation systems are scarcer in the
ir abilities, such as the impossibility of working with non-IID data distributions, and reduced the perform
ance of the models under the influence of privacy-saving processes. As a way of managing those problems, th

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e necessity to shift to more developed federated learning models and secure a high rank of dealing with dif
ferent data characteristics and high-quality recommendations and high privacy policies. Development of fram
eworks that can ensure the impartiality and fairness of the recommendations, privacy of the users and discl
osure of the outcomes is significant in improving the future recommender systems. In particular, the introd
uction of large language models and foundation models to federated recommendation systems introduces the ne
w paradigm in an attempt to increase the scope of the generalization of such systems, as well as the qualit
y of its recommendations, which are limited by the classical approaches to collaborative filtering. It is t
hrough this integration that the foundation models possess vast pre-trained knowledge that allows it to be
capable of solving data sources sparsity and cold-start problems and federated learning provides privacy re
assuring form of personalized recommendations. (Azezew, 2025; Bakhtiari, 2025a; Bhalla, 2025; Carrasco Ramí
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 with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
 a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi
ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized m
odel parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another
 potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
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Online Federated Learning Framework for Travel Time Estimation\",\"author\":[{\"family\":\"Zhang\",\"given\

":\"Zhiwen\"},{\"family\":\"Wang\",\"given\":\"Hongjun\"},{\"family\":\"Chen\",\"given\":\"Jiyuan\"},{\"family\":\"Fan\",\"given\":\"Zipei\"},{\"family\":\"Song\",\"given\":\"Xuan\"},{\"family\":\"Shibasaki\",\"given\":\"Ryosuke\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":765,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XDSP8U6R\"],\"itemData\":{\"id\":765,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.24191886\",\"title\":\"Privacy-Preserving Fine-Tuning of Artificial Intelligence (AI) Foundation Models With Federated Learning, Differential Privacy, Offsite Tuning, and Parameter-Efficient Fine-Tuning (PEFT)\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Jun\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively, with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devices. One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client devices which typically leads to diminished model performance in classical federated systems of recommendation is another issue of the field importance. More research on the ability of such models to be effectively tuned to decentralized data and the fact that their privacy processes do not attract the attention of the quality of recommendations and fairness can also be conducted. The Federation Foundation Models development presupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sensitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. Zhang, Yang, et al., 2023; J. Zhao, 2023)",
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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively,
 with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
 a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi
ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized m
odel parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another
 potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
resupposes these issues with the help of collaborative learning in federated structures toward the improved
 model performance and data privacy as well as the possibility to develop more personalized and context-sen
sitive models. It has the possibility of increasing data and has a smaller computational requirement, with
the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor
tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-
preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID
data, combining parameter-efficient and personalization models to decrease the communication and computatio
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One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client devices which typically leads to diminished model performance in classical federated systems of recommendation is another issue of the field importance. More research on the ability of such models to be effectively tuned to decentralized data and the fact that their privacy processes do not attract the attention of the quality of recommendations and fairness can also be conducted. The Federation Foundation Models development presupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sensitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. 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with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi
ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized m
odel parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another
potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
resupposes these issues with the help of collaborative learning in federated structures toward the improved
model performance and data privacy as well as the possibility to develop more personalized and context-sen
sitive models. It has the possibility of increasing data and has a smaller computational requirement, with
the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor
tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-
preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID
data, combining parameter-efficient and personalization models to decrease the communication and computatio
n expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al
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5a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al.,
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One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. 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It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al.,

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 with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
 a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi

ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client devices which typically leads to diminished model performance in classical federated systems of recommendation is another issue of the field importance. More research on the ability of such models to be effectively tuned to decentralized data and the fact that their privacy processes do not attract the attention of the quality of recommendations and fairness can also be conducted. The Federation Foundation Models development presupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sensitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. Zhang, Yang, et al., 2023; J. Zhao, 2023)",


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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively,
 with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
 a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi
ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized m
odel parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another
 potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
resupposes these issues with the help of collaborative learning in federated structures toward the improved
 model performance and data privacy as well as the possibility to develop more personalized and context-sen
sitive models. It has the possibility of increasing data and has a smaller computational requirement, with
the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor
tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-
preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID
data, combining parameter-efficient and personalization models to decrease the communication and computatio
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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively,
 with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
 a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi
ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized m
odel parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another
 potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
resupposes these issues with the help of collaborative learning in federated structures toward the improved
 model performance and data privacy as well as the possibility to develop more personalized and context-sen
sitive models. It has the possibility of increasing data and has a smaller computational requirement, with
the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor
tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-
preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID
data, combining parameter-efficient and personalization models to decrease the communication and computatio
n expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al
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5a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al.,
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Privately\",\"author\":[{\"family\":\"Wu\",\"given\":\"Huiwen\"},{\"family\":\"Zhang\",\"given\":\"Deyi\"},{\"family\":\"Li\",\"given\":\"Xiaohan\"},{\"family\":\"Xu\",\"given\":\"Xiaogang\"},{\"family\":\"Wu\",\"given\":\"Jiafei\"},{\"family\":\"Liu\",\"given\":\"Zhe\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":753,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/IJJVY2IK\"],\"itemData\":{\"id\":753,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2302.04870\",\"title\":\"Offsite-Tuning: Transfer Learning Without Full Model\",\"author\":[{\"family\":\"Xiao\",\"given\":\"Guangxuan\"},{\"family\":\"Lin\",\"given\":\"Ji\"},{\"family\":\"Han\",\"given\":\"Song\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":759,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/RZF7KWJZ\"],\"itemData\":{\"id\":759,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.04601\",\"title\":\"FedSRD: Sparsify-Reconstruct-Decompose for Communication-Efficient Federated 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Estimation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Zhiwen\"},{\"family\":\"Wang\",\"given\":\"Hongjun\"},{\"family\":\"Chen\",\"given\":\"Jiyuan\"},{\"family\":\"Fan\",\"given\":\"Zipei\"},{\"family\":\"Song\",\"given\":\"Xuan\"},{\"family\":\"Shibasaki\",\"given\":\"Ryosuke\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":765,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XDSP8U6R\"],\"itemData\":{\"id\":765,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.24191886\",\"title\":\"Privacy-Preserving Fine-Tuning of Artificial Intelligence (AI) Foundation Models With Federated Learning, Differential Privacy, Offsite Tuning, and Parameter-Efficient Fine-Tuning (PEFT)\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Jun\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively, with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devices. One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client devices which typically leads to diminished model performance in classical federated systems of recommendation is another issue of the field importance. More research on the ability of such models to be effectively tuned to decentralized data and the fact that their privacy processes do not attract the attention of the quality of recommendations and fairness can also be conducted. The Federation Foundation Models development presupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sen

sitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. Zhang, Yang, et al., 2023; J. Zhao, 2023)",
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Estimation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Zhiwen\"},{\"family\":\"Wang\",\"given\":\"Hongjun\"},{\"family\":\"Chen\",\"given\":\"Jiyuan\"},{\"family\":\"Fan\",\"given\":\"Zipei\"},{\"family\":\"Song\",\"given\":\"Xuan\"},{\"family\":\"Shibasaki\",\"given\":\"Ryosuke\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":765,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XDSP8U6R\"],\"itemData\":{\"id\":765,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.24191886\",\"title\":\"Privacy-Preserving Fine-Tuning of Artificial Intelligence (AI) Foundation Models With Federated Learning, Differential Privacy, Offsite Tuning, and Parameter-Efficient Fine-Tuning (PEFT)\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Jun\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively, with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devices. One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another

 potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
resupposes these issues with the help of collaborative learning in federated structures toward the improved
 model performance and data privacy as well as the possibility to develop more personalized and context-sen
sitive models. It has the possibility of increasing data and has a smaller computational requirement, with
the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor
tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-
preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID
data, combining parameter-efficient and personalization models to decrease the communication and computatio
n expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al
., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2
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5a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al.,
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One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client devices which typically leads to diminished model performance in classical federated systems of recommendation is another issue of the field importance. More research on the ability of such models to be effectively tuned to decentralized data and the fact that their privacy processes do not attract the attention of the quality of recommendations and fairness can also be conducted. The Federation Foundation Models development presupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sensitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. 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It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. 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Estimation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Zhiwen\"},{\"family\":\"Wang\",\"given\":\"Hongjun\"},{\"family\":\"Chen\",\"given\":\"Jiyuan\"},{\"family\":\"Fan\",\"given\":\"Zipei\"},{\"family\":\"Song\",\"given\":\"Xuan\"},{\"family\":\"Shibasaki\",\"given\":\"Ryosuke\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":765,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XDSP8U6R\"],\"itemData\":{\"id\":765,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.24191886\",\"title\":\"Privacy-Preserving Fine-Tuning of Artificial Intelligence (AI) Foundation Models With Federated Learning, Differential Privacy, Offsite Tuning, and Parameter-Efficient Fine-Tuning (PEFT)\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Jun\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively, with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devices. One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalization. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computational demands of large-scale models with privacy guarantees provided by the federated learning in the situation of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client devices which typically leads to diminished model performance in classical federated systems of recommendation is another issue of the field importance. More research on the ability of such models to be effectively tuned to decentralized data and the fact that their privacy processes do not attract the attention of the quality of recommendations and fairness can also be conducted. The Federation Foundation Models development presupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sensitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shortage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID

data, combining parameter-efficient and personalization models to decrease the communication and computation expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 2025a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. Zhang, Yang, et al., 2023; J. Zhao, 2023)",
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Estimation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Zhiwen\"},{\"family\":\"Wang\",\"given\":\"Hongjun\"},{\"family\":\"Chen\",\"given\":\"Jiyuan\"},{\"family\":\"Fan\",\"given\":\"Zipei\"},{\"family\":\"Song\",\"given\":\"Xuan\"},{\"family\":\"Shibasaki\",\"given\":\"Ryosuke\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":765,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XDSP8U6R\"],\"itemData\":{\"id\":765,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.24191886\",\"title\":\"Privacy-Preserving Fine-Tuning of Artificial Intelligence (AI) Foundation Models With Federated Learning, Differential Privacy, Offsite Tuning, and Parameter-Efficient Fine-Tuning (PEFT)\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Jun\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively, with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devices. One significant benefit of the approach is that the communication overhead of exchanging large-sized model parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another potential solution to the suggested system enhancement suggests implementing large language models into the federated learning systems because large language models have an opportunity to leverage their large pre-trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio

n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de vices which typically leads to diminished model performance in classical federated systems of recommendatio n is another issue of the field importance. More research on the ability of such models to be effectively t uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p resupposes these issues with the help of collaborative learning in federated structures toward the improved model performance and data privacy as well as the possibility to develop more personalized and context-sen sitive models. It has the possibility of increasing data and has a smaller computational requirement, with the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID data, combining parameter-efficient and personalization models to decrease the communication and computatio n expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al ., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2 025a; Solat, 2025; G. Sun et al., 2022; Thuau, 2025; Y. Wang, 2025; Z. Wang et al., 2022; H. Wu et al., 202 5a, 2025c; Xiao et al., 2023; G. Yan, 2025b; Yashwanth, 2025; C. Zhang et al., 2024; M. Zhang, Ma, et al., 2025; Z. Zhang, Wang, et al., 2022; Z. Zhang, Yang, et al., 2023; J. Zhao, 2023)",
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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Indicatively,
 with an example, federated adaptation mechanisms can be employed to permit desire -shift variations amidst
 a user in an agreeable and privacy conscious means with featherweight customized adjustment on client devi
ces. One significant benefit of the approach is that the communication overhead of exchanging large-sized m
odel parameters is reduced and, therefore, improves resource-limited mobile environments. Moreover, another
 potential solution to the suggested system enhancement suggests implementing large language models into th
e federated learning systems because large language models have an opportunity to leverage their large pre-
trainings and generalization capabilities to address the issue of data scarcity and increase personalizatio
n. This remedy however comes at a cost in that new techniques are necessary to interdiscipline the computat
ional demands of large-scale models with privacy guarantees provided by the federated learning in the situa
tion of resource-limited environments. The decrease of the data sparsity and heterogeneity of the client de
vices which typically leads to diminished model performance in classical federated systems of recommendatio
n is another issue of the field importance. More research on the ability of such models to be effectively t
uned to decentralized data and the fact that their privacy processes do not attract the attention of the qu
ality of recommendations and fairness can also be conducted. The Federation Foundation Models development p
resupposes these issues with the help of collaborative learning in federated structures toward the improved
 model performance and data privacy as well as the possibility to develop more personalized and context-sen
sitive models. It has the possibility of increasing data and has a smaller computational requirement, with
the pre-training and data augmentation techniques, and serves as an option to the problem, such as the shor
tage of data, variability in computation, etc. More so, federated foundation models offer a robust privacy-
preserving scheme to collaboratively fine-tune models with varying disparate and decentralized and non-IID
data, combining parameter-efficient and personalization models to decrease the communication and computatio
n expenses. (Abdel-Sater \u0026 Hamza, 2024; Bakhtiari, 2025b; Bandara, 2025; S. Chen, 2024; Z. Jiang et al
., 2022; Jin et al., 2022; X.-Y. Liu et al., 2025; Y. Liu et al., 2025; Mdhaffar et al., 2021; Sangeetha, 2
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Approach to Preventing Diabetes-Related Plantar Forefoot Ulcers and Complications\",\"author\":[{\"family\":\"Ahmed\",\"given\":\"Sayed\"},{\"family\":\"Ashad Kabir\",\"given\":\"Muhammad\"},{\"family\":\"E.H. Chowdhury\",\"given\":\"Muhammad\"},{\"family\":\"Nancarrow\",\"given\":\"Susan\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":781,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NDZ4AKI2\"],\"itemData\":{\"id\":781,\"type\":\"article-journal\",\"abstract\":\"With the explosion of delay-sensitive and computation-intensive vehicular applications, traditional cloud computing has encountered enormous challenges. Vehicular edge computing, as an emerging computing paradigm, has provided powerful support for vehicular networks. However, vehicle mobility and time-varying characteristics of communication channels have further complicated the design and implementation of vehicular network systems, leading to increased delays and energy consumption. 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Byzantine-Robust Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Qun\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in which data is spread and there are privacy concerns involving the sharing of the data of users. Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, communication cost and system utility as a whole. This is composed in the formulation of successful methods of measurement to the extent to which privacy preserving methods including privacy and secure aggregation, safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions obtained. In practice, this is typically a very fine balancing act because on one hand, the provision of increased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven

ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying 
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile 
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, 
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of 
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob

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tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of 
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in which data is spread and there are privacy concerns involving the sharing of the data of users. Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, communication cost and system utility as a whole. This is composed in the formulation of successful methods of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,

safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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al., 2020; Khanna et al., 2022; Z. Lian et al., 2022; Z. Liang \u0026 Chen, 2025d; Mandal, 2020; Matsumoto
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying 
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile 
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of 
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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2022; Y. Tang et al., 2023; Wassan et al., 2025a; C. Wu et al., 2022; Z. Wu \u0026 Wang, 2024; Xue et al.,
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob

tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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Byzantine-Robust Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Qun\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in which data is spread and there are privacy concerns involving the sharing of the data of users. Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,

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communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of 
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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uage/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in whic
h data is spread and there are privacy concerns involving the sharing of the data of users. Getting over wi
th this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying
models with adaptability to new users or items based on prior knowledge learnt during the execution of a my
riad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The
 other is data augmentation which artificially creates additional data to be used in training or the effect
 of current data to be augmented to enhance the learning process to counter the sparsity and augment the ro
bustness of the model. A combination of these methods can allow this system to produce valuable recommendat
ions even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile
environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage,
communication cost and system utility as a whole. This is composed in the formulation of successful methods
 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, communication cost and system utility as a whole. This is composed in the formulation of successful methods of measurement to the extent to which privacy preserving methods including privacy and secure aggregation, safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions obtained. In practice, this is typically a very fine balancing act because on one hand, the provision of increased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiveness of the model and on the other end, greater respect towards the utility might also increase the cost of privacy of the systems - the architectural design of federated learning systems in mobile recommendation systems is therefore a matter of significant concern to balance these trade-offs. The main points are provision of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterogeneity and incentives to produce client sustained participation. More importantly, the system must uphold the abilities of the effectiveness personalisation where the desirability of recommendations is not subdivided when the constraints of the occurrence of local computation and privacy regulations are incorporated within consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implementation because this has a direct effect to user trust, engagement and overall adoption of federated learning based mobile recommendation services. in short, even though the combination of federated learning and recommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisation, it also demands the radical solutions to the problem of resource limitation, cold starting and the interrelation between privacy and utility. To deal with these problems, highly developed model design, optimisation strategies and integrated evaluation structures are necessary in order to make the most out of potential of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023; W. Chen et al., 2021; Fares \u0026 Sertba■, 2024; Y. Hu et al., 2020; Hussien et al., 2023b; L. U. Khan et al., 2020; Khanna et al., 2022; Z. Lian et al., 2022; Z. Liang \u0026 Chen, 2025d; Mandal, 2020; Matsumoto et al., 2025; Mugunthan et al., 2020; Otoum et al., 2022a; Qi et al., 2023a; Rokade, 2024a; Silva et al., 2022; Y. Tang et al., 2023; Wassan et al., 2025a; C. Wu et al., 2022; Z. Wu \u0026 Wang, 2024; Xue et al.,

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privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
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Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, communication cost and system utility as a whole. This is composed in the formulation of successful methods of measurement to the extent to which privacy preserving methods including privacy and secure aggregation, safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions obtained. In practice, this is typically a very fine balancing act because on one hand, the provision of increased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiveness of the model and on the other end, greater respect towards the utility might also increase the cost of privacy of the systems - the architectural design of federated learning systems in mobile recommendation systems is therefore a matter of significant concern to balance these trade-offs. The main points are provision of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterogeneity and incentives to produce client sustained participation. More importantly, the system must uphold the abilities of the effectiveness personalisation where the desirability of recommendations is not subdivided when the constraints of the occurrence of local computation and privacy regulations are incorporated within consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implementation because this has a direct effect to user trust, engagement and overall adoption of federated learning based mobile recommendation services. in short, even though the combination of federated learning and recommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisation, it also demands the radical solutions to the problem of resource limitation, cold starting and the interrelation between privacy and utility. To deal with these problems, highly developed model design, optimisation strategies and integrated evaluation structures are necessary in order to make the most out of potential of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023; W. Chen et al., 2021; Fares \u0026 Sertba■, 2024; Y. Hu et al., 2020; Hussien et al., 2023b; L. U. Khan et al., 2020; Khanna et al., 2022; Z. Lian et al., 2022; Z. Liang \u0026 Chen, 2025d; Mandal, 2020; Matsumoto et al., 2025; Mugunthan et al., 2020; Otoum et al., 2022a; Qi et al., 2023a; Rokade, 2024a; Silva et al., 2022; Y. Tang et al., 2023; Wassan et al., 2025a; C. Wu et al., 2022; Z. Wu \u0026 Wang, 2024; Xue et al., 2023; H. Yan et al., 2023a; X. Yang \u0026 Xing, 2024a; Y. Yang, 2023; Q. Zhou, 2024a) (refer to Figure 6)"
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Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, communication cost and system utility as a whole. This is composed in the formulation of successful methods of measurement to the extent to which privacy preserving methods including privacy and secure aggregation, safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions obtained. In practice, this is typically a very fine balancing act because on one hand, the provision of increased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiveness of the model and on the other end, greater respect towards the utility might also increase the cost of privacy of the systems - the architectural design of federated learning systems in mobile recommendation systems is therefore a matter of significant concern to balance these trade-offs. The main points are provision of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterogeneity and incentives to produce client sustained participation. More importantly, the system must uphold the abilities of the effectiveness personalisation where the desirability of recommendations is not subdivided when the constraints of the occurrence of local computation and privacy regulations are incorporated within consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implementation because this has a direct effect to user trust, engagement and overall adoption of federated learning based mobile recommendation services. in short, even though the combination of federated learning and recommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisation, it also demands the radical solutions to the problem of resource limitation, cold starting and the interrelation between privacy and utility. To deal with these problems, highly developed model design, optimisation strategies and integrated evaluation structures are necessary in order to make the most out of potential of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023; W. Chen et al., 2021; Fares \u0026 Sertba■, 2024; Y. Hu et al., 2020; Hussien et al., 2023b; L. U. Khan et al., 2020; Khanna et al., 2022; Z. Lian et al., 2022; Z. Liang \u0026 Chen, 2025d; Mandal, 2020; Matsumoto et al., 2025; Mugunthan et al., 2020; Otoum et al., 2022a; Qi et al., 2023a; Rokade, 2024a; Silva et al.,

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Mobile Computing\",\"DOI\":\"10.1155/2024/8138644\",\"title\":\"Federated Medical Learning Framework Based on Blockchain and Homomorphic Encryption\",\"author\":[{\"family\":\"Yang\",\"given\":\"Xiaohui\"},{\"family\":\"Xing\",\"given\":\"Chongbo\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":785,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/MZPYL5BW\"],\"itemData\":{\"id\":785,\"type\":\"article-journal\",\"container-title\":\"Highlights in Science Engineering and Technology\",\"DOI\":\"10.54097/hset.v34i.5446\",\"title\":\"The Improvement of Federated Learning in Communication Efficiency\",\"author\":[{\"family\":\"Yang\",\"given\":\"Yimeng\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":798,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/86MJFHZB\"],\"itemData\":{\"id\":798,\"type\":\"article-journal\",\"container-title\":\"Frontiers in Computing and Intelligent Systems\",\"DOI\":\"10.54097/6jamgy43\",\"title\":\"PPBRFL: Privacy-Preserving Byzantine-Robust Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Qun\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} This is more so when it comes to federated environments in which data is spread and there are privacy concerns involving the sharing of the data of users. Getting over with this implies that they employ complex techniques, like meta-learning and data enhancement. By modifying models with adaptability to new users or items based on prior knowledge learnt during the execution of a myriad of tasks, meta-learning can help new users or items achieve better generalisation on limited data. The other is data augmentation which artificially creates additional data to be used in training or the effect of current data to be augmented to enhance the learning process to counter the sparsity and augment the robustness of the model. A combination of these methods can allow this system to produce valuable recommendations even in data sparse environments. Moreover, testing of federated recommendation systems in the mobile environment should move beyond the conventional metrics of accuracy to include testing of privacy leakage, communication cost and system utility as a whole. This is composed in the formulation of successful methods

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 of measurement to the extent to which privacy preserving methods including privacy and secure aggregation,
 safeguard the privacy of individual user data to the cost of subtracting the performance of suggestions ob
tained. In practice, this is typically a very fine balancing act because on one hand, the provision of incr
eased privacy by the means of noise, cryptographic or other tools may compromise the quality or expressiven
ess of the model and on the other end, greater respect towards the utility might also increase the cost of 
privacy of the systems - the architectural design of federated learning systems in mobile recommendation sy
stems is therefore a matter of significant concern to balance these trade-offs. The main points are provisi
on of combination of adaptive privacy budgets, dynamic aggregation techniques by considering device heterog
eneity and incentives to produce client sustained participation. More importantly, the system must uphold t
he abilities of the effectiveness personalisation where the desirability of recommendations is not subdivid
ed when the constraints of the occurrence of local computation and privacy regulations are incorporated wit
hin consideration. The trade-offs involved and ensuring that this balance is paramount to real-world implem
entation because this has a direct effect to user trust, engagement and overall adoption of federated learn
ing based mobile recommendation services. in short, even though the combination of federated learning and r
ecommender systems in mobile setting has a lot of a potential in preserving privacy of the user personalisa
tion, it also demands the radical solutions to the problem of resource limitation, cold starting and the in
terrelation between privacy and utility. To deal with these problems, highly developed model design, optimi
sation strategies and integrated evaluation structures are necessary in order to make the most out of poten
tial of this beneficial paradigm and to reveal its full potential in practical scenarios. (Abdollahi, 2023;
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 et al., 2025; Mugunthan et al., 2020; Otoum et al., 2022a; Qi et al., 2023a; Rokade, 2024a; Silva et al.,
2022; Y. Tang et al., 2023; Wassan et al., 2025a; C. Wu et al., 2022; Z. Wu \u0026 Wang, 2024; Xue et al.,
2023; H. Yan et al., 2023a; X. Yang \u0026 Xing, 2024a; Y. Yang, 2023; Q. Zhou, 2024a) (refer to Figure 6)"
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10453\",\"title\":\"Fine-Tuning Graph Neural Networks via Graph Topology Induced Optimal Transport\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jiying\"},{\"family\":\"Xiao\",\"given\":\"Xi\"},{\"family\":\"Huang\",\"given\":\"Long-Kai\"},{\"family\":\"Rong\",\"given\":\"Yu\"},{\"family\":\"Bian\",\"given\":\"Yatao\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":831,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/FLAXX9RH\"],\"itemData\":{\"id\":831,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs17071272\",\"title\":\"Parameter-Efficient Fine-Tuning for Individual Tree Crown Detection and Species Classification Using UAV-Acquired Imagery\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jiuyu\"},{\"family\":\"Lei\",\"given\":\"Fan\"},{\"family\":\"Fan\",\"given\":\"Xijian\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Second, regularization term enhances a successful knowledge transfer through ensuring that the difference between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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g\",\"given\":\"Long-Kai\"},{\"family\":\"Rong\",\"given\":\"Yu\"},{\"family\":\"Bian\",\"given\":\"Yatao\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":831,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/FLAXX9RH\"],\"itemData\":{\"id\":831,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs17071272\",\"title\":\"Parameter-Efficient Fine-Tuning for Individual Tree Crown Detection and Species Classification Using UAV-Acquired Imagery\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jiuyu\"},{\"family\":\"Lei\",\"given\":\"Fan\"},{\"family\":\"Fan\",\"given\":\"Xijian\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Second, regularization term enhances a successful knowledge transfer through ensuring that the difference between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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nal Networks\",\"author\":[{\"family\":\"Li\",\"given\":\"Xuhong\"},{\"family\":\"Grandvalet\",\"given\":\"Yves\"},{\"family\":\"Davoine\",\"given\":\"Franck\"}],\"issued\":{\"date-parts\":[[\"2018\"]]}}},{\"id\":812,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ISXYUN55\"],\"itemData\":{\"id\":812,\"type\":\"article-journal\",\"container-title\":\"Image and Vision Computing\",\"DOI\":\"10.1016/j.imavis.2019.103853\",\"title\":\"Transfer Learning in Computer Vision Tasks: Remember Where You Come from\",\"author\":[{\"family\":\"Li\",\"given\":\"Xuhong\"},{\"family\":\"Grandvalet\",\"given\":\"Yves\"},{\"family\":\"Davoine\",\"given\":\"Franck\"},{\"family\":\"Cheng\",\"given\":\"Jingchun\"},{\"family\":\"Cui\",\"given\":\"Yin\"},{\"family\":\"Zhang\",\"given\":\"Hang\"},{\"family\":\"Belongie\",\"given\":\"Serge\"},{\"family\":\"Tsai\",\"given\":\"Yi-Hsuan\"},{\"family\":\"Yang\",\"given\":\"Ming–Hsuan\"}],\"issued\":{\"date-parts\":[[\"2020\"]]}}},{\"id\":814,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VKH9BLC2\"],\"itemData\":{\"id\":814,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2206.13559\",\"title\":\"ST-Adapter: Parameter-Efficient Image-to-Video Transfer Learning\",\"author\":[{\"family\":\"Pan\",\"given\":\"Junting\"},{\"family\":\"Lin\",\"given\":\"Ziyi\"},{\"family\":\"Zhu\",\"given\":\"Xiatian\"},{\"family\":\"Shao\",\"given\":\"Jing\"},{\"family\":\"Li\",\"given\":\"Hongsheng\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":833,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/2PCBLMCA\"],\"itemData\":{\"id\":833,\"type\":\"article-journal\",\"container-title\":\"Micromachines\",\"DOI\":\"10.3390/mi10090616\",\"title\":\"Investigation on the Edge Chipping in Ultrasonic Assisted Sawing of Monocrystalline Silicon\",\"author\":[{\"family\":\"Shen\",\"given\":\"Jianyun\"},{\"family\":\"Zhu\",\"given\":\"Xu\"},{\"family\":\"Chen\",\"given\":\"Jianbin\"},{\"family\":\"Tao\",\"given\":\"Ping\"},{\"family\":\"Wu\",\"given\":\"Xian\"}],\"issued\":{\"date-parts\":[[\"2019\"]]}}},{\"id\":822,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YP5P4M5X\"],\"itemData\":{\"id\":822,\"type\":\"article-journal\",\"container-title\":\"International Journal for Research in Applied Science and Engineering Technology\",\"DOI\":\"10.22214/ijraset.2025.73930\",\"title\":\"Decoding Efficiency: A Comprehensive Review of Knowledge Distillation Techniques in Large Language Model\",\"author\":[{\"family\":\"Soni\",\"given\":\"Goldi\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":830,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JMBPHDM2\"],\"itemData\":{\"id\":830,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.19575760.v2\",\"title\":\"Feature Distribution Matching for Federated Domain Generalization\",\"author\":[{\"family\":\"Sun\",\"given\":\"Yuwei\"},{\"family\":\"T. 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C.\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":816,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/MG5LSSKD\"],\"itemData\":{\"id\":816,\"type\":\"article-journal\",\"DOI\":\"10.24963/ijcai.2022/515\",\"title\":\"Improved Pure Exploration in Linear Bandits With No-Regret Learning\",\"author\":[{\"family\":\"Zaki\",\"given\":\"Mohammadi\"},{\"family\":\"Mohan\",\"given\":\"Avi\"},{\"family\":\"Gopalan\",\"given\":\"Aditya\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":815,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/D9PGCTDU\"],\"itemData\":{\"id\":815,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2203.10453\",\"title\":\"Fine-Tuning Graph Neural Networks via Graph Topology Induced Optimal Transport\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jiying\"},{\"family\":\"Xiao\",\"given\":\"Xi\"},{\"family\":\"Huang\",\"given\":\"Long-Kai\"},{\"family\":\"Rong\",\"given\":\"Yu\"},{\"family\":\"Bian\",\"given\":\"Yatao\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":831,\"uris\":[\"http://zotero.org/users/local/M0u71wY

g/items/FLAXX9RH\"],\"itemData\":{\"id\":831,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs17071272\",\"title\":\"Parameter-Efficient Fine-Tuning for Individual Tree Crown Detection and Species Classification Using UAV-Acquired Imagery\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jiuyu\"},{\"family\":\"Lei\",\"given\":\"Fan\"},{\"family\":\"Fan\",\"given\":\"Xijian\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Second, regularization term enhances a successful knowledge transfer through ensuring that the difference between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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that the difference between the fine-tuned parameters and the foundation model is controlled. It enables th
e model to take advantage of abundant semantic representations and inductive biases held in the foundation
model and also learn to make recommendations to every user individually. Such a tradeoff is so important in
 federated recommendation systems, where one can suit very well to the needs of a specific user, without th
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Detection and Species Classification Using UAV-Acquired Imagery\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Jiuyu\"},{\"family\":\"Lei\",\"given\":\"Fan\"},{\"family\":\"Fan\",\"given\":\"Xijian\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Second, regularization term enhances a successful knowledge transfer through ensuring that the difference between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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,\"given\":\"Franck\"},{\"family\":\"Cheng\",\"given\":\"Jingchun\"},{\"family\":\"Cui\",\"given\":\"Yin\"},{\"family\":\"Zhang\",\"given\":\"Hang\"},{\"family\":\"Belongie\",\"given\":\"Serge\"},{\"family\":\"Tsai\",\"given\":\"Yi-Hsuan\"},{\"family\":\"Yang\",\"given\":\"Ming–Hsuan\"}],\"issued\":{\"date-parts\":[[\"2020\"]]}}},{\"id\":814,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VKH9BLC2\"],\"itemData\":{\"id\":814,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2206.13559\",\"title\":\"ST-Adapter: Parameter-Efficient Image-to-Video Transfer Learning\",\"author\":[{\"family\":\"Pan\",\"given\":\"Junting\"},{\"family\":\"Lin\",\"given\":\"Ziyi\"},{\"family\":\"Zhu\",\"given\":\"Xiatian\"},{\"family\":\"Shao\",\"given\":\"Jing\"},{\"family\":\"Li\",\"given\":\"Hongsheng\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":833,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/2PCBLMCA\"],\"itemData\":{\"id\":833,\"type\":\"article-journal\",\"container-title\":\"Micromachines\",\"DOI\":\"10.3390/mi10090616\",\"title\":\"Investigation on the Edge Chipping in Ultrasonic Assisted Sawing of Monocrystalline Silicon\",\"author\":[{\"family\":\"Shen\",\"given\":\"Jianyun\"},{\"family\":\"Zhu\",\"given\":\"Xu\"},{\"family\":\"Chen\",\"given\":\"Jianbin\"},{\"family\":\"Tao\",\"given\":\"Ping\"},{\"family\":\"Wu\",\"given\":\"Xian\"}],\"issued\":{\"date-parts\":[[\"2019\"]]}}},{\"id\":822,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YP5P4M5X\"],\"itemData\":{\"id\":822,\"type\":\"article-journal\",\"container-title\":\"International Journal for Research in Applied Science and Engineering Technology\",\"DOI\":\"10.22214/ijraset.2025.73930\",\"title\":\"Decoding Efficiency: A Comprehensive Review of Knowledge Distillation Techniques in Large Language Model\",\"author\":[{\"family\":\"Soni\",\"given\":\"Goldi\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":830,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JMBPHDM2\"],\"itemData\":{\"id\":830,\"type\":\"article-journal\",\"DOI\":\"10.36227/techrxiv.19575760.v2\",\"title\":\"Feature Distribution Matching for Federated Domain Generalization\",\"author\":[{\"family\":\"Sun\",\"given\":\"Yuwei\"},{\"family\":\"T. 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that the difference between the fine-tuned parameters and the foundation model is controlled. It enables th
e model to take advantage of abundant semantic representations and inductive biases held in the foundation

model and also learn to make recommendations to every user individually. Such a tradeoff is so important in
 federated recommendation systems, where one can suit very well to the needs of a specific user, without th
e overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q.
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It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",

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that the difference between the fine-tuned parameters and the foundation model is controlled. It enables th
e model to take advantage of abundant semantic representations and inductive biases held in the foundation
model and also learn to make recommendations to every user individually. Such a tradeoff is so important in
 federated recommendation systems, where one can suit very well to the needs of a specific user, without th
e overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q.
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It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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that the difference between the fine-tuned parameters and the foundation model is controlled. It enables th
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model and also learn to make recommendations to every user individually. Such a tradeoff is so important in
 federated recommendation systems, where one can suit very well to the needs of a specific user, without th
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that the difference between the fine-tuned parameters and the foundation model is controlled. It enables th
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model and also learn to make recommendations to every user individually. Such a tradeoff is so important in
 federated recommendation systems, where one can suit very well to the needs of a specific user, without th
e overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q.
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between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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between the fine-tuned parameters and the foundation model is controlled. It enables the model to take advantage of abundant semantic representations and inductive biases held in the foundation model and also learn to make recommendations to every user individually. Such a tradeoff is so important in federated recommendation systems, where one can suit very well to the needs of a specific user, without the overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q. Huang, 2025; Z. Huang et al., 2025; H. Jiang, 2019; Jonker et al., 2024; Kalluri, 2024; Kucharavy, 2024; H. Li et al., 2019; X. Li et al., 2018, 2020; J. Pan et al., 2022; J. Shen et al., 2019; Soni, 2025; Y. Sun et al., 2022; Tian et al., 2023; A. Wang et al., 2021; L. Wang et al., 2021; A. Wu \u0026 Kwon, 2023; H. Wu et al., 2025b; X. Yang et al., 2021, 2023; Zaki et al., 2022; J. Zhang et al., 2022; J. Zhang, Lei, et al., 2025)",
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that the difference between the fine-tuned parameters and the foundation model is controlled. It enables th
e model to take advantage of abundant semantic representations and inductive biases held in the foundation
model and also learn to make recommendations to every user individually. Such a tradeoff is so important in
 federated recommendation systems, where one can suit very well to the needs of a specific user, without th
e overall intelligences that the foundation model represents. (Feng et al., 2022; A. Gupta et al., 2019; Q.
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f this is an interesting solution introduced by the highly learned foundation models with high-level semant
ic knowledge that can be utilized to generate good representations of users and items and thus allows zero-
shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providi
ng an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the
 federated learning environment without subjecting it to a heavy consumption of computationally intensive l
oad. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing
 of the model structure and training algorithms to be executed using hardcore budgets of resources. It will
 involve the creation of new solutions in order to quantize and prune optimized to execute on-device infere
nce and federated learning approaches that are able to assist the capabilities of different hardware settin
gs and network circumstances. Such a complicated interaction implies that both the architectural aspects of
 the foundation models and the limitations of communication of federated mobile environments should be shar
ed broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and eff
ective deployment strategies are great in the environmentally friendly platform of the federated foundation
 model on the wireless field. Speaking more specifically, by using the foundation models and the system of
federated learning, we will be capable of performing a more personalized approach and an efficient system o
f recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data amo
ng the consumers. However, with the integration, new complexities are attached including an effective perso
nalization modelling because of the variety of preferences of the separated clients and effective fusion be
tween collective knowledge of previously trained foundation models and personalization of the users. These
complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally
 a trade-off between the computation problems of large models and privacy protecting on-device learning and
 efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficien
t fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily do
ing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni

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ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods 
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount 
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation. 
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and 
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
obile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 
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:\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these large models on federated networks, which can tackle the technicalities related to the on-device execution. The mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount of data that is transmitted between clients and central server which in turn increases the practicability of executing more compilated recommender systems in the background of mobile that has bandwidth limitation. The federated learning models could be enhanced by adding the foundation models to help improve collaborative filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and that has been acquired by examining many massive datasets to provide guidance to more common adaptable components of the recommendation system. Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. 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Optimization\",\"author\":[{\"family\":\"Hee\",\"given\":\"Kim Eun\"},{\"family\":\"Maros\",\"given\":\"Máté E.\"},{\"family\":\"Siegel\",\"given\":\"Fabian\"},{\"family\":\"Ganslandt\",\"given\":\"Thomas\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":851,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/8MW7XCET\"],\"itemData\":{\"id\":851,\"type\":\"article-journal\",\"DOI\":\"10.1101/2024.12.09.627458\",\"title\":\"Tissue-Specific Cell Type Annotation With Supervised Representation Learning Using Split Vector Quantization and Its Comparisons With Single-Cell Foundation Models\",\"author\":[{\"family\":\"Heryanto\",\"given\":\"Yusri 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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommen

dation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these large models on federated networks, which can tackle the technicalities related to the on-device execution. The mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount of data that is transmitted between clients and central server which in turn increases the practicability of executing more compilated recommender systems in the background of mobile that has bandwidth limitation. The federated learning models could be enhanced by adding the foundation models to help improve collaborative filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and that has been acquired by examining many massive datasets to provide guidance to more common adaptable components of the recommendation system. Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out o
f this is an interesting solution introduced by the highly learned foundation models with high-level semant
ic knowledge that can be utilized to generate good representations of users and items and thus allows zero-
shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providi
ng an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the
 federated learning environment without subjecting it to a heavy consumption of computationally intensive l
oad. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing
 of the model structure and training algorithms to be executed using hardcore budgets of resources. It will
 involve the creation of new solutions in order to quantize and prune optimized to execute on-device infere
nce and federated learning approaches that are able to assist the capabilities of different hardware settin
gs and network circumstances. Such a complicated interaction implies that both the architectural aspects of
 the foundation models and the limitations of communication of federated mobile environments should be shar
ed broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and eff
ective deployment strategies are great in the environmentally friendly platform of the federated foundation
 model on the wireless field. Speaking more specifically, by using the foundation models and the system of
federated learning, we will be capable of performing a more personalized approach and an efficient system o
f recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data amo
ng the consumers. However, with the integration, new complexities are attached including an effective perso
nalization modelling because of the variety of preferences of the separated clients and effective fusion be
tween collective knowledge of previously trained foundation models and personalization of the users. These
complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally
 a trade-off between the computation problems of large models and privacy protecting on-device learning and
 efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficien
t fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily do
ing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni
ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
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The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen

t quantization approaches like AWQ, can also become a potential solution to running and training these large models on federated networks, which can tackle the technicalities related to the on-device execution. The mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount of data that is transmitted between clients and central server which in turn increases the practicability of executing more compilated recommender systems in the background of mobile that has bandwidth limitation. The federated learning models could be enhanced by adding the foundation models to help improve collaborative filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and that has been acquired by examining many massive datasets to provide guidance to more common adaptable components of the recommendation system. Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)\",\"noteIndex\":0},\"citationItems\":[{\"id\":855,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ABW57SRU\"],\"itemData\":{\"id\":855,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.1707.01209\",\"title\":\"Model Compression as Constrained Optimization, With Application to Neural Nets. Part I: General Framework\",\"author\":[{\"family\":\"Carreira-Perpiñán\",\"given\":\"Miguel Á.\"}],\"issued\":{\"date-parts\":[[\"2017\"]]}}},{\"id\":853,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VXZA2GCB\"],\"itemData\":{\"id\":853,\"type\":\"article-journal\",\"container-title\":\"Mmaa\",\"DOI\":\"10.54097/4qsv1565\",\"title\":\"YOLO V8 Network Lightweight Method Based on Pruning and Quantization\",\"author\":[{\"family\":\"Cheng\",\"given\":\"Yu\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":846,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7H2KA42Z\"],\"itemData\":{\"id\":846,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2209.09577\",\"title\":\"Understanding Real-World Threats to Deep Learning Models in Android Apps\",\"author\":[{\"family\":\"Deng\",\"given\":\"Zizhuang\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":860,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VQY2G7II\"],\"itemData\":{\"id\":860,\"type\":\"article-journal\",\"container-title\":\"Sensors\",\"DOI\":\"10.3390/s25154612\",\"title\":\"Optical Sensor-Based Approaches in Obesity Detection: A Literature Review of Gait Analysis, Pose Estimation, and Human Voxel Modeling\",\"author\":[{\"family\":\"Dhaouadi\",\"given\":\"Sabrine\"},{\"family\":\"Ben Khelifa\",\"given\":\"Mohamed Moncef\"},{\"family\":\"Balti\",\"given\":\"Ala\"},{\"family\":\"Duché\",\"given\":\"Pascale\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":859,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/EPNK92D2\"],\"itemData\":{\"id\":859,\"type\":\"article-journal\",\"DOI\":\"10.17918/00011095\",\"title\":\"Know When to Trust\",\"author\":[{\"family\":\"Duan\",\"given\":\"Jinhao\"},{\"family\":\"Xu\",\"given\":\"Kaidi\"}]}},{\"id\":844,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/KG8DUJV6\"],\"itemData\":{\"id\":844,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.1510.00149\",\"title\":\"Deep Compression: Compressing Deep Neural Networks With Pruning, Trained Quantization and Huffman Coding\",\"author\":[{\"family\":\"Han\",\"given\":\"Song\"},{\"family\":\"Mao\",\"given\":\"Huizi\"},{\"family\":\"Dally\",\"given\":\"William J.\"}],\"issued\":{\"date-parts\":[[\"2015\"]]}}},{\"id\":842,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/MH5VBMIB\"],\"itemData\":{\"id\":842,\"type\":\"article-journal\",\"container-title\":\"Biomedicines\",\"DOI\":\"10.3390/biomedicines10112808\",\"title\":\"Rapid Convolutional Neural Networks for Gram-Stained Image Classification at Inference Time on Mobile Devices: Empirical Study From Transfer Learning to Optimization\",\"author\":[{\"family\":\"Hee\",\"given\":\"Kim Eun\"},{\"family\":\"Maros\",\"given\":\"Máté E.\"},{\"family\":\"Siegel\",\"given\":\"Fabian\"},{\"family\":\"Ganslandt\",\"given\":\"Thomas\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":851,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/8MW7XCET\"],\"itemData\":{\"id\":851,\"type\":\"article-journal\",\"DOI\":\"10.1101/2024.12.09.627458\",\"title\":\"Tissue-Specific Cell Type Annotation With Supervised Representation Learning Using Split Vector Quantization and Its Comparisons With Single-Cell Foundation Models\",\"author\":[{\"family\":\"Heryanto\",\"given\":\"Yusri Dwi\"},{\"family\":\"Zhang\",\"given\":\"Yaozhong\"},{\"family\":\"Imoto\",\"given\":\"Seiya\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":854,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/CDV8BDPQ\"],\"itemData\":{\"id\":854,\"type\":\"article-journal\",\"container-title\":\"Drones\",\"DOI\":\"10.3390/drones8120707\",\"title\":\"Construction of a Deep Learning Model for Unmanned Aerial Vehicle-Assisted Safe Lightweight Industrial Quality Inspection in Complex Environments\",\"author\":[{\"family\":\"Jing\",\"given\":\"Zhongyuan\"},{\"family\":\"Wang\",\"given\":\"Ruyan\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":836,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/

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M\"},{\"family\":\"Navarro\",\"given\":\"Jorge\"},{\"family\":\"Zhang\",\"given\":\"Lihua\"},{\"family\":\"Farouk\",\"given\":\"Omar\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":852,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/PUIH8CFF\"],\"itemData\":{\"id\":852,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2207.12779\",\"title\":\"Reconciling Security and Communication Efficiency in Federated Learning\",\"author\":[{\"family\":\"Prasad\",\"given\":\"Karthik\"},{\"family\":\"Ghosh\",\"given\":\"Sayan\"},{\"family\":\"Cormode\",\"given\":\"Graham\"},{\"family\":\"Mironov\",\"given\":\"Ilya\"},{\"family\":\"Yousefpour\",\"given\":\"Ashkan\"},{\"family\":\"Stock\",\"given\":\"Pierre\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":856,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NYW6XK6H\"],\"itemData\":{\"id\":856,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.1908.11250\",\"title\":\"Smaller Models, Better 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f this is an interesting solution introduced by the highly learned foundation models with high-level semant
ic knowledge that can be utilized to generate good representations of users and items and thus allows zero-
shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providi
ng an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the
 federated learning environment without subjecting it to a heavy consumption of computationally intensive l
oad. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing
 of the model structure and training algorithms to be executed using hardcore budgets of resources. It will
 involve the creation of new solutions in order to quantize and prune optimized to execute on-device infere
nce and federated learning approaches that are able to assist the capabilities of different hardware settin
gs and network circumstances. Such a complicated interaction implies that both the architectural aspects of
 the foundation models and the limitations of communication of federated mobile environments should be shar
ed broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and eff
ective deployment strategies are great in the environmentally friendly platform of the federated foundation
 model on the wireless field. Speaking more specifically, by using the foundation models and the system of
federated learning, we will be capable of performing a more personalized approach and an efficient system o
f recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data amo
ng the consumers. However, with the integration, new complexities are attached including an effective perso
nalization modelling because of the variety of preferences of the separated clients and effective fusion be
tween collective knowledge of previously trained foundation models and personalization of the users. These
complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally
 a trade-off between the computation problems of large models and privacy protecting on-device learning and
 efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficien
t fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily do
ing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni
ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
obile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these larg

e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
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",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. 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Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. 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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out o
f this is an interesting solution introduced by the highly learned foundation models with high-level semant
ic knowledge that can be utilized to generate good representations of users and items and thus allows zero-
shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providi
ng an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the
 federated learning environment without subjecting it to a heavy consumption of computationally intensive l
oad. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing
 of the model structure and training algorithms to be executed using hardcore budgets of resources. It will
 involve the creation of new solutions in order to quantize and prune optimized to execute on-device infere
nce and federated learning approaches that are able to assist the capabilities of different hardware settin
gs and network circumstances. Such a complicated interaction implies that both the architectural aspects of
 the foundation models and the limitations of communication of federated mobile environments should be shar
ed broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and eff
ective deployment strategies are great in the environmentally friendly platform of the federated foundation
 model on the wireless field. Speaking more specifically, by using the foundation models and the system of
federated learning, we will be capable of performing a more personalized approach and an efficient system o
f recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data amo
ng the consumers. However, with the integration, new complexities are attached including an effective perso
nalization modelling because of the variety of preferences of the separated clients and effective fusion be
tween collective knowledge of previously trained foundation models and personalization of the users. These
complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally
 a trade-off between the computation problems of large models and privacy protecting on-device learning and
 efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficien
t fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily do
ing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni
ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in

 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
obile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026
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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out o
f this is an interesting solution introduced by the highly learned foundation models with high-level semant
ic knowledge that can be utilized to generate good representations of users and items and thus allows zero-
shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providi
ng an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the
 federated learning environment without subjecting it to a heavy consumption of computationally intensive l
oad. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing
 of the model structure and training algorithms to be executed using hardcore budgets of resources. It will
 involve the creation of new solutions in order to quantize and prune optimized to execute on-device infere
nce and federated learning approaches that are able to assist the capabilities of different hardware settin
gs and network circumstances. Such a complicated interaction implies that both the architectural aspects of
 the foundation models and the limitations of communication of federated mobile environments should be shar
ed broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and eff
ective deployment strategies are great in the environmentally friendly platform of the federated foundation
 model on the wireless field. Speaking more specifically, by using the foundation models and the system of
federated learning, we will be capable of performing a more personalized approach and an efficient system o
f recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data amo
ng the consumers. However, with the integration, new complexities are attached including an effective perso
nalization modelling because of the variety of preferences of the separated clients and effective fusion be
tween collective knowledge of previously trained foundation models and personalization of the users. These
complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally
 a trade-off between the computation problems of large models and privacy protecting on-device learning and
 efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficien
t fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily do
ing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni
ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
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The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni

ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
obile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026
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Pruning\",\"author\":[{\"family\":\"Song\",\"given\":\"Yingjie\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":839,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XH6W7XYI\"],\"itemData\":{\"id\":839,\"type\":\"article-journal\",\"container-title\":\"Interantional Journal of Scientific Research in Engineering and Management\",\"DOI\":\"10.55041/ijsrem43474\",\"title\":\"Developing New AI Model Compression Techniques\",\"author\":[{\"family\":\"Soundararajan\",\"given\":\"Balaji\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":848,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5CCXKJ6L\"],\"itemData\":{\"id\":848,\"type\":\"article-journal\",\"container-title\":\"Applied and Computational Engineering\",\"DOI\":\"10.54254/2755-2721/107/20241215\",\"title\":\"Optimizing Federated Learning Efficiency: A Comparative Analysis of Model Compression Techniques for Communication 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It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these large models on federated networks, which can tackle the technicalities related to the on-device execution. The mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount of data that is transmitted between clients and central server which in turn increases the practicability of executing more compilated recommender systems in the background of mobile that has bandwidth limitation. The federated learning models could be enhanced by adding the foundation models to help improve collaborative filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and that has been acquired by examining many massive datasets to provide guidance to more common adaptable components of the recommendation system. Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. 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dation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these large models on federated networks, which can tackle the technicalities related to the on-device execution. The mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount of data that is transmitted between clients and central server which in turn increases the practicability of executing more compilated recommender systems in the background of mobile that has bandwidth limitation. The federated learning models could be enhanced by adding the foundation models to help improve collaborative filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and that has been acquired by examining many massive datasets to provide guidance to more common adaptable components of the recommendation system. Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2

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The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. 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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these larg

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e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\

",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these large models on federated networks, which can tackle the technicalities related to the on-device execution. The mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount of data that is transmitted between clients and central server which in turn increases the practicability of executing more compilated recommender systems in the background of mobile that has bandwidth limitation. The federated learning models could be enhanced by adding the foundation models to help improve collaborative filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and that has been acquired by examining many massive datasets to provide guidance to more common adaptable components of the recommendation system. Not only in making good initial recommendation is it addressing the problem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed mobile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",

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The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen

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t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out o
f this is an interesting solution introduced by the highly learned foundation models with high-level semant
ic knowledge that can be utilized to generate good representations of users and items and thus allows zero-
shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providi
ng an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the
 federated learning environment without subjecting it to a heavy consumption of computationally intensive l
oad. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing
 of the model structure and training algorithms to be executed using hardcore budgets of resources. It will
 involve the creation of new solutions in order to quantize and prune optimized to execute on-device infere
nce and federated learning approaches that are able to assist the capabilities of different hardware settin
gs and network circumstances. Such a complicated interaction implies that both the architectural aspects of
 the foundation models and the limitations of communication of federated mobile environments should be shar
ed broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and eff
ective deployment strategies are great in the environmentally friendly platform of the federated foundation
 model on the wireless field. Speaking more specifically, by using the foundation models and the system of
federated learning, we will be capable of performing a more personalized approach and an efficient system o
f recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data amo
ng the consumers. However, with the integration, new complexities are attached including an effective perso
nalization modelling because of the variety of preferences of the separated clients and effective fusion be
tween collective knowledge of previously trained foundation models and personalization of the users. These
complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally
 a trade-off between the computation problems of large models and privacy protecting on-device learning and
 efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficien
t fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily do
ing considerable customization to the large ground model itself. It is a paradigm of teacher-student learni
ng that can be effectively deployed to small devices that belong to clients to utilize the amount of knowle
dge about the domain of the enormous foundation model. It is a method that is often supported with methods
such as knowledge distillation applied in the student model in order to realize causally similar performanc
e with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in
 a mobile environment when the computing resources present is minimal; in this case one can have a recommen
dation system that possesses high capability at levels that do not necessarily overload the available capac
ity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplemen
t quantization approaches like AWQ, can also become a potential solution to running and training these larg
e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
obile dataset. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":845,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/AN5SZM9F\"],\"itemData\":{\"id\":845,\"type\":\"article-journal\",\"DOI\":\"10.23919/date51398.2021.9474031\",\"title\":\"Activation Density Based Mixed-Precision Quantization for Energy Efficient Neural Networks\",\"author\":[{\"family\":\"Vasquez\",\"given\":\"Karina\"},{\"family\":\"Venkatesha\",\"given\":\"Yeshwanth\"},{\"family\":\"Bhattacharjee\",\"given\":\"Abhiroop\"},{\"family\":\"Moitra\",\"given\":\"Abhishek\"},{\"family\":\"Panda\",\"given\":\"Priyadarshini\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":838,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7YYJ5KU4\"],\"itemData\":{\"id\":838,\"type\":\"article-journal\",\"container-title\":\"Journal of Physics Conference Series\",\"DOI\":\"10.1088/1742-6596/2303/1/012086\",\"title\":\"Exploration and Research About Key Technologies Concerning Deep Learning Models Targeting Mobile Terminals\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yang\"},{\"family\":\"Yi\",\"given\":\"Bohai\"},{\"family\":\"Yu\",\"given\":\"Kexin\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":841,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LV793E5Y\"],\"itemData\":{\"id\":841,\"type\":\"article-journal\",\"container-title\":\"International Journal of Intelligent Systems\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} The way out of this is an interesting solution introduced by the highly learned foundation models with high-level semantic knowledge that can be utilized to generate good representations of users and items and thus allows zero-shot and few-shot learning to simplify the process. The capacity is accompanied by the necessity of providing an adequate transfer of knowledge within the framework of a basis model, which are being loaded into the federated learning environment without subjecting it to a heavy consumption of computationally intensive load. The foundation of web adapt on mobile federated learning is also elaborated to incorporate a balancing of the model structure and training algorithms to be executed using hardcore budgets of resources. It will involve the creation of new solutions in order to quantize and prune optimized to execute on-device inference and federated learning approaches that are able to assist the capabilities of different hardware settings and network circumstances. Such a complicated interaction implies that both the architectural aspects of the foundation models and the limitations of communication of federated mobile environments should be shared broadly to achieve the right accuracy privacy trade-offs. Further bonus plans to attract clients and effective deployment strategies are great in the environmentally friendly platform of the federated foundation model on the wireless field. Speaking more specifically, by using the foundation models and the system of federated learning, we will be capable of performing a more personalized approach and an efficient system of recommendations, by dealing with such concerns as the data sparsity and non-IID distributions of data among the consumers. However, with the integration, new complexities are attached including an effective personalization modelling because of the variety of preferences of the separated clients and effective fusion between collective knowledge of previously trained foundation models and personalization of the users. These complexities lead to issues, and these issues are best addressed by a complex resolution, which is normally a trade-off between the computation problems of large models and privacy protecting on-device learning and efficient knowledge transfer relations in a federated scheme. These methods are usually parameter efficient fine-tuning methods, which can be adapted to the specifics of the individual user, without necessarily doing considerable customization to the large ground model itself. It is a paradigm of teacher-student learning that can be effectively deployed to small devices that belong to clients to utilize the amount of knowledge about the domain of the enormous foundation model. It is a method that is often supported with methods such as knowledge distillation applied in the student model in order to realize causally similar performance with reduced computation costs as opposed to a larger teacher model. It also can be potentially useful in a mobile environment when the computing resources present is minimal; in this case one can have a recommendation system that possesses high capability at levels that do not necessarily overload the available capacity of the device. Besides, parameter-efficient fine-tuning approaches, which can be suggested to supplement quantization approaches like AWQ, can also become a potential solution to running and training these larg

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e models on federated networks, which can tackle the technicalities related to the on-device execution. The
 mechanism is additionally effective as it reduces communication overheads in terms of reducing the amount
of data that is transmitted between clients and central server which in turn increases the practicability o
f executing more compilated recommender systems in the background of mobile that has bandwidth limitation.
The federated learning models could be enhanced by adding the foundation models to help improve collaborati
ve filtering through exploiting the immense inductive biases and external knowledge exemplified in FMs and
that has been acquired by examining many massive datasets to provide guidance to more common adaptable comp
onents of the recommendation system. Not only in making good initial recommendation is it addressing the pr
oblem of cold-start, but also in addressing the problem of data sparsity and heterogeneity of distributed m
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"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless, the federated adaptation is only successful in hybrid approach the art of biological continuous updating and amalgamating the already obtained user embeddings. They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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\",\"DOI\":\"10.1155/int/5813659\",\"title\":\"CSI Acquisition in Internet of Vehicle Network: Federated Edge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless, the federated adaptation is only successful in hybrid approach the art of biological continuous updating and amalgamating the already obtained user embeddings. They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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Optimization\",\"author\":[{\"family\":\"Hee\",\"given\":\"Kim Eun\"},{\"family\":\"Maros\",\"given\":\"Máté E.\"},{\"family\":\"Siegel\",\"given\":\"Fabian\"},{\"family\":\"Ganslandt\",\"given\":\"Thomas\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":851,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/8MW7XCET\"],\"itemData\":{\"id\":851,\"type\":\"article-journal\",\"DOI\":\"10.1101/2024.12.09.627458\",\"title\":\"Tissue-Specific Cell Type Annotation With Supervised Representation Learning Using Split Vector Quantization and Its Comparisons With Single-Cell Foundation Models\",\"author\":[{\"family\":\"Heryanto\",\"given\":\"Yusri Dwi\"},{\"family\":\"Zhang\",\"given\":\"Yaozhong\"},{\"family\":\"Imoto\",\"given\":\"Seiya\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":854,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/CDV8BDPQ\"],\"itemData\":{\"id\":854,\"type\":\"article-journal\",\"container-title\":\"Drones\",\"DOI\":\"10.3390/drones8120707\",\"title\":\"Construction of a Deep Learning Model for Unmanned Aerial Vehicle-Assisted Safe Lightweight Industrial Quality Inspection in Complex Environments\",\"author\":[{\"family\":\"Jing\",\"given\":\"Zhongyuan\"},{\"family\":\"Wang\",\"given\":\"Ruyan\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":836,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ZRGMJJXU\"],\"itemData\":{\"id\":836,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-3825246/v1\",\"title\":\"An Adaptive Quantization Model Compression Method for Heterogeneous Federated 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2; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang
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They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. 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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless, the federated adaptation is only successful in hybrid approach the art of biological continuous updating and amalgamating the already obtained user embeddings. They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng e

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 the federated adaptation is only successful in hybrid approach the art of biological continuous updating a
nd amalgamating the already obtained user embeddings. They are commonly constructed when the information th
at is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce
 misguided suggestions, or even cause a performance decline because behaviour or interests of a user in var
ious aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-eff
icient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific insta
nce, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. li
ghtweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communicat

ion and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it min
imizes the number of information that is exchanged among client-specific devices and central servers, as we
ll, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. C
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\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless, the federated adaptation is only successful in hybrid approach the art of biological continuous updating and amalgamating the already obtained user embeddings. They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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 the federated adaptation is only successful in hybrid approach the art of biological continuous updating a
nd amalgamating the already obtained user embeddings. They are commonly constructed when the information th
at is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce
 misguided suggestions, or even cause a performance decline because behaviour or interests of a user in var
ious aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-eff
icient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific insta
nce, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. li
ghtweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communicat
ion and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it min
imizes the number of information that is exchanged among client-specific devices and central servers, as we
ll, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. C
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 the federated adaptation is only successful in hybrid approach the art of biological continuous updating a
nd amalgamating the already obtained user embeddings. They are commonly constructed when the information th
at is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce
 misguided suggestions, or even cause a performance decline because behaviour or interests of a user in var
ious aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-eff
icient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific insta
nce, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. li
ghtweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communicat
ion and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it min
imizes the number of information that is exchanged among client-specific devices and central servers, as we
ll, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. C
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 the federated adaptation is only successful in hybrid approach the art of biological continuous updating a
nd amalgamating the already obtained user embeddings. They are commonly constructed when the information th
at is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce
 misguided suggestions, or even cause a performance decline because behaviour or interests of a user in var
ious aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-eff
icient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific insta
nce, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. li
ghtweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communicat
ion and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it min
imizes the number of information that is exchanged among client-specific devices and central servers, as we
ll, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. C
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They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. 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"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless,
 the federated adaptation is only successful in hybrid approach the art of biological continuous updating a
nd amalgamating the already obtained user embeddings. They are commonly constructed when the information th
at is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce
 misguided suggestions, or even cause a performance decline because behaviour or interests of a user in var
ious aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-eff
icient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific insta
nce, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. li
ghtweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communicat
ion and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it min
imizes the number of information that is exchanged among client-specific devices and central servers, as we
ll, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. C
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ce Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless, the federated adaptation is only successful in hybrid approach the art of biological continuous updating and amalgamating the already obtained user embeddings. They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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From Transfer Learning to Optimization\",\"author\":[{\"family\":\"Hee\",\"given\":\"Kim Eun\"},{\"family\":\"Maros\",\"given\":\"Máté E.\"},{\"family\":\"Siegel\",\"given\":\"Fabian\"},{\"family\":\"Ganslandt\",\"given\":\"Thomas\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":851,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/8MW7XCET\"],\"itemData\":{\"id\":851,\"type\":\"article-journal\",\"DOI\":\"10.1101/2024.12.09.627458\",\"title\":\"Tissue-Specific Cell Type Annotation With Supervised Representation Learning Using Split Vector Quantization and Its Comparisons With Single-Cell Foundation Models\",\"author\":[{\"family\":\"Heryanto\",\"given\":\"Yusri Dwi\"},{\"family\":\"Zhang\",\"given\":\"Yaozhong\"},{\"family\":\"Imoto\",\"given\":\"Seiya\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":854,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/CDV8BDPQ\"],\"itemData\":{\"id\":854,\"type\":\"article-journal\",\"container-title\":\"Drones\",\"DOI\":\"10.3390/drones8120707\",\"title\":\"Construction of a Deep Learning Model for Unmanned Aerial Vehicle-Assisted Safe Lightweight Industrial Quality Inspection in Complex Environments\",\"author\":[{\"family\":\"Jing\",\"given\":\"Zhongyuan\"},{\"family\":\"Wang\",\"given\":\"Ruyan\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":836,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ZRGMJJXU\"],\"itemData\":{\"id\":836,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-3825246/v1\",\"title\":\"An Adaptive Quantization Model Compression Method for Heterogeneous Federated 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Pruning\",\"author\":[{\"family\":\"Song\",\"given\":\"Yingjie\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":839,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/XH6W7XYI\"],\"itemData\":{\"id\":839,\"type\":\"article-journal\",\"container-title\":\"Interantional Journal of Scientific Research in Engineering and Management\",\"DOI\":\"10.55041/ijsrem43474\",\"title\":\"Developing New AI Model Compression Techniques\",\"author\":[{\"family\":\"Soundararajan\",\"given\":\"Balaji\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":848,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5CCXKJ6L\"],\"itemData\":{\"id\":848,\"type\":\"article-journal\",\"container-title\":\"Applied and Computational Engineering\",\"DOI\":\"10.54254/2755-2721/107/20241215\",\"title\":\"Optimizing Federated Learning Efficiency: A Comparative Analysis of Model Compression Techniques for Communication 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ge Learning With Model Pruning and Vector Quantization\",\"author\":[{\"family\":\"Wang\",\"given\":\"Yi\"},{\"family\":\"Zhi\",\"given\":\"Junlei\"},{\"family\":\"Mei\",\"given\":\"Linsheng\"},{\"family\":\"Huang\",\"given\":\"Wei\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":857,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/73VAUYC5\"],\"itemData\":{\"id\":857,\"type\":\"article-journal\",\"container-title\":\"Journal of Intelligent \u0026 Fuzzy Systems\",\"DOI\":\"10.3233/jifs-232213\",\"title\":\"Cattle Face Detection Method Based on Channel Pruning YOLOv5 Network and Mobile Deployment\",\"author\":[{\"family\":\"Weng\",\"given\":\"Zhi\"},{\"family\":\"Liu\",\"given\":\"Ke\"},{\"family\":\"Zheng\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":847,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/LFGMZAHX\"],\"itemData\":{\"id\":847,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.3078229\",\"title\":\"Efficient Training of Deep Neural Networks on Resource-Constrained Devices for Industrial Applications\",\"author\":[{\"family\":\"Yarashov\",\"given\":\"Shokhrukh\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":840,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3IL6TSKK\"],\"itemData\":{\"id\":840,\"type\":\"article-journal\",\"container-title\":\"Remote Sensing\",\"DOI\":\"10.3390/rs15164015\",\"title\":\"Deep Hybrid Compression Network for Lidar Point Cloud Classification and Segmentation\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Zhi\"},{\"family\":\"Ma\",\"given\":\"Yanxin\"},{\"family\":\"Xu\",\"given\":\"Ke\"},{\"family\":\"Wan\",\"given\":\"Jianwei\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Nevertheless, the federated adaptation is only successful in hybrid approach the art of biological continuous updating and amalgamating the already obtained user embeddings. They are commonly constructed when the information that is being used is an inhomogeneous set of users, and when exploited irresponsibly can be known to produce misguided suggestions, or even cause a performance decline because behaviour or interests of a user in various aspects are not matched. To mitigate such risks, new federated systems are based on high parameter-efficient ahead of time tuning, including adapter-based tuning and prompt-based tuning. In this specific instance, these special techniques are designed to only scale the smaller percentage of parameter model (e.g. lightweight adapters, or prompt vectors), and thus cause an enormous decrease in both the costs of communication and computation. It is specifically applicable in privacy-sensitive federated solutions whereby, it minimizes the number of information that is exchanged among client-specific devices and central servers, as well, it also renders the user information more secure on an individual basis. (Carreira-Perpiñán, 2017; Y. Cheng, 2024; Z. Deng, 2022; Dhaouadi et al., 2025; Duan \u0026 Xu, n.d.; S. Han et al., 2015; Hee et al., 2022; Heryanto et al., 2024; Jing \u0026 Wang, 2024; li et al., 2024; Marion et al., 2025; Prasad et al., 2022; M. Sharma et al., 2019; T. Sheng et al., 2018; Y. Song, 2024; Soundararajan, 2025; Y. Sun, 2024; W. Tang et al., 2020; Tatarnikova \u0026 Raskopina, 2025; Vasquez et al., 2021; Y. Wang et al., 2022, 2025; Weng et al., 2023; Yarashov, 2025; Z. Zhao et al., 2023)",
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with Differential Privacy (DP-FL), Federated Transfer Learning (FTL), Personalized FL (PFL) and Lightweight FL (LFL).We also discuss the principles of FL, its integration with deep learning models, and its advantages in healthcare specially for heart diseases detection. Additionally, we review recent advancements, challenges, and future directions in this field. 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This paper explores the application of Deep Federated Learning (DFL) algorithms in the detection of heart disease namely Federated Averaging (FedAvg), Federated Learning with Differential Privacy (DP-FL), Federated Transfer Learning (FTL), Personalized FL (PFL) and Lightweight FL (LFL).We also discuss the principles of FL, its integration with deep learning models, and its advantages in healthcare specially for heart diseases detection. Additionally, we review recent advancements, challenges, and future directions in this field. Finally, it is observed that Personalized Federated Learning (PFL) achieves the highest performance with 96.2% accuracy, 96.4% F1-score that demonstrating the benefits of adapting models to individual client data.\",\"container-title\":\"Journal of Information Systems Engineering and Management\",\"DOI\":\"10.52783/jisem.v10i25s.4035\",\"issue\":\"25s\",\"journalAbbreviation\":\"Journal of Information Systems Engineering and Management\",\"page\":\"399-408\",\"title\":\"Detection of Heart Disease Using Deep Federated Learning Algorithms\",\"volume\":\"10\",\"author\":[{\"family\":\"Kshirod Sarmah\",\"given\":\"None\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":916,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/8DFG74BQ\"],\"itemData\":{\"id\":916,\"type\":\"article-journal\",\"abstract\":\"Federated learning (FL), which enables collaborative learning across distributed nodes, confronts a significant heterogeneity challenge, primarily including resource heterogeneity induced by different hardware platforms, and statistical heterogeneity originating from non-IID private data distributions among clients. Neural architecture search (NAS), particularly one-shot NAS, holds great promise for automatically designing optimal personalized models tailored to such heterogeneous scenarios. However, the coexistence of both resource and statistical heterogeneity destabilizes the training of the one-shot supernet, impairs the evaluation of candidate architectures, and ultimately hinders the discovery of optimal personalized models. To address this problem, we propose a heterogeneity-aware personalized federated NAS (HAPFNAS) method. First, we leverage lightweight knowledge models to distill knowledge from clients to server-side supernet, thereby effectively mitigating the effects of heterogeneity and enhancing the training stability. Then, we build random-forest-based personalized performance predictors to enable the efficient evaluation of candidate architectures across clients. Furthermore, we develop a model-heterogeneous FL algorithm called heteroFedAvg to facilitate collaborative model training for the discovered personalized models. Comprehensive experiments on CIFAR-10/100 and Tiny-ImageNet classification datasets demonstrate the effectiveness of our HAPFNAS, compared to state-of-the-art federated NAS methods.\",\"container-title\":\"Entropy\",\"DOI\":\"10.3390/e27070759\",\"issue\":\"7\",\"journalAbbreviation\":\"Entropy\",\"page\":\"759\",\"title\":\"Heterogeneity-Aware Personalized Federated Neural Architecture Search\",\"volume\":\"27\",\"author\":[{\"family\":\"Yang\",\"given\":\"An\"},{\"family\":\"Liu\",\"given\":\"Ying\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":914,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/CLI96AAH\"],\"itemData\":{\"id\":914,\"type\":\"article-journal\",\"abstract\":\"Lightweight training and distributed tiny data storage in local model will lead to the severe challenge of convergence for tiny federated learning (FL). Achieving fast convergence in tiny FL is crucial for many emerging applications in Internet of Unmanned Aerial Vehicles (IUAVs) networks. Excessive information exchange between UAVs and IoT devices could lead to security risks and data breaches, while insufficient information can slow down the learning process and negatively system performance experience due to significant computational and communication constraints in tiny FL hardware system. This paper proposes a trusting, low latency, and energy-efficient tiny wireless FL framework with blockchain (TBWFL) for IUAV systems. We develop a quantifiable model to determine the trustworthiness of IoT devices in IUAV networks. This model incorporates the time spent in communication, computation, and block production with a decay function in each round of FL at the UAVs. Then it combines the trust information from different UAVs, considering their credibility of trust recommendation. We formulate the TBWFL as an optimization problem that balances trustworthiness, learning speed, and energy consumption for IoT devices with diverse computing and energy capabilities. We decompose the complex optimization problem into three sub-problems for improved local accuracy, fast learning, trust verification, and energy efficiency of IoT devices. Our extensive experiments show that TBWFL offers higher trustworthiness, faster convergence, and lower energy consumption than the existing state-of-the-art FL scheme.\",\"container-title\":\"IEEE Internet of Things Journal\",\"DOI\":\"10.1109/jiot.2024.3363443\",\"issue\":\"12\",\"journalAbbreviation\":\"IEEE Internet of Things Journal\",\"page\":\"21046-21060\",\"title\":\"Trust Management of Tiny Federated Learning in Internet of Unmanned Aerial Vehicles\",\"volume\":\"11\",\"author\":[{\"family\":\"Zheng\",\"given\":\"Jie\"},{\"family\":\"Xu\",\"given\":\"Jipeng\"},{\"family\":\"Du\",\"given\":\"Hongyang\"},{\"family\":\"Niyato\",\"given\":\"Dusit\"},{\"family\":\"Kang\",\"given\":\"Jiawen\"},{\"family\":\"Nie\",\"given\":\"Jiangtian\"},{\"family\":\"Wang\",\"given\":\"Zheng\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} On another attempt to achieve the balance between the foundation models strength and the practicability of mobile federated learning is a trend of development of more and more personalised adapters. They are tiny modules which ought to have the capability of learning and encoding preferences in individual users without putting much strain on the computing and memory. The customisation adapters allow offering finer-grained and adaptive suggestion services to trade-offs between mixed and universal knowledge and consider the presence of user-specific personalisation. Better still, they do so without invading the privacy of the users either as sensitive information about individual aspects of behaviour is not sent anywhere because it is stored in one of the local storage of the user to the device itself and is never transmitted to the servers out there. (Colombo et al., 2023; Kshirod Sarmah, 2025; A. Yang \u0026 Liu, 2025; J. Zheng et al., 2024)",
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s\":[[\"2023\"]]}}},{\"id\":926,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/GBPUEDT5\"],\"itemData\":{\"id\":926,\"type\":\"article-journal\",\"container-title\":\"Acm Transactions on Information Systems\",\"DOI\":\"10.1145/3779442\",\"title\":\"Beyond Similarity: Personalized Federated Recommendation With Composite Aggregation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Honglei\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}}},{\"id\":927,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VWYMNZRH\"],\"itemData\":{\"id\":927,\"type\":\"article-journal\",\"DOI\":\"10.4271/2013-01-1351\",\"title\":\"Test Process and Correction of Automotive Wind Tunnel in Jilin University China\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yingchao\"},{\"family\":\"Zhang\",\"given\":\"Zhe\"},{\"family\":\"Li\",\"given\":\"Jie\"},{\"family\":\"Shao\",\"given\":\"Shuxin\"},{\"family\":\"Guo\",\"given\":\"Sai\"}],\"issued\":{\"date-parts\":[[\"2013\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Moreover, with the ability to optimize the resources by tuning the fine details of the resource based devices, several methods like heterogeneous Low-Rank Adaptation will arrive at the optimal solution of operation only by consuming the minimum communication resources without overfitting simultaneously. Besides this, further research on federated prompt tuning and continuous learning in these models can also be an efficient direction, to make them more personal and offer context-sensitivity, by allowing them to spend more computational power on optimizing the models with new information generated privately. More specific attention needs to be paid to the architectural design and privacy mechanisms of the said federated recommender systems since they are the most important elements to influence the say-off between the accuracy of the recommendation model and privacy of user information. This incorporates a subtle understanding of the dissimilarly secretive protocol and secure summation algorithms which can effortlessly blend so as to result in the forecasting capability of elevated foundation models being eroded to an unreasonable scope. Thus a subsequent study should be aimed at optimization of these privacy enhancing technologies such that it can be in a position to ensure high recommendability, as much as can be, and absolute safety of the sensitive information knowledge of the user with the distributed learning paradigm. This optimisation refers to the search of other ways to compress gradient and asynchronous updating protocols that can bring the cost of communication and ability to cover heterogeneity of clients to a bearable threshold, and will be fundamental in the future expansion of federated learning to large user bases and multifaceted foundation architecture. In addition, the adaptive regularization policies and the client-adapted federated learning algorithms also can be examined to enhance the performance further in the model and alleviate client drift in conditions where the data distribution is highly non-IID, which is characteristic to mobile feedback recommendation under realistic settings. (Chuan et al., 2025; Haddadpour et al., 2020; Hai et al., 2022; Harasic et al., 2023; J. Li \u0026 Wu, 2024; L. Li et al., 2025a; R. Li et al., 2024c; Luo et al., 2024; X. Sheng et al., 2024; Y. Song et al., 2024; C. Xu, 2024; Y.-R. Yang et al., 2023; H. Zhang, 2026b; Y. Zhang et al., 2013)",
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resources without overfitting simultaneously. Besides this, further research on federated prompt tuning and continuous learning in these models can also be an efficient direction, to make them more personal and offer context-sensitivity, by allowing them to spend more computational power on optimizing the models with new information generated privately. More specific attention needs to be paid to the architectural design and privacy mechanisms of the said federated recommender systems since they are the most important elements to influence the say-off between the accuracy of the recommendation model and privacy of user information. This incorporates a subtle understanding of the dissimilarly secretive protocol and secure summation algorithms which can effortlessly blend so as to result in the forecasting capability of elevated foundation models being eroded to an unreasonable scope. Thus a subsequent study should be aimed at optimization of these privacy enhancing technologies such that it can be in a position to ensure high recommendability, as much as can be, and absolute safety of the sensitive information knowledge of the user with the distributed learning paradigm. This optimisation refers to the search of other ways to compress gradient and asynchronous updating protocols that can bring the cost of communication and ability to cover heterogeneity of clients to a bearable threshold, and will be fundamental in the future expansion of federated learning to large user bases and multifaceted foundation architecture. In addition, the adaptive regularization policies and the client-adapted federated learning algorithms also can be examined to enhance the performance further in the model and alleviate client drift in conditions where the data distribution is highly non-IID, which is characteristic to mobile feedback recommendation under realistic settings. 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\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Moreover, with the ability to optimize the resources by tuning the fine details of the resource based devices, several methods like heterogeneous Low-Rank Adaptation will arrive at the optimal solution of operation only by consuming the minimum communication resources without overfitting simultaneously. Besides this, furth

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ly by consuming the minimum communication resources without overfitting simultaneously. Besides this, furth
er research on federated prompt tuning and continuous learning in these models can also be an efficient dir
ection, to make them more personal and offer context-sensitivity, by allowing them to spend more computatio
nal power on optimizing the models with new information generated privately. More specific attention needs 
to be paid to the architectural design and privacy mechanisms of the said federated recommender systems sin
ce they are the most important elements to influence the say-off between the accuracy of the recommendation
 model and privacy of user information. This incorporates a subtle understanding of the dissimilarly secret
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 ensure high recommendability, as much as can be, and absolute safety of the sensitive information knowledg
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to compress gradient and asynchronous updating protocols that can bring the cost of communication and abili
ty to cover heterogeneity of clients to a bearable threshold, and will be fundamental in the future expansi
on of federated learning to large user bases and multifaceted foundation architecture. In addition, the ada
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enhance the performance further in the model and alleviate client drift in conditions where the data distri
bution is highly non-IID, which is characteristic to mobile feedback recommendation under realistic setting
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resources without overfitting simultaneously. Besides this, further research on federated prompt tuning and continuous learning in these models can also be an efficient direction, to make them more personal and offer context-sensitivity, by allowing them to spend more computational power on optimizing the models with new information generated privately. More specific attention needs to be paid to the architectural design and privacy mechanisms of the said federated recommender systems since they are the most important elements to influence the say-off between the accuracy of the recommendation

 model and privacy of user information. This incorporates a subtle understanding of the dissimilarly secret
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resources without overfitting simultaneously. Besides this, further research on federated prompt tuning and continuous learning in these models can also be an efficient direction, to make them more personal and offer context-sensitivity, by allowing them to spend more computational power on optimizing the models with new information generated privately. More specific attention needs to be paid to the architectural design and privacy mechanisms of the said federated recommender systems since they are the most important elements to influence the say-off between the accuracy of the recommendation model and privacy of user information. This incorporates a subtle understanding of the dissimilarly secretive protocol and secure summation algorithms which can effortlessly blend so as to result in the forecasting capability of elevated foundation models being eroded to an unreasonable scope. Thus a subsequent study should be aimed at optimization of these privacy enhancing technologies such that it can be in a position to ensure high recommendability, as much as can be, and absolute safety of the sensitive information knowledge of the user with the distributed learning paradigm. This optimisation refers to the search of other ways to compress gradient and asynchronous updating protocols that can bring the cost of communication and ability to cover heterogeneity of clients to a bearable threshold, and will be fundamental in the future expansi

on of federated learning to large user bases and multifaceted foundation architecture. In addition, the adaptive regularization policies and the client-adapted federated learning algorithms also can be examined to enhance the performance further in the model and alleviate client drift in conditions where the data distribution is highly non-IID, which is characteristic to mobile feedback recommendation under realistic settings. (Chuan et al., 2025; Haddadpour et al., 2020; Hai et al., 2022; Harasic et al., 2023; J. Li \u0026 Wu, 2024; L. Li et al., 2025a; R. Li et al., 2024c; Luo et al., 2024; X. Sheng et al., 2024; Y. Song et al., 2024; C. Xu, 2024; Y.-R. Yang et al., 2023; H. Zhang, 2026b; Y. Zhang et al., 2013)",
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ly\":\"Xu\",\"given\":\"Chuanzhi\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}},{\"id\":929,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NXCW6XZS\"],\"itemData\":{\"id\":929,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2303.05206\",\"title\":\"FedREP: A Byzantine-Robust, Communication-Efficient and Privacy-Preserving Framework for Federated Learning\",\"author\":[{\"family\":\"Yang\",\"given\":\"Yi-Rui\"},{\"family\":\"Wang\",\"given\":\"Kun\"},{\"family\":\"Li\",\"given\":\"Wu-Jun\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}},{\"id\":926,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/GBPUEDT5\"],\"itemData\":{\"id\":926,\"type\":\"article-journal\",\"container-title\":\"Acm Transactions on Information Systems\",\"DOI\":\"10.1145/3779442\",\"title\":\"Beyond Similarity: Personalized Federated Recommendation With Composite Aggregation\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Honglei\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}},{\"id\":927,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VWYMNZRH\"],\"itemData\":{\"id\":927,\"type\":\"article-journal\",\"DOI\":\"10.4271/2013-01-1351\",\"title\":\"Test Process and Correction of Automotive Wind Tunnel in Jilin University China\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Yingchao\"},{\"family\":\"Zhang\",\"given\":\"Zhe\"},{\"family\":\"Li\",\"given\":\"Jie\"},{\"family\":\"Shao\",\"given\":\"Shuxin\"},{\"family\":\"Guo\",\"given\":\"Sai\"}],\"issued\":{\"date-parts\":[[\"2013\"]]}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Moreover, with the ability to optimize the resources by tuning the fine details of the resource based devices, several methods like heterogeneous Low-Rank Adaptation will arrive at the optimal solution of operation only by consuming the minimum communication resources without overfitting simultaneously. Besides this, further research on federated prompt tuning and continuous learning in these models can also be an efficient direction, to make them more personal and offer context-sensitivity, by allowing them to spend more computational power on optimizing the models with new information generated privately. More specific attention needs to be paid to the architectural design and privacy mechanisms of the said federated recommender systems since they are the most important elements to influence the say-off between the accuracy of the recommendation model and privacy of user information. This incorporates a subtle understanding of the dissimilarly secretive protocol and secure summation algorithms which can effortlessly blend so as to result in the forecasting capability of elevated foundation models being eroded to an unreasonable scope. Thus a subsequent study should be aimed at optimization of these privacy enhancing technologies such that it can be in a position to ensure high recommendability, as much as can be, and absolute safety of the sensitive information knowledge of the user with the distributed learning paradigm. This optimisation refers to the search of other ways to compress gradient and asynchronous updating protocols that can bring the cost of communication and ability to cover heterogeneity of clients to a bearable threshold, and will be fundamental in the future expansion of federated learning to large user bases and multifaceted foundation architecture. In addition, the adaptive regularization policies and the client-adapted federated learning algorithms also can be examined to enhance the performance further in the model and alleviate client drift in conditions where the data distribution is highly non-IID, which is characteristic to mobile feedback recommendation under realistic settings. (Chuan et al., 2025; Haddadpour et al., 2020; Hai et al., 2022; Harasic et al., 2023; J. Li \u0026 Wu, 2024; L. Li et al., 2025a; R. Li et al., 2024c; Luo et al., 2024; X. Sheng et al., 2024; Y. Song et al., 2024; C. Xu, 2024; Y.-R. Yang et al., 2023; H. Zhang, 2026b; Y. Zhang et al., 2013)",
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Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et al., 2019; Tricomi et al., 2024; Urblik et al., 2023; Y. 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As These Models Evolve, Context Awareness Has Become a Crucial Factor in Improving Their Adaptability, Coherence, and Decision-Making Capabilities. This Paper Explores the Evolution of LLMs From Rule-Based Systems to Deep Learning Architectures, Emphasizing Advancements in Contextu\",\"author\":[{\"family\":\"Ali\",\"given\":\"Arshad\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":1000,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/D92XF6L3\"],\"itemData\":{\"id\":1000,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics12092068\",\"title\":\"Blockchain-Based Trusted Federated Learning With Pre-Trained Models for COVID-19 Detection\",\"author\":[{\"family\":\"Bian\",\"given\":\"Genqing\"},{\"family\":\"Qu\",\"given\":\"Wenjing\"},{\"family\":\"Shao\",\"given\":\"Bilin\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}},{\"id\":997,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/EEGZTN47\"],\"itemData\":{\"id\":997,\"type\":\"article-journal\",\"DOI\":\"10.1145/1048935.1050192\",\"title\":\"A Self-Organizing Flock of 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Network\",\"author\":[{\"family\":\"Cheng\",\"given\":\"Julian\"},{\"family\":\"Li\",\"given\":\"Qilei\"},{\"family\":\"Chen\",\"given\":\"Jinyong\"},{\"family\":\"Gao\",\"given\":\"Mingliang\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}},{\"id\":985,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VU85X6JB\"],\"itemData\":{\"id\":985,\"type\":\"article-journal\",\"container-title\":\"International Journal of Advanced Research in Science Communication and Technology\",\"DOI\":\"10.48175/ijarsct-17097\",\"title\":\"Optimizing Latency and Intelligence Trade-Offs in AI-Driven Games: Edge-Cloud Architectures, Scheduling Policies, and Observability Frameworks Aravind Chinnaraju\",\"author\":[{\"family\":\"Chinnaraju\",\"given\":\"Aravind\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}},{\"id\":1002,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/K2GH7ZPY\"],\"itemData\":{\"id\":1002,\"type\":\"article-journal\",\"DOI\":\"10.1109/wicom.2009.5304178\",\"title\":\"P2p-Based VoIP 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,{\"family\":\"Papcun\",\"given\":\"P.\"},{\"family\":\"Zolotová\",\"given\":\"Iveta\"}],\"issued\":{\"date
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ning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fa
ult tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calc
ulating and transferring data that, specifically, on the basis of massive foundation models will demand the
 new mechanisms of resource management and offloading of tasks. These techniques play a very important role
 in the trade-off between cost of communication, and the cost of computation locally, which turns crucial i
n the instance of fine-tuning at scale of large models when prediction is required to predict applications
 such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions o
f neural networks, which can successfully be implemented on portable devices, and which can attract fewer e
nergy, and less resources. Further reduction of the model footprint and the computing needs could also be a
ccomplished by pruning and adaptive quantisation - federated learning can be made available even on a broad
er range of mobile devices. It is also hectic that engineers and designers face in the implementation of th
e foundation models of these such systems especially the real-time performance and scalability in the indus
trial applications. The computation amounts to and communication efficiency maximisation in federated learn
ing systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often
 involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in t
raining and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al.,
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Cloud\",\"author\":[{\"family\":\"Wu\",\"given\":\"Jiagao\"},{\"family\":\"Yuan\",\"given\":\"Hang\"},{\"family\":\"He\",\"given\":\"Ying\"},{\"family\":\"Zou\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2014\"]]}}},{\"id\":990,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/X3WJYRKM\"],\"itemData\":{\"id\":990,\"type\":\"article-journal\",\"container-title\":\"Diagnostics\",\"DOI\":\"10.3390/diagnostics13111964\",\"title\":\"Federated Machine Learning for Skin Lesion Diagnosis: An Asynchronous and Weighted Approach\",\"author\":[{\"family\":\"Yaqoob\",\"given\":\"Muhammad Mateen\"},{\"family\":\"Alsulami\",\"given\":\"Musleh\"},{\"family\":\"Khan\",\"given\":\"Muhammad Amir\"},{\"family\":\"Alsadie\",\"given\":\"Deafallah\"},{\"family\":\"Jilani Saudagar\",\"given\":\"Abdul Khader\"},{\"family\":\"Alkhathami\",\"given\":\"Mohammed\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Higher security and fault tolerance: Decentralised federated learning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fault tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often

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 involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in t
raining and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al.,
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Failure Prevention and Management: A Systematic Review and Prospects\",\"author\":[{\"family\":\"Shumba\",\"given\":\"Angela Tafadzwa\"},{\"family\":\"Montanaro\",\"given\":\"Teodoro\"},{\"family\":\"Sergi\",\"given\":\"Ilaria\"},{\"family\":\"Bramanti\",\"given\":\"Alessia\"},{\"family\":\"Ciccarelli\",\"given\":\"Michele\"},{\"family\":\"Rispoli\",\"given\":\"Antonella\"},{\"family\":\"Carrizzo\",\"given\":\"Albino\"},{\"family\":\"Vittorio\",\"given\":\"Massimo De\"},{\"family\":\"Patrono\",\"given\":\"Luigi\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1007,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/W2NTQJJJ\"],\"itemData\":{\"id\":1007,\"type\":\"article-journal\",\"container-title\":\"Symmetry\",\"DOI\":\"10.3390/sym15061264\",\"title\":\"AI-Enabled Consensus Algorithm in Human-Centric Collaborative Computing for Internet of 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CPU\",\"author\":[{\"family\":\"Tao\",\"given\":\"Yunzheng\"},{\"family\":\"You\",\"given\":\"Changsheng\"},{\"family\":\"Zhang\",\"given\":\"Ping\"},{\"family\":\"Huang\",\"given\":\"Kaibin\"}],\"issued\":{\"date-parts\":[[\"2019\"]]}}},{\"id\":994,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VBC8IY7D\"],\"itemData\":{\"id\":994,\"type\":\"article-journal\",\"container-title\":\"Sensors\",\"DOI\":\"10.3390/s24103108\",\"title\":\"Smart City as Cooperating Smart Areas: On the Way of Symbiotic Cyber–Physical Systems Environment\",\"author\":[{\"family\":\"Tricomi\",\"given\":\"Giuseppe\"},{\"family\":\"Giacobbe\",\"given\":\"Maurizio\"},{\"family\":\"Ficili\",\"given\":\"Ilenia\"},{\"family\":\"Peditto\",\"given\":\"Nicola\"},{\"family\":\"Puliafito\",\"given\":\"Antonio\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":1001,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/RRMMRAJZ\"],\"itemData\":{\"id\":1001,\"type\":\"article-journal\",\"container-title\":\"Sensors\",\"DOI\":\"10.3390/s23177662\",\"title\":\"A Modular Framework for Data Processing at the Edge: Design and 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Cloud\",\"author\":[{\"family\":\"Wu\",\"given\":\"Jiagao\"},{\"family\":\"Yuan\",\"given\":\"Hang\"},{\"family\":\"He\",\"given\":\"Ying\"},{\"family\":\"Zou\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2014\"]]}}},{\"id\":990,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/X3WJYRKM\"],\"itemData\":{\"id\":990,\"type\":\"article-journal\",\"container-title\":\"Diagnostics\",\"DOI\":\"10.3390/diagnostics13111964\",\"title\":\"Federated Machine Learning for Skin Lesion Diagnosis: An Asynchronous and Weighted Approach\",\"author\":[{\"family\":\"Yaqoob\",\"given\":\"Muhammad Mateen\"},{\"family\":\"Alsulami\",\"given\"

:\"Musleh\"},{\"family\":\"Khan\",\"given\":\"Muhammad Amir\"},{\"family\":\"Alsadie\",\"given\":\"Deafallah\"},{\"family\":\"Jilani Saudagar\",\"given\":\"Abdul Khader\"},{\"family\":\"Alkhathami\",\"given\":\"Mohammed\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Higher security and fault tolerance: Decentralised federated learning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fault tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et al., 2019; Tricomi et al., 2024; Urblik et al., 2023; Y. Wang et al., 2020; J. Wu et al., 2014; Yaqoob et al., 2023)",
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ning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fa
ult tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calc
ulating and transferring data that, specifically, on the basis of massive foundation models will demand the
 new mechanisms of resource management and offloading of tasks. These techniques play a very important role
 in the trade-off between cost of communication, and the cost of computation locally, which turns crucial i
n the instance of fine-tuning at scale of large models when prediction is required to predict applications 
such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions o
f neural networks, which can successfully be implemented on portable devices, and which can attract fewer e
nergy, and less resources. Further reduction of the model footprint and the computing needs could also be a
ccomplished by pruning and adaptive quantisation - federated learning can be made available even on a broad
er range of mobile devices. It is also hectic that engineers and designers face in the implementation of th
e foundation models of these such systems especially the real-time performance and scalability in the indus
trial applications. The computation amounts to and communication efficiency maximisation in federated learn
ing systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often
 involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in t
raining and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al.,
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ning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fa
ult tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calc
ulating and transferring data that, specifically, on the basis of massive foundation models will demand the
 new mechanisms of resource management and offloading of tasks. These techniques play a very important role
 in the trade-off between cost of communication, and the cost of computation locally, which turns crucial i
n the instance of fine-tuning at scale of large models when prediction is required to predict applications
such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions o
f neural networks, which can successfully be implemented on portable devices, and which can attract fewer e
nergy, and less resources. Further reduction of the model footprint and the computing needs could also be a
ccomplished by pruning and adaptive quantisation - federated learning can be made available even on a broad
er range of mobile devices. It is also hectic that engineers and designers face in the implementation of th
e foundation models of these such systems especially the real-time performance and scalability in the indus
trial applications. The computation amounts to and communication efficiency maximisation in federated learn
ing systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often
 involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in t
raining and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al.,
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Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et al., 2019; Tricomi et al., 2024; Urblik et al., 2023; Y. 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CPU\",\"author\":[{\"family\":\"Tao\",\"given\":\"Yunzheng\"},{\"family\":\"You\",\"given\":\"Changsheng\"},{\"family\":\"Zhang\",\"given\":\"Ping\"},{\"family\":\"Huang\",\"given\":\"Kaibin\"}],\"issued\":{\"date-parts\":[[\"2019\"]]}}},{\"id\":994,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/VBC8IY7D\"],\"itemData\":{\"id\":994,\"type\":\"article-journal\",\"container-title\":\"Sensors\",\"DOI\":\"10.3390/s24103108\",\"title\":\"Smart City as Cooperating Smart Areas: On the Way of Symbiotic Cyber–Physical Systems 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Cloud\",\"author\":[{\"family\":\"Wu\",\"given\":\"Jiagao\"},{\"family\":\"Yuan\",\"given\":\"Hang\"},{\"family\":\"He\",\"given\":\"Ying\"},{\"family\":\"Zou\",\"given\":\"Zhiqiang\"}],\"issued\":{\"date-parts\":[[\"2014\"]]}}},{\"id\":990,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/X3WJYRKM\"],\"itemData\":{\"id\":990,\"type\":\"article-journal\",\"container-title\":\"Diagnostics\",\"DOI\":\"10.3390/diagnostics13111964\",\"title\":\"Federated Machine Learning for Skin Lesion Diagnosis: An Asynchronous and Weighted Approach\",\"author\":[{\"family\":\"Yaqoob\",\"given\":\"Muhammad Mateen\"},{\"family\":\"Alsulami\",\"given\":\"Musleh\"},{\"family\":\"Khan\",\"given\":\"Muhammad Amir\"},{\"family\":\"Alsadie\",\"given\":\"Deafallah\"},{\"family\":\"Jilani Saudagar\",\"given\":\"Abdul Khader\"},{\"family\":\"Alkhathami\",\"given\":\"Mohammed\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Higher security and fault tolerance: Decentralised federated learning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fault tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et al., 2019; Tricomi et al., 2024; Urblik et al., 2023; Y. 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Khader\"},{\"family\":\"Alkhathami\",\"given\":\"Mohammed\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Higher security and fault tolerance: Decentralised federated learning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fault tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et a

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ning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fa

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ult tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calc
ulating and transferring data that, specifically, on the basis of massive foundation models will demand the
 new mechanisms of resource management and offloading of tasks. These techniques play a very important role
 in the trade-off between cost of communication, and the cost of computation locally, which turns crucial i
n the instance of fine-tuning at scale of large models when prediction is required to predict applications 
such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions o
f neural networks, which can successfully be implemented on portable devices, and which can attract fewer e
nergy, and less resources. Further reduction of the model footprint and the computing needs could also be a
ccomplished by pruning and adaptive quantisation - federated learning can be made available even on a broad
er range of mobile devices. It is also hectic that engineers and designers face in the implementation of th
e foundation models of these such systems especially the real-time performance and scalability in the indus
trial applications. The computation amounts to and communication efficiency maximisation in federated learn
ing systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often
 involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in t
raining and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al.,
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Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et al., 2019; Tricomi et al., 2024; Urblik et al., 2023; Y. 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nguage Understanding, Generation, and Reasoning. As These Models Evolve, Context Awareness Has Become a Crucial Factor in Improving Their Adaptability, Coherence, and Decision-Making Capabilities. This Paper Explores the Evolution of LLMs From Rule-Based Systems to Deep Learning Architectures, Emphasizing Advancements in Contextu\",\"author\":[{\"family\":\"Ali\",\"given\":\"Arshad\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1000,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/D92XF6L3\"],\"itemData\":{\"id\":1000,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics12092068\",\"title\":\"Blockchain-Based Trusted Federated Learning With Pre-Trained Models for COVID-19 Detection\",\"author\":[{\"family\":\"Bian\",\"given\":\"Genqing\"},{\"family\":\"Qu\",\"given\":\"Wenjing\"},{\"family\":\"Shao\",\"given\":\"Bilin\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":997,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/EEGZTN47\"],\"itemData\":{\"id\":997,\"type\":\"article-journal\",\"DOI\":\"10.1145/1048935.1050192\",\"title\":\"A Self-Organizing Flock of 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ning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fa
ult tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calc
ulating and transferring data that, specifically, on the basis of massive foundation models will demand the
 new mechanisms of resource management and offloading of tasks. These techniques play a very important role
 in the trade-off between cost of communication, and the cost of computation locally, which turns crucial i
n the instance of fine-tuning at scale of large models when prediction is required to predict applications
such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions o
f neural networks, which can successfully be implemented on portable devices, and which can attract fewer e
nergy, and less resources. Further reduction of the model footprint and the computing needs could also be a
ccomplished by pruning and adaptive quantisation - federated learning can be made available even on a broad
er range of mobile devices. It is also hectic that engineers and designers face in the implementation of th
e foundation models of these such systems especially the real-time performance and scalability in the indus
trial applications. The computation amounts to and communication efficiency maximisation in federated learn
ing systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often
 involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in t
raining and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al.,
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Khader\"},{\"family\":\"Alkhathami\",\"given\":\"Mohammed\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Higher security and fault tolerance: Decentralised federated learning is a peer-to-peer communication framework that can be enhanced with regard to superior security and fault tolerance. Meanwhile, the cross-device federated learning structure will unlock the bottlenecks of calculating and transferring data that, specifically, on the basis of massive foundation models will demand the new mechanisms of resource management and offloading of tasks. These techniques play a very important role in the trade-off between cost of communication, and the cost of computation locally, which turns crucial in the instance of fine-tuning at scale of large models when prediction is required to predict applications such as movie recommender. To address these issues, we need to develop lean and mobile optimised versions of neural networks, which can successfully be implemented on portable devices, and which can attract fewer energy, and less resources. Further reduction of the model footprint and the computing needs could also be accomplished by pruning and adaptive quantisation - federated learning can be made available even on a broader range of mobile devices. It is also hectic that engineers and designers face in the implementation of the foundation models of these such systems especially the real-time performance and scalability in the industrial applications. The computation amounts to and communication efficiency maximisation in federated learning systems, in more detail of complex multi-modal models such as GPT-4o is an excellent problem, and often involves distributed strategies of computation to manage the colossal resource consumption, intrinsic in training and fine-tuning such models over time with massive compute clusters. (Abbaszadi, 2025; Akon et al., 2008; A. Ali, 2025; Bian et al., 2023; Butt et al., 2003; L. Chen, 2025; J. Cheng et al., 2025; Chinnaraju, 2024; Fu \u0026 Wang, 2009; Kumar et al., 2020; Z. Li et al., 2022; Ma et al., 2014; Maher \u0026 Nasr, 2

021; Matsubara, 2020; Panisson et al., 2006; Park, 2025; Shumba et al., 2023; C. Sun et al., 2023; Tao et al., 2019; Tricomi et al., 2024; Urblik et al., 2023; Y. Wang et al., 2020; J. Wu et al., 2014; Yaqoob et al., 2023)",
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/csl-citation.json\"} Personalisation, in the event of federated learning, particularly by the recommender systems, must be done in a manner that will not jeopardize privacy of the users. This is not simply encryption procedure of the data, and aims at sophisticated mechanisms. One of the techniques studied nowadays is called an advanced privacy or secure multiparty computation, to make sure that the data of the user remain secure, not only during the training of the models, but also during inference. Homomorphic encryption and zero-knowledge overwhelming, e.g., are schemes of ensuring that there exist efficient structures of maintaining privacy without exposing the utility of aggregated updates in a model. Providing personalised federated learning methods would allow creating strong and understandable user representations and ensuring that privacy remains the priority. It is especially relevant since it views the issues in relation to sparse data and statistical difference between the various individuals on which it is hard to conclusively ascertain about the finesse of the users in the proper sense. Also, since user preferences may shift with time, there is always the necessity of models changing with of a changing data distribution. What cannot be defined as being thus deemed to fall under the category of “personality leakage” and what can be done to be held to proper measures on the categorizing of becoming less important is one of the greatest problems that remain unresolved fully, yet (sec. 4). So, it is extremely important to develop improved objective functions and optimisation of strategies with the stated goal of preserving personality during the federation of foundation models refinement. It is also vital to generate mechanisms of the adequate quantification of such trade-offs between the accuracy of personalisation, and the risk of leakage of personality, especially of multidimensional and intricate representations of a consumer. Another issue that is of the particular importance is that advanced models will extract sensitive user traits like personality off of seemingly innocent user-interactions and that is the ethical issue of the misuse of the data and bias of algorithms. To meet these considerations, it is necessary that there exist mechanisms that allow transparent auditing, personal governance such that inferred user features are treated in an irresponsible manner. Moreover, federated foundation models would have to overcome current distribution jumps to achieve an ongoing execution, as traditional risks might lack the ability to address distribution changes, as it must being able to educate all of the parameters and is delicate to variations in the user information. (Z. Guo et al., 2024; Priyadarsini \u0026 Raja, 2024; Rokade, 2024c; C. Shen et al., 2024; W.-H. Shen et al., 2024; W. Wang, 2024b; Wassan et al., 2025c; Woubie et al., 2024; X. Yang \u0026 Xing, 2024b; L. Zhang et al., 2024; X. Zhang, Zhou, et al., 2025; Q. Zhou, 2024b)",
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This is not simply encryption procedure of the data, and aims at sophisticated mechanisms. One of the techniques studied nowadays is called an advanced privacy or secure multiparty computation, to make sure that the data of the user remain secure, not only during the training of the models, but also during inference. Homomorphic encryption and zero-knowledge overwhelming, e.g., are schemes of ensuring that there exist efficient structures of maintaining privacy without exposing the utility of aggregated updates in a model. Providing personalised federated learning methods would allow creating strong and understandable user representations and ensuring that privacy remains the priority. It is especially relevant since it views the issues in relation to sparse data and statistical difference between the various individuals on which it is hard to conclusively ascertain about the finesse of the users in the proper sense. Also, since user preferences may shift with time, there is always the necessity of models changing with of a changing data distribution. What cannot be defined as being thus deemed to fall under the category of “personality leakage” and what can be done to be held to proper measures on the categorizing of becoming less important is one of the greatest problems that remain unresolved fully, yet (sec. 4). So, it is extremely important to develop improved objective functions and optimisation of strategies with the stated goal of preserving personality during the federation of foundation models refinement. It is also vital to generate mechanisms of the adequate quantification of such trade-offs between the accuracy of personalisation, and the risk of leakage of personality, especially of multidimensional and intricate representations of a consumer. Another issue that is of the particular importance is that advanced models will extract sensitive user traits like personality off of seemingly innocent user-interactions and that is the ethical issue of the misuse of the data and bias of algorithms. To meet these considerations, it is necessary that there exist mechanisms that allow transparent auditing, personal governance such that inferred user features are treated in an irresponsible manner. Moreover, federated foundation models would have to overcome current distribution jumps to achieve an ongoing execution, as traditional risks might lack the ability to address distribution changes, as it must being able to educate all of the parameters and is delicate to variations in the user information. (Z. Guo et al., 2024; Priyadarsini \u0026 Raja, 2024; Rokade, 2024c; C. Shen et al., 2024; W.-H. Shen et al., 2024; W. Wang, 2024b; Wassan et al., 2025c; Woubie et al., 2024; X. Yang \u0026 Xing, 2024b; L. Zhang et al., 2024; X. 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w/master/csl-citation.json\"} More so, the current study of federated unlearning methods also can potential
ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation 
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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C.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1052,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/CPINNTF9\"],\"itemData\":{\"id\":1052,\"type\":\"article-journal\",\"container-title\":\"Ieee Access\",\"DOI\":\"10.1109/access.2022.3197671\",\"title\":\"Explainable AI for Healthcare 5.0: Opportunities and Challenges\",\"author\":[{\"family\":\"Saraswat\",\"given\":\"Deepti\"},{\"family\":\"Bhattacharya\",\"given\":\"Pronaya\"},{\"family\":\"Verma\",\"given\":\"Ashwin\"},{\"family\":\"Prasad\",\"given\":\"Vivek Kumar\"},{\"family\":\"Tanwar\",\"given\":\"Sudeep\"},{\"family\":\"Sharma\",\"given\":\"Gulshan\"},{\"family\":\"Bokoro\",\"given\":\"Pitshou N.\"},{\"family\":\"Sharma\",\"given\":\"Ravi\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":1035,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/N5RPASSC\"],\"itemData\":{\"id\":1035,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2511.00406\",\"title\":\"Quantum Machine Unlearning: Foundations, Mechanisms, and Taxonomy\",\"author\":[{\"family\":\"Shaik\",\"given\":\"Thanveer\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1047,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/2Q48GFX4\"],\"itemData\":{\"id\":1047,\"type\":\"article-journal\",\"container-title\":\"World Journal of Advanced Research and Reviews\",\"DOI\":\"10.30574/wjarr.2025.27.1.2541\",\"title\":\"Enhancing Malware Detection Using Federated Learning and Explainable AI for Privacy-Preserving Threat Intelligence\",\"author\":[{\"family\":\"Shallom\",\"given\":\"Kigbu\"},{\"family\":\"Ikemefuna\",\"given\":\"Chukwujekwu Damian\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1039,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/DXVG6K68\"],\"itemData\":{\"id\":1039,\"type\":\"article-journal\",\"container-title\":\"Security and Privacy\",\"DOI\":\"10.1002/spy2.70094\",\"title\":\"A Comprehensive Review of Integrating \u003cscp\u003eAI\u003c/Scp\u003e and Blockchain Security: Innovations, Challenges, and Future Directions\",\"author\":[{\"family\":\"Zebari\",\"given\":\"Gheyath Mustafa\"},{\"family\":\"Musalhi\",\"given\":\"Nasser Al\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1040,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5AEPQT2K\"],\"itemData\":{\"id\":1040,\"type\":\"article-journal\",\"container-title\":\"Revista Ibero-Americana De Humanidades Ciências E Educação\",\"DOI\":\"10.51891/rease.v10i8.15346\",\"title\":\"Harnessing the Power of Ai and Machine Learning for Next-Generation Sequencing Data Analysis: A Comprehensive Review of Applications, Challenges, and Future Directions in Precision Oncology\",\"author\":[{\"family\":\"Zeraik Viduedo\",\"given\":\"Pedro Henrique\"},{\"family\":\"Urquiza Candeia\",\"given\":\"Victor César\"},{\"family\":\"Legname Martins\",\"given\":\"Victor Balceiro\"},{\"family\":\"Destefani\",\"given\":\"Afrânio Côgo\"},{\"family\":\"Destefani\",\"given\":\"Vinícius Côgo\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} More so, the current study of federated unlearning methods also can potentially promise the existence of dynamic elimination of sensitive data of shared models after training and, therefore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make certain that federated learning systems that act as recommendation engines are adaptive to evolving user requirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic

h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary solutions that encompass advances in machine learning, cyber security and responsible AI are necessary to develop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; Alshammari et al., 2024; Aswani, 2025; Bieniek et al., 2025; Dhanawat et al., 2025; Divya, 2025; Garg et al., 2024; Keerthana et al., 2025; Kejriwal, Goel, et al., 2023; Kejriwal, Pujari, et al., 2023; Komalasari, 2024; Leelavathi, 2025; Nassar \u0026 Al-Mashagba, 2025; Pandey, 2025; Pasupuleti, 2024; C. Peng et al., 2025; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shaik, 2025; Shallom \u0026 Ikemefuna, 2025; Zebari \u0026 Musalhi, 2025; Zeraik Viduedo et al., 2024)",
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Preserving Anomaly Detection Using Federated Learning and Explainable AI\",\"author\":[{\"family\":\"Divya\",\"given\":\"R.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1057,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/CPV4QXSQ\"],\"itemData\":{\"id\":1057,\"type\":\"article-journal\",\"DOI\":\"10.4018/979-8-3693-5718-7.ch013\",\"title\":\"Explainable AI, Federated Learning, and AI Ethics in E-Commerce\",\"author\":[{\"family\":\"Garg\",\"given\":\"Gunjan\"},{\"family\":\"Prabha\",\"given\":\"Chander\"},{\"family\":\"Ahuja\",\"given\":\"Bhavisha\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":1048,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/B5H3ZNFH\"],\"itemData\":{\"id\":1048,\"type\":\"article-journal\",\"DOI\":\"10.4018/979-8-3693-9725-1.ch004\",\"title\":\"AI Technologies for Predicting Susceptibility and Outcomes of Immunological Disorders\",\"author\":[{\"family\":\"Keerthana\",\"given\":\"R.\"},{\"family\":\"Savitha\",\"given\":\"S.\"},{\"family\":\"Logeswaran\",\"given\":\"K.\"},{\"family\":\"Rajivkannan\",\"given\":\"A.\"},{\"family\":\"Namasivayam\",\"given\":\"M.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1058,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/I85LHUMJ\"],\"itemData\":{\"id\":1058,\"type\":\"article-journal\",\"container-title\":\"Iosr Journal of Computer Engineering\",\"DOI\":\"10.9790/0661-2506033243\",\"title\":\"Adversarial AI in Federated Learning: Threats, Robust Defenses, and the Role of Explain Ability for Trustworthy Distributed AI\",\"author\":[{\"family\":\"Kejriwal\",\"given\":\"Deepak\"},{\"family\":\"Goel\",\"given\":\"Anshul\"},{\"family\":\"Pujari\",\"given\":\"Tejakumar Dattatray\"},{\"family\":\"Pakina\",\"given\":\"Anil Kumar\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1059,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/PRQDTR9K\"],\"itemData\":{\"id\":1059,\"type\":\"article-journal\",\"container-title\":\"International Journal Science and Technology\",\"DOI\":\"10.56127/ijst.v2i3.1960\",\"title\":\"Adversarial AI in Federated Learning: Threats, Robust Defenses, and the Role of Explainability for Trustworthy Distributed AI\",\"author\":[{\"family\":\"Kejriwal\",\"given\":\"Deepak\"},{\"family\":\"Pujari\",\"given\":\"Tejaskumar\"},{\"family\":\"Goel\",\"given\":\"Anshul\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1046,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/BFKWZ9L9\"],\"itemData\":{\"id\":1046,\"type\":\"article-journal\",\"DOI\":\"10.4018/979-8-3693-1874-4.ch018\",\"title\":\"Secure and Privacy-Preserving Federated Learning With Explainable Artificial Intelligence for Smart Healt

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cy-Preserving Threat Intelligence\",\"author\":[{\"family\":\"Shallom\",\"given\":\"Kigbu\"},{\"family\":\"Ikemefuna\",\"given\":\"Chukwujekwu Damian\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1039,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/DXVG6K68\"],\"itemData\":{\"id\":1039,\"type\":\"article-journal\",\"container-title\":\"Security and Privacy\",\"DOI\":\"10.1002/spy2.70094\",\"title\":\"A Comprehensive Review of Integrating \u003cscp\u003eAI\u003c/Scp\u003e and Blockchain Security: Innovations, Challenges, and Future Directions\",\"author\":[{\"family\":\"Zebari\",\"given\":\"Gheyath Mustafa\"},{\"family\":\"Musalhi\",\"given\":\"Nasser Al\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1040,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5AEPQT2K\"],\"itemData\":{\"id\":1040,\"type\":\"article-journal\",\"container-title\":\"Revista Ibero-Americana De Humanidades Ciências E Educação\",\"DOI\":\"10.51891/rease.v10i8.15346\",\"title\":\"Harnessing the Power of Ai and Machine Learning for Next-Generation Sequencing Data Analysis: A Comprehensive Review of Applications, Challenges, and Future Directions in Precision Oncology\",\"author\":[{\"family\":\"Zeraik Viduedo\",\"given\":\"Pedro Henrique\"},{\"family\":\"Urquiza Candeia\",\"given\":\"Victor César\"},{\"family\":\"Legname Martins\",\"given\":\"Victor Balceiro\"},{\"family\":\"Destefani\",\"given\":\"Afrânio Côgo\"},{\"family\":\"Destefani\",\"given\":\"Vinícius Côgo\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} More so, the current study of federated unlearning methods also can potentially promise the existence of dynamic elimination of sensitive data of shared models after training and, therefore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make certain that federated learning systems that act as recommendation engines are adaptive to evolving user requirements, not simply, but also engaged in averting rendering sensitive personality characteristics to which exposing sensitive personality characteristics have increasingly been of a growing concern as foundation models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary solutions that encompass advances in machine learning, cyber security and responsible AI are necessary to develop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; Alshammari et al., 2024; Aswani, 2025; Bieniek et al., 2025; Dhanawat et al., 2025; Divya, 2025; Garg et al., 2024; Keerthana et al., 2025; Kejriwal, Goel, et al., 2023; Kejriwal, Pujari, et al., 2023; Komalasari, 2024; Leelavathi, 2025; Nassar \u0026 Al-Mashagba, 2025; Pandey, 2025; Pasupuleti, 2024; C. Peng et al., 2025; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shaik, 2025; Shallom \u0026 Ikemefuna, 2025; Zebari \u0026 Musalhi, 2025; Zeraik Viduedo et al., 2024)",
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ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation 
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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5; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shai
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ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation 
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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w/master/csl-citation.json\"} More so, the current study of federated unlearning methods also can potential
ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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, 2024; Keerthana et al., 2025; Kejriwal, Goel, et al., 2023; Kejriwal, Pujari, et al., 2023; Komalasari, 2
024; Leelavathi, 2025; Nassar \u0026 Al-Mashagba, 2025; Pandey, 2025; Pasupuleti, 2024; C. Peng et al., 202
5; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shai
k, 2025; Shallom \u0026 Ikemefuna, 2025; Zebari \u0026 Musalhi, 2025; Zeraik Viduedo et al., 2024)",
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ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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These innovations are also needed to make certain that federated learning systems that act as recommendation engines are adaptive to evolving user requirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic

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h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
lshammari et al., 2024; Aswani, 2025; Bieniek et al., 2025; Dhanawat et al., 2025; Divya, 2025; Garg et al.
, 2024; Keerthana et al., 2025; Kejriwal, Goel, et al., 2023; Kejriwal, Pujari, et al., 2023; Komalasari, 2
024; Leelavathi, 2025; Nassar \u0026 Al-Mashagba, 2025; Pandey, 2025; Pasupuleti, 2024; C. Peng et al., 202
5; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shai
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w/master/csl-citation.json\"} More so, the current study of federated unlearning methods also can potential ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A lshammari et al., 2024; Aswani, 2025; Bieniek et al., 2025; Dhanawat et al., 2025; Divya, 2025; Garg et al. , 2024; Keerthana et al., 2025; Kejriwal, Goel, et al., 2023; Kejriwal, Pujari, et al., 2023; Komalasari, 2 024; Leelavathi, 2025; Nassar \u0026 Al-Mashagba, 2025; Pandey, 2025; Pasupuleti, 2024; C. Peng et al., 202 5; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shai k, 2025; Shallom \u0026 Ikemefuna, 2025; Zebari \u0026 Musalhi, 2025; Zeraik Viduedo et al., 2024)",
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ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation 
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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w/master/csl-citation.json\"} More so, the current study of federated unlearning methods also can potential
ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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024; Leelavathi, 2025; Nassar \u0026 Al-Mashagba, 2025; Pandey, 2025; Pasupuleti, 2024; C. Peng et al., 202
5; A. Rahman et al., 2024; Raj, 2025a; Rane et al., 2024; Sandhya et al., 2025; Saraswat et al., 2022; Shai
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ly promise the existence of dynamic elimination of sensitive data of shared models after training and, ther
efore, provide upper-notch and versatile protection of privacy. These innovations are also needed to make c
ertain that federated learning systems that act as recommendation engines are adaptive to evolving user req
uirements, not simply, but also engaged in averting rendering sensitive personality characteristics to whic
h exposing sensitive personality characteristics have increasingly been of a growing concern as foundation
models have become increasingly prolific. It is a multi-layered and dynamic issue, and interdisciplinary so
lutions that encompass advances in machine learning, cyber security and responsible AI are necessary to dev
elop certified and compliant systems that are ready to adapt to new regulations. (Abimannan et al., 2023; A
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hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
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pecifically targeting large language models, to ensure that federated foundation models are secure and resi
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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DOI\":\"10.48550/arxiv.2012.01791\",\"title\":\"FAT: Federated Adversarial Training\",\"author\":[{\"family\":\"Zizzo\",\"given\":\"Giulio\"},{\"family\":\"Rawat\",\"given\":\"Ambrish\"},{\"family\":\"Sinn\",\"given\":\"Mathieu\"},{\"family\":\"Buesser\",\"given\":\"Beat\"}],\"issued\":{\"date-parts\":[[\"2020\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} A specific relevance of this adaptive approach is specifically considering the fact that representatives of the opposite side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inference and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-enhancing technologies. It is thus essential that in future studies, efforts be made to strictly test federated foundation models concerning the emerging threats, such as evasion methods that have the capability of specifically targeting large language models, to ensure that federated foundation models are secure and resilient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich \u0026 Eshete, 2021; Aparna et al., 2025; Baji et al., 2024; Bondok et al., 2023; Borah, 2025; Briouya et al., 2025b; Calzavara et al., 2020; Chernikova \u0026 Oprea, 2022; Cools, 2025; Dziubi■ski et al., 2016; Giri et al., 2024; A. K. Gupta et al., 2024; Koike et al., 2024; Natarajan, 2025a; Pestana \u0026 Sofou, 2024; M. A. Rahman et al., 2021; Sekti, 2025; Srikanth Vutukuru, 2024; Teryak et al., 2023; Umejiaku \u0026 Sheng, 2024; X. Zhang et al., 2024; Zizzo et al., 2020)",
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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It is thus essential that in future studies, efforts be made to strictly test federated foundation models concerning the emerging threats, such as evasion methods that have the capability of specifically targeting large language models, to ensure that federated foundation models are secure and resilient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich \u0026 Eshete, 2021; Aparna et al., 2025; Baji et al., 2024; Bondok et al., 2023; Borah, 2025; Briouya et al., 2025b; Calzavara et al., 2020; Chernikova \u0026 Oprea, 2022; Cools, 2025; Dziubi■ski et al., 2016; Giri et al., 2024; A. K. Gupta et al., 2024; Koike et al., 2024; Natarajan, 2025a; Pestana \u0026 Sofou, 2024; M. A. Rahman et al., 2021; Sekti, 2025; Srikanth Vutukuru, 2024; Teryak et al., 2023; Umejiaku \u0026 Sheng, 2024; X. Zhang et al., 2024; Zizzo et al., 2020)",
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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Training\",\"author\":[{\"family\":\"Zizzo\",\"given\":\"Giulio\"},{\"family\":\"Rawat\",\"given\":\"Ambrish\"},{\"family\":\"Sinn\",\"given\":\"Mathieu\"},{\"family\":\"Buesser\",\"given\":\"Beat\"}],\"issued\":{\"date-parts\":[[\"2020\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} A specific relevance of this adaptive approach is specifically considering the fact that representatives of the opposite side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inference and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-enhancing technologies. It is thus essential that in future studies, efforts be made to strictly test federated foundation models concerning the emerging threats, such as evasion methods that have the capability of specifically targeting large language models, to ensure that federated foundation models are secure and resilient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich \u0026 Eshete, 2021; Aparna et al., 2025; Baji et al., 2024; Bondok et al., 2023; Borah, 2025; Briouya et al., 2025b; Calzavara et al., 2020; Chernikova \u0026 Oprea, 2022; Cools, 2025; Dziubi■ski et al., 2016; Giri et al., 2024; A. K. Gupta et al., 2024; Koike et al., 2024; Natarajan, 2025a; Pestana \u0026 Sofou, 2024; M. A. Rahman et al., 2021; Sekti, 2025; Srikanth Vutukuru, 2024; Teryak et al., 2023; Umejiaku \u0026 Sheng, 2024; X. Zhang et al., 2024; Zizzo et al., 2020)",
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021; Sekti, 2025; Srikanth Vutukuru, 2024; Teryak et al., 2023; Umejiaku \u0026 Sheng, 2024; X. 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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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levance of this adaptive approach is specifically considering the fact that representatives of the opposite
 side may capitalize on weaknesses in order to make an esoteric set of attacks, such as membership inferenc
e and gradient leaks, and in the process, to provoke the shortcomings of the currently available privacy-en
hancing technologies. It is thus essential that in future studies, efforts be made to strictly test federat
ed foundation models concerning the emerging threats, such as evasion methods that have the capability of s
pecifically targeting large language models, to ensure that federated foundation models are secure and resi
lient in practical applications. (S. Ali et al., 2025; Aljanabi et al., 2024; Alruwaili et al., 2024; Amich
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and complex models. On the same note, the privacy offered with such technologies as secure multiparty compu
tation and homomorphic encryption is excellent and yet they are computationally intensive to an extent of n
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e most essential problem when developing the effective federated learning system in order to suggest the mo
vies during mobile devices use is to find the right balance between privacy to the users and the model unde
r question. Although federated learning offers the protection of privacy inherent in that the method does n
ot store the raw data in a remote location, this does not mean that the approach is not vulnerable to newer
 attacks, including model inversion and data poisoning. There is the need as such to incorporate more priva
cy-saving techniques. However, these solutions have the propensity of introducing their own form of trade-o
ffs such as worsened performance of the models or increased computer hardware requirements and thereby comp
ounding the trade-offs of the accuracy and privacy even more. An example is differentiational privacy, whic
h provides quantifiable privacy guarantees but with converse benefits of accuracy particularly in low-data
and complex models. On the same note, the privacy offered with such technologies as secure multiparty compu
tation and homomorphic encryption is excellent and yet they are computationally intensive to an extent of n
ot being applicable to real-time mobile recommendation systems. (Natarajan, 2025b; H. Zhang, Xie, et al., 2
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n.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is hig
hly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as
high as possible. However, because of the trade off When too much noise is added to hide the data, the inac
curacy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so th
at the recommendations become useless. Now, second problem is that if the clinical data and the recommendat
ions of the clinical data are to be displayed on mobile system, because the mobile system are computational
ly weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even dif
ficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due
 to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we em
ploy, especially on mobile recommendation systems along with other platforms like your hosting server etc.
has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mo
bile phone, it is different from the one that is being recommended on a hosting website. This is because th
e nature of threats and attacks keep changing from time to time. Similar is the case with data distribution
. For example, in the field of medical science, new diseases keep coming in. New devices are launched that
may have completely different hardware and computational power, so that the privacy mechanism that worked o
n previous versions of the device may prove null and void in its purpose when employed in this newly launch
ed device. This situation may put us into a situation where we are fighting with old disease with old techn
ologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapid
ly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the
 problem that comes is to find a balance between protecting the data using privacy mechanism and the accura
cy which the recommendation system as a consequence shows to the users in their device. The problem gets bi
gger and severe in more complex machine learning models like speech recognition and medical image analysis,
 where this balance between privacy and accuracy, if biased towards privacy, will result in even severely i
naccurate recommended contents. For example, in speech recognition, if we add more noise to train the syste
m with strong privacy, this extra noise when sent out to work in real environment where user will speak wo
rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted b
y the system. This will make the system useless for daily communication and usage. Similarly, if we apply a
 similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection),
The final result when employee to real life medical applications may cause the model to miss critical featu
res in X ray or CT scan or completely false positive reports. And even worse is that missing out on feature
s in the images which point towards underlying disease may result in false negatives and the patient may be
 advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Ha
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l., 2024; T. Li, 2019; M. Liang, 2025; Mahmood et al., 2025; Osamika et al., 2024; Patel, 2025; Paulson \u0
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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1128,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5DZ7WEEX\"],\"itemData\":{\"id\":1128,\"type\":\"article-journal\",\"DOI\":\"10.71443/9789349552036-13\",\"title\":\"Federated Learning and Data Privacy in Connected Healthcare Devices\",\"author\":[{\"family\":\"Thanikasalam\",\"given\":\"A\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}}},{\"id\":1124,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7VUDUDUG\"],\"itemData\":{\"id\":1124,\"type\":\"article-journal\",\"container-title\":\"Itm Web of Conferences\",\"DOI\":\"10.1051/itmconf/20257804003\",\"title\":\"Privacy Protection Optimization in Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Xinyi\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1129,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NZLIGNY2\"],\"itemData\":{\"id\":1129,\"type\":\"article-journal\",\"container-title\":\"Open Information and Computer Integrated Technologies\",\"DOI\":\"10.32620/oikit.2025.105.19\",\"title\":\"■■■■■■■ ■■■■■■■■■■■ ■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■ ■■■■■■■■■ «■■■■■■■■■ ■■■■■■■■»: ■■■■■■■■■■■■■, ■■■■■■■■ ■■ ■■■■■■■■■■■■\",\"author\":[{\"family\":\"■■■■■■■■■■\",\"given\":\"■■■■■■ ■■■■■■■■■■■■■■\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Hamilton, 2025; B. T. Hasan, 2025; N. 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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1128,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5DZ7WEEX\"],\"itemData\":{\"id\":1128,\"type\":\"article-journal\",\"DOI\":\"10.71443/9789349552036-13\",\"title\":\"Federated Learning and Data Privacy in Connected Healthcare Devices\",\"author\":[{\"family\":\"Thanikasalam\",\"given\":\"A\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}}},{\"id\":1124,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7VUDUDUG\"],\"itemData\":{\"id\":1124,\"type\":\"article-journal\",\"container-title\":\"Itm Web of Conferences\",\"DOI\":\"10.1051/itmconf/20257804003\",\"title\":\"Privacy Protection Optimization in Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Xinyi\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1129,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NZLIGNY2\"],\"itemData\":{\"id\":1129,\"type\":\"article-journal\",\"container-title\":\"Open Information and Computer Integrated Technologies\",\"DOI\":\"10.32620/oikit.2025.105.19\",\"title\":\"■■■■■■■ ■■■■■■■■■■ ■■■■■■■■ ■ ■■■■■■■■■■■■■■■■ ■■■■■■■■■ «■■■■■■■■■ ■■■■■■■■»: ■■■■■■■■■■■■■■, ■■■■■■■■■ ■■ ■■■■■■■■■■■■■\",\"author\":[{\"family\":\"■■■■■■■■■■\",\"given\":\"■■■■■■ ■■■■■■■■■■■■■■\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak wo

rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Hamilton, 2025; B. T. Hasan, 2025; N. Hasan et al., 2022; Kalejaiye et al., 2025; Z. Khan, 2025; Kvochko et al., 2024; T. Li, 2019; M. Liang, 2025; Mahmood et al., 2025; Osamika et al., 2024; Patel, 2025; Paulson \u0026 Elvis, 2025; Pittala, 2024; Raj, 2025b; Reddy, 2025; Shahsavari et al., 2025; Shakeer \u0026 Babu, 2025; Siva \u0026 Anbarasi, 2025; Thanikasalam, 2026; X. Zhou, 2025; [redacted], 2025)",
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\"itemData\":{\"id\":1129,\"type\":\"article-journal\",\"container-title\":\"Open Information and Computer Integrated Technologies\",\"DOI\":\"10.32620/oikit.2025.105.19\",\"title\":\"[redacted]\
",\"author\":[{\"family\":\"[redacted]\",\"given\":\"[redacted]\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Hamilton, 2025; B. T. Hasan, 2025; N. Hasan et al., 2022; Kalejaiye et al., 2025; Z. Khan, 2025; Kvochko et al., 2024; T. Li, 2019; M. Liang, 2025; Mahmood et al., 2025; Osamika et al., 2024; Patel, 2025; Paulson \u0026 Elvis, 2025; Pittala, 2024; Raj, 2025b; Reddy, 2025; Shahsavari et al., 2025; Shakeer \u0026 Babu, 2025; Siva \u0026 Anbarasi, 2025; Thanikasalam, 2026; X. Zhou, 2025; [redacted], 2025)",
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n.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is hig
hly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as
high as possible. However, because of the trade off When too much noise is added to hide the data, the inac
curacy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so th
at the recommendations become useless. Now, second problem is that if the clinical data and the recommendat
ions of the clinical data are to be displayed on mobile system, because the mobile system are computational
ly weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even dif
ficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due
 to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we em
ploy, especially on mobile recommendation systems along with other platforms like your hosting server etc.
has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mo
bile phone, it is different from the one that is being recommended on a hosting website. This is because th
e nature of threats and attacks keep changing from time to time. Similar is the case with data distribution
. For example, in the field of medical science, new diseases keep coming in. New devices are launched that
may have completely different hardware and computational power, so that the privacy mechanism that worked o
n previous versions of the device may prove null and void in its purpose when employed in this newly launch
ed device. This situation may put us into a situation where we are fighting with old disease with old techn
ologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapid
ly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the
 problem that comes is to find a balance between protecting the data using privacy mechanism and the accura
cy which the recommendation system as a consequence shows to the users in their device. The problem gets bi
gger and severe in more complex machine learning models like speech recognition and medical image analysis,
 where this balance between privacy and accuracy, if biased towards privacy, will result in even severely i
naccurate recommended contents. For example, in speech recognition, if we add more noise to train the syste
m with strong  privacy, this extra noise when sent out to work in real environment where user will speak wo
rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted b
y the system. This will make the system useless for daily communication and usage. Similarly, if we apply a
 similar noisy machine learning model (i.e. A model trend on too much noisy data for  privacy protection),
The final result when employee to real life medical applications may cause the model to miss critical featu
res in X ray or CT scan or completely false positive reports. And even worse is that missing out on feature
s in the images which point towards underlying disease may result in false negatives and the patient may be
 advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Ha
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",\"author\":[{\"family\":\"[redacted]\",\"given\":\"[redacted]\"}],\"issued\":{\"date-par
ts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citatio
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n.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so th

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at the recommendations become useless. Now, second problem is that if the clinical data and the recommendat
ions of the clinical data are to be displayed on mobile system, because the mobile system are computational
ly weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even dif
ficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due
 to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we em
ploy, especially on mobile recommendation systems along with other platforms like your hosting server etc.
has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mo
bile phone, it is different from the one that is being recommended on a hosting website. This is because th
e nature of threats and attacks keep changing from time to time. Similar is the case with data distribution
. For example, in the field of medical science, new diseases keep coming in. New devices are launched that
may have completely different hardware and computational power, so that the privacy mechanism that worked o
n previous versions of the device may prove null and void in its purpose when employed in this newly launch
ed device. This situation may put us into a situation where we are fighting with old disease with old techn
ologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapid
ly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the
 problem that comes is to find a balance between protecting the data using privacy mechanism and the accura
cy which the recommendation system as a consequence shows to the users in their device. The problem gets bi
gger and severe in more complex machine learning models like speech recognition and medical image analysis,
 where this balance between privacy and accuracy, if biased towards privacy, will result in even severely i
naccurate recommended contents. For example, in speech recognition, if we add more noise to train the syste
m with strong  privacy, this extra noise when sent out to work in real environment where user will speak wo
rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted b
y the system. This will make the system useless for daily communication and usage. Similarly, if we apply a
 similar noisy machine learning model (i.e. A model trend on too much noisy data for  privacy protection),
The final result when employee to real life medical applications may cause the model to miss critical featu
res in X ray or CT scan or completely false positive reports. And even worse is that missing out on feature
s in the images which point towards underlying disease may result in false negatives and the patient may be
 advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Ha
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n.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution . For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Hamilton, 2025; B. T. Hasan, 2025; N. Hasan et al., 2022; Kalejaiye et al., 2025; Z. Khan, 2025; Kvochko et al., 2024; T. Li, 2019; M. Liang, 2025; Mahmood et al., 2025; Osamika et al., 2024; Patel, 2025; Paulson \u0026 Elvis, 2025; Pittala, 2024; Raj, 2025b; Reddy, 2025; Shahsavari et al., 2025; Shakeer \u0026 Babu, 2025; Siva \u0026 Anbarasi, 2025; Thanikasalam, 2026; X. Zhou, 2025; [redacted], 2025)",

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S.\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1128,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/5DZ7WEEX\"],\"itemData\":{\"id\":1128,\"type\":\"article-journal\",\"DOI\":\"10.71443/9789349552036-13\",\"title\":\"Federated Learning and Data Privacy in Connected Healthcare Devices\",\"author\":[{\"family\":\"Thanikasalam\",\"given\":\"A\"}],\"issued\":{\"date-parts\":[[\"2026\"]]}}},{\"id\":1124,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/7VUDUDUG\"],\"itemData\":{\"id\":1124,\"type\":\"article-journal\",\"container-title\":\"Itm Web of Conferences\",\"DOI\":\"10.1051/itmconf/20257804003\",\"title\":\"Privacy Protection Optimization in Federated Learning\",\"author\":[{\"family\":\"Zhou\",\"given\":\"Xinyi\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1129,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/NZLIGNY2\"],\"itemData\":{\"id\":1129,\"type\":\"article-journal\",\"container-title\":\"Open Information and Computer Integrated Technologies\",\"DOI\":\"10.32620/oikit.2025.105.19\",\"title\":\"■■■■■■■ ■■■■■■■■■■ ■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■■■ ■■■■■■■■■ «■■■■■■■■■ ■■■■■■■■»: ■■■■■■■■■■■■■, ■■■■■■■■■ ■■ ■■■■■■■■■■■■\",\"author\":[{\"family\":\"■■■■■■■■■■■\",\"given\":\"■■■■■■■ ■■■■■■■■■■■■■■■\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be

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In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we em

ploy, especially on mobile recommendation systems along with other platforms like your hosting server etc.
has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mo
bile phone, it is different from the one that is being recommended on a hosting website. This is because th
e nature of threats and attacks keep changing from time to time. Similar is the case with data distribution
. For example, in the field of medical science, new diseases keep coming in. New devices are launched that
may have completely different hardware and computational power, so that the privacy mechanism that worked o
n previous versions of the device may prove null and void in its purpose when employed in this newly launch
ed device. This situation may put us into a situation where we are fighting with old disease with old techn
ologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapid
ly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the
 problem that comes is to find a balance between protecting the data using privacy mechanism and the accura
cy which the recommendation system as a consequence shows to the users in their device. The problem gets bi
gger and severe in more complex machine learning models like speech recognition and medical image analysis,
 where this balance between privacy and accuracy, if biased towards privacy, will result in even severely i
naccurate recommended contents. For example, in speech recognition, if we add more noise to train the syste
m with strong privacy, this extra noise when sent out to work in real environment where user will speak wo
rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted b
y the system. This will make the system useless for daily communication and usage. Similarly, if we apply a
 similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection),
The final result when employee to real life medical applications may cause the model to miss critical featu
res in X ray or CT scan or completely false positive reports. And even worse is that missing out on feature
s in the images which point towards underlying disease may result in false negatives and the patient may be
 advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Ha
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In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Hamilton, 2025; B. T. Hasan, 2025; N. Hasan et al., 2022; Kalejaiye et al., 2025; Z. Khan, 2025; Kvochko et al., 2024; T. Li, 2019; M. Liang, 2025; Mahmood et al., 2025; Osamika et al., 2024; Patel, 2025; Paulson \u0026 Elvis, 2025; Pittala, 2024; Raj, 2025b; Reddy, 2025; Shahsavari et al., 2025; Shakeer \u0026 Babu, 2025

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In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bi

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gger and severe in more complex machine learning models like speech recognition and medical image analysis,
 where this balance between privacy and accuracy, if biased towards privacy, will result in even severely i
naccurate recommended contents. For example, in speech recognition, if we add more noise to train the syste
m with strong  privacy, this extra noise when sent out to work in real environment where user will speak wo
rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted b
y the system. This will make the system useless for daily communication and usage. Similarly, if we apply a
 similar noisy machine learning model (i.e. A model trend on too much noisy data for  privacy protection),
The final result when employee to real life medical applications may cause the model to miss critical featu
res in X ray or CT scan or completely false positive reports. And even worse is that missing out on feature
s in the images which point towards underlying disease may result in false negatives and the patient may be
 advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Ha
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n.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is hig
hly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as
high as possible. However, because of the trade off When too much noise is added to hide the data, the inac
curacy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so th
at the recommendations become useless. Now, second problem is that if the clinical data and the recommendat
ions of the clinical data are to be displayed on mobile system, because the mobile system are computational
ly weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even dif
ficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due
 to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we em
ploy, especially on mobile recommendation systems along with other platforms like your hosting server etc.
has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mo
bile phone, it is different from the one that is being recommended on a hosting website. This is because th
e nature of threats and attacks keep changing from time to time. Similar is the case with data distribution
. For example, in the field of medical science, new diseases keep coming in. New devices are launched that
may have completely different hardware and computational power, so that the privacy mechanism that worked o
n previous versions of the device may prove null and void in its purpose when employed in this newly launch
ed device. This situation may put us into a situation where we are fighting with old disease with old techn
ologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapid
ly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the
 problem that comes is to find a balance between protecting the data using privacy mechanism and the accura
cy which the recommendation system as a consequence shows to the users in their device. The problem gets bi
gger and severe in more complex machine learning models like speech recognition and medical image analysis,
 where this balance between privacy and accuracy, if biased towards privacy, will result in even severely i
naccurate recommended contents. For example, in speech recognition, if we add more noise to train the syste
m with strong  privacy, this extra noise when sent out to work in real environment where user will speak wo
rds and sentences for the machine to listen to and interpret will result in wrong words being interpreted b
y the system. This will make the system useless for daily communication and usage. Similarly, if we apply a
 similar noisy machine learning model (i.e. A model trend on too much noisy data for  privacy protection),
The final result when employee to real life medical applications may cause the model to miss critical featu
res in X ray or CT scan or completely false positive reports. And even worse is that missing out on feature
s in the images which point towards underlying disease may result in false negatives and the patient may be
 advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Ha
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Rajasekhara\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1131,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/74XF9Z54\"],\"itemData\":{\"id\":1131,\"type\":\"article-journal\",\"container-title\":\"International Journal on Artificial Intelligence Tools\",\"DOI\":\"10.1142/s0218213025500095\",\"title\":\"A Federated Graph-Based Multimodal AI Framework for Privacy-Preserving and Explainable Osteoporosis Detection\",\"author\":[{\"family\":\"Siva\",\"given\":\"O. Venkata\"},{\"family\":\"Anbarasi\",\"given\":\"M. 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In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be

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Rajasekhara\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1131,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/74XF9Z54\"],\"itemData\":{\"id\":1131,\"type\":\"article-journal\",\"container-title\":\"International Journal on Artificial Intelligence Tools\",\"DOI\":\"10.1142/s0218213025500095\",\"title\":\"A Federated Graph-Based Multimodal AI Framework for Privacy-Preserving and Explainable Osteoporosis Detection\",\"author\":[{\"family\":\"Siva\",\"given\":\"O. Venkata\"},{\"family\":\"Anbarasi\",\"given\":\"M. 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ts\":[[\"2025\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citatio
n.json\"} One major field where the challenge is enormous is the clinical AI. In the clinical AI, It is highly Essential to hide the data And secure the data by adding noise An insured that privacy is intact as as high as possible. However, because of the trade off When too much noise is added to hide the data, the inaccuracy in the recommendations increases rapidly. Sometimes the inaccuracy goes over the roof, so much so that the recommendations become useless. Now, second problem is that if the clinical data and the recommendations of the clinical data are to be displayed on mobile system, because the mobile system are computationally weaker than any other system for example Laptop or desktop, decoding of the hidden data becomes even difficult. Devices like mobile phones prove almost useless in decoding the hidden data for the clinical AI Due to their computationally weak nature. It has to be thought about the fact that any privacy mechanism we employ, especially on mobile recommendation systems along with other platforms like your hosting server etc. has to be created dynamic in nature. Which means that if a recommendation system is being recommended on mobile phone, it is different from the one that is being recommended on a hosting website. This is because the nature of threats and attacks keep changing from time to time. Similar is the case with data distribution. For example, in the field of medical science, new diseases keep coming in. New devices are launched that may have completely different hardware and computational power, so that the privacy mechanism that worked on previous versions of the device may prove null and void in its purpose when employed in this newly launched device. This situation may put us into a situation where we are fighting with old disease with old technologies and privacy mechanism, while the new more powerful disease comes up and does damage even more rapidly than any disease we previously encountered with our older privacy protection mechanisms. Then again, the problem that comes is to find a balance between protecting the data using privacy mechanism and the accuracy which the recommendation system as a consequence shows to the users in their device. The problem gets bigger and severe in more complex machine learning models like speech recognition and medical image analysis, where this balance between privacy and accuracy, if biased towards privacy, will result in even severely inaccurate recommended contents. For example, in speech recognition, if we add more noise to train the system with strong privacy, this extra noise when sent out to work in real environment where user will speak words and sentences for the machine to listen to and interpret will result in wrong words being interpreted by the system. This will make the system useless for daily communication and usage. Similarly, if we apply a similar noisy machine learning model (i.e. A model trend on too much noisy data for privacy protection), The final result when employee to real life medical applications may cause the model to miss critical features in X ray or CT scan or completely false positive reports. And even worse is that missing out on features in the images which point towards underlying disease may result in false negatives and the patient may be advised to not take any serious action. (Ahluwalia, 2021; Almsallat, 2025; Batista, 2025; Graham \u0026 Hamilton, 2025; B. T. Hasan, 2025; N. Hasan et al., 2022; Kalejaiye et al., 2025; Z. Khan, 2025; Kvochko et al., 2024; T. Li, 2019; M. Liang, 2025; Mahmood et al., 2025; Osamika et al., 2024; Patel, 2025; Paulson \u0026 Elvis, 2025; Pittala, 2024; Raj, 2025b; Reddy, 2025; Shahsavari et al., 2025; Shakeer \u0026 Babu, 2025; Siva \u0026 Anbarasi, 2025; Thanikasalam, 2026; X. Zhou, 2025; [redacted], 2025)",
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This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated learning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et al., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al., 2024d; Maheshwari et al., 2022; Padala et al., 2021b; C. Pan et al., 2024; Pentyala et al., 2022b; Rodríguez-Gálvez et al., 2021b; Salem et al., 2022; K. Sun, 2025c; Vijay, 2025b; J. Xu et al., 2021; Yaghini et al., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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org/users/local/M0u71wYg/items/J3TKBPCC\"],\"itemData\":{\"id\":1152,\"type\":\"article-journal\",\"DOI\":\"10.18653/v1/2022.findings-emnlp.514\",\"title\":\"Fair NLP Models With Differentially Private Text Encoders\",\"author\":[{\"family\":\"Maheshwari\",\"given\":\"Gaurav\"},{\"family\":\"Denis\",\"given\":\"Pascal\"},{\"family\":\"Keller\",\"given\":\"Mikaela\"},{\"family\":\"Bellet\",\"given\":\"Aurélien\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":1154,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/PK9EYFWD\"],\"itemData\":{\"id\":1154,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2108.09932\",\"title\":\"Federated Learning Meets Fairness and Differential Privacy\",\"author\":[{\"family\":\"Padala\",\"given\":\"Manisha\"},{\"family\":\"Damle\",\"given\":\"Sankarshan\"},{\"family\":\"Gujar\",\"given\":\"Sujit\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":1161,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/36U2NC8M\"],\"itemData\":{\"id\":1161,\"type\":\"article-journal\",\"container-title\":\"Proceedings of the Aaai Conference on Artificial Intelligence\",\"DOI\":\"10.1609/aaai.v38i13.29365\",\"title\":\"Towards Fair Graph Federated Learning via Incentive Mechanisms\",\"author\":[{\"family\":\"Pan\",\"given\":\"Chenglu\"},{\"family\":\"Xu\",\"given\":\"Jiarong\"},{\"family\":\"Yu\",\"given\":\"Yue\"},{\"family\":\"Yang\",\"given\":\"Ziqi\"},{\"family\":\"Wu\",\"given\":\"Qingbiao\"},{\"family\":\"Wang\",\"given\":\"Chunping\"},{\"family\":\"Chen\",\"given\":\"Lei\"},{\"family\":\"Yang\",\"given\":\"Yang\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}},{\"id\":1146,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/A9YRERHV\"],\"itemData\":{\"id\":1146,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2205.11584\",\"title\":\"PrivFairFL: Privacy-Preserving Group Fairness in Federated Learning\",\"author\":[{\"family\":\"Pentyala\",\"given\":\"Sikha\"},{\"family\":\"Neophytou\",\"given\":\"Nicola\"},{\"family\":\"A. Nascimento\",\"given\":\"Anderson C.\"},{\"family\":\"Cock\",\"given\":\"Martine De\"},{\"family\":\"Farnadi\",\"given\":\"Golnoosh\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":1149,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ZADPIDIJ\"],\"itemData\":{\"id\":1149,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2109.08604\",\"title\":\"Enforcing Fairness in Private Federated Learning via the Modified Method of Differential Multipliers\",\"author\":[{\"family\":\"Rodríguez-Gálvez\",\"given\":\"Borja\"},{\"family\":\"Granqvist\",\"given\":\"Filip\"},{\"family\":\"Dalen\",\"given\":\"Rogier\",\"dropping-particle\":\"van\"},{\"family\":\"Seigel\",\"given\":\"Matt\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":1151,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/N3DN3XMU\"],\"itemData\":{\"id\":1151,\"type\":\"article-journal\",\"DOI\":\"10.1145/3498361.3538788\",\"title\":\"Quantifying Fairness of Federated Learning LPPM Models\",\"author\":[{\"family\":\"Salem\",\"given\":\"Amina Ben\"},{\"family\":\"Khalfoun\",\"given\":\"Besma\"},{\"family\":\"Mokhtar\",\"given\":\"Sonia Ben\"},{\"family\":\"Mashhadi\",\"given\":\"Afra\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":1139,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/84LICIG4\"],\"itemData\":{\"id\":1139,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.26841\",\"title\":\"Accurate Target Privacy Preserving Federated Learning Balancing Fairness and Utility\",\"author\":[{\"family\":\"Sun\",\"given\":\"Kangkang\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1159,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/KCVWT8EH\"],\"itemData\":{\"id\":1159,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-7128890/v1\",\"title\":\"Federated Reinforcement Learning for Distributed MAC Optimization in IEEE 802.11bn Networks\",\"author\":[{\"family\":\"Vijay\",\"given\":\"B T\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1147,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JWHTG6QV\"],\"itemData\":{\"id\":1147,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2106.12086\",\"title\":\"A Federated Data-Driven Evolutionary Algorithm for Expensive Multi/Many-Objective Optimization\",\"author\":[{\"family\":\"Xu\",\"given\":\"Jinjin\"},{\"family\":\"Jin\",\"given\":\"Yaochu\"},{\"family\":\"Du\",\"given\":\"Wenli\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":1155,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/Z9YR4JUU\"],\"itemData\":{\"id\":1155,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2302.09183\",\"title\":\"Learning With Impartiality to Walk on the Pareto Frontier of Fairness, Privacy, and Utility\",\"author\":[{\"family\":\"Yaghini\",\"given\":\"Mohammad\"},{\"family\":\"Liu\",\"given\":\"Patty\"},{\"family\":\"Boenisch\",\"given\":\"Franziska\"},{\"family\":\"Papernot\",\"given\":\"Nicolas\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1141,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3AGXPBI8\"],\"itemData\":{\"id\":1141,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.2684744\",\"title\":\"A Privacy-Enhanced Federated Learning Model Training Method\",\"author\":[{\"family\":\"Yan\",\"given\":\"Hong\"},{\"family\":\"Huang\",\"given\":\"Zhiqing\"},{\"family\":\"Zhang\",\"given\":\"Chenyang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1157,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YWCFSAAB\"],\"itemData\":{\"id\":1157,\"type\":\"article-journal\",\"container-title\":\"Engineering Reports\",\"DOI\":\"10.1002/eng2.70429\",\"title\":\"Federated Learning With Differential Privacy Based on Summary Statistics\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Peng\"},{\"family\":\"Liu\",\"given\":\"Pingqing\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1148,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JP2MV2TT\"],\"itemData\":{\"id\":1148,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics13112023\",\"title\":\"A Federated-Learning Algorithm Based on Client Sampling and Gradient Projection for the Smart Grid\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Ruifeng\"},{\"family\":\"Lu\",\"given\":\"Jiangang\"},{\"family\":\"Liu\",\"given\":\"Zewei\"},{\"family\":\"Wang\",\"given\":\"Tianqi\"},{\"family\":\"Guo\",\"given\":\"Wenxin\"},{\"family\":\"Lan\",\"given\":\"Tian\"},{\"family\":\"Hu\",\"given\":\"Chunqiang\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact

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 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve 
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that 
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 
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t al., 2024b)\",\"plainCitation\":\"(Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al.,
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,\"given\":\"Pingqing\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1148,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JP2MV2TT\"],\"itemData\":{\"id\":1148,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics13112023\",\"title\":\"A Federated-Learning Algorithm Based on Client Sampling and Gradient Projection for the Smart Grid\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Ruifeng\"},{\"family\":\"Lu\",\"given\":\"Jiangang\"},{\"family\":\"Liu\",\"given\":\"Zewei\"},{\"family\":\"Wang\",\"given\":\"Tianqi\"},{\"family\":\"Guo\",\"given\":\"Wenxin\"},{\"family\":\"Lan\",\"given\":\"Tian\"},{\"family\":\"Hu\",\"given\":\"Chunqiang\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated learning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et al., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al., 2024d; Maheshwari et al., 2022; Padala et al., 2021b; C. Pan et al., 2024; Pentyala et al., 2022b; Rodríguez-Gálvez et al., 2021b; Salem et al., 2022; K. Sun, 2025c; Vijay, 2025b; J. Xu et al., 2021; Yaghini et al., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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he noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means t
hat if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up
discriminating between different user groups or users data. In other words, the noise which is added to the
 data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
 rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat
a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
l., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al.
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\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy

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he noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means t
hat if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up
discriminating between different user groups or users data. In other words, the noise which is added to the
 data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
 rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat
a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
l., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al.
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This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique na

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med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
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\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated learning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et al., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al., 2024d; Maheshwari et al., 2022; Padala et al., 2021b; C. Pan et al., 2024; Pentyala et al., 2022b; Rodríguez-Gálvez et al., 2021b; Salem et al., 2022; K. Sun, 2025c; Vijay, 2025b; J. Xu et al., 2021; Yaghini et al., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated lear

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ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
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he noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means t
hat if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up
discriminating between different user groups or users data. In other words, the noise which is added to the
data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t

he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated learning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et al., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al., 2024d; Maheshwari et al., 2022; Padala et al., 2021b; C. Pan et al., 2024; Pentyala et al., 2022b; Rodríguez-Gálvez et al., 2021b; Salem et al., 2022; K. Sun, 2025c; Vijay, 2025b; J. Xu et al., 2021; Yaghini et al., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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0.48550/arxiv.2302.09183\",\"title\":\"Learning With Impartiality to Walk on the Pareto Frontier of Fairness, Privacy, and Utility\",\"author\":[{\"family\":\"Yaghini\",\"given\":\"Mohammad\"},{\"family\":\"Liu\",\"given\":\"Patty\"},{\"family\":\"Boenisch\",\"given\":\"Franziska\"},{\"family\":\"Papernot\",\"given\":\"Nicolas\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1141,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3AGXPBI8\"],\"itemData\":{\"id\":1141,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.2684744\",\"title\":\"A Privacy-Enhanced Federated Learning Model Training Method\",\"author\":[{\"family\":\"Yan\",\"given\":\"Hong\"},{\"family\":\"Huang\",\"given\":\"Zhiqing\"},{\"family\":\"Zhang\",\"given\":\"Chenyang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1157,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YWCFSAAB\"],\"itemData\":{\"id\":1157,\"type\":\"article-journal\",\"container-title\":\"Engineering Reports\",\"DOI\":\"10.1002/eng2.70429\",\"title\":\"Federated Learning With Differential Privacy Based on Summary Statistics\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Peng\"},{\"family\":\"Liu\",\"given\":\"Pingqing\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1148,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JP2MV2TT\"],\"itemData\":{\"id\":1148,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics13112023\",\"title\":\"A Federated-Learning Algorithm Based on Client Sampling and Gradient Projection for the Smart Grid\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Ruifeng\"},{\"family\":\"Lu\",\"given\":\"Jiangang\"},{\"family\":\"Liu\",\"given\":\"Zewei\"},{\"family\":\"Wang\",\"given\":\"Tianqi\"},{\"family\":\"Guo\",\"given\":\"Wenxin\"},{\"family\":\"Lan\",\"given\":\"Tian\"},{\"family\":\"Hu\",\"given\":\"Chunqiang\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useless recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated learning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et al., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al., 2024d; Maheshwari et al., 2022; Padala et al., 2021b; C. Pan et al., 2024; Pentyala et al., 2022b; Rodríguez-Gálvez et al., 2021b; Salem et al., 2022; K. Sun, 2025c; Vijay, 2025b; J. Xu et al., 2021; Yaghini et al., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from t
he noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means t
hat if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up
discriminating between different user groups or users data. In other words, the noise which is added to the
 data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
 rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat
a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
l., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al.
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he noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means t
hat if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up
discriminating between different user groups or users data. In other words, the noise which is added to the
 data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
 rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat
a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
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discriminating between different user groups or users data. In other words, the noise which is added to the
data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat

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a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
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\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into the model the data set may start collecting into groups of small sub data (as in unsupervised learning) and the whole data set may end up looking like a collection of lumps of data at many different places on the graph. Here lies another problem. User who contribute rare type of data, for example some users may like very rare genre of movies. If we apply equitable noise to the data set of all the users these rare contributions may for sure will get clocked in too much of noise and the system may not learn much about these rare data set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data of the user, it is therefore almost impossible to tell whether the system is recommending content in all fairness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming minority group of users. Especially though group of users who contribute rare genre of data set. To resolve this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy protection is tailored to each user according to the data the user generates. This helps in keeping sensitive information of each user safe while at the same time system works efficiently and reliably from for each user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy tag of work and create a balance between all the groups so that no group of users is left behind with useles

s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, privacy and fairness, all at the same time. There are also some technical challenges like a privacy technique named Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU. This will exclude users who possess older phones and hence will collect correct data from only users that have high end specifications mobile phones and devices especially when they are new. Consequently the unfairness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while creating a privacy system without bringing into the game fairness, then the resulting model of Federated learning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al., 2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et al., 2020; Geng et al., 2024b; Hosseini et al., 2023b; Ju et al., 2023c; M. J. Khan et al., 2024; Kim et al., 2024d; Maheshwari et al., 2022; Padala et al., 2021b; C. Pan et al., 2024; Pentyala et al., 2022b; Rodríguez-Gálvez et al., 2021b; Salem et al., 2022; K. Sun, 2025c; Vijay, 2025b; J. Xu et al., 2021; Yaghini et al., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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he noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means t
hat if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up
discriminating between different user groups or users data. In other words, the noise which is added to the
 data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
 rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat
a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
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l., 2023b; H. Yan et al., 2023b; P. Zhang \u0026 Liu, 2025; R. Zhao et al., 2024b)",
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Nascimento\",\"given\":\"Anderson C.\"},{\"family\":\"Cock\",\"given\":\"Martine De\"},{\"family\":\"Farnadi\",\"given\":\"Golnoosh\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":1149,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/ZADPIDIJ\"],\"itemData\":{\"id\":1149,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2109.08604\",\"title\":\"Enforcing Fairness in Private Federated Learning via the Modified Method of Differential Multipliers\",\"author\":[{\"family\":\"Rodríguez-Gálvez\",\"given\":\"Borja\"},{\"family\":\"Granqvist\",\"given\":\"Filip\"},{\"family\":\"Dalen\",\"given\":\"Rogier\",\"dropping-particle\":\"van\"},{\"family\":\"Seigel\",\"given\":\"Matt\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":1151,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/N3DN3XMU\"],\"itemData\":{\"id\":1151,\"type\":\"article-journal\",\"DOI\":\"10.1145/3498361.3538788\",\"title\":\"Quantifying Fairness of Federated Learning LPPM Models\",\"author\":[{\"family\":\"Salem\",\"given\":\"Amina Ben\"},{\"family\":\"Khalfoun\",\"given\":\"Besma\"},{\"family\":\"Mokhtar\",\"given\":\"Sonia Ben\"},{\"family\":\"Mashhadi\",\"given\":\"Afra\"}],\"issued\":{\"date-parts\":[[\"2022\"]]}}},{\"id\":1139,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/84LICIG4\"],\"itemData\":{\"id\":1139,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2510.26841\",\"title\":\"Accurate Target Privacy Preserving Federated Learning Balancing Fairness and Utility\",\"author\":[{\"family\":\"Sun\",\"given\":\"Kangkang\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1159,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/KCVWT8EH\"],\"itemData\":{\"id\":1159,\"type\":\"article-journal\",\"DOI\":\"10.21203/rs.3.rs-7128890/v1\",\"title\":\"Federated Reinforcement Learning for Distributed MAC Optimization in IEEE 802.11bn Networks\",\"author\":[{\"family\":\"Vijay\",\"given\":\"B T\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1147,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JWHTG6QV\"],\"itemData\":{\"id\":1147,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2106.12086\",\"title\":\"A Federated Data-Driven Evolutionary Algorithm for Expensive Multi/Many-Objective Optimization\",\"author\":[{\"family\":\"Xu\",\"given\":\"Jinjin\"},{\"family\":\"Jin\",\"given\":\"Yaochu\"},{\"family\":\"Du\",\"given\":\"Wenli\"}],\"issued\":{\"date-parts\":[[\"2021\"]]}}},{\"id\":1155,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/Z9YR4JUU\"],\"itemData\":{\"id\":1155,\"type\":\"article-journal\",\"DOI\":\"10.48550/arxiv.2302.09183\",\"title\":\"Learning With Impartiality to Walk on the Pareto Frontier of Fairness, Privacy, and Utility\",\"author\":[{\"family\":\"Yaghini\",\"given\":\"Mohammad\"},{\"family\":\"Liu\",\"given\":\"Patty\"},{\"family\":\"Boenisch\",\"given\":\"Franziska\"},{\"family\":\"Papernot\",\"given\":\"Nicolas\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1141,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/3AGXPBI8\"],\"itemData\":{\"id\":1141,\"type\":\"article-journal\",\"DOI\":\"10.1117/12.2684744\",\"title\":\"A Privacy-Enhanced Federated Learning Model Training Method\",\"author\":[{\"family\":\"Yan\",\"given\":\"Hong\"},{\"family\":\"Huang\",\"given\":\"Zhiqing\"},{\"family\":\"Zhang\",\"given\":\"Chenyang\"}],\"issued\":{\"date-parts\":[[\"2023\"]]}}},{\"id\":1157,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/YWCFSAAB\"],\"itemData\":{\"id\":1157,\"type\":\"article-journal\",\"container-title\":\"Engineering Reports\",\"DOI\":\"10.1002/eng2.70429\",\"title\":\"Federated Learning With Differential Privacy Based on Summary Statistics\",\"author\":[{\"family\":\"Zhang\",\"given\":\"Peng\"},{\"family\":\"Liu\",\"given\":\"Pingqing\"}],\"issued\":{\"date-parts\":[[\"2025\"]]}}},{\"id\":1148,\"uris\":[\"http://zotero.org/users/local/M0u71wYg/items/JP2MV2TT\"],\"itemData\":{\"id\":1148,\"type\":\"article-journal\",\"container-title\":\"Electronics\",\"DOI\":\"10.3390/electronics13112023\",\"title\":\"A Federated-Learning Algorithm Based on Client Sampling and Gradient Projection for the Smart Grid\",\"author\":[{\"family\":\"Zhao\",\"given\":\"Ruifeng\"},{\"family\":\"Lu\",\"given\":\"Jiangang\"},{\"family\":\"Liu\",\"given\":\"Zewei\"},{\"family\":\"Wang\",\"given\":\"Tianqi\"},{\"family\":\"Guo\",\"given\":\"Wenxin\"},{\"family\":\"Lan\",\"given\":\"Tian\"},{\"family\":\"Hu\",\"given\":\"Chunqiang\"}],\"issued\":{\"date-parts\":[[\"2024\"]]}}}],\"schema\":\"https://github.com/citation-style-language/schema/raw/master/csl-citation.json\"} Apart from the noise accuracy trade-off in privacy, there is one more problem known as “poor gets poorer”. This means that if privacy mechanisms are not designed with proper balance and buyers and equity, the model may end up discriminating between different user groups or users data. In other words, the noise which is added to the

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 data may be proportional to the magnitude of the data point. So if this kind of noise is propagated into t
he model the data set may start collecting into groups of small sub data (as in unsupervised learning) and
the whole data set may end up looking like a collection of lumps of data at many different places on the gr
aph. Here lies another problem. User who contribute rare type of data, for example some users may like very
 rare genre of movies. If we apply equitable noise to the data set of all the users these rare contribution
s may for sure will get clocked in too much of noise and the system may not learn much about these rare dat
a set types and hence perform poorly. Because the whole point of Federated learning is to hide the raw data
 of the user, it is therefore almost impossible to tell whether the system is recommending content in all f
airness to all users or not. Privacy and fairness cannot go hand in hand at some instances. The simple fact
 that if privacy mechanisms and measures are not designed with fairness in mind, they can lead to harming m
inority group of users. Especially though group of users who contribute rare genre of data set. To resolve
this dilemma, we can deploy what is called “personalised privacy”. In this what happens is that the privacy
 protection is tailored to each user according to the data the user generates. This helps in keeping sensit
ive information of each user safe while at the same time system works efficiently and reliably from for eac
h user. A new strategy is proposed known as “multi objective optimization” to handle fairness and privacy t
ag of work and create a balance between all the groups so that no group of users is left behind with useles
s recommendations on their devices. This strategy has a goal to find an optimal point among accuracy, priva
cy and fairness, all at the same time. There are also some technical challenges like a privacy technique na
med Poisson subsampling which can put too much load on the low end mobile phones or hardware and their CPU.
 This will exclude users who possess older phones and hence will collect correct data from only users that
have high end specifications mobile phones and devices especially when they are new. Consequently the unfai
rness. This leads us to a warning named Matthew effect warning. It says that privacy if focused on while cr
eating a privacy system without bringing into the game fairness, then the resulting model of Federated lear
ning will reinforce or amplify already existing inequalities making the “poor even poorer”. (Badar et al.,
2024b; Binns, 2017; Choudhury et al., 2024b; Chu et al., 2021b; Culnan, 2000; J. Ding et al., 2020; Du et a
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MT\"],\"itemData\":{\"id\":1179,\"type\":\"article-journal\",\"container-title\":\"Neurocomputing\",\"DOI\"
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n, 2023; Reshi et al., 2023; Sangeeta \u0026 Nam, 2023; Satilmisoglu \u0026 Keskin, 2023; S. Sharma et al.,
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```


File: clean_citations.py

```
import logging

# Configure logging
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')

def clean_document(file_path):
    try:
        from docx import Document
    except ImportError:
        logging.error("python-docx is not installed. Please install it with 'pip install python-docx'.")
        return

    doc = Document(file_path)

    bad_citations = [
        # Block 1 - Energy/P2P/Old Wireless (Context: GPT-4o / Foundation Models)
        "Abbaszadi, 2025",
        "Akon et al., 2008",
        "Butt et al., 2003",
        "Fu & Wang, 2009",
        "Ma et al., 2014",
        "Panisson et al., 2006",
        "Wu et al., 2014",
        "Maher & Nasr, 2021",
        "Urblik et al., 2023",
        "Chen, 2025", # Assuming SP-MoE is irrelevant? No, SP-MoE is relevant to LLMs. Check plan.

        # Block 2 - Wireless/IoV/Fog (Context: LLMs)
        "Zhang et al., 2023", # Verify exact citation text match
        "Zhou et al., 2024",
        "Song et al., 2022",
        "Hosseinalipour et al., 2020",

        # Block 4 - IoT/Grid Attacks (Context: LLM/RecSys Attacks?)
```

```
        "S. Ali et al., 2025",
        "Alruwaili et al., 2024",
        "Bondok et al., 2023",
        "Teryak et al., 2023",
        "Aparna et al., 2025", # HoneyFed/MANET
        "Baji et al., 2024", # SDN

        # Block 5 - Model Compression (Context: LLM Compression)
        "Dhaouadi et al., 2025", # Obesity
        "Jing & Wang, 2024", # UAV
        "Weng et al., 2023", # Cattle Face
        "Heryanto et al., 2024", # Tissue
        "Deng, 2022", # Android Apps (maybe keep?) - Keep for safety if Android is relevant.

        # Block 6 - Smart Contracts (Context: General SC?)
        "Adekola & Dada, 2024", # Pharma
        "Herzog & Herzog, 2024", # Water
        "J. Yang, 2024", # Real Estate
        "Satilmisoglu & Keskin, 2023", # Water
        "Naher & Uddin, 2023", # Finance (Maybe relevant if fintech) - Keep

        # Block 19 - Health (Context: Health) - These might be relevant if Health is the domain.
        # Check plan. Plan says verify context.
        # The user said "remove garbage citations".
        # I will stick to the explicitly identified ones like Cattle/Water.
    ]

    replacements_made = 0

    for paragraph in doc.paragraphs:
        original_text = paragraph.text
        modified_text = original_text

        # Simple string replacement for now.
        # Note: This is fragile if citations are split across runs or have varying whitespace.
        # But for a first pass cleanup, it's efficient.

        clean_count = 0
        for bad in bad_citations:
            # Try variations: "; Bad, YYYY" or "Bad, YYYY; "

            # Case 1: Middle of list: "; Bad, YYYY;"
            if f"; {bad}" in modified_text:
                 modified_text = modified_text.replace(f"; {bad}", "")
                 clean_count += 1

            # Case 2: Start of list: "(Bad, YYYY;"
            elif f"({bad}; " in modified_text:
                 modified_text = modified_text.replace(f"({bad}; ", "(")
                 clean_count += 1

            # Case 3: End of list: "; Bad, YYYY)"
            elif f"; {bad})" in modified_text:
                 modified_text = modified_text.replace(f"; {bad})", ")")
                 clean_count += 1

            # Case 4: Only citation: "(Bad, YYYY)"
            elif f"({bad})" in modified_text:
                 modified_text = modified_text.replace(f"({bad})", "")
                 clean_count += 1

        if clean_count > 0:
            # We need to apply this change to the runs.
            # Replacing paragraph.text destroys formatting.
            # A safer way to preserve *some* formatting is to clear runs and add new text,
            # but that loses bold/italic.
            # Given the request "Edit... and save", text correctness > perfect complex formatting for now.
            # BUT, citations are often in their own runs.

            # Let's try to set paragraph.text and hope for the best.
            # Or better: don't.
            # Use strict replacement ONLY if we are sure.

            paragraph.text = modified_text
```

```
            replacements_made += clean_count
            logging.info(f"Cleaned {clean_count} citations in paragraph starting: {original_text[:30]}...")

    doc.save(file_path)
    logging.info(f"Document saved. Total removals: {replacements_made}")

if __name__ == "__main__":
    clean_document("Empirical Analysis of Accuracy.docx")
```

File: client.py

```
------------------------------------------------------------------------------------------------------------
"""
Federated Learning Client for Mobile Movie Recommendation Systems
Using AIJack for federated learning with DP-SGD support.
"""

import base64
import json
import logging
from typing import List, Optional, Dict, Tuple
import numpy as np
import pandas as pd
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
import requests
from dataclasses import dataclass

# Configure logging
logging.basicConfig(level=logging.INFO)
logger = logging.getLogger(__name__)


@dataclass
class ClientConfig:
    """Client configuration"""
    client_id: str
    server_url: str
    num_users: int
    num_items: int
    embedding_dim: int = 16
    local_epochs: int = 3  # Increased from 1 to 3 for better local convergence
    learning_rate: float = 0.01
    batch_size: int = 32

    # Differential Privacy parameters
    use_dp: bool = False
    dp_sigma: float = 1.0  # Noise multiplier
    dp_clip_norm: float = 1.0  # Gradient clipping norm
    dp_delta: float = 1e-5  # Delta for (ε, δ)-DP


class MovieLensDataset(Dataset):
    """MovieLens dataset for recommendation"""

    def __init__(self, interactions: List[tuple]):
        """
        Args:
            interactions: List of (user_id, item_id, rating) tuples
        """
        self.interactions = interactions

    def __len__(self):
        return len(self.interactions)

    def __getitem__(self, idx):
        user_id, item_id, rating = self.interactions[idx]
        return torch.LongTensor([user_id]), torch.LongTensor([item_id]), torch.FloatTensor([rating])


def load_ratings_csv(csv_path: str, binarize: bool = True, threshold: float = 4.0) -> Tuple[List[tuple], in
t, int, Dict[int, int], Dict[int, int]]:
    """
```

```
    Load ratings from CSV file and preprocess according to expose requirements.

    Args:
        csv_path: Path to ratings.csv file (columns: userId, movieId, rating, timestamp)
        binarize: If True, convert ratings to binary (≥threshold → 1.0, <threshold → 0.0)
        threshold: Rating threshold for binarization (default 4.0 as per expose)

    Returns:
        interactions: List of (user_id, item_id, rating) tuples with 0-indexed IDs
        num_users: Number of unique users
        num_items: Number of unique items
        user_id_map: Mapping from original userId to 0-indexed user_id
        item_id_map: Mapping from original movieId to 0-indexed item_id
    """
    logger.info(f"Loading ratings from {csv_path}...")

    # Load CSV file
    df = pd.read_csv(csv_path)

    logger.info(f"Loaded {len(df)} ratings from CSV")

    # Get unique users and items, create mapping to 0-indexed
    unique_users = sorted(df['userId'].unique())
    unique_items = sorted(df['movieId'].unique())

    user_id_map = {original_id: idx for idx, original_id in enumerate(unique_users)}
    item_id_map = {original_id: idx for idx, original_id in enumerate(unique_items)}

    num_users = len(unique_users)
    num_items = len(unique_items)

    logger.info(f"Found {num_users} unique users and {num_items} unique items")

    # Map to 0-indexed and process ratings
    interactions = []
    for _, row in df.iterrows():
        user_idx = user_id_map[row['userId']]
        item_idx = item_id_map[row['movieId']]

        if binarize:
            # Binarize rating: ≥threshold → 1.0, <threshold → 0.0
            rating = 1.0 if row['rating'] >= threshold else 0.0
        else:
            rating = float(row['rating'])

        interactions.append((user_idx, item_idx, rating))

    logger.info(f"Preprocessed {len(interactions)} interactions")
    if binarize:
        positive_count = sum(1 for _, _, r in interactions if r >= 0.5)  # Count binarized positive ratings
        logger.info(f"Positive ratings (≥{threshold}): {positive_count}, Negative: {len(interactions) - pos
itive_count}")

    return interactions, num_users, num_items, user_id_map, item_id_map


def split_train_test(interactions: List[tuple], test_ratio: float = 0.2, random_seed: int = 42) -> Tuple[Li
st[tuple], List[tuple]]:
    """
    Split interactions into train and test sets.

    Args:
        interactions: List of (user_id, item_id, rating) tuples
        test_ratio: Ratio of test data (default 0.2 = 20%)
        random_seed: Random seed for reproducibility

    Returns:
        train_interactions: Training set
        test_interactions: Test set
    """
    np.random.seed(random_seed)
    indices = np.random.permutation(len(interactions))

    split_idx = int(len(interactions) * (1 - test_ratio))
```

```
    train_indices = indices[:split_idx]
    test_indices = indices[split_idx:]

    train_interactions = [interactions[i] for i in train_indices]
    test_interactions = [interactions[i] for i in test_indices]

    logger.info(f"Split data: {len(train_interactions)} training, {len(test_interactions)} test")

    return train_interactions, test_interactions


def create_matrix_factorization_model(num_users: int, num_items: int, embedding_dim: int = 16):
    """
    Create a Matrix Factorization model (same as server).

    Args:
        num_users: Number of users
        num_items: Number of items
        embedding_dim: Embedding dimension

    Returns:
        PyTorch model
    """
    class MatrixFactorization(nn.Module):
        def __init__(self, num_users, num_items, embedding_dim):
            super().__init__()
            self.user_embedding = nn.Embedding(num_users, embedding_dim)
            self.item_embedding = nn.Embedding(num_items, embedding_dim)

            nn.init.normal_(self.user_embedding.weight, std=0.1)  # Increased from 0.01 to 0.1
            nn.init.normal_(self.item_embedding.weight, std=0.1)  # Increased from 0.01 to 0.1

        def forward(self, user_ids, item_ids):
            user_emb = self.user_embedding(user_ids)
            item_emb = self.item_embedding(item_ids)
            return (user_emb * item_emb).sum(dim=1)  # Predict rating directly (no sigmoid)

        def predict(self, user_ids, item_ids):
            return self.forward(user_ids, item_ids)

    return MatrixFactorization(num_users, num_items, embedding_dim)


class ModelParamsEncoder:
    """Encode/decode model parameters to/from JSON format"""

    @staticmethod
    def model_to_json_params(model: nn.Module) -> List[Dict]:
        """Convert PyTorch model to portable JSON format"""
        params = []
        state_dict = model.state_dict()

        for name, tensor in state_dict.items():
            np_array = tensor.detach().cpu().numpy().astype(np.float32)
            data_bytes = np_array.tobytes()
            data_b64 = base64.b64encode(data_bytes).decode('utf-8')

            params.append({
                "name": name,
                "shape": list(np_array.shape),
                "dtype": "float32",
                "data": data_b64
            })

        return params

    @staticmethod
    def json_params_to_model(params: List[Dict], model: nn.Module) -> nn.Module:
        """Load parameters from JSON format into model"""
        state_dict = {}

        for param in params:
            data_bytes = base64.b64decode(param["data"])
```

```
            np_array = np.frombuffer(data_bytes, dtype=np.float32).reshape(param["shape"])
            tensor = torch.from_numpy(np_array.copy())
            state_dict[param["name"]] = tensor

        model.load_state_dict(state_dict)
        return model


class DPNoise:
    """Differential Privacy noise addition utilities for DP-SGD"""

    @staticmethod
    def clip_gradients(model: nn.Module, clip_norm: float):
        """
        Clip gradients of model parameters in-place.

        Args:
            model: PyTorch model
            clip_norm: Maximum gradient norm
        """
        # Collect all gradients
        gradients = []
        for param in model.parameters():
            if param.grad is not None:
                gradients.append(param.grad.data.flatten())

        if not gradients:
            return

        # Compute total norm
        total_norm = torch.norm(torch.cat(gradients))

        # Clip if necessary
        if total_norm > clip_norm:
            clip_coef = clip_norm / (total_norm + 1e-6)
            for param in model.parameters():
                if param.grad is not None:
                    param.grad.data.mul_(clip_coef)

    @staticmethod
    def add_gaussian_noise(model: nn.Module, sigma: float, clip_norm: float):
        """
        Add Gaussian noise to gradients for DP-SGD (after clipping).

        Args:
            model: PyTorch model
            sigma: Noise multiplier
            clip_norm: Gradient clipping norm (for noise scaling)
        """
        for param in model.parameters():
            if param.grad is not None:
                noise = torch.normal(
                    0.0,
                    sigma * clip_norm,
                    size=param.grad.shape,
                    device=param.grad.device,
                    dtype=param.grad.dtype
                )
                param.grad.data.add_(noise)


class FederatedClient:
    """Federated Learning Client with DP-SGD support"""

    def __init__(self, config: ClientConfig, local_data: List[tuple]):
        """
        Initialize federated client.

        Args:
            config: Client configuration
            local_data: Local training data as (user_id, item_id, rating) tuples
        """
        self.config = config
        self.local_data = local_data
```

```
        self.model = create_matrix_factorization_model(
            config.num_users, config.num_items, config.embedding_dim
        )

        # Create dataset and dataloader (only if we have training data)
        self.dataset = MovieLensDataset(local_data)
        self.dataloader = None
        if len(local_data) > 0:
            self.dataloader = DataLoader(
                self.dataset,
                batch_size=config.batch_size,
                shuffle=True
            )

        self.encoder = ModelParamsEncoder()
        self.dp_noise = DPNoise()

    def register(self):
        """Register with the server"""
        url = f"{self.config.server_url}/register"
        response = requests.post(url, json={"client_id": self.config.client_id})
        response.raise_for_status()
        logger.info(f"Client {self.config.client_id} registered")
        return response.json()

    def fetch_global_params(self):
        """Fetch global model parameters from server"""
        url = f"{self.config.server_url}/global-params-json"
        response = requests.get(url)
        response.raise_for_status()

        data = response.json()
        params = data["params"]

        # Load into local model
        self.model = self.encoder.json_params_to_model(params, self.model)

        logger.info(f"Fetched global params for round {data.get('round', 0)}")
        return data

    def train_local(self) -> Dict:
        """
        Train model locally using local data.
        Supports DP-SGD if enabled in config.

        Returns:
            Training metrics
        """
        if self.dataloader is None or len(self.local_data) == 0:
            logger.warning("No training data available. Skipping local training.")
            return {
                "loss": 0.0,
                "samples": 0,
                "epochs": 0
            }

        self.model.train()
        optimizer = optim.SGD(
            self.model.parameters(),
            lr=self.config.learning_rate,
            weight_decay=1e-5  # L2 regularization for stability
        )
        criterion = nn.MSELoss()  # MSE loss for rating prediction

        total_loss = 0.0
        num_batches = 0

        for epoch in range(self.config.local_epochs):
            epoch_loss = 0.0

            for batch_idx, (user_ids, item_ids, ratings) in enumerate(self.dataloader):
                user_ids = user_ids.squeeze(1)  # Squeeze only dim 1, preserve batch dimension
                item_ids = item_ids.squeeze(1)
                ratings = ratings.squeeze(1)
```

```
                # Forward pass
                predictions = self.model(user_ids, item_ids)
                loss = criterion(predictions, ratings)

                # Backward pass
                optimizer.zero_grad()
                loss.backward()

                # Apply DP-SGD if enabled
                if self.config.use_dp:
                    # Clip gradients first
                    self.dp_noise.clip_gradients(self.model, self.config.dp_clip_norm)
                    # Then add Gaussian noise
                    self.dp_noise.add_gaussian_noise(
                        self.model,
                        self.config.dp_sigma,
                        self.config.dp_clip_norm
                    )

                optimizer.step()

                epoch_loss += loss.item()
                num_batches += 1

            total_loss += epoch_loss
            logger.info(f"Epoch {epoch+1}/{self.config.local_epochs}, Loss: {epoch_loss/len(self.dataloader
):.4f}")

        avg_loss = total_loss / (num_batches + 1e-6)
        return {
            "loss": avg_loss,
            "samples": len(self.local_data),
            "epochs": self.config.local_epochs
        }

    def upload_params(self):
        """Upload local model parameters to server"""
        params = self.encoder.model_to_json_params(self.model)

        payload = {
            "client_id": self.config.client_id,
            "params": params,
            "sample_count": len(self.local_data)
        }

        url = f"{self.config.server_url}/upload-params-json"
        response = requests.post(url, json=payload)
        response.raise_for_status()

        logger.info(f"Uploaded params to server (samples: {len(self.local_data)})")
        return response.json()

    def run_training_round(self) -> Dict:
        """
        Execute one complete training round:
        1. Fetch global parameters
        2. Train locally
        3. Upload parameters

        Returns:
            Round metrics
        """
        # Fetch global model
        global_info = self.fetch_global_params()

        # Train locally
        train_metrics = self.train_local()

        # Upload parameters
        upload_info = self.upload_params()

        return {
            "round": global_info.get("round", 0),
```

```
            **train_metrics,
            **upload_info
        }

    def evaluate(self, test_data: List[tuple]) -> Dict:
        """
        Evaluate model on test data.

        Args:
            test_data: List of (user_id, item_id, rating) tuples for testing

        Returns:
            Evaluation metrics (MSE, MAE, accuracy for binary classification)
        """
        self.model.eval()
        criterion = nn.MSELoss()

        if len(test_data) == 0:
            return {"mse": 0.0, "mae": 0.0, "accuracy": 0.0}

        # Create test dataset and dataloader
        test_dataset = MovieLensDataset(test_data)
        test_loader = DataLoader(test_dataset, batch_size=64, shuffle=False)

        total_loss = 0.0
        total_mae = 0.0
        correct_predictions = 0
        total_samples = 0

        with torch.no_grad():
            for user_ids, item_ids, ratings in test_loader:
                user_ids = user_ids.squeeze()
                item_ids = item_ids.squeeze()
                ratings = ratings.squeeze()

                predictions = self.model(user_ids, item_ids)
                loss = criterion(predictions, ratings)

                total_loss += loss.item() * len(ratings)
                total_mae += torch.abs(predictions - ratings).sum().item()

                # For binary classification accuracy (assuming binarized ratings)
                pred_binary = (predictions > 0.5).float()
                correct_predictions += (pred_binary == ratings).sum().item()
                total_samples += len(ratings)

        mse = total_loss / total_samples
        mae = total_mae / total_samples
        accuracy = correct_predictions / total_samples if total_samples > 0 else 0.0

        return {
            "mse": mse,
            "mae": mae,
            "accuracy": accuracy,
            "samples": total_samples
        }


def evaluate_model(model: nn.Module, test_data: List[tuple], batch_size: int = 64) -> Dict:
    """
    Standalone function to evaluate a model on test data.

    Args:
        model: PyTorch model to evaluate
        test_data: List of (user_id, item_id, rating) tuples for testing
        batch_size: Batch size for evaluation

    Returns:
        Evaluation metrics (MSE, MAE, accuracy for binary classification)
    """
    model.eval()
    criterion = nn.MSELoss()

    if len(test_data) == 0:
```

```
        return {"mse": 0.0, "mae": 0.0, "accuracy": 0.0, "samples": 0}

    # Create test dataset and dataloader
    test_dataset = MovieLensDataset(test_data)
    test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False)

    total_loss = 0.0
    total_mae = 0.0
    correct_predictions = 0
    total_samples = 0

    with torch.no_grad():
        for user_ids, item_ids, ratings in test_loader:
            user_ids = user_ids.squeeze()
            item_ids = item_ids.squeeze()
            ratings = ratings.squeeze()

            predictions = model(user_ids, item_ids)
            loss = criterion(predictions, ratings)

            total_loss += loss.item() * len(ratings)
            total_mae += torch.abs(predictions - ratings).sum().item()

            # For binary classification accuracy (assuming binarized ratings)
            pred_binary = (predictions > 0.5).float()
            correct_predictions += (pred_binary == ratings).sum().item()
            total_samples += len(ratings)

    mse = total_loss / total_samples if total_samples > 0 else 0.0
    mae = total_mae / total_samples if total_samples > 0 else 0.0
    accuracy = correct_predictions / total_samples if total_samples > 0 else 0.0

    return {
        "mse": mse,
        "mae": mae,
        "accuracy": accuracy,
        "samples": total_samples
    }


def create_non_iid_split(interactions: List[tuple], num_clients: int, alpha: float = 0.5):
    """
    Create non-IID data split using Dirichlet distribution.

    Args:
        interactions: List of (user_id, item_id, rating) tuples
        num_clients: Number of clients
        alpha: Dirichlet concentration parameter (lower = more heterogeneous)

    Returns:
        List of client data splits
    """
    # Group interactions by user
    user_interactions = {}
    for user_id, item_id, rating in interactions:
        if user_id not in user_interactions:
            user_interactions[user_id] = []
        user_interactions[user_id].append((user_id, item_id, rating))

    user_ids = list(user_interactions.keys())
    num_users = len(user_ids)

    # Sample Dirichlet distribution for each user
    np.random.seed(42)  # For reproducibility
    proportions = np.random.dirichlet([alpha] * num_clients, size=num_users)

    # Assign interactions to clients
    client_splits = [[] for _ in range(num_clients)]

    for user_idx, user_id in enumerate(user_ids):
        user_data = user_interactions[user_id]
        # Sample client assignment for each interaction
        for interaction in user_data:
            client_idx = np.random.choice(num_clients, p=proportions[user_idx])
```

```
            client_splits[client_idx].append(interaction)

    return client_splits


# Example usage
if __name__ == "__main__":
    # Example configuration
    # In practice, load MovieLens 100K data here
    # interactions = load_movielens_data("path/to/ml-100k/u.data")

    # For demonstration, create synthetic data
    num_users = 943
    num_items = 1682

    # Create synthetic interactions (replace with real data loading)
    interactions = []
    for _ in range(1000):
        user_id = np.random.randint(0, num_users)
        item_id = np.random.randint(0, num_items)
        rating = np.random.choice([1, 2, 3, 4, 5])
        interactions.append((user_id, item_id, float(rating)))

    # Create client configuration
    config = ClientConfig(
        client_id="client_1",
        server_url="http://localhost:8000",
        num_users=num_users,
        num_items=num_items,
        embedding_dim=16,
        local_epochs=1,
        learning_rate=0.01,
        batch_size=32,
        use_dp=True,  # Enable DP-SGD
        dp_sigma=1.0,
        dp_clip_norm=1.0,
        dp_delta=1e-5
    )

    # Create client with local data
    client = FederatedClient(config, interactions)

    # Register with server
    try:
        client.register()

        # Run training round
        metrics = client.run_training_round()
        logger.info(f"Training round completed: {metrics}")
    except requests.exceptions.ConnectionError:
        logger.error("Cannot connect to server. Make sure server is running on http://localhost:8000")
```

File: compare_results.py

```
"""
Compare Experiment Results

This script compares newly generated results with published results
to verify reproducibility.
"""

import json
import numpy as np
from pathlib import Path
from typing import Dict, List


def load_experiment_results(results_dir: str = "results") -> Dict:
    """Load all experiment results from JSON files."""
    results = {}
    results_path = Path(results_dir)

    if not results_path.exists():
```

```
        print(f"Results directory not found: {results_dir}")
        return results

    for json_file in results_path.glob("*.json"):
        if json_file.name == "attack_evaluation_summary.json":
            continue  # Skip attack summary for now

        with open(json_file) as f:
            data = json.load(f)
            results[json_file.stem] = data

    return results


def extract_metrics(results: Dict) -> Dict:
    """Extract key metrics from experiment results."""
    metrics = {}

    # Group by configuration
    configurations = {}

    for exp_id, data in results.items():
        if exp_id == "centralized_baseline":
            metrics["centralized"] = {
                "NDCG@10": data.get("final_metrics", {}).get("NDCG@10", 0),
                "Hit@10": data.get("final_metrics", {}).get("Hit@10", 0),
            }
            continue

        # Parse experiment ID: dp_X_alpha_Y_dim_Z_clients_N_seed_S
        # Remove the seed part to group configurations
        parts = exp_id.split("_seed_")
        if len(parts) == 2:
            config_key = parts[0]
        else:
            # Fallback: remove last part if it looks like a number
            parts = exp_id.split("_")
            if parts[-1].isdigit():
                config_key = "_".join(parts[:-2])  # Remove "seed_XXX"
            else:
                config_key = exp_id

        if config_key not in configurations:
            configurations[config_key] = []

        configurations[config_key].append({
            "NDCG@10": data.get("final_metrics", {}).get("NDCG@10", 0),
            "Hit@10": data.get("final_metrics", {}).get("Hit@10", 0),
        })

    # Compute statistics across seeds
    for config_key, runs in configurations.items():
        ndcg_values = [r["NDCG@10"] for r in runs]
        hit_values = [r["Hit@10"] for r in runs]

        metrics[config_key] = {
            "NDCG@10": {
                "mean": np.mean(ndcg_values),
                "std": np.std(ndcg_values),
                "values": ndcg_values
            },
            "Hit@10": {
                "mean": np.mean(hit_values),
                "std": np.std(hit_values),
                "values": hit_values
            },
            "num_runs": len(runs)
        }

    return metrics


def compare_metrics(published: Dict, new: Dict, tolerance: float = 0.02):
    """Compare published results with new results."""
```

```
    print("=" * 80)
    print("RESULTS COMPARISON")
    print("=" * 80)

    all_configs = set(published.keys()) | set(new.keys())

    differences = []

    for config in sorted(all_configs):
        print(f"\n{config}:")
        print("-" * 80)

        if config not in published:
            print("  ■■  NOT FOUND in published results")
            continue

        if config not in new:
            print("  ■■  NOT FOUND in new results")
            continue

        pub = published[config]
        new_res = new[config]

        # Compare NDCG@10
        if isinstance(pub, dict) and "NDCG@10" in pub:
            if isinstance(pub["NDCG@10"], dict):
                pub_ndcg = pub["NDCG@10"]["mean"]
                new_ndcg = new_res["NDCG@10"]["mean"]
                diff_ndcg = abs(pub_ndcg - new_ndcg)

                status = "■" if diff_ndcg < tolerance else "■■"
                print(f"  NDCG@10: Published={pub_ndcg:.4f}±{pub['NDCG@10']['std']:.4f}, "
                      f"New={new_ndcg:.4f}±{new_res['NDCG@10']['std']:.4f}, "
                      f"Diff={diff_ndcg:.4f} {status}")

                if diff_ndcg >= tolerance:
                    differences.append((config, "NDCG@10", diff_ndcg))
            else:
                pub_ndcg = pub["NDCG@10"]
                new_ndcg = new_res["NDCG@10"]
                diff_ndcg = abs(pub_ndcg - new_ndcg)

                status = "■" if diff_ndcg < tolerance else "■■"
                print(f"  NDCG@10: Published={pub_ndcg:.4f}, New={new_ndcg:.4f}, "
                      f"Diff={diff_ndcg:.4f} {status}")

                if diff_ndcg >= tolerance:
                    differences.append((config, "NDCG@10", diff_ndcg))

        # Compare Hit@10
        if isinstance(pub, dict) and "Hit@10" in pub:
            if isinstance(pub["Hit@10"], dict):
                pub_hit = pub["Hit@10"]["mean"]
                new_hit = new_res["Hit@10"]["mean"]
                diff_hit = abs(pub_hit - new_hit)

                status = "■" if diff_hit < tolerance else "■■"
                print(f"  Hit@10:  Published={pub_hit:.4f}±{pub['Hit@10']['std']:.4f}, "
                      f"New={new_hit:.4f}±{new_res['Hit@10']['std']:.4f}, "
                      f"Diff={diff_hit:.4f} {status}")

                if diff_hit >= tolerance:
                    differences.append((config, "Hit@10", diff_hit))
            else:
                pub_hit = pub["Hit@10"]
                new_hit = new_res["Hit@10"]
                diff_hit = abs(pub_hit - new_hit)

                status = "■" if diff_hit < tolerance else "■■"
                print(f"  Hit@10:  Published={pub_hit:.4f}, New={new_hit:.4f}, "
                      f"Diff={diff_hit:.4f} {status}")

                if diff_hit >= tolerance:
                    differences.append((config, "Hit@10", diff_hit))
```

```
    print("\n" + "=" * 80)
    print("SUMMARY")
    print("=" * 80)

    if not differences:
        print("\n■ All results match published values within tolerance!")
        print(f"   Tolerance: ±{tolerance:.4f}")
    else:
        print(f"\n■■  Found {len(differences)} significant differences:")
        for config, metric, diff in differences:
            print(f"   - {config} ({metric}): diff = {diff:.4f}")
        print(f"\n   Tolerance: ±{tolerance:.4f}")
        print("\n   Note: Some variation is expected due to:")
        print("   - Random initialization")
        print("   - Floating-point precision")
        print("   - Hardware differences")
        print("   - PyTorch version differences")


def main():
    # Define published results (from EXPERIMENT_STATUS.md)
    published_results = {
        "centralized": {
            "NDCG@10": 0.2250,
            "Hit@10": 0.3800
        },
        "dp_inf_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0539, "std": 0.0108},
            "Hit@10": {"mean": 0.0633, "std": 0.0205}
        },
        "dp_8_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0534, "std": 0.0131},
            "Hit@10": {"mean": 0.0600, "std": 0.0082}
        },
        "dp_4_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0479, "std": 0.0091},
            "Hit@10": {"mean": 0.0600, "std": 0.0082}
        },
        "dp_2_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0456, "std": 0.0095},
            "Hit@10": {"mean": 0.0567, "std": 0.0094}
        },
        "dp_1_alpha_0.5_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0467, "std": 0.0121},
            "Hit@10": {"mean": 0.0533, "std": 0.0047}
        },
        "dp_inf_alpha_0.1_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0538, "std": 0.0108},
            "Hit@10": {"mean": 0.0633, "std": 0.0205}
        },
        "dp_inf_alpha_1.0_dim_64_clients_100": {
            "NDCG@10": {"mean": 0.0539, "std": 0.0108},
            "Hit@10": {"mean": 0.0633, "std": 0.0205}
        }
    }

    # Load new results
    print("Loading new experiment results...")
    new_results_raw = load_experiment_results("results")
    new_results = extract_metrics(new_results_raw)

    print(f"Loaded {len(new_results)} configurations\n")

    # Compare
    compare_metrics(published_results, new_results, tolerance=0.02)


if __name__ == "__main__":
    main()
```

File: compile_all_code_into_txt.py

```
import os

# Root directory to scan (change if needed)
ROOT_DIR = "."

# Output file
OUTPUT_FILE = "compiled_text_output.txt"

# File extensions considered text
TEXT_EXTENSIONS = {
    ".py", ".txt", ".md", ".json", ".csv", ".yaml", ".yml",
    ".xml", ".html", ".css", ".js", ".ts",
    ".java", ".c", ".cpp", ".h", ".hpp",
    ".sh", ".bat", ".ini", ".cfg", ".conf"
}

def is_text_file(filepath):
    _, ext = os.path.splitext(filepath)
    return ext.lower() in TEXT_EXTENSIONS


def read_file(filepath):
    try:
        with open(filepath, "r", encoding="utf-8", errors="ignore") as f:
            return f.read()
    except Exception as e:
        return f"[ERROR READING FILE: {e}]"


def main():
    with open(OUTPUT_FILE, "w", encoding="utf-8") as output:

        for root, dirs, files in os.walk(ROOT_DIR):

            # Skip common unwanted folders
            dirs[:] = [d for d in dirs if d not in {
                ".git", "__pycache__", "node_modules", ".idea", ".vscode"
            }]

            for file in files:

                filepath = os.path.join(root, file)

                if is_text_file(filepath):

                    print(f"Processing: {filepath}")

                    content = read_file(filepath)

                    output.write("\n")
                    output.write("=" * 80 + "\n")
                    output.write(f"FILE: {filepath}\n")
                    output.write("=" * 80 + "\n\n")

                    output.write(content)
                    output.write("\n\n")

    print(f"\nCompilation complete.")
    print(f"Output file: {OUTPUT_FILE}")


if __name__ == "__main__":
    main()
```

File: comprehensive_analysis.py

```
"""
Comprehensive Analysis Script for All Experiments

Loads and analyzes results from:
- Centralized baseline
- DP sweep experiments
```

```
- Heterogeneity sweep experiments
- Generates publication-quality figures
"""

import json
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from pathlib import Path
import glob
from typing import Dict, List, Optional
import seaborn as sns

# Set style
try:
    plt.style.use('seaborn-v0_8-darkgrid')
except:
    try:
        plt.style.use('seaborn-darkgrid')
    except:
        plt.style.use('default')

plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['font.size'] = 12
plt.rcParams['axes.labelsize'] = 14
plt.rcParams['axes.titlesize'] = 16
plt.rcParams['xtick.labelsize'] = 12
plt.rcParams['ytick.labelsize'] = 12
plt.rcParams['legend.fontsize'] = 11


def load_baseline() -> Optional[Dict]:
    """Load centralized baseline results."""
    baseline_path = Path("results/centralized_baseline.json")
    if baseline_path.exists():
        with open(baseline_path, 'r') as f:
            return json.load(f)
    return None


def load_dp_sweep_results() -> Dict:
    """Load all DP sweep experiment results."""
    results = {}
    # Try both dim_64 and dim_16 patterns for backwards compatibility
    for dim in [64, 16]:
        pattern = f"results/dp_*_alpha_0.5_dim_{dim}_clients_100_seed_*.json"
        for filepath in glob.glob(pattern):
            with open(filepath, 'r') as f:
                data = json.load(f)
                config = data.get('config', {})
                epsilon = config.get('dp_epsilon')
                seed = config.get('seed', 42)

                if epsilon is None:
                    epsilon = float('inf')

                if epsilon not in results:
                    results[epsilon] = []

                results[epsilon].append({
                    'seed': seed,
                    'final_metrics': data.get('final_metrics', {}),
                    'rounds': data.get('rounds', [])
                })

    return results


def load_heterogeneity_results() -> Dict:
    """Load all heterogeneity sweep experiment results."""
    results = {}
    # Try both dim_64 and dim_16 patterns for backwards compatibility
```

```
    for dim in [64, 16]:
        pattern = f"results/dp_*_alpha_*_dim_{dim}_clients_100_seed_*.json"
        for filepath in glob.glob(pattern):
            with open(filepath, 'r') as f:
                data = json.load(f)
                config = data.get('config', {})
                alpha = config.get('alpha')
                epsilon = config.get('dp_epsilon')
                seed = config.get('seed', 42)

                # Only include non-DP experiments for heterogeneity analysis
                if epsilon is None or epsilon == float('inf'):
                    if alpha not in results:
                        results[alpha] = []

                    results[alpha].append({
                        'seed': seed,
                        'final_metrics': data.get('final_metrics', {}),
                        'rounds': data.get('rounds', [])
                    })

    return results


def plot_accuracy_vs_epsilon(dp_results: Dict, baseline: Optional[Dict], save_path: str = "figures/accuracy
_vs_epsilon.png"):
    """Plot accuracy metrics vs DP budget (epsilon)."""
    epsilons = sorted([e for e in dp_results.keys() if e != float('inf')] + [float('inf')])

    ndcg_values = []
    hit_values = []
    ndcg_stds = []
    hit_stds = []

    for epsilon in epsilons:
        if epsilon in dp_results:
            results = dp_results[epsilon]
            ndcg_vals = [r['final_metrics'].get('NDCG@10', 0) for r in results]
            hit_vals = [r['final_metrics'].get('Hit@10', 0) for r in results]

            ndcg_values.append(np.mean(ndcg_vals))
            hit_values.append(np.mean(hit_vals))
            ndcg_stds.append(np.std(ndcg_vals))
            hit_stds.append(np.std(hit_vals))
        else:
            ndcg_values.append(0)
            hit_values.append(0)
            ndcg_stds.append(0)
            hit_stds.append(0)

    # Convert inf to string for plotting
    epsilon_labels = [str(e) if e != float('inf') else '∞' for e in epsilons]

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    # NDCG@10
    ax1.errorbar(range(len(epsilons)), ndcg_values, yerr=ndcg_stds,
                 marker='o', linewidth=2, markersize=8, capsize=5, label='Federated')

    if baseline:
        baseline_ndcg = baseline.get('final_metrics', {}).get('NDCG@10', 0)
        ax1.axhline(y=baseline_ndcg, color='r', linestyle='--', linewidth=2, label='Centralized Baseline')

    ax1.set_xlabel('DP Budget (ε)')
    ax1.set_ylabel('NDCG@10')
    ax1.set_title('NDCG@10 vs DP Budget')
    ax1.set_xticks(range(len(epsilons)))
    ax1.set_xticklabels(epsilon_labels)
    ax1.grid(True, alpha=0.3)
    ax1.legend()

    # Hit@10
    ax2.errorbar(range(len(epsilons)), hit_values, yerr=hit_stds,
                 marker='s', linewidth=2, markersize=8, capsize=5, label='Federated', color='green')
```

```
    if baseline:
        baseline_hit = baseline.get('final_metrics', {}).get('Hit@10', 0)
        ax2.axhline(y=baseline_hit, color='r', linestyle='--', linewidth=2, label='Centralized Baseline')

    ax2.set_xlabel('DP Budget (ε)')
    ax2.set_ylabel('Hit@10')
    ax2.set_title('Hit@10 vs DP Budget')
    ax2.set_xticks(range(len(epsilons)))
    ax2.set_xticklabels(epsilon_labels)
    ax2.grid(True, alpha=0.3)
    ax2.legend()

    plt.tight_layout()
    Path(save_path).parent.mkdir(parents=True, exist_ok=True)
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    print(f"Saved: {save_path}")
    plt.close()


def plot_accuracy_vs_alpha(heterogeneity_results: Dict, save_path: str = "figures/accuracy_vs_alpha.png"):
    """Plot accuracy metrics vs data heterogeneity (alpha)."""
    alphas = sorted(heterogeneity_results.keys())

    ndcg_values = []
    hit_values = []
    ndcg_stds = []
    hit_stds = []

    for alpha in alphas:
        if alpha in heterogeneity_results:
            results = heterogeneity_results[alpha]
            ndcg_vals = [r['final_metrics'].get('NDCG@10', 0) for r in results]
            hit_vals = [r['final_metrics'].get('Hit@10', 0) for r in results]

            ndcg_values.append(np.mean(ndcg_vals))
            hit_values.append(np.mean(hit_vals))
            ndcg_stds.append(np.std(ndcg_vals))
            hit_stds.append(np.std(hit_vals))
        else:
            ndcg_values.append(0)
            hit_values.append(0)
            ndcg_stds.append(0)
            hit_stds.append(0)

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    # NDCG@10
    ax1.errorbar(range(len(alphas)), ndcg_values, yerr=ndcg_stds,
                 marker='o', linewidth=2, markersize=8, capsize=5, color='purple')
    ax1.set_xlabel('Data Heterogeneity (α)')
    ax1.set_ylabel('NDCG@10')
    ax1.set_title('NDCG@10 vs Data Heterogeneity')
    ax1.set_xticks(range(len(alphas)))
    ax1.set_xticklabels([str(a) for a in alphas])
    ax1.grid(True, alpha=0.3)

    # Hit@10
    ax2.errorbar(range(len(alphas)), hit_values, yerr=hit_stds,
                 marker='s', linewidth=2, markersize=8, capsize=5, color='orange')
    ax2.set_xlabel('Data Heterogeneity (α)')
    ax2.set_ylabel('Hit@10')
    ax2.set_title('Hit@10 vs Data Heterogeneity')
    ax2.set_xticks(range(len(alphas)))
    ax2.set_xticklabels([str(a) for a in alphas])
    ax2.grid(True, alpha=0.3)

    plt.tight_layout()
    Path(save_path).parent.mkdir(parents=True, exist_ok=True)
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    print(f"Saved: {save_path}")
    plt.close()
```

```
def plot_accuracy_loss_vs_epsilon(dp_results: Dict, baseline: Optional[Dict], save_path: str = "figures/acc
uracy_loss_vs_epsilon.png"):
    """Plot relative accuracy loss vs DP budget."""
    if not baseline:
        print("No baseline available. Skipping accuracy loss plot.")
        return

    baseline_ndcg = baseline.get('final_metrics', {}).get('NDCG@10', 0)
    baseline_hit = baseline.get('final_metrics', {}).get('Hit@10', 0)

    if baseline_ndcg == 0 or baseline_hit == 0:
        print("Baseline metrics are zero. Skipping accuracy loss plot.")
        return

    epsilons = sorted([e for e in dp_results.keys() if e != float('inf')] + [float('inf')])

    ndcg_losses = []
    hit_losses = []
    ndcg_loss_stds = []
    hit_loss_stds = []

    for epsilon in epsilons:
        if epsilon in dp_results:
            results = dp_results[epsilon]
            ndcg_vals = [r['final_metrics'].get('NDCG@10', 0) for r in results]
            hit_vals = [r['final_metrics'].get('Hit@10', 0) for r in results]

            # Calculate relative loss
            ndcg_loss_vals = [((baseline_ndcg - val) / baseline_ndcg) * 100 for val in ndcg_vals]
            hit_loss_vals = [((baseline_hit - val) / baseline_hit) * 100 for val in hit_vals]

            ndcg_losses.append(np.mean(ndcg_loss_vals))
            hit_losses.append(np.mean(hit_loss_vals))
            ndcg_loss_stds.append(np.std(ndcg_loss_vals))
            hit_loss_stds.append(np.std(hit_loss_vals))
        else:
            ndcg_losses.append(0)
            hit_losses.append(0)
            ndcg_loss_stds.append(0)
            hit_loss_stds.append(0)

    epsilon_labels = [str(e) if e != float('inf') else '∞' for e in epsilons]

    fig, ax = plt.subplots(figsize=(10, 6))

    ax.errorbar(range(len(epsilons)), ndcg_losses, yerr=ndcg_loss_stds,
                marker='o', linewidth=2, markersize=8, capsize=5, label='NDCG@10 Loss')
    ax.errorbar(range(len(epsilons)), hit_losses, yerr=hit_loss_stds,
                marker='s', linewidth=2, markersize=8, capsize=5, label='Hit@10 Loss')

    # Target threshold line
    ax.axhline(y=5, color='r', linestyle='--', linewidth=2, label='Target (≤5%)')

    ax.set_xlabel('DP Budget (ε)')
    ax.set_ylabel('Relative Accuracy Loss (%)')
    ax.set_title('Accuracy Loss vs DP Budget')
    ax.set_xticks(range(len(epsilons)))
    ax.set_xticklabels(epsilon_labels)
    ax.grid(True, alpha=0.3)
    ax.legend()

    plt.tight_layout()
    Path(save_path).parent.mkdir(parents=True, exist_ok=True)
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    print(f"Saved: {save_path}")
    plt.close()


def generate_summary_table(dp_results: Dict, heterogeneity_results: Dict, baseline: Optional[Dict], save_pa
th: str = "figures/summary_table.csv"):
    """Generate summary table of all results."""
    rows = []

    # Baseline
```

```
    if baseline:
        rows.append({
            'Experiment': 'Centralized Baseline',
            'Epsilon': 'N/A',
            'Alpha': 'N/A',
            'NDCG@10': baseline.get('final_metrics', {}).get('NDCG@10', 0),
            'Hit@10': baseline.get('final_metrics', {}).get('Hit@10', 0),
            'Accuracy': baseline.get('final_metrics', {}).get('accuracy', 0)
        })

    # DP sweep results
    for epsilon in sorted(dp_results.keys()):
        results = dp_results[epsilon]
        ndcg_vals = [r['final_metrics'].get('NDCG@10', 0) for r in results]
        hit_vals = [r['final_metrics'].get('Hit@10', 0) for r in results]
        acc_vals = [r['final_metrics'].get('accuracy', 0) for r in results]

        rows.append({
            'Experiment': 'DP Sweep',
            'Epsilon': str(epsilon) if epsilon != float('inf') else '∞',
            'Alpha': '0.5',
            'NDCG@10': f"{np.mean(ndcg_vals):.4f} ± {np.std(ndcg_vals):.4f}",
            'Hit@10': f"{np.mean(hit_vals):.4f} ± {np.std(hit_vals):.4f}",
            'Accuracy': f"{np.mean(acc_vals):.4f} ± {np.std(acc_vals):.4f}"
        })

    # Heterogeneity results
    for alpha in sorted(heterogeneity_results.keys()):
        results = heterogeneity_results[alpha]
        ndcg_vals = [r['final_metrics'].get('NDCG@10', 0) for r in results]
        hit_vals = [r['final_metrics'].get('Hit@10', 0) for r in results]
        acc_vals = [r['final_metrics'].get('accuracy', 0) for r in results]

        rows.append({
            'Experiment': 'Heterogeneity Sweep',
            'Epsilon': '∞',
            'Alpha': str(alpha),
            'NDCG@10': f"{np.mean(ndcg_vals):.4f} ± {np.std(ndcg_vals):.4f}",
            'Hit@10': f"{np.mean(hit_vals):.4f} ± {np.std(hit_vals):.4f}",
            'Accuracy': f"{np.mean(acc_vals):.4f} ± {np.std(acc_vals):.4f}"
        })

    df = pd.DataFrame(rows)
    Path(save_path).parent.mkdir(parents=True, exist_ok=True)
    df.to_csv(save_path, index=False)
    print(f"Saved: {save_path}")

    return df


def plot_convergence(dp_results: Dict, save_path: str = "figures/convergence.png"):
    """Plot training convergence curves for different DP budgets."""
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    colors = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728', '#9467bd']
    epsilons = sorted([e for e in dp_results.keys() if e != float('inf')] + [float('inf')])

    for idx, epsilon in enumerate(epsilons):
        if epsilon not in dp_results:
            continue
        results = dp_results[epsilon]
        # Average across seeds
        all_losses = []
        all_ndcg = []
        for r in results:
            losses = [rd.get('train_loss', 0) for rd in r['rounds']]
            ndcg = [rd.get('test_metrics', {}).get('NDCG@10', 0) for rd in r['rounds']]
            all_losses.append(losses)
            all_ndcg.append(ndcg)

        if not all_losses:
            continue

        max_len = max(len(l) for l in all_losses)
```

```
        avg_losses = np.mean([l + [l[-1]]*(max_len-len(l)) for l in all_losses], axis=0)
        avg_ndcg = np.mean([n + [n[-1]]*(max_len-len(n)) for n in all_ndcg], axis=0)
        rounds = list(range(1, len(avg_losses) + 1))

        label = f'ε={epsilon}' if epsilon != float('inf') else 'ε=∞ (no DP)'
        color = colors[idx % len(colors)]

        ax1.plot(rounds, avg_losses, marker='o', markersize=4, linewidth=2, label=label, color=color)
        ax2.plot(rounds, avg_ndcg, marker='s', markersize=4, linewidth=2, label=label, color=color)

    ax1.set_xlabel('Round')
    ax1.set_ylabel('Training Loss')
    ax1.set_title('Training Loss Convergence')
    ax1.grid(True, alpha=0.3)
    ax1.legend()

    ax2.set_xlabel('Round')
    ax2.set_ylabel('NDCG@10')
    ax2.set_title('NDCG@10 Convergence')
    ax2.grid(True, alpha=0.3)
    ax2.legend()

    plt.tight_layout()
    Path(save_path).parent.mkdir(parents=True, exist_ok=True)
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    print(f"Saved: {save_path}")
    plt.close()


def plot_attack_results(save_path: str = "figures/attack_evaluation.png"):
    """Plot attack evaluation results across DP budgets."""
    attack_path = Path("results/attack_evaluation_summary.json")
    if not attack_path.exists():
        print("No attack evaluation results found. Skipping attack plot.")
        return

    with open(attack_path) as f:
        attack_data = json.load(f)

    if not attack_data:
        return

    epsilons = []
    mia_aucs = []
    mia_accs = []
    inv_accs = []

    for eps_str in sorted(attack_data.keys(), key=lambda x: float(x) if x != 'inf' else float('inf')):
        epsilons.append('∞' if eps_str == 'inf' else eps_str)
        mia = attack_data[eps_str].get('mia', {})
        inv = attack_data[eps_str].get('inversion', {})
        mia_aucs.append(mia.get('auc', 0))
        mia_accs.append(mia.get('accuracy', 0))
        inv_accs.append(inv.get('top_k_accuracy', 0))

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    x = range(len(epsilons))
    ax1.bar([i - 0.2 for i in x], mia_aucs, 0.4, label='MIA AUC', color='#d62728', alpha=0.8)
    ax1.bar([i + 0.2 for i in x], mia_accs, 0.4, label='MIA Accuracy', color='#ff7f0e', alpha=0.8)
    ax1.axhline(y=0.5, color='gray', linestyle='--', linewidth=1, label='Random Baseline')
    ax1.set_xlabel('DP Budget (ε)')
    ax1.set_ylabel('Score')
    ax1.set_title('Membership Inference Attack')
    ax1.set_xticks(list(x))
    ax1.set_xticklabels(epsilons)
    ax1.legend()
    ax1.grid(True, alpha=0.3, axis='y')

    ax2.bar(x, inv_accs, 0.5, color='#9467bd', alpha=0.8)
    ax2.set_xlabel('DP Budget (ε)')
    ax2.set_ylabel('Top-K Accuracy')
    ax2.set_title('Model Inversion Attack')
    ax2.set_xticks(list(x))
```

```
    ax2.set_xticklabels(epsilons)
    ax2.grid(True, alpha=0.3, axis='y')

    plt.tight_layout()
    Path(save_path).parent.mkdir(parents=True, exist_ok=True)
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    print(f"Saved: {save_path}")
    plt.close()


def main():
    """Main analysis function"""
    print("="*60)
    print("Comprehensive Experiment Analysis")
    print("="*60)

    # Load results
    print("\nLoading results...")
    baseline = load_baseline()
    dp_results = load_dp_sweep_results()
    heterogeneity_results = load_heterogeneity_results()

    print(f"  Baseline: {'Found' if baseline else 'Not found'}")
    print(f"  DP sweep experiments: {len(dp_results)} epsilon values")
    print(f"  Heterogeneity experiments: {len(heterogeneity_results)} alpha values")

    # Generate plots
    print("\nGenerating plots...")

    if dp_results:
        plot_accuracy_vs_epsilon(dp_results, baseline)
        plot_accuracy_loss_vs_epsilon(dp_results, baseline)
        plot_convergence(dp_results)

    if heterogeneity_results:
        plot_accuracy_vs_alpha(heterogeneity_results)

    # Attack evaluation plot
    plot_attack_results()

    # Generate summary table
    print("\nGenerating summary table...")
    summary_df = generate_summary_table(dp_results, heterogeneity_results, baseline)

    print("\n" + "="*60)
    print("Analysis complete!")
    print("="*60)
    print("\nGenerated files:")
    print("  - figures/accuracy_vs_epsilon.png")
    print("  - figures/accuracy_loss_vs_epsilon.png")
    print("  - figures/accuracy_vs_alpha.png")
    print("  - figures/convergence.png")
    print("  - figures/attack_evaluation.png")
    print("  - figures/summary_table.csv")
    print("\nSummary Table:")
    print(summary_df.to_string(index=False))


if __name__ == "__main__":
    main()
```

File: convert_all_dart_files_into_pdf.py

```
import os

def create_txt_from_content(combined_content, txt_file_path):
    """Creates a text file from the given content.

    Args:
        combined_content (str): The combined text content of the Dart files.
        txt_file_path (str): Absolute path where the text file should be saved.
    """
    try:
```

```
        # Create the text file
        with open(txt_file_path, 'w', encoding='utf-8') as f:
            f.write(combined_content)

        print(f"Successfully created text file: '{os.path.basename(txt_file_path)}'")

    except Exception as e:
        print(f"Error creating text file '{os.path.basename(txt_file_path)}': {e}")

def main(lib_directory_path, output_path):
    """Walks through the lib directory, concatenates Dart files, and creates a single text file.

    Args:
        lib_directory_path (str): Absolute path to the 'lib' directory.
        output_path (str): Absolute path to the folder where the text file should be saved.
    """
    if not os.path.exists(output_path):
        os.makedirs(output_path)

    combined_dart_code = ""
    dart_files_found = False

    for root, _, files in os.walk(lib_directory_path):
        for file in files:
            if file.endswith('.dart') or file.endswith('.php') or file.endswith('.md'):
                dart_file_path = os.path.join(root, file)
                try:
                    with open(dart_file_path, 'r', encoding='utf-8') as f:
                        dart_code = f.read()
                        combined_dart_code += f"```\nContent of {os.path.relpath(dart_file_path, lib_direct
ory_path)}\n```\n"
                        combined_dart_code += dart_code + "\n\n"  # Add separator
                        dart_files_found = True
                except Exception as e:
                    print(f"Error reading '{dart_file_path}': {e}")

    if dart_files_found:
        txt_file_name = input("Enter the name of the text file (without extension): ")
        txt_file_path = os.path.join(output_path, f"{txt_file_name}.txt")
        create_txt_from_content(combined_dart_code, txt_file_path)
    else:
        print(f"No Dart files found in '{lib_directory_path}' or its subdirectories.")

if __name__ == "__main__":
    lib_directory = input("Enter the absolute path to the 'lib' directory: ")
    text_output_directory = input("Enter the absolute path to the folder where you want to save the text fi
le: ")

    if not os.path.isdir(lib_directory):
        print(f"Error: '{lib_directory}' is not a valid directory.")
    elif not os.path.isdir(text_output_directory):
        print(f"Error: '{text_output_directory}' is not a valid directory.")
    else:
        main(lib_directory, text_output_directory)
```

File: convert_results.py

```
from pdf2docx import Converter
from docx import Document
import os

def convert_results():
    pdf_file = "Results_Section.pdf"
    docx_file = "Results_Section.docx"

    print(f"Converting {pdf_file} to {docx_file}...")
    try:
        cv = Converter(pdf_file)
        cv.convert(docx_file, start=0, end=None)
        cv.close()
        print("Conversion successful.")

        # Apply Arial Font
```

```
        print("Applying Arial font...")
        doc = Document(docx_file)
        style = doc.styles['Normal']
        font = style.font
        font.name = 'Arial'

        for p in doc.paragraphs:
            for r in p.runs:
                r.font.name = 'Arial'

        doc.save(docx_file)
        print(f"Saved {docx_file} with Arial font.")

    except Exception as e:
        print(f"Error: {e}")

if __name__ == "__main__":
    convert_results()
```

File: create_title_page.py

```
from docx import Document
from docx.shared import Pt, Cm
from docx.enum.text import WD_ALIGN_PARAGRAPH

def set_font(run, font_name, font_size, bold=False):
    run.font.name = font_name
    run.font.size = Pt(font_size)
    run.bold = bold

def create_title_page():
    doc = Document()

    # Set margins to 2cm as per guidelines
    section = doc.sections[0]
    section.top_margin = Cm(2)
    section.bottom_margin = Cm(2)
    section.left_margin = Cm(2)
    section.right_margin = Cm(2)

    # Helper to add empty lines
    def add_empty_lines(count):
        for _ in range(count):
            doc.add_paragraph()

    # --- Header / Top Section ---
    # "EXAMINATION OFFICE" / "IU.ORG" based on the guideline visual
    # usually top left or right, let's put it top left as plain text for now or standard layout
    # Guidelines text: "page 1 of 5 EXAMINATION OFFICE IU.ORG" seems to be header content.
    # We will simulate a clean professional header.

    # p = doc.add_paragraph()
    # run = p.add_run("EXAMINATION OFFICE\nIU.ORG")
    # set_font(run, "Arial", 11)
    # p.alignment = WD_ALIGN_PARAGRAPH.RIGHT

    add_empty_lines(15) # Adjust spacing to push title to center-ish

    # --- Thesis Type ---
    p = doc.add_paragraph()
    run = p.add_run("MASTER THESIS")
    set_font(run, "Arial", 16, bold=True)
    p.alignment = WD_ALIGN_PARAGRAPH.CENTER

    add_empty_lines(2)

    # --- Title ---
    title_text = "Empirical Analysis of Accuracy–Privacy Trade-offs in Federated Learning for Mobile Movie 
Recommendation Systems"
    p = doc.add_paragraph()
    run = p.add_run(title_text)
    set_font(run, "Arial", 20, bold=True)
    p.alignment = WD_ALIGN_PARAGRAPH.CENTER
```

```
    add_empty_lines(8)

    # --- Course Details ---
    # Usually in a block

    details = [
        ("Course of Study:", "M.Sc. in Artificial Intelligence (120 ECTS)"),
        ("Course Name:", "Master Thesis"), # Using 'Master Thesis' as course name if not specified otherwis
e, but user said 'M.Sc...' is Course Name? User input: "Course Name is M.Sc. in Artificial Intelligence (12
0 ECTS)". Wait, usually Course Name is the module (e.g. 'Master Thesis') and Course of Study is the degree.
 User put the same? Re-reading: "Course Name is M.Sc. in Artificial Intelligence". Guidelines say: "course
name, course of study". I will put specific inputs.
        # User input:
        # Course Name: M.Sc. in Artificial Intelligence (120 ECTS)
        # Author: Jonathan John
        # Matriculation: 92126351
        # Tutor: Duong-Trung, Nghia, Dr.
        # Date: 2026-02-18
    ]

    # Using a table for alignment is often cleaner for the details section
    table = doc.add_table(rows=0, cols=2)
    table.autofit = False
    # Set column widths?
    # Let's just use paragraphs for simplicity and standard look

    def add_detail_line(label, value):
        p = doc.add_paragraph()
        run_label = p.add_run(f"{label} ")
        set_font(run_label, "Arial", 12, bold=True)
        if value:
            run_val = p.add_run(value)
            set_font(run_val, "Arial", 12)
        p.alignment = WD_ALIGN_PARAGRAPH.LEFT

    # Course Name according to user input
    add_detail_line("Course of Study:", "M.Sc. in Artificial Intelligence (120 ECTS)")
    # add_detail_line("Course Name:", "Master Thesis") # Often explicit? Let's assume standard if not provi
ded separately or implicit. User said "Course Name IS M.Sc...". I will follow user input strictly?
    # Actually, guidelines say "course name, course of study". User gave "Course Name is M.Sc...". I will l
abel it 'Course:'

    # add_detail_line("Course:", "M.Sc. in Artificial Intelligence (120 ECTS)")
    # Wait, usually it is:
    # Course of Study: M.Sc. Artificial Intelligence
    # Course: Master Thesis (or similar module name)
    # User input might have combined them. I will list what was provided.

    # Let's try to infer standard structure:
    # Author
    add_detail_line("Author:", "Jonathan John")
    add_detail_line("Matriculation Number:", "92126351")

    add_empty_lines(1)

    add_detail_line("Tutor:", "Duong-Trung, Nghia, Dr.")

    add_empty_lines(1)

    add_detail_line("Date:", "2026-02-18")

    doc.save("Title_Page.docx")
    print("Title_Page.docx created successfully.")

if __name__ == "__main__":
    create_title_page()
```

File: dp_sweep_experiment.py

```
"""
Differential Privacy Budget Sweep Experiment

Runs federated learning experiments with different DP budgets (ε ∈ {∞, 8, 4, 2, 1})
```

```
to answer RQ1: How does model accuracy degrade across DP budgets?
"""

import requests
import time
import sys
import os
from client import (
    FederatedClient,
    ClientConfig,
    create_non_iid_split,
    load_ratings_csv,
    split_train_test,
    evaluate_model,
    create_matrix_factorization_model,
    ModelParamsEncoder
)
import torch
import numpy as np
from scripts.metrics_collector import MetricsCollector, create_experiment_id
from scripts.recommendation_metrics import evaluate_recommendations_simple
from scripts.rdp_accountant import compute_sigma_for_target_epsilon, compute_epsilon_for_sigma
import json
from pathlib import Path


SERVER_URL = "http://localhost:8000"
MAX_WAIT_TIME = 30


def wait_for_server(url: str, max_wait: int = MAX_WAIT_TIME) -> bool:
    """Wait for the server to become available."""
    print(f"Waiting for server at {url}...")
    start_time = time.time()

    while time.time() - start_time < max_wait:
        try:
            response = requests.get(f"{url}/healthz", timeout=2)
            if response.status_code == 200:
                print(f"[OK] Server is available!")
                return True
        except (requests.exceptions.ConnectionError, requests.exceptions.Timeout):
            pass

        print(".", end="", flush=True)
        time.sleep(1)

    print(f"\n[ERROR] Server not available after {max_wait} seconds")
    return False


def initialize_model(url: str, num_users: int, num_items: int, embedding_dim: int):
    """Initialize the server model."""
    print("Initializing server model...")
    try:
        response = requests.post(
            f"{url}/init-model",
            params={
                "num_users": num_users,
                "num_items": num_items,
                "embedding_dim": embedding_dim
            },
            timeout=10
        )
        response.raise_for_status()
        print(f"[OK] Model initialized: {response.json()}")
        return True
    except requests.exceptions.RequestException as e:
        print(f"[ERROR] Failed to initialize model: {e}")
        return False


def load_centralized_baseline():
    """Load centralized baseline metrics for comparison."""
```

```
    baseline_path = Path("results/centralized_baseline.json")
    if baseline_path.exists():
        with open(baseline_path, 'r') as f:
            baseline_data = json.load(f)
            return baseline_data.get('final_metrics', {})
    else:
        print("[WARNING] Centralized baseline not found. Run centralized_baseline.py first.")
        return {}


def run_dp_experiment(epsilon: float,
                     train_interactions,
                     test_interactions,
                     num_users,
                     num_items,
                     embedding_dim,
                     num_clients,
                     alpha,
                     num_rounds,
                     local_epochs,
                     learning_rate,
                     batch_size,
                     seed):
    """
    Run a single federated learning experiment with specified DP budget.

    Args:
        epsilon: Target DP budget (use float('inf') for no DP)
        ... other parameters ...

    Returns:
        Experiment results dictionary
    """
    print(f"\n{'='*60}")
    print(f"Running experiment with ε = {epsilon}")
    print(f"{'='*60}")

    # Calculate DP parameters
    use_dp = epsilon != float('inf')
    dp_sigma = 0.0
    dp_clip_norm = 1.0

    if use_dp:
        # Estimate average samples per client
        avg_samples = len(train_interactions) // num_clients

        # Find sigma for target epsilon
        dp_sigma = compute_sigma_for_target_epsilon(
            target_epsilon=epsilon,
            num_rounds=num_rounds,
            samples_per_client=avg_samples,
            batch_size=batch_size,
            clip_norm=dp_clip_norm
        )

        if dp_sigma is None:
            print(f"[WARNING] Could not find sigma for ε={epsilon}. Using default sigma=1.0")
            dp_sigma = 1.0
        else:
            print(f"  Computed sigma={dp_sigma:.4f} for target ε={epsilon}")
    else:
        print("  No DP (ε=∞)")

    # Create experiment ID
    experiment_id = create_experiment_id(
        dp_epsilon=epsilon if use_dp else None,
        alpha=alpha,
        embedding_dim=embedding_dim,
        num_clients=num_clients,
        seed=seed
    )

    # Initialize metrics collector
    collector = MetricsCollector(experiment_id=experiment_id, results_dir="results")
```

```
    experiment_config = {
        "num_users": num_users,
        "num_items": num_items,
        "embedding_dim": embedding_dim,
        "num_clients": num_clients,
        "alpha": alpha,
        "dp_epsilon": epsilon if use_dp else None,
        "use_dp": use_dp,
        "dp_sigma": dp_sigma,
        "dp_clip_norm": dp_clip_norm,
        "num_rounds": num_rounds,
        "local_epochs": local_epochs,
        "learning_rate": learning_rate,
        "batch_size": batch_size,
        "seed": seed
    }
    collector.set_config(experiment_config)

    # Create non-IID data splits
    print(f"\nSplitting data across {num_clients} clients (α={alpha})...")
    client_data_splits = create_non_iid_split(train_interactions, num_clients, alpha=alpha)

    # Run federated learning rounds
    for round_num in range(num_rounds):
        print(f"\n--- Round {round_num + 1}/{num_rounds} ---")

        clients = []
        client_metrics_list = []
        total_train_loss = 0.0

        # Train all clients
        for client_id in range(num_clients):
            config = ClientConfig(
                client_id=f"client_{client_id}",
                server_url=SERVER_URL,
                num_users=num_users,
                num_items=num_items,
                embedding_dim=embedding_dim,
                local_epochs=local_epochs,
                learning_rate=learning_rate,
                batch_size=batch_size,
                use_dp=use_dp,
                dp_sigma=dp_sigma,
                dp_clip_norm=dp_clip_norm,
                dp_delta=1e-5
            )

            client = FederatedClient(config, client_data_splits[client_id])
            clients.append(client)

            try:
                client.register()
                metrics = client.run_training_round()

                if client_id < 3 or (client_id + 1) % 20 == 0:
                    print(f"  Client {client_id}: loss={metrics.get('loss', 0):.4f}")

                client_metrics_list.append({
                    "client_id": f"client_{client_id}",
                    "loss": metrics.get('loss', 0.0),
                    "samples": metrics.get('samples', 0),
                    "epochs": metrics.get('epochs', 0)
                })

                total_train_loss += metrics.get('loss', 0.0)

            except Exception as e:
                print(f"  [ERROR] Client {client_id} error: {e}")

        avg_train_loss = total_train_loss / len(client_metrics_list) if client_metrics_list else 0.0

        # Aggregate
        print("Aggregating parameters...")
```

```
        try:
            response = requests.post(f"{SERVER_URL}/aggregate", timeout=120)
            response.raise_for_status()
            aggregation_info = response.json()
        except requests.exceptions.RequestException as e:
            print(f"[ERROR] Aggregation failed: {e}")
            aggregation_info = {}

        # Evaluate on test set
        print("Evaluating model...")
        try:
            test_model = create_matrix_factorization_model(num_users, num_items, embedding_dim=embedding_di
m)
            encoder = ModelParamsEncoder()

            response = requests.get(f"{SERVER_URL}/global-params-json")
            response.raise_for_status()
            data = response.json()
            params = data["params"]
            test_model = encoder.json_params_to_model(params, test_model)

            basic_metrics = evaluate_model(test_model, test_interactions)
            rec_metrics = evaluate_recommendations_simple(
                test_model, test_interactions, num_users, num_items, k=10
            )

            test_metrics = {**basic_metrics, **rec_metrics}

            print(f"  NDCG@10: {test_metrics.get('NDCG@10', 0):.4f}, "
                  f"Hit@10: {test_metrics.get('Hit@10', 0):.4f}")

            collector.add_round_metrics(
                round_num=round_num + 1,
                train_loss=avg_train_loss,
                test_metrics=test_metrics,
                aggregation_info=aggregation_info,
                client_metrics=client_metrics_list
            )

        except Exception as e:
            print(f"[ERROR] Evaluation failed: {e}")
            collector.add_round_metrics(
                round_num=round_num + 1,
                train_loss=avg_train_loss,
                test_metrics={},
                aggregation_info=aggregation_info,
                client_metrics=client_metrics_list
            )

        time.sleep(1)

    # Final evaluation
    print("\nFinal evaluation...")
    try:
        test_model = create_matrix_factorization_model(num_users, num_items, embedding_dim=embedding_dim)
        encoder = ModelParamsEncoder()

        response = requests.get(f"{SERVER_URL}/global-params-json")
        response.raise_for_status()
        data = response.json()
        params = data["params"]
        test_model = encoder.json_params_to_model(params, test_model)

        basic_metrics = evaluate_model(test_model, test_interactions)
        rec_metrics = evaluate_recommendations_simple(
            test_model, test_interactions, num_users, num_items, k=10
        )

        final_metrics = {**basic_metrics, **rec_metrics}
        collector.add_final_metrics(final_metrics)

    except Exception as e:
        print(f"[ERROR] Final evaluation failed: {e}")
        collector.add_final_metrics({})
```

```
    # Save results
    json_path = collector.save_json()
    csv_path = collector.save_csv_summary()

    print(f"\nResults saved:")
    print(f"  JSON: {json_path}")
    print(f"  CSV:  {csv_path}")

    # Compare with baseline
    baseline_metrics = load_centralized_baseline()
    if baseline_metrics:
        final_metrics = collector.get_summary()
        baseline_ndcg = baseline_metrics.get('NDCG@10', 0)
        baseline_hit = baseline_metrics.get('Hit@10', 0)

        federated_ndcg = final_metrics.get('final_ndcg@10', 0)
        federated_hit = final_metrics.get('final_hit@10', 0)

        if baseline_ndcg > 0:
            ndcg_loss = ((baseline_ndcg - federated_ndcg) / baseline_ndcg) * 100
            hit_loss = ((baseline_hit - federated_hit) / baseline_hit) * 100

            print(f"\nComparison with centralized baseline:")
            print(f"  NDCG@10: Baseline={baseline_ndcg:.4f}, Federated={federated_ndcg:.4f}, Loss={ndcg_los
s:.2f}%")
            print(f"  Hit@10: Baseline={baseline_hit:.4f}, Federated={federated_hit:.4f}, Loss={hit_loss:.2
f}%")
            print(f"  Target: ≤5% accuracy loss")
            print(f"  Status: {'✓ Meets target' if ndcg_loss <= 5 and hit_loss <= 5 else '✗ Exceeds target'
}")

    return collector.get_summary()


def main():
    """Main function to run DP sweep experiments"""

    # Check server
    if not wait_for_server(SERVER_URL):
        print("\nPlease start the server first: python server.py")
        sys.exit(1)

    # Load data
    csv_path = "ratings.csv"
    if not os.path.exists(csv_path):
        print(f"[ERROR] Dataset file not found: {csv_path}")
        sys.exit(1)

    print("Loading MovieLens 100K dataset...")
    all_interactions, num_users, num_items, user_id_map, item_id_map = load_ratings_csv(
        csv_path, binarize=False  # Use original ratings (1-5) for regression
    )

    train_interactions, test_interactions = split_train_test(all_interactions, test_ratio=0.2)

    # Initialize server model
    embedding_dim = 64  # Increased from 16 to 64 for better item representation
    if not initialize_model(SERVER_URL, num_users, num_items, embedding_dim):
        sys.exit(1)

    # Experiment configuration
    DP_EPSILONS = [float('inf'), 8, 4, 2, 1]  # As per expose
    num_clients = 100
    alpha = 0.5
    num_rounds = 10
    local_epochs = 3  # Increased from 1 to 3 for better local convergence
    learning_rate = 0.01
    batch_size = 32
    seeds = [42, 123, 456]  # 3 seeds for statistical significance

    print(f"\n{'='*60}")
    print("DP Budget Sweep Experiment")
    print(f"{'='*60}")
```

```
    print(f"DP Budgets: {DP_EPSILONS}")
    print(f"Seeds: {seeds}")
    print(f"Total experiments: {len(DP_EPSILONS)} × {len(seeds)} = {len(DP_EPSILONS) * len(seeds)}")
    print(f"{'='*60}")

    # Run experiments
    all_results = {}

    for epsilon in DP_EPSILONS:
        epsilon_results = []

        for seed in seeds:
            # Reset server model for each experiment
            if not initialize_model(SERVER_URL, num_users, num_items, embedding_dim):
                print(f"[ERROR] Failed to reset model. Skipping ε={epsilon}, seed={seed}")
                continue

            result = run_dp_experiment(
                epsilon=epsilon,
                train_interactions=train_interactions,
                test_interactions=test_interactions,
                num_users=num_users,
                num_items=num_items,
                embedding_dim=embedding_dim,
                num_clients=num_clients,
                alpha=alpha,
                num_rounds=num_rounds,
                local_epochs=local_epochs,
                learning_rate=learning_rate,
                batch_size=batch_size,
                seed=seed
            )

            epsilon_results.append(result)
            time.sleep(2)  # Brief pause between experiments

        all_results[epsilon] = epsilon_results

    # Summary
    print(f"\n{'='*60}")
    print("DP Sweep Summary")
    print(f"{'='*60}")

    for epsilon in DP_EPSILONS:
        if epsilon in all_results:
            results = all_results[epsilon]
            ndcg_values = [r.get('final_ndcg@10', 0) for r in results]
            hit_values = [r.get('final_hit@10', 0) for r in results]

            avg_ndcg = np.mean(ndcg_values)
            avg_hit = np.mean(hit_values)
            std_ndcg = np.std(ndcg_values)
            std_hit = np.std(hit_values)

            print(f"\nε = {epsilon}:")
            print(f"  NDCG@10: {avg_ndcg:.4f} ± {std_ndcg:.4f}")
            print(f"  Hit@10: {avg_hit:.4f} ± {std_hit:.4f}")

    print(f"\n{'='*60}")
    print("All experiments completed!")


if __name__ == "__main__":
    main()
```

File: dump_pdf_text.py

```
import pypdf

def dump_text(pdf_path, output_path):
    print(f"Dumping text from {pdf_path} to {output_path}...")
    try:
```

```
        reader = pypdf.PdfReader(pdf_path)
        full_text = ""
        for i, page in enumerate(reader.pages):
            full_text += f"\n--- Page {i+1} ---\n"
            full_text += page.extract_text() + "\n"

        with open(output_path, "w") as f:
            f.write(full_text)

    except Exception as e:
        print(f"Error reading PDF: {e}")

if __name__ == "__main__":
    dump_text("Guidelines_Written Assignment.pdf", "pdf_content.txt")
```

File: extract_citations.py

```
import re

def extract_citations_with_context(file_path):
    with open(file_path, 'r', encoding='utf-8') as f:
        text = f.read()

    # Regex for citations like (Author, Year) or (Author et al., Year)
    # Handling multiple citations in one block: (Author, Year; Author2, Year)
    citation_pattern = r'\((?:[A-Za-z\s\&\.\-]+,\s\d{4}[a-z]?(?:;\s)?)+\)'

    # Split by paragraphs
    paragraphs = text.split('\n')

    citations_found = []

    for i, para in enumerate(paragraphs):
        if not para.strip():
            continue

        matches = re.findall(citation_pattern, para)
        if matches:
            for match in matches:
                # Clean up the citation string
                clean_match = match.strip('()')
                # Split multiple citations
                individual_citations = [c.strip() for c in clean_match.split(';')]

                for cit in individual_citations:
                    citations_found.append({
                        "citation": cit,
                        "paragraph_sample": para[:200] + "...", # First 200 chars for context
                        "full_paragraph": para
                    })

    # Deduplicate based on citation string, but keep one context
    unique_citations = {}
    for item in citations_found:
        if item['citation'] not in unique_citations:
            unique_citations[item['citation']] = item

    return list(unique_citations.values())

if __name__ == "__main__":
    results = extract_citations_with_context("thesis_content.txt")
    print(f"Found {len(results)} unique citations.")

    # Print a few samples to check
    for i, res in enumerate(results[:5]):
        print(f"--- Citation {i+1} ---")
        print(f"Cit: {res['citation']}")
        print(f"Context: {res['paragraph_sample']}")
        print("-" * 20)
```

File: extract_pdf_rules.py

```
import pypdf
import re

def extract_rules(pdf_path):
    print(f"Extracting text from {pdf_path}...")
    try:
        reader = pypdf.PdfReader(pdf_path)
        full_text = ""
        for page in reader.pages:
            full_text += page.extract_text() + "\n"

        # Search for keywords
        keywords = ["caption", "figure", "diagram", "image", "label", "table"]

        print("\n--- Usage of Keywords ---")
        lines = full_text.split('\n')
        for i, line in enumerate(lines):
            if any(k in line.lower() for k in keywords):
                # Print context
                print(f"[{i}] {line.strip()}")

    except Exception as e:
        print(f"Error reading PDF: {e}")

if __name__ == "__main__":
    extract_rules("Guidelines_Written Assignment.pdf")
```

File: extract_text_for_analysis.py

```
import os
import sys

def extract_pdf(pdf_path, output_path):
    text = ""
    try:
        import pypdf
        reader = pypdf.PdfReader(pdf_path)
        for page in reader.pages:
            text += page.extract_text() + "\n"
    except ImportError:
        try:
            import PyPDF2
            reader = PyPDF2.PdfReader(pdf_path)
            for page in reader.pages:
                text += page.extract_text() + "\n"
        except ImportError:
            print(f"Error: Neither pypdf nor PyPDF2 is installed.")
            return False
    except Exception as e:
        print(f"Error reading PDF: {e}")
        return False

    with open(output_path, "w", encoding="utf-8") as f:
        f.write(text)
    print(f"Successfully extracted PDF to {output_path}")
    return True

def extract_docx(docx_path, output_path):
    try:
        from docx import Document
        document = Document(docx_path)
        text = "\n".join([para.text for para in document.paragraphs])
    except ImportError:
        print("Error: python-docx is not installed.")
        return False
    except Exception as e:
        print(f"Error reading DOCX: {e}")
        return False

    with open(output_path, "w", encoding="utf-8") as f:
```

```
        f.write(text)
    print(f"Successfully extracted DOCX to {output_path}")
    return True

if __name__ == "__main__":
    base_dir = "/Users/jonathanjohn/StudioProjects/master_thesis"
    pdf_file = "Guidelines_Written Assignment.pdf"
    docx_file = "Empirical Analysis of Accuracy.docx"

    pdf_path = os.path.join(base_dir, pdf_file)
    docx_path = os.path.join(base_dir, docx_file)

    pdf_out = "guidelines_content.txt"
    docx_out = "thesis_content.txt"

    print(f"Processing {pdf_path}...")
    extract_pdf(pdf_path, pdf_out)

    print(f"Processing {docx_path}...")
    extract_docx(docx_path, docx_out)
```

File: find_clusters.py

```
import json
from collections import defaultdict

def find_citation_clusters(json_path):
    try:
        with open(json_path, 'r', encoding='utf-8') as f:
            data = json.load(f)
    except FileNotFoundError:
        print(f"Error: File not found at {json_path}")
        return

    # Group citations by context
    context_clusters = defaultdict(list)
    for entry in data:
        context = entry.get('context', '').strip()
        if context:
            context_clusters[context].append(entry)

    print(f"Found {len(context_clusters)} unique contexts.")

    # Filter for clusters with many citations (likely stuffing)
    citation_stuffing_threshold = 5

    stuffed_contexts = []
    for context, items in context_clusters.items():
        if len(items) >= citation_stuffing_threshold:
            stuffed_contexts.append({
                'context': context,
                'count': len(items),
                'citations': [f"{item.get('author', ['Unknown'])[0]} ({item.get('year', '????')})" for item
 in items]
            })

    # Sort by count descending
    stuffed_contexts.sort(key=lambda x: x['count'], reverse=True)

    print(f"\nFound {len(stuffed_contexts)} paragraphs with >= {citation_stuffing_threshold} citations.\n")

    for i, item in enumerate(stuffed_contexts, 1):
        print(f"--- Block {i} (Count: {item['count']}) ---")
        # Print first 200 chars of context to identify the paragraph
        print(f"Paragraph Start: {item['context'][:200]}...")
        print(f"Citations: {', '.join(item['citations'][:5])} ... and {len(item['citations'])-5} more")
        print("\n")

if __name__ == "__main__":
    find_citation_clusters(r'c:\Users\jonat\OneDrive\Documents\GitHub\master_thesis\citations_to_verify.jso
n')
```

File: fix_captions.py

```
import docx
import re
from docx.oxml import OxmlElement
from docx.text.paragraph import Paragraph

def delete_paragraph(paragraph):
    p = paragraph._element
    p.getparent().remove(p)
    p._p = p._element = None

def fix_captions(doc_path, output_path):
    doc = docx.Document(doc_path)

    # Step 1: Clean up dragging captions at the end
    # We suspect the last ~30 paragraphs might contain our accidentally added "Figure X"
    # We iterate backwards
    print("Cleaning up misplaced captions at the end...")
    paras = doc.paragraphs
    removed_count = 0
    for i in range(len(paras) - 1, len(paras) - 50, -1):
        if i < 0: break
        p = paras[i]
        # Match exactly "Figure \d+" with no punctuation (our script generated "Figure X")
        if re.match(r"^Figure\s+\d+$", p.text.strip()):
            print(f"Removing misplaced: '{p.text}'")
            delete_paragraph(p)
            removed_count += 1

    print(f"Removed {removed_count} misplaced captions.")

    # Step 2: Proper Captioning
    # We need to identify images again.
    # We use list(doc.paragraphs) to get a snapshot of paragraphs to iterate
    # But we need to be careful: if we removed paragraphs, the list might be stale?
    # No, we removed from end. The top part is stable.
    # We will re-read doc.paragraphs to be safe.

    all_paras = list(doc.paragraphs)

    image_count = 0

    for i, p in enumerate(all_paras):
        # 1. Check for image
        has_image = False
        if 'w:drawing' in p._element.xml:
            has_image = True

        if has_image:
            image_count += 1
            expected_caption = f"Figure {image_count}"

            # Check if next paragraph is ALREADY a caption
            # We need to find the current valid next sibling in the live document
            # all_paras[i+1] is the *original* next paragraph.
            # If we haven't inserted anything yet, it is still the next paragraph.

            if i + 1 >= len(all_paras):
                # Image is at end of doc?
                # We should append caption.
                doc.add_paragraph(expected_caption)
                print(f"Appended {expected_caption} at end.")
                continue

            next_p = all_paras[i+1]

            # Check if it looks like a caption (Figure \d+)
            # Note: We already renamed existing captions to "Figure X"
            # So "Figure 8" should be there.

            if re.match(r"^Figure\s+\d+", next_p.text.strip()):
                # It is a caption.
```

```
                # Check if number matches.
                # If we have "Figure 8" and we expect "Figure 8", good.
                # If we have "Figure 8" and we expect something else?
                # The count should align if our first pass was sequential.
                # First pass said "Found 27 images".
                pass
            else:
                # No caption found. This is where we failed last time.
                print(f"Inserting {expected_caption} after image {image_count}")
                next_p.insert_paragraph_before(expected_caption)

                # Add text reference to next_p (which is now after the caption)
                # Adjust wording: " (refer to Figure X)"
                # Skip if next_p contains an image (don't add text to an image paragraph)
                if next_p.text.strip() and 'w:drawing' not in next_p._element.xml:
                     next_p.add_run(f" (refer to {expected_caption})")

    doc.save(output_path)
    print(f"Saved fixed document to {output_path}")

if __name__ == "__main__":
    fix_captions("Empirical Analysis of Accuracy.docx", "Empirical Analysis of Accuracy.docx")
```

File: fix_references.py

```
import docx
import re

def fix_references(doc_path, output_path):
    doc = docx.Document(doc_path)

    # Identify protected paragraphs (Captions)
    # We scanned 27 images.
    # We expect captions at specific places.
    # Heuristic: Paragraphs that start with "Figure \d+" AND are typically adjacent to images
    # OR the ones we know are captions.
    # Better: Scan for images. The very next paragraph (or the one after) checking for "Figure X" matches.

    protected_indices = set()
    protected_texts = []

    # Map Image Index -> Caption Paragraph Index
    img_idx = 0
    for i, p in enumerate(doc.paragraphs):
        if 'w:drawing' in p._element.xml:
            img_idx += 1
            # Check next few paragraphs for caption
            # We inserted them immediately after.
            # So usually i+1
            if i + 1 < len(doc.paragraphs):
                next_p = doc.paragraphs[i+1]
                if re.match(r"^Figure\s+\d+", next_p.text.strip()):
                    protected_indices.add(i+1)
                    protected_texts.append(next_p.text[:30])

    print(f"Protected {len(protected_indices)} caption paragraphs.")

    # Define Mapping Rule: Old X -> New (X+7)
    # Range 1 to 8.
    mapping = {x: x + 7 for x in range(1, 9)} # 1..8

    # Apply replacements in REVERSE order of keys (8 down to 1)
    # to avoid chaining issues (e.g. 1->8, then 8->15) if we had overlaps in the source side.
    # Here, 8->15. 1->8.
    # If we replace 1->8 first, "Figure 1" becomes "Figure 8".
    # Then if we replace 8->15, that "Figure 8" becomes "Figure 15". WRONG.
    # So we MUST replace 8->15 FIRST. Then 1->8.

    sorted_keys = sorted(mapping.keys(), reverse=True)

    count = 0
    for i, p in enumerate(doc.paragraphs):
        if i in protected_indices:
```

```
            continue

        # Optimization: Skip if no "Figure" in text
        if "Figure" not in p.text:
            continue

        original_text = p.text
        new_text = original_text

        for old_num in sorted_keys:
            new_num = mapping[old_num]
            # Replace complete word "Figure X"
            pattern = r"\bFigure\s+" + str(old_num) + r"\b"
            if re.search(pattern, new_text):
                 replacement = f"Figure {new_num}"
                 new_text = re.sub(pattern, replacement, new_text)

        if new_text != original_text:
            p.text = new_text
            count += 1
            print(f"Updated Para {i}: ...{original_text[:40]}... -> ...{new_text[:40]}...")

    doc.save(output_path)
    print(f"Updated {count} text references. Saved to {output_path}")

if __name__ == "__main__":
    fix_references("Empirical Analysis of Accuracy.docx", "Empirical Analysis of Accuracy.docx")
```